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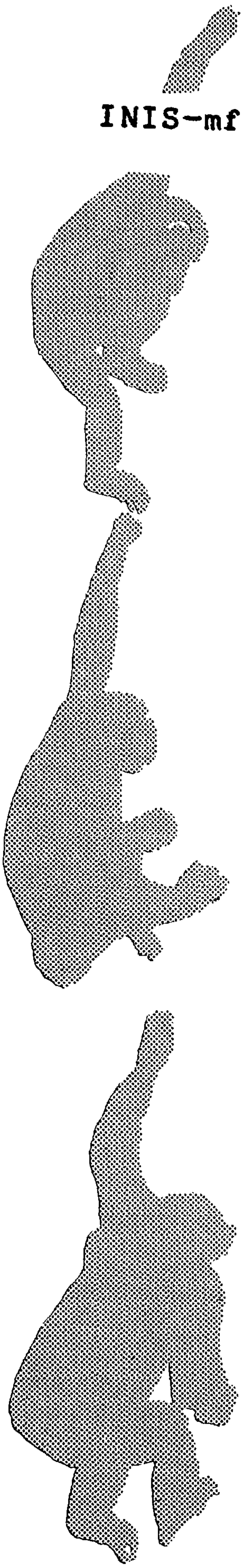

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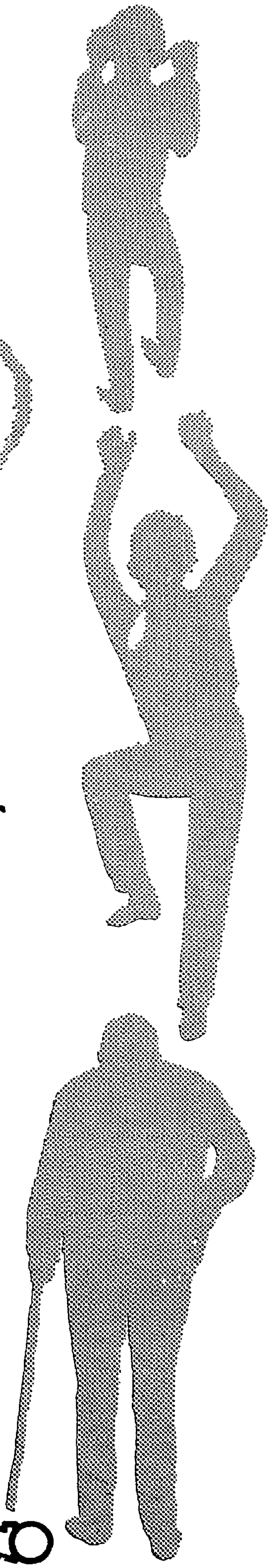

DIVISION FOR HEALTH RESEARCH TNO

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MANAGEMENT IN BIOMEDICAL RESEARCH

Management has its origin in business where the primary objective is to make profits. The great majority of business is engaged in the manufacture and sale of products or in the provision of services for a fee. When a larger number of people becomes involved in a certain business, the level of its originality tends to decline, which introduces competition. In the face of competition, the demand for effectivity and efficiency increases, because this will ultimately determine the margin of profits of the business. Since biomedical research has expanded from the enclosure of the single scientist's laboratory room, and complicated equipment requiring teams of operators has replaced the microscope and the Warburg manometer, the management of research groups has become a matter of concern to the people who are responsible for the financing of these activities. When reading the statements of leading authorities in the art (or science) of management there can be no doubt that the requirements for adequate management are not fundamentally different in research as compared to business. "In a time of unpredecented change and intensified competition for physical and human resources on all levels, there is a reason to ask how we can energize dormant talent, control without stifling, resolve conflicts that can frustate the ablest of men, inspire personal commitment to constructive lines of action, and supply a managerial leadership worthy of the challenges facing all forms of organized enterprise today". (E.M. Fox and L. Urwick: Introduction to second edition of Dynamic Administration, the collected papers of Mary Parker Follett, Hippocrene Books Inc., New York, 1973). The conflicts which have emerged between scientists and administrators about research management are obviously not due to a disagreement over the necessity of management, but solely concerned with the type of management which might be optimal for research activities. Is the type of leadership required of the kind provided by the highly original investigator who is also able to inspire and guide his collaborators into research activities of high quality, or of the kind provided by top administrators who are able to provide the controllers with detailed information on the amount of time and money spent by each member of the group on every project and who manage to complete all the required applications and reports in good time and order? Is it important to instruct each member of the group as to what achievements are expected from this person and when, with a detailed planning of his or her career in the organisation, or is it more effective to let the researchers develop their own opportunities and merely provide some guidance when the necessity arises? Is it effective to have scientists record the hours they spend on each of their various activities or is it better to leave them to decide for their own whether to work or dream and merely evaluate their achievements at appropriate intervals?

The answers to such important questions of management are directly related to the objectives of the research. If the objective is the highest number of publications in international journals, the management should select subjects for investigation that yield a maximum of publishable results. If the objective is money, the management should seek the maximum of funding for the cheapest possible activities. If the objective is original contributions which lead to progress in the field, the management will be at a loss, as innovative research can only be recognized after it has been carried out and because there is no formula available for the planning of innovations.

In the biomedical sciences the main function of research has to do with problem solving. As most of the problems which might be solved quickly and easily have been solved by now, we are left with the problems that require much time and effort. Perhaps the most important function of the management then is to establish conditions which allow researchers to concentrate for a period of several years on their subject; to promote stability while preventing inflexibility; to provide maximal challenge without introducing the sense of insecurity and to prepare for changes in a setting of continuity. The directors of the REP-Institutes have attempted to increase the efficiency of their research by adopting a joint management of their technical, biotechnical and administrative services, while keeping the direction of the research programs of the individual institutes separate. This system of management has been rewarding in that it saves time and money in the non-research parts of the operations. The system has also strongly stimulated the interaction and collaboration between the researchers from the different institutes.

Clearly, the successful execution of such a policy depends much on the personal commitment of each of the directors. In 1982 the director of the Primate Center, Prof.Dr. H. Balner, had to retire from his position to take up his responsibilities with the Commission on Biology, Radioprotection and Medical Research at the Headquarters of the European Communities in Brussels. Hans Balner has been at Rijswijk since 1961; first with the Radiobiological

Left: Prof.Dr. H. Balner, director of the Primate Center from 1972 to 1982. Right: Dr. A.A. van Es, the present director.

Institute and since 1972 as Director of the Primate Center, which he helped to establish and which grew into its present size and scientific status under his leadership. Hans Balner managed to pursue his own research interest to unravel the MHC of monkeys and chimpanzees along with his many responsibilities as a director. In 1980 he was appointed to a newly created chair at the Medical Faculty of Leiden University to teach on the subject of immunogenetics. His merits for primate research in the Netherlands were reviewed masterly by Prof.Dr. J.J. van Rood from Leiden University in a lecture entitled "The significance of the Primate Center TNO for the Health Care" at a symposium on the occasion of Prof.Dr. H. Balner's retirement from TNO. Copies of this lecture can be obtained from the REP-office upon request. On that occasion the appointment was announced of Dr. A.A. van Es as the new director of the Primate Center. Arthur van Es carried out research on kidney transplantation in Rhesus monkeys between 1971 and 1978 at the Primate Center under Balner and left the Center for surgical training in Maastricht and Rotterdam. All of us are delighted to welcome the new director of the Primate Center, whose intricate knowledge of the Center and its research areas adds confidence in the future of the REP-Institutes. The management of the Primate Center is further strengthened by the appointment of Dr. Huub Schellekens as deputy director. Dr. Schellekens is a medical microbiologist and engaged in interferon and hepatitis research at the Primate Center.

Great demands on managerial capacities are also made when a large international congress has to be organised. It was a great challenge for the Primate Center to have the Second International TNO Meeting on the Biology of the Interferon System, April 18-22, 1983 in Rotterdam with Dr. Schellekens as Scientific Secretary. Also in 1983 the Netherlands will host the 7th International Congress of Radiation Research, July 1983 in Amsterdam with an expected 1500 members. Dr. Johan Broerse of the Radiobiological Institute has been appointed Secretary-General of the Congress. The staff of the Radiobiological Institute is looking forward to welcome many of their colleagues in July in Amsterdam or on July 5 on the occasion of the organised visit for congress-participants to the institute in Rijswijk.

D.W. van Bekkum

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RADIOBIOLOGY

RBE OF NEUTRONS FOR INDUCTION OF CELL REPRODUCTIVE DEATH AND CHROMOSOME ABERRATIONS IN THREE CELL LINES

J. Zoetelief, W.C. Kuijpers, A. Baten-Wittwer and G.W. Barendsen

The relations among different types of cellular effects caused by ionizing radiation, i.e., cell reproductive death, chromosomal aberrations and malignant transformation are insufficiently known. The significance of studies on the induction of cell reproductive death and chromosome aberrations is based in part on the suggestion that the chromosome aberrations represent a major cause of cell reproductive death. Similarities in the dose-effect relations for induction of these types of damage led several investigators (e.g., Lloyd et al., 1976) to the conclusion that all impairment of clonogenic capacity is directly related to gross chromosome aberrations (di- and polycentrics, centric rings and acentric fragments).

It has been suggested that chromosome aberrations are important in relation to cancer induction. It is commonly assumed that cells containing a dicentric or a centric ring are not viable and will therefore not play an important role in carcinogenesis. However, their symmetrical counterparts, reciprocal translocations and pericentric inversions, are not expected to induce cell reproductive death and it is not excluded that these types of chromosome aberrations play a part in tumour induction. The frequencies for induction of pericentric inversions and reciprocal translocations are expected to be equal to those for centric rings and dicentrics. It seems worthwhile therefore to compare the RBE values for induction of dicentrics and centric rings with those for cell inactivation and with the mean or effective quality factors (Q) recommended for radiation protection.

The induction of cell reproductive death and chromosome aberrations (dicentrics and centric rings) has been investigated in plateau phase cultures of established lines of a rat rhabdomyosarcoma (R-1, M), a rat ureter carcinoma (RUC-2) and Chinese hamster (V-79) cells (containing fractions of cells in the G_1 phase of about 0.75, 0.8 and 0.7, respectively) for single doses of 300 kV X rays and 0.5, 4.2 and 15 MeV neutrons. The studies were restric-

Table I

RBE OF NEUTRONS AS A FUNCTION OF DOSE OF 300 kV X-RAYS AVERAGE FOR INDUCTION OF CELL INACTIVATION AND CHROMOSOME DAMAGE*

	dose of X-rays (Gy)	15 MeV neutrons		4.2 MeV neutrons		0.5 MeV neutrons	
R-1, M cells	0.001	2.5	4.7 / 1.2	2.9	6.1 / 1.2	7.8	13 / 4.4
	1	2.3	4.3 / 1.2	2.6	5.5 / 1.2	7.1	12 / 4.4
	2	2.1	3.9 / 1.2	2.4	5.0 / 1.2	6.5	11 / 4.4
	5	1.7	3.1 / 1.2	1.9	4.0 / 1.2	5.2	8.8 / 4.1
RUC-2 cells	0.001	6.5	47 / 3.1	6.0	50 / 2.7	12	110 / 4.4
	1	5.1	13 / 3.7	4.7	10 / 3.4	10	32 / 5.8
	2	4.2	8.3 / 3.1	3.9	7.1 / 2.9	8.4	21 / 4.9
	5	2.8	4.1 / 2.2	2.6	4.0 / 2.0	5.6	10 / 3.4
V-79 cells	0.001	4.9	14 / 2.5	6.7	25 / 3.2	12	38 / 6.3
	1	3.9	8.6 / 2.1	5.3	15 / 2.5	9.2	23 / 4.6
	2	3.3	6.8 / 2.1	4.4	10 / 2.4	7.9	16 / 4.5
	5	2.3	4.2 / 1.5	3.0	5.5 / 2.0	5.5	9.5 / 3.9

*The results presented are the means with maximum and minimum values corresponding to the standard deviations.

ted to chromosome-type aberrations, which are mainly produced in the G_0 and G_1 phases of the cell cycle. This endpoint is of importance, since the cell population at risk in humans is not rapidly proliferating, i.e., most cells will be in the G_0 or prolonged G_1 phase of the cell cycle.

The different cell lines show considerable variations in sensitivity for induction of cell inactivation as well as for induction of chromosome aberrations. With regard to cell inactivation, R-1,M cells show the greatest sensitivity to all types of radiation; however, for induction of chromosome aberrations, these cells are most sensitive to the photon irradiation but not to neutrons at all energies. The fraction of cells showing cell reproductive death is higher than that showing dicentrics and centric rings by factors of about 7, 3.5 and 3 for R-1,M, V-79 and RUC-2 cells, respectively. Assuming that the number of dicentrics and centric rings represent only half the number of gross chromosome aberrations, a direct correlation between fractions of cells which have lost their clonogenic capacity and those showing gross chromosome aberrations cannot be derived; other types of damage must play important roles in the induction frequency of cell reproductive death.

The RBE values for neutrons obtained for both biological endpoints are presented in Table I for the different cell lines employed. The results for levels of effect corresponding to different doses of 300 kV X rays are given. Corrections for the contributions of the photon components of the neutron beams are not included in this table. It can be seen from the table that RBE decreases with increasing dose of X rays. No significant differences are observed between the RBE values for the different biological endpoints. The largest RBE values (average for cell death and chromosome aberrations) are found for 0.5 MeV neutrons, which show values of about 8, 12 and 12 at the low dose levels for R-1,M, RUC-2 and V-79 cells, respectively. These RBE values are in reasonable agreement with the Q value of about 10. For 4.2 MeV neutrons, the mean RBE values at low dose levels are about 3, 6 and 7 for R-1,M, RUC-2 and V-79 cells, respectively. However, in the experiments with such neutrons, the relative photon contribution to the total dose was rather high (26 per cent). Correction for the photon contribution will increase the RBE values to about 4, 8 and 9 for 4.2 MeV neutrons alone; the relevant Q value is about 8. The mean RBE values at low doses for 15 MeV neutrons are about 2.5, 6 and 5 for R-1,M, RUC-2 and V-79 cells, respectively, and these values must be compared with Q values of about 7.

It can be concluded that the mean RBE values for R-1,M cells are lower than those for the relatively resistant RUC-2 and V-79 cells, which are similar. The RBE values for the latter lines extrapolated to levels corresponding to low doses of X rays are in good agreement with the quoted Q values. However, due to the large uncertainties in the RBE values, especially at the low dose levels, a variation by a factor of 2 to 3 cannot be excluded.

Lloyd, D.C. et al., Int. J. Radiat. Biol. 29 (1976) 169.

COMPARISON OF THE EFFECTIVENESS OF DIFFERENT RADIATIONS FOR THE INDUCTION OF REPRODUCTIVE DEATH, CHROMOSOME ABERRATIONS, MORPHOLOGICAL TRANSFORMATIONS AND SPECIFIC MUTATIONS IN CULTURED MAMMALIAN CELLS

G.W. Barendsen

Ionizing radiations can induce a variety of changes in cultured mammalian cells, many of which are initiated by damage to the chromosomes. Important examples are cell reproductive death, chromosome structural changes, morphological transformation and specific gene mutations. Several types of lesions have been demonstrated in the DNA of irradiated cells: single strand breaks, double strand breaks, base damage and cross linking of DNA to DNA or to other macromolecules. All of these changes might have observable cellular effects, but the effectiveness per unit dose is likely to depend on the type of lesion assessed and the region of DNA involved.

If the primary mechanisms of damage at the molecular level are similar, it can be expected that dose-effect relationships for the different cellular responses should exhibit common characteristics. A comparison of dose-effect relationships has been made for published data on several types of cells treated with radiations of different Linear Energy Transfer (LET) and assessed with respect to two or more endpoints (Table 1). This analysis was based on a few general assumptions concerning biophysical mechanisms. 1) If single events can cause sufficient primary damage for the induction of a given type of cellular effect, then the fraction of cells showing a response is expected to increase as a linear function of the dose. This applies to low LET radiation and even more to high LET radiation. 2) In addition damage may also be caused by interaction of sublesions or accumulation of subeffective damage. This is consistent with the assumption that the frequency of many types of cellular effects can be described to a first approximation by:

$$F\ (D) = a_1\ D + a_2\ D^2 \qquad (1)$$

in which a_1 and a_2 are constants with values depending on the effect studied, on the cell type investigated and on the conditions of exposure to a dose D (Barendsen, 1979).

The data on lethal lesions were derived from cell survival curves described by:

$$S(D)/S(O) = \exp - (a_1\ D + a_2\ D^2) \qquad (2)$$

in which a_1 and a_2 have the same meaning as in (1) (Barendsen, 1979; Gilbert et al., 1980).

Comparison of the different dose-effect relationships for lethal lesions shows that various types of cells have different sensitivities to low LET as well as to high LET radiation. Values of a_1 for cell reproductive death have been shown to range from about 10^{-1} to $1\ Gy^{-1}$ in the case of photons and between about 1 and $3\ Gy^{-1}$ in the case of the most effective fast neutrons (mean energy 0.4 to 1.0 MeV) (Barendsen, 1979). Values of a_2 for low LET radiation also vary considerably (range of about 10^{-2} to $10^{-1}\ Gy^{-2}$). The a_1/a_2 values (for both types of radiation) vary somewhat less, with a range of 2 to 10 Gy (Barendsen, 1982).

Results of many types of experiments suggest that reproductive death induced by ionizing radiation is due mainly to structural chromosome abnormalities, although not all of these aberrations are readily detectable. Relative to the incidence of reproductive death, the commonly scored chromosome aberrations, dicentrics and centric rings, are induced with a frequency which is lower by a factor ranging from 2 to about 10. It is evident that dicentrics and centric rings constitute only a proportion of all aberration types. Deletions are also potentially lethal, but the minimal size needed for lethality is not known.

Data on morphological transformation of cells in culture by different radiations are available for only two cell types. The values for a_1 show that this effect is much less frequently induced than reproductive death or chromosome structural abnormalities i.e., by a factor of about 50 less for Hamster embryo cells and by one of about 10^3 less for C3H 10T1/2 cells. The results on mutations, although limited for high LET radiations, show a_1 values which are a factor of 10^4 to 10^5 smaller than for reproductive death. In the UNSCEAR 1982 report, an average value of 4×10^{-5} mutations per lethal event is mentioned, which is similar to that which can be deduced from the data in Table I. With ionizing radiation, specific mutations are less frequently induced than morphological cell transformation by a factor of 30 to 1000.

With respect to the ratio a_1/a_2, which represents the dose at which the quadratic term contributes equally to the total frequency as does the linear term, no general differences can be deduced between the different types of effect.

That various cellular effects are induced at different frequencies per unit dose has important implications with respect to the mechanisms involved. Cell reproductive death and chromosome aberrations can presumably be induced as a result of damage in any one of the chromosomes. Chromosome breaks leading to deletions may occur at many sites. The probability of breaks may not be uniform along chromosomes, but this is difficult to establish because not all aberrations are observable with the presently available techniques. Compared to lethal events, specific gene mutations, for

Table I

PARAMETERS OF DOSE-EFFECT RELATIONSHIPS FOR THE INDUCTION OF CHANGES IN CULTURED MAMMALIAN CELLS BY DIFFERENT TYPES OF IONIZING RADIATION

Frequency $F = a_1 D + a_2 D^2$

Lethal lesions	a_1 (Gy^{-1})	a_2 (Gy^{-2})	a_1/a_2 (Gy)	Ref.
Cell type	($\times 10^{-1}$)	($\times 10^{-2}$)		
CH-V79				
photons*	1.5	2	7	1,2
alpha-part.**	10	1	10	1
neutrons	14	-	-	2
C3H 10T1/2				
photons	1.4	3.6	4	3,4,5
neutrons	7	1.1	6	3,6
human ly.				
photons	6	15	4	7
neutrons	20	-	-	7
hamster embryo				
photons	2	3	7	4
neutrons	25	-	-	4

Chromosome aberrations***	a_1 (Gy^{-1})	a_2 (Gy^{-2})	a_1/a_2 (Gy)	Ref.
Cell type	($\times 10^{-1}$)	($\times 10^{-2}$)		
CH-V79				
photons	0.3	0.6	5	2
neutrons	3	-	-	2
C3H 10T1/2				
photons	2	7	3	9
human ly.				
photons	0.4	4	1.0	8
neutrons	8	-	-	8

Transformation	a_1	a_2	a_1/a_2	Ref.
Cell type	($\times 10^{-4}$)	($\times 10^{-4}$)		
C3H 10T1/2				
photons	0.8	1	0.8	3,4
neutrons	20	-	-	6
	8	5	1.6	3
	6	-	-	5
hamster embryo				
photons	40	40	1.0	4
neutrons	500	-	-	4

Mutations¤	a_1	a_2	a_1/a_2	Ref.
Cell type	($\times 10^{-6}$)	($\times 10^{-6}$)		
CH-V79				
photons	3.5	0.9	4	1
alpha-part.	60	40	1.5	1

* No distinction is made between X rays and gamma rays, although differences up to a factor of 2 have been demonstrated for some effects at low doses.

** When no data on neutrons were available, data on alpha particles with LET values of 50 to 100 keV/μm in tissue were analysed.

***Dicentrics + centric rings for Chinese Hamster cells and human lymphocytes, deletions for C3H 10T1/2 cells.

¤ Resistance to 6-thioguanine (6-TG^R).

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which changes at a specific site are required are less frequently induced by radiation by a factor of 10^4 to 10^5.

Because cell transformation is more frequently (30 to 1000 times) induced by ionizing radiations than specific gene mutations, it may be inferred that many, if not all, chromosomes contain one or more sites with genes which, if damaged, deleted, or transposed to another site, may cause morphological malignant transformation. A hypothesis which can account for many features of the induction and development of malignant tumours may be based on the assumption that the primary chromosomal changes induced by radiation do not affect transforming genes (oncogenes) directly, but that the primary changes occur at other sites or in other genes which are present at multiple sites on all chromosomes. These primary changes are likely to be associated with functions which are necessarily performed by all chromosomes. A notable representative of such a function is the replication process of chromosomes during DNA synthesis, which is under control of several genes present on each chromosome. If one of these genes is damaged or changed in its expression, then during subsequent cell cycles an instability or fragility of chromosomes could develop, which may result in gene transposition. These secondary steps in turn could lead to the derepression or activation of oncogenes; in addition, the development of further mutations may be promoted in later stages. This hypothesis might also provide an explanation for the observation that the expression of cell transformation can be modified during subsequent cell generations. Transformation does not need to be the uniquely determined consequence of the primary induced chromosome fragility, however (Strauss, 1981).

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SKIN REACTIONS OF RAT FEET EXPOSED TO MULTIPLE FRACTIONS OF X RAYS PER DAY

E. van Rongen

Acute reactions of rodent foot skin have been extensively used as a model of normal tissue reactions to radiation treatment. Although skin reactions are no longer dose-limiting in radiotherapy because of the use of high energy X or gamma rays, the skin can be used as a model for an acutely responding, fast regenerating normal tissue, e.g., to compare various fractionation schemes.

The right hind feet of female WAG/Rij rats were irradiated with multiple fractions per day (4 x 2 Gy daily), with daily fractions of 3 Gy and with single doses of 300 kV X-rays. The rats were anaesthetized with an Ethrane/O_2 mixture. Acute reactions were scored 5 times per week until the reactions disappeared. A refined modification of the scoring system devised by Kal and Gaiser (1977) was used. In this system, a score of 0.07 is given for dry desquamation, 0.1 for one small area of moist desquamation and 0.9 for complete moist desquamation. Scores between 0.1 and 0.9 are equivalent with the extent of the moist desquamation. Mean scores were derived from the 7-day period encompassing the peak reaction.

The mean scores for the acute reactions are presented in Fig. 1. Tolerance doses, defined as the total doses resulting in one small area of moist desquamation (corresponding to a score of 0.1) in 50% of the animals, as calculated by probit analysis are 55.7 $\pm$ 0.7 Gy for the 4 x 2 Gy per day scheme and 67.3 $\pm$ 2.9 Gy for the 3 Gy per day scheme. It can be seen from Fig. 1 that, when increasing the total dose above the tolerance dose, the mean reaction to the 3 Gy per day regimen reaches a plateau (corresponding to 10 to 40% moist desquamation of the sole of the foot), while the reaction to the 4 x 2 Gy per day regimen increases to the maximum, i.e., moist desquamation of the entire foot. These differences can be attributed to differences in repair of sublethal damage and repopulation.

Repair of sublethal damage in mouse feet is exponential (Douglas, 1982) and, for doses of 3.4 Gy, it is almost complete in 6 hours (Douglas and Fowler, 1976). Hence, it can be assumed that repair was complete in the 24-hour interval in our 3 Gy per day regimen. It was probably not complete in the 2-hour interval between subsequent fractions of 2 Gy given 4 times per day in a 6-hour period. Hence, an accumulation of this type of damage might occur.

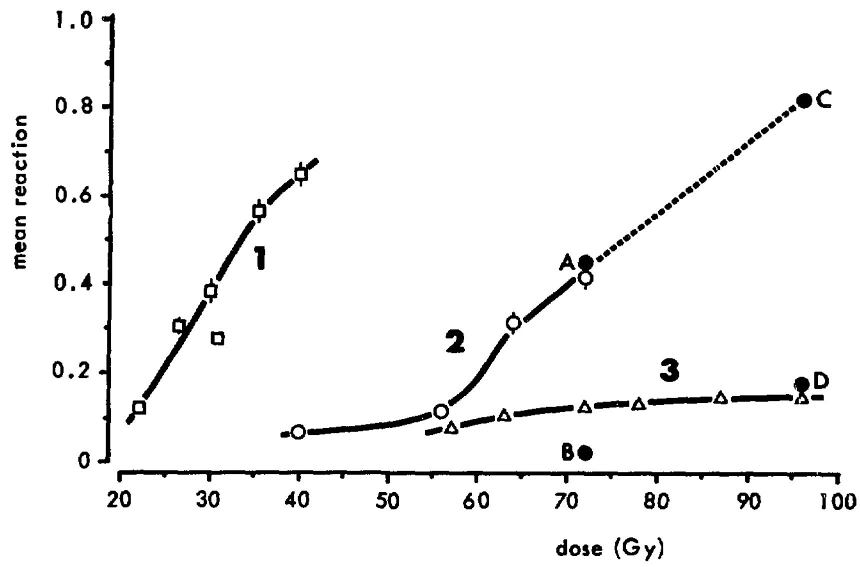

Figure 1: Mean foot skin reactions calculated over the 7-day period encompassing the peak reaction.
Curve 1: acute single doses.
Curve 2: 4 x 2 Gy daily.
Curve 3: 3 Gy daily.
Point A: 5 x (4x2 Gy), 9-day interval, 4 x (4x2 Gy).
Point B: 5 x (4x2 Gy), 16-day interval, 4 x (4x2 Gy).
Point C: 6 x (4x2 Gy), 9-day interval, 6 x (4x2 Gy).
Point D: 6 x (4x2 Gy), 16-day interval, 6 x (4x2 Gy).
All regimens were started on Monday. No fractions were given during weekends.

Repopulation during fractionated irradiations in mouse skin is of minor importance during the first 2 weeks after the start of the irradiation (Denekamp, 1973; Douglas and Fowler, 1976). After this period, it becomes increasingly important. In the present rat study, it can be seen from the 3 Gy daily curve in Fig. 1 that, after about 5 weeks of irradiation (total dose ca. 75 Gy), each additional daily dose of 3 Gy is completely compensated for by repair and repopulation. In the 4 x 2 Gy daily regimen, repopulation is also evident when an interval is introduced about halfway during the treatment course. A total dose of 72 Gy administered in 17 days in which an interval of 9 days between the first 5 and the last 4 days of treatment was introduced (point A in Fig. 1) did not result in a lower mean reaction as compared with the same total dose administered in a period of 11 days with an interval of only 2 days (the 72 Gy point of curve 2). However, the same treatment with an interval of 16 days resulted in a great reduction in the mean reaction (point B), demonstrating that, following a total dose of 40 Gy, repopulation becomes significant from about 2 weeks after the start of the treatment. In another experiment with the same 4 x 2 Gy daily regimen, it was demonstrated that, following a total dose of 48 Gy, repopulation is not important up to 2 to 3 weeks after the first dose (point C in Fig. 1). After 3 to 4 weeks following the first dose, repopulation becomes significant (point D).

The conclusion is that, with a multiple fractions per day regimen, one series of fractions up to the tolerance dose can be given to the skin; then, an interval of about 3 weeks must be introduced, after which another series of fractions up to the tolerance dose can be given without causing excessive skin damage. So, the overall treatment time is about the same as in a conventional fractionation regimen (6 to 7 weeks). The number of treatment days, however, is reduced.

Similar findings were reported recently for clinical pilot studies using multiple fractions per day in the treatment of prostatic and bladder cancer (Ang and Van der Schueren, 1982), gliomas (Ang et al., 1982) and head and neck tumours (Van den Bogaert et al., 1982). From these studies, it was concluded that these treatment regimens might offer a therapeutic gain over conventional treatments.

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Bogaert, W. van den et al., Int. J. Radiation Oncology Biol. Phys. 8 (1982) 1649.

ABSORBED DOSES DUE TO MAMMOGRAPHY IN VARIOUS DUTCH HOSPITALS

C. Zuur, J. Zoetelief, A.G. Visser* and J.J. Broerse

Because of the relatively high incidence of mammary cancer in The Netherlands (5000 new cases per 7 million women per year), it is presently being considered to implement screening programs for breast cancer. In this connection, it is of interest to determine the absolute dose and dose distributions in functional mammography installations. It must be considered, however, that in most of the hospitals in The Netherlands mammography is applied only upon medical indication. The mammography procedures in twelve hospitals have been compared with respect to the dose and the dose distribution in an acrylic plastic phantom (10.2 x 10.2 x 4.9 cm^3) simulating the breast. Dose determinations were made at 4.95, 24.5 and 44.05 mm depth with a Baldwin-Farmer ionisation chamber (BF-IC) connected to a Keithley 616 digital electrometer and with thermoluminescent dosimeters (TLD). The TLD were supplied and read out by the Radiological Service TNO, Arnhem, The Netherlands. The BF-IC was especially calibrated for this low energy range by Hofmeester et al. from the National Institute for Health Care, Bilthoven, The Netherlands. The measurements were made under conditions similar to those in routine mammography using the automatic phototimers. The diagnosticimage quality at the mammographic units was investigated by the use of a RMI Mammographic Random Phantom (Zuur et al., 1982). A score of "3.6" can be considered as a criterion for a "good" radiograph and a score of "3" as "reasonable". The radiographs of the test phantom were examined independently by four radiologists.

The doses received per mammoradiograph as derived from the ionisation chamber differed greatly among the different hospitals (Table I): between 2 and 21 mGy for the entrance dose, 0.1 and 0.3 mGy for the exit dose and 0.8 and 4 mGy for the mean tissue dose. The mean absorbed dose in the breast per investigation varies from 2 to 9 mGy. Information on the image quality is also given in Table I. The score of image quality should expected to be positively correlated with the mean absorbed dose, but no correlation was found. This implies that a significant reduction could be achieved at several hospitals without loss of image quality.

The variations in mean absorbed dose at the same mammographic installation due to the use of a grid and changes in the photo-

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Table I

ABSORBED DOSES PER RADIOGRAPH, MEAN ABSORBED DOSES PER INVESTIGATION AND NUMBER OF WOMEN INVESTIGATED PER YEAR AT DIFFERENT MAMMOGRAPHY INSTALLATIONS

code number hospital	entrance dose (mGy)	mid dose (mGy)	exit dose (mGy)	mean dose (mGy)	score image quality	mean dose in one breast* (mGy)	number of women
1	8.9	1.0	0.14	1.9	2.8 ± 0.3	3.7	500
2a	5.9	0.7	0.12	1.4	2.8 + 0.2	2.8	1000
2b	5.6	0.8	0.15	1.4	3.0 + 0.4	2.9	
3	11.2	1.3	0.19	2.4	3.2 + 0.6	5.0	1750
4	4.8	0.7	0.11	1.2	3.1 + 0.3	2.6	110
5	5.7	0.7	0.13	1.3	2.5 ± 0.2	2.9	1000
6	21.1	1.9	0.23	4.2	3.0 ± 0.3	8.8	1300
7a	19.9	2.3	0.34	4.4	3.8 ± 0.3	9.2	1500
7b	8.6	1.0	0.15	1.9	2.7 ± 0.3	4.0	
8	10.2	1.3	0.21	2.3	3.0 ± 0.4	5.4	370
9a	8.0	0.9	0.14	1.8	3.1 ± 0.4	3.7	1000
9b	9.8	1.2	0.19	2.2		4.6	
10	3.6	1.2	0.44	1.5		3.0	10,000
11a	3.4	0.4	0.08	0.8	3.4 ± 0.4	1.8	
11b	4.0	0.5	0.09	0.9	3.3 ± 0.5	2.2	
11c	6.9	1.0	0.20	1.7		4.0	
11d	4.2	0.5	0.09	1.0		2.2	1850
12	3.9	0.5	0.07	0.9	3.6 + 0.3	2.0	
13a	4.9	0.5	0.08	1.1	3.5 + 0.3	2.5	
13b	6.9	0.8	0.12	1.5		3.5	
14a	4.7	0.5	0.07	1.0	3.6**	2.1	
14b	3.5	0.4	0.05	0.8	3.6**	1.5	2,500
14c	2.8	0.3	0.04	0.6	3.3**	1.3	

The dose values have an estimated uncertainty of ± 10%.

* Mean dose in one breast per investigation, including failures and the number of radiographs per investigation.

**Radiographs examined by two radiologists.

graphic material are given in Table II. The application of a grid causes an increase in the received dose by a factor of 2 to 3 unless the high voltage is increased. Employing a cassette instead of a vacuum envelope results in an about 30% higher mean absorbed dose. A 15% higher density in the radiographs results in a 25% higher absorbed dose. TLD were employed at about the same site and in the same time as the BF-IC.

A comparison of the two dosimetry methods revealed that the doses derived from TLD were considerable lower than those from the BF-IC. The mean ratios of the absorbed dose values (TLD/BF-IC) determined with the two dosimetric methods on the surface and at 4.95, 24.5 and 49 mm depth in the phantom were equal to 0.6, 0.85, 0.80 and 0.75, respectively. The reasons for these differences are still under investigation. The discrepancy in the measurements on the surface between the TLD and the BF-IC is of special importance and should be explored in detail because the dosimetry in mammography and in other diagnostic examinations is generally performed with TLD on the skin.

Table II

VARIATIONS IN RELATIVE MEAN ABSORBED DOSE VALUES UNDER DIFFERENT CONDITIONS FOR THE SAME MAMMOGRAPHY EQUIPMENT

Grid	
without grid	100
with a grid	233
with a grid + higher kVp (30 $\rightarrow$ 32 kVp)	172
with a grid + higher kVp (30 $\rightarrow$ 35 kVp)	100
Vacuum envelope/cassette	
vacuum envelope	100
cassette	138
Density	
normal density	100
ca. 15% more density	125
Film screen combination	
NMB-R	100
3M/alpha 2M	75
3M/-R	100*

*Thyssen, M.A.O., personal communication.

The absorbed dose values determined for the Dutch mammography installations are higher than those obtained by other groups. This might be due to an underestimation of the dose as measured by TLD but also the the use of different conversion factors from exposure in air (R) to the mean absorbed dose in the breast (rad). Most of the investigators in mammography dosimetry use a factor of 0.10 $rad.R^{-1}$ (Hammerstein et al., 1979). This value, however, is valid only under special circumstances. Our studies on the dose distribution in the phantom showed values of 0.16 to 0.23 $rad.R^{-1}$ for this conversion. It has to be concluded that the factor derived by Hammerstein et al. (1979) should not be rashly used to calculate the mean dose in mammography from the exposure in air or from the entrance dose.

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Zuur, C. et al., abstract 19.26. In: Proc. of the World Congress on Medical Physics and Biomedical Engineering 1982. 13th Int. Conf. on Medical and Biological Engineering, Sept. 5-11, 1982, Hamburg, BRD.

LATE EFFECTS OF RADIATION ON THE RAT LIVER

A.J. van der Kogel, D.L. Knook, H.A. Sissingh,
C.F.A. van Bezooijen and C. Zurcher

For many abdominal tumours, it is the relative radiosensitivity of the liver that limits administration of a therapeutically effective dose of radiation. Combined treatment with chemotherapeutic agents may increase the risk of development of late complications. Quantitative data on the factors determining the radiation tolerance of the liver as well as its modification by drugs are not available.

It is known from previous studies in rats (Jackson et al., 1977) that after partial liver irradiation, only high doses of radiation (50 Gy) resulted in impairment of hepatocyte function. A recent study in rats on the in vivo survival of hepatocytes (Jirtle et al., 1982) showed that the intrinsic radiosensitivity was less (larger D_0) as compared with other normal tissues, whereas the capacity for repair of sublethal damage was strongly reduced. If this observation is of relevance to the tolerance of the liver, it indicates that a relatively high single dose will be tolerated but that fractionated irradiation will not result in a significant increase in tolerance. In addition, clinical data suggest that the tolerance of the liver greatly depends on the volume irradiated. The aim of the present study is to develop a clinically relevant animal model in which the different aspects of liver tolerance to radiation and chemotherapeutic drugs can be evaluated.

In an initial series of experiments, the whole liver of 13 to 15-week-old male WAG/Rij rats was irradiated bilaterally with 300 kV X rays. Single doses in the range of 10 to 25 Gy were given. After irradiation the animals were weighed regularly and the first assays of liver function were performed after 6 to 12 months. This included the determination of the serum transaminases (SGOT and SGPT) and the retention of bromsulfophthalein (BSP).

Within 1 to 2 weeks after irradiation, a dose-dependent weight loss due to acute damage in stomach and duodenum was observed. This acute gastrointestinal damage and its sequelae was the major cause of death during the first 20 weeks after irradiation. This is also reflected in the survival curves (Fig. 1). Median survival decreased significantly from 90 weeks after a dose of 10 Gy to 62 weeks after 20 Gy and 34 weeks after 22.5 Gy.

Tests of liver function were performed most extensively in the groups irradiated with 15 and 20 Gy, while the 10 Gy group was

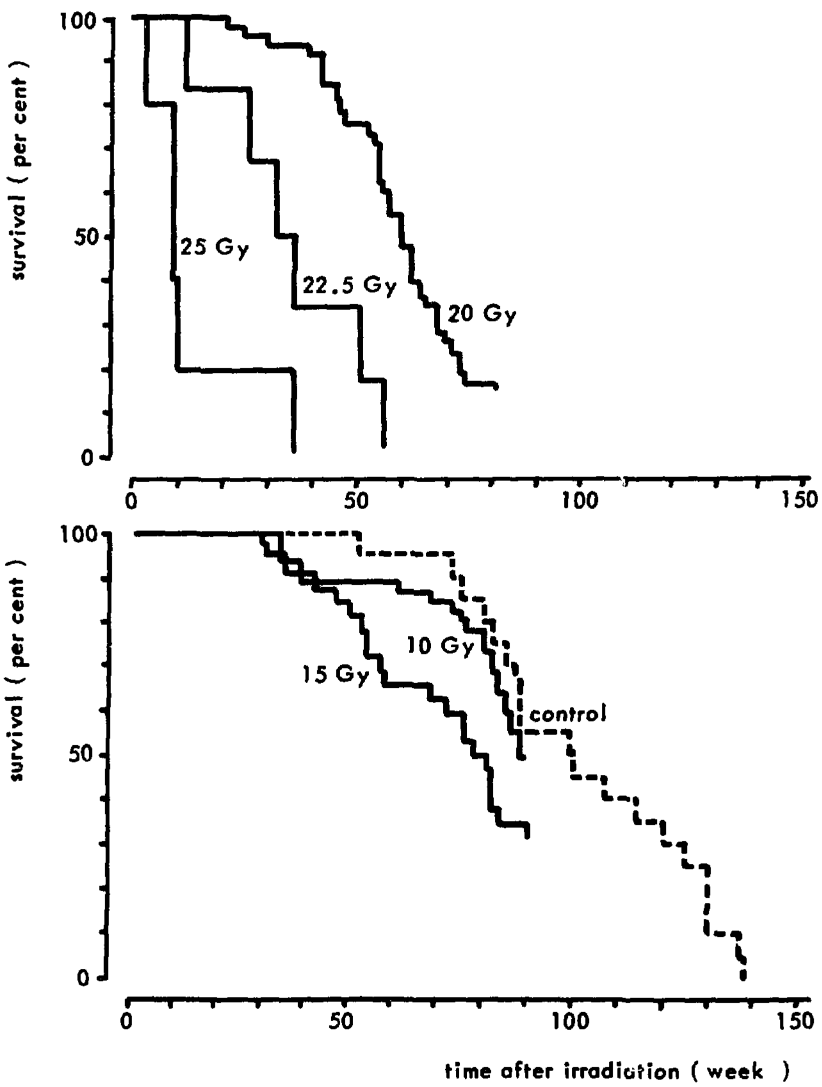

Figure 1: Survival of male WAG/Rij rats after whole liver irradiation with single doses of 300 kV X rays.

tested from 75 weeks after irradiation. BSP retention, which is an indication of the excretory function of the total hepatocyte population, was increased significantly in about 50% of the rats at 27 weeks after irradiation with 20 Gy; but, at longer intervals after irradiation, the percentage of animals showing abnormal values did not increase. In the rats irradiated with 10 or 15 Gy, increased BSP retention was observed only from 70 weeks after irradiation. A similar pattern was observed for the serum transaminases, with over 50% of the rats showing abnormal SGOT and SGPT values at 26 weeks after a dose of 20 Gy; after 15 Gy, abnorr..l functions were detected only after more than one year.

Histopathological examination of the liver and surrounding organs was carried out on all animals that died or were killed when seriously ill. Several sections were taken from all liver lobes and the incidence and severity of various lesions were scored as presented in Table I. A brief description of hepatic lesions is given elsewhere in this report (p. 29) in a study on the hepatic effects of low doses of total body irradiation and estrogens. Some lesions that do not occur or are infrequently observed in nonirradiated animals, e.g., necrosis, spongiosis, bile duct proliferation and portal vascular changes, show the existence of a dose-effect relationship. Especially the incidence of coagulative necrosis increases significantly with dose. The incidence of biliary cysts, bile duct proliferation and vascular lesions shows a maximum at 15 Gy, which might be related to the shorter survival times after 20 Gy. A lower incidence with higher dose is also seen for areas and foci of cellular change and regenerative nodular hyperplasia, lesions which also occur in untreated controls but at a lower incidence and at older ages than in the irradiated animals (Zurcher et al., 1982).

Table I

INCIDENCE (PER CENT) OF HEPATIC LESIONS IN MALE WAG/Rij RATS AFTER WHOLE LIVER IRRADIATION WITH 300 kV X RAYS

liver lesion	dose (Gy) 10		15		20	
	all	++	all	++	all	++
foci/areas	100	63	88	25	44	13
nodular regeneration	88	0	75	19	38	19
coagulative necrosis	13	0	50	25	75	25
spongiosis	50	13	50	19	25	6
biliary cysts	50	25	88	25	38	6
bile duct proliferation	13	0	50	19	38	1
sinusoidal dilatation	100	63	63	6	50	0
portal fibrosis	75	13	69	6	81	13
vascular lesions	25	0	94	44	69	38
no. rats	8		16		16	
age range (weeks)*	65-80		35-98		35-82	
mean age (weeks)*	74		62		51	

all: includes all animals with lesions.
++ : only animals with moderate to severe lesions.
*time after irradiation (at the age of 13 to 15 weeks).

For those lesions that show a clear-cut dose-effect relationship, it might be speculated that this is based on damage to specific cell populations. For example, a) coagulative necrosis might be an expression of late damage in hepatocytes or it may occur as a result of ischemia; b) bile duct proliferation may be a regenerative reaction to bile duct epithelial damage or to intra- or extrahepatic bile duct obstruction; c) spongiosis, which has been attributed to damage in fat-storing perisinusoidal cells (Bannasch et al., 1981). This lesion has an incidence of 50% even after a dose of 10 Gy. It is of interest to note that, in small groups of rats in which irradiation was preceded by administration of actinomycin-D, the incidence of spongiosis was 100% (6/6) after 10 Gy and 67% (4/6) after 15 Gy of X rays.

It can be concluded that irradiation of the rat liver with doses in the therapeutic range results in late functional and morphological damage. Specific histological lesions showed different time dose relationships. The correlation of these lesions with the impairment of specific liver functions as a result of late radiation or drug induced damage will be evaluated in further studies.

Bannasch, P. et al., Lab. Invest. 44 (1981) 252.

Jackson, K.L. et al., p. 23. In: Annual Report of REP-TNO (1977).

Jirtle, R.L. et al., Br. J. Radiol. 55 (1982) 847.

Zurcher, C. et al., p. 19. In: Liver and Aging (ed. K. Kitani) Elsevier Biomedical Press, Amsterdam (1982).

AMMONIA CONCENTRATION IN AN ANIMAL RADON EXPOSURE FACILITY

B. Hogeweg and J.S. Groen

Radon and thoron are produced by the decay of radium-222, -224 and -226 isotopes which belong to the natural radioactive elements of the actinium, thorium and uranium series. The presence of these natural radioactive elements in all mineral sources in combination with the noble gas characteristics of radon and thoron results in concentrations of these gases in outdoor as well as in indoor air. Because of this occurrence of radon and thoron in nature, man has always been exposed, mainly through inhalation, to their daughter products.

The application of industrial rest products with increased concentrations of naturally radioactive nuclides in building materials as well as energy conserving measures, such as decreasing the ventilation rate, can result in increased levels of radon and thoron concentrations in the indoor air. Consequently, these increased levels can result in an increased potential risk for lung tumour induction. For the quantification of this additional risk to the public, risk factors which are derived from epidemiological data of uranium miners are applied. The validity of the application of such risk factors to the public is questionable, since conditions, levels and exposure of these miners were greatly different from those of residents.

An experimental and dosimetric study on lung tumour induction by inhalation of radon daughters under indoor conditions has been started with the aim of analysing the various factors which determine the risk due to this inhalation. In the experimental part of the study, WAG/Rij rats will be exposed daily for a period of about 18 months in an exposure chamber to a relatively low concentration of radon daughter products. With regard to equilibrium conditions of the radon daughters and to aerosol distribution, conditions in the exposure chamber are comparable with the situation in indoor air.

As generally accepted, the potential risk of radon inhalation results from the deposition of inhaled aerosols to which one of the alpha particle emitting radon daughters is attached. The energy of the emitted alpha particles will be deposited locally with high density and, correspondingly, biological effects will be produced with relative high efficiency. The amount of energy which can be deposited depends on the ratio of activities of the alpha emitting isotopes Rn, RaA, RaB and RaC. It can be demonstrated that the required condition of indoor air corresponds to a ventilation rate of the exposure chamber of about 1 h^{-1}.

In chronic exposure chambers, ammonia is produced by the action of urease-positive bacteria in urine and faeces. There are reports showing that, in such exposure chambers, the ammonia concentrations will rise beyond acceptable levels. Since exposure of rats for 4 to 6 weeks to levels of 25 to 250 ppm of ammonia increased the severity of rhinitis, otitis media, tracheitis and pneumonia characteristic of murine respiratory mycoplasmosis (Broderson et al., 1976), it is evident that, for the study of lung tumour induction, the concentration of ammonia is of interest.

In order to study the ammonia concentration as a function of time in the exposure chamber, this concentration was measured at regular intervals. In addition, the influence of the bacteriostatic cage board impregnated with Neomycin and the addition of Aqualloy in the form of granules or foam on the ammonia concentra-

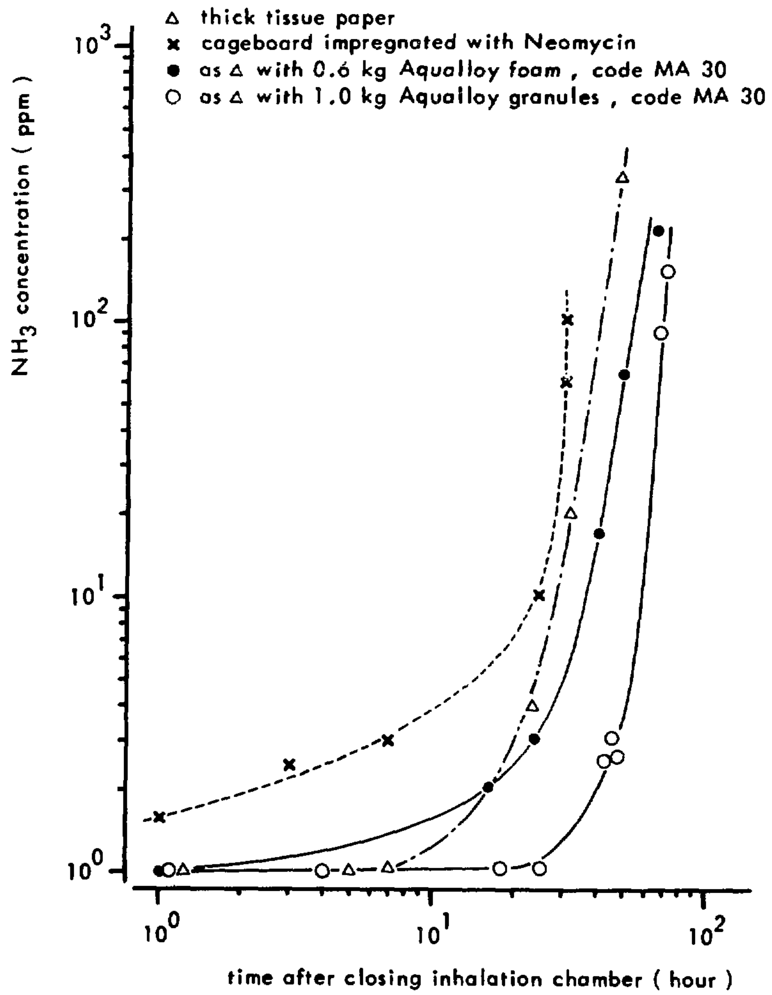

Figure 1: Ammonia concentration in the radon exposure chamber for different types of cage board additives.

tion was measured. Aqualloy is the trade mark of a synthetic hydrogel developed by KRI-TNO; this hydrogel is a mixture of styrene-maleicanhydride and polyvinylacetate and is reported to be an ammonia reducing additive (Van Bekkum et al., in press).

In the exposure chamber with a volume of 2.5 m^3, 192 WAG/Rij male rats (the maximum number for which the chamber was designed) were exposed to fresh air at a ventilation rate of the chamber of 1 h^{-1}. The ammonia concentration was measured by using Dräger tubes type Ammonia 5/a. The results of these measurements under the different conditions are presented in Fig. 1. This figure shows that, for the unprepared tissue paper, the bacteriostatic cage board and the Aqualloy foam (total amount 0.6 kg), the ammonia concentration gradually increases with time and that there is a steep rise in the concentration after about 30 to 40 hours. For Aqualloy granules (total amount 1.0 kg), the figure shows that the concentration remains constant at a level of about 1 ppm for a period of about 40 hours and rises steeply after that period.

The results demonstrate that, with the addition of Aqualloy granules, the ammonia concentration can be reduced over a long period of time to a level for which no cooperative action with the radon exposure is to be expected.

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Broderson, J.R. et al., Am. J. Pathol. 85 (1976) 115.

TUMOUR INDUCTION

MAMMARY CARCINOGENESIS IN DIFFERENT RAT STRAINS AFTER SINGLE AND FRACTIONATED IRRADIATIONS

J.J. Broerse, L.A. Hennen, M.J. van Zwieten and C.F. Hollander

A number of successive programs on radiation induced mammary cancer in the rat have been initiated at the REP Institutes TNO for single and fractionated irradiations with X rays and monoenergetic neutrons of three energies. The aims of the programs are the investigation of the nature of the dose-effect relationships, the determination of the relative biological effectiveness (RBE) of neutrons and the assessment of the effects of fractionated irradiation with low doses.

The results on mammary carcinogenesis after single dose irradiation have been presented earlier (Broerse et al., 1978; Van Bekkum et al., 1979); however, these reports dealt primarily with dose-effect relations for all mammary tumours regardless of type. The histological classification of the mammary tumours arising in the irradiated rats allows the separate investigation of the dose-effect relationships for the benign and the malignant neoplasms. The present contribution is concerned with the dose-effect relations for benign and malignant mammary neoplasms in selected groups of irradiated rats.

Details on the background and breeding history of the three rat strains, WAG/Rij, BN/BiRij and Sprague Dawley, used in these studies are provided elsewhere (Van Zwieten et al., 1982). The rats were irradiated with 0.5, 4 and 15 MeV neutrons produced with a double-belt van de Graaff accelerator through the p+T, d+D and d+T reactions, respectively. The doses delivered to the Sprague-Dawley rats (X rays: 0.10 to 2.0 Gy; neutrons: 0.03 to 0.55 Gy) were generally lower than those applied to WAG/Rij and BN/BiRij rats (X rays: 0.25 to 4.0 Gy; neutrons: 0.06 to 1.6 Gy) in view of the expected lower susceptibility of the latter two strains to mammary radiation carcinogenesis (Broerse et al., 1978; van Bekkum et al., 1979).

All animals were inspected weekly for the presence of mammary tumours. The animals were kept during their full lifespans and only killed when moribund. The tumour incidence data were based on histological confirmation of neoplasms palpated during life as well as those found at necropsy. The probability of animals surviving without evidence of tumour was computed according to a life table analysis. The proportion of animals surviving past time t_k without known tumour was recursively calculated from:

$$P(t_k) = P(t_{k-1}) \cdot \frac{n_k - \delta_k}{n_k} \qquad (1)$$

For this calculation, distinct observation times, t_k, are considered, $P(t_0) = 1$, n_k is the number of animals at risk during the time interval (t_{k-1}, t_k) and is the number of animals alive without diagnosed tumour at t_{k-1}. δ_k is the number of animals showing a first tumour in the interval (t_{k-1}, t_k).

The life table analysis provides an adequate means of correcting for intercurrent disease and loss of animals during the total observation period; however, the probability curves produced by this analysis inevitably show discontinuities in the course of time. It was anticipated that the probability curves would show a common shape and it was therefore decided to smoothen the actuarial survival data with Weibull functions:

$$P(t) = e^{-(\frac{t-\gamma}{\alpha})^{\beta}} \qquad (2)$$

where α is the time scale parameter, β is the shape parameter and γ is the location parameter.

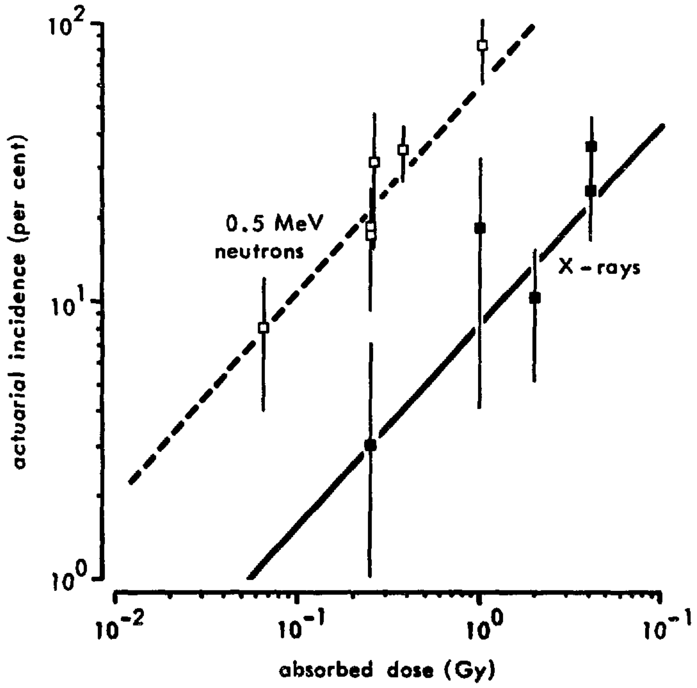

Figure 1: Actuarial incidence of mammary carcinomas in 95-week-old WAG/Rij rats after irradiation with X rays and 0.5 MeV neutrons.

The actuarial survival data have been used to derive dose-effect relationships by two connected procedures. First, the proportions of animals surviving without known tumour were connected at a time when none or a relatively small number of the nonirradiated control animals had developed tumours. Second, the curves of probability of survival without known tumour show a forward shift in time of appearance of the lesions with increasing dose.

The latency periods for mammary neoplasms in nonirradiated controls were generally in excess of 95 weeks. We have observed considerable differences in susceptibility for tumorigenesis among the WAG/Rij, Sprague-Dawley and BN/BiRij rat strains. Malignant tumours developed infrequently in the Sprague-Dawley and the BN/BiRij strains but were quite common in the WAG/Rij rats (Van Zwieten et al., 1982). The actuarial incidence, previously, designated as cumulative prevalence was calculated as 1 - P(t). Dose-effect relations for actuarial incidence at 95 weeks are shown in Fig. 1 for carcinomas in WAG/Rij rats. The error bars refer to standard deviations calculated as the square root of the variances derived from the life table analyses according to (1). It must be realized that the smoothing of the curves by Weibull fits (2) will reduce the error limits; however, the impact of this refinement cannot be quantified at present.

The dose-effect relationships were used to calculate the effectiveness of the different neutron beams relative to that of X rays for mammary tumorigenesis. The RBE values for 0.5 MeV neutrons are summarized in Table I. The actuarial incidence curves for induction of carcinomas after irradiation with X rays and 0.5 MeV neutrons are parallel, which implies a nearly constant RBE at the different dose levels.

Table I

RBE VALUES FOR MAMMARY TUMORIGENESIS IN INTACT FEMALE RATS

absorbed dose of 0.5 MeV neutrons (Gy)	WAG/Rij benign	WAG/Rij malignant	Sprague-Dawley benign	BN/BiRij benign
0.05	25	14	5	12
0.2	10	14	5	7

The RBE values calculated from the present studies are lower in general than those observed by other groups for neutron beams of comparable energy. The studies on the effects of fractionated versus single dose irradiation in WAG/Rij rats indicate that, for equal total absorbed dose, carcinomas appeared at approximately the same age. Additional studies are necessary to investigate whether this is due to a greater efficacy of fractionated irradiation or to a change in susceptibility for mammary carcinogenesis during the first year of life of the animals.

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PATHOLOGICAL CHANGES IN THE LIVERS OF IRRADIATED AND HORMONE TREATED FEMALE WAG/Rij RATS

M.J. van Zwieten, C. Zurcher and C.F. Hollander

The role of estrogens in the etiology of certain hepatic lesions in women remains uncertain. Oral contraceptive compounds have been linked epidemiologically with two main categories of liver lesions: 1) tumours and tumour-like lesions, including hepatocellular adenoma, carcinoma, focal nodular hyperplasia and nodular transformation; and 2) vascular changes including sinusoidal dilatation, peliosis hepatis and Budd-Chiari syndrome (hepatic vein thrombosis) (Ishak, 1981). Considering the widespread use of these hormonal preparations by women, experimental data clarifying the role of estrogens in liver disease are needed.

Recent studies in rats (Wanless and Medline, 1982; Yager and Yager, 1980) have suggested that estrogens may promote rather than initiate hepatic neoplasms and foci of cellular change. These investigators used diethylnitrosamine as the initiating agent. Our previous preliminary study in estrogen treated rats not exposed to an initiator (Nooteboom and Van Zwieten, 1981) tends to support that suggestion, since no effect of estrogen on the development of liver neoplasms and foci was found. Therefore, examination of liver sections from rats administered estrogen and exposed to whole body ionizing radiation was considered relevant to this question. Animal experiments employing irradiation as the carcinogenic agent in such studies are pertinent in view of the increasing awareness of the detrimental effects of exposure to ionizing radiation in our society from industrial and medical sources.

The rats used for this survey of liver lesions were part of long-term studies on mammary radiation carcinogenesis (Broerse et al., 1982; van Zwieten et al., 1982) and received single, whole-body doses of X-rays and 15 MeV neutrons in the range of 0.5 to 4.0 Gy. The rats were irradiated at 8 weeks of age and received 2 mg estradiol-17β subcutaneously in a paraffin cholesterol base at 6 to 7 weeks of age. All rats lived out their lifespans and were necropsied when found dead or when killed in extremis. A related study in rats on the hepatic effects of radiation doses in the range used in radiotherapy is presented elsewhere in this Annual Report (p. 17).

This histopathological survey was carried out on liver sections from 228 rats divided over 10 experimental groups (Table I). Generally, 2 or 3 sections of liver (range, 1 to 5) per rat were

Table I

INCIDENCE (PER CENT) OF HEPATIC LESIONS IN IRRADIATED AND ESTROGEN (E2)-TREATED FEMALE WAG/Rij RATS

	experimental groups									
	-E2					+E2				
liver lesion / rad. type	-	X	X	N	N	-	X	X	N	N
dose (Gy)	0	1.0	4.0	0.5	1.5	0	1.0	4.0	0.5	1.5
foci (acidoph/clear cell)	89	94	97	94	100	65	74	82	89*	90*
areas (acidoph/clear cell)	57	35	70	71	67	13	42*	59*	21	30
baso.foci/areas	51	53	79*	65	72	6	16	47*	16	55*
regen. nod. hyp.	14	12	21	35	44*	3	26*	12	5	10
neoplastic nod.	3	6	0	0	11	0	0	6	0	0
hepatocell. CA	0	0	0	0	6	0	0	0	0	0
spongiosis	14	6	27	18	22	3	5	18	0	0
sinusoidal dilat.	59	88*	52	71	89*	68	74	53	53	30
cystic sinusoid. dilat.	3	6	21*	12	11	3	5	24*	5	0
biliary cysts	14	18	45*	53*	50*	0	0	6	11	15
bile duct prolif.	5	6	21	0	11	0	5	0	0	10
portal fibrosis	8	18	24	24	22	0	11	24*	11	0
parench. fibrosis	0	6	15*	18*	11	0	5	24*	5	0
no. rats	37	17	33	17	18	31	19	17	19	20
mean age (mo)	28	23	22	24	21	18	19	14	15	14
range (mo)	21-36	11-30	14-29	16-28	15-30	13-25	15-21	5-20	8-18	9-19

Abbreviations: X, X-ray; N, neutrons (15 MeV); E2, Estradiol-17β.
*significantly different from controls ($p < 0.05$, one-tailed Fisher exact test).

available for study. The lesions tabulated in this survey included: a) foci and areas of hepatocellular alteration, which includes changes in tinctorial or morphological properties of hepatocytes which distinguish them from the surrounding normal liver cells [acidophilic, clear cell and basophilic types; the latter is considered likely to be a preneoplastic lesion (Squire and Levitt, 1975)]; b) regenerative nodular hyperplasia; c) neoplastic nodule (also termed "hepatocellular adenoma"); d) hepatocellular carcinoma; e) spongiosis hepatis [a cystic degenerative lesion associated with an accumulation of alcianophilic material in cysts, possibly resulting from a disturbance of the perisinusoidal liver cells (Bannasch et al., 1981)]; f) sinusoidal dilatation; g) cystic sinusoidal dilatation; h) biliary cysts; i) bile duct proliferation; j) portal fibrosis and k) parenchymal fibrosis (central-to-central, or central-to-portal zone fibrosis).

The results of this survey are given in Table I. As can be seen, most lesions occurred more frequently in irradiated rats than in nonirradiated controls, although in many cases the relative increase was small due to high corresponding values in the control animals. For a few lesions, for example, foci and areas (all types), regenerative nodular hyperplasia, spongiosis and biliary cysts, a more or less firm dose-related increase in incidence appeared to exist, especially in the neutron irradiated rats, but, in some instances, also in the X-irradiated rats. Also, with a few minor exceptions, estrogen administration did not appear to enhance the development of the lesions studied. It must be noted, however, that the mean ages of estrogen treated rats at death were considerably lower than in the corresponding groups without hormones. This was due, in general, to the common occurrence of other tumours, notably mammary neoplasms, in the estrogen treated animals. It is essential, therefore, to compare these preliminary data on liver lesions on the basis of age-specific incidences.

The relative extent and severity of liver involvement by these various lesions was compared among the groups and no obvious differences were evident. All lesions scored tended to be mild to moderate in severity and generally involved a minor portion of a representative cross section of a liver lobe. The few examples of severe or extensive involvement by a particular lesion, primarily biliary cysts and basophilic foci and areas, were found in the higher radiation dose groups.

In summary, it can be stated that data on the hepatic effects of estrogen combined with irradiation in the low to mid-range doses studied are inconclusive at present and require further analysis. In the non-estrogen-treated groups, most lesions showed an increase in frequency with increasing dose of both X rays and neutrons, although this was sometimes marginal due to the high background levels in control animals. These findings may suggest that the doses studied are at the threshold of being effective in terms of producing clear-cut increases in frequency of neoplastic and nonneoplastic liver lesions. Data from another study (p. 17) employing higher doses of X-irradiation indicate that this may be the case.

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INDUCTION OF LUNG TUMOURS IN THE RAT BY RADIATION EMITTED FROM IMPLANTED ^{192}Ir WIRES OR ^{125}I SEEDS

H.B. Kal, C. Zurcher and D.W. van Bekkum

In a previous report, the method of implantation of the radioisotope ^{192}Ir into the lungs of WAG/Rij rats and the results obtained were described (Kal et al., 1979). This method of implantation was also used for ^{125}I seeds (3M Company) with an activity of 0.4 mCi at the time of implantation. WAG/Rij rats were implanted at the age of 4 weeks. They were anaesthetized with ether. The skin of the thorax was shaved and an incision between two ribs of about 1 cm was made. A lobe of the lung was brought outside the thoracic cavity and the isotope was implanted by using a hollow needle of 10 cm with an inner diameter of 0.9 mm. After implantation, the lobe was replaced into the thoracic cavity and the skin was closed by two or three clamps. The animals were killed when moribund and a complete necropsy including histological examination of those tissues which showed abnormalities at gross examination was performed.

In the observation period of 15 months, 30 of the 40 animals implanted with ^{192}Ir wires developed tumours in or close to the thoracic cavity. Lung tumours were observed only in the implanted lung. Malignant hemangioendotheliomas occurred with a frequency of 50 per cent. In 12 lungs, a squamous cell carcinoma was found. Several other types of tumours were found. In 8 lungs, two types of tumours and in one three different types of tumours were identified. The frequency of tumours derived from bronchial epithelial tissue was 42.5 per cent. Four animals were lost due to massive bleeding in the implanted lung without signs of tumour development and 6 others due to other causes. The frequency of the tumours induced as a function of time after implantation is shown in Fig. 1A.

In the observation period of 18 months, only one of 20 animals implanted with a ^{125}I seed developed a lung tumour, a squamous cell carcinoma in the implanted lung (Fig. 1B) and two others with a tumour that could not be identified. Six animals were lost due to other causes, with no macroscopic signs of neoplasms in the lungs. Of these, one animal had a massive bleeding in the lung. Eleven animals were still alive at the end of this observation period.

Tumour fragments were transplanted subcutaneously into the flanks of syngeneic hosts for propagation of the tumours. Fragments of tumours obtained after at least 13 passages in the rat

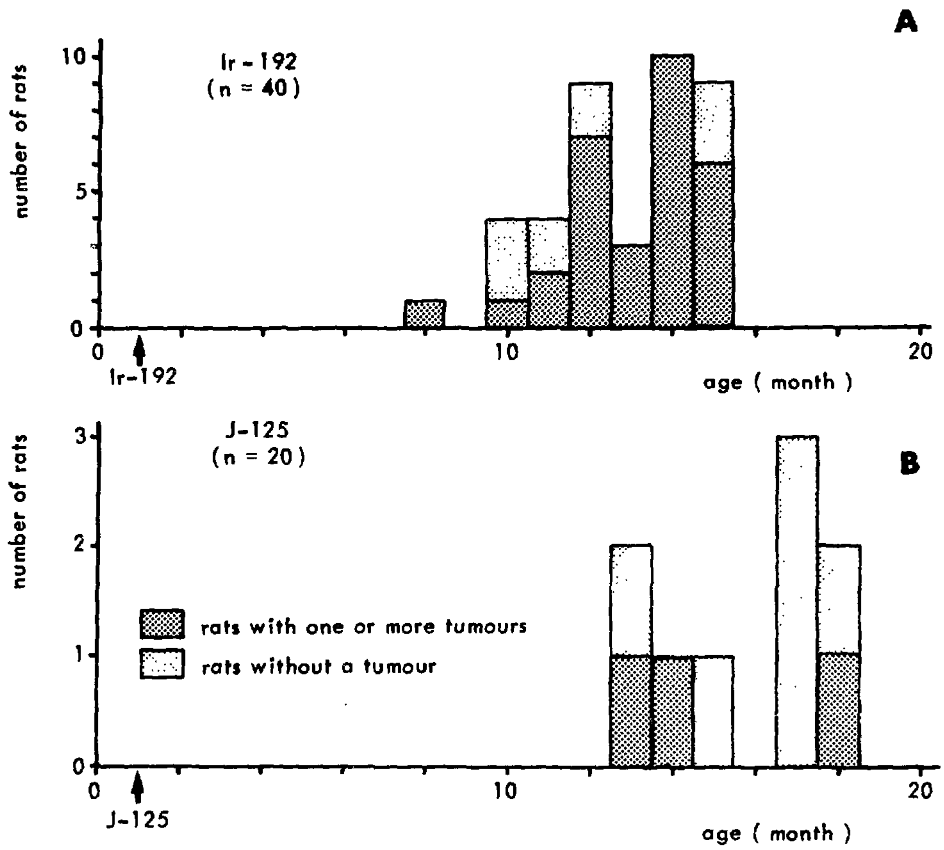

Figure 1: Tumour development as a function of age in WAG/Rij rats implanted with ^{192}Ir (A) or ^{125}I (B) in the lungs.

were transplanted into nude mice; the take rate was 100 per cent. The tumours growing in the nude mice all had the same histological appearance as the tumours from which they were derived.

The calculated dose rates and doses at several distances from the radioisotopes are presented in Table I. The dose administered in the first two months (i.e., in the first half-life of the isotopes) is about one half of the total doses given in Table I. This dose may be relevant for tumour induction. From these data, it can be concluded that in the observation period of 18 months the frequency of induced tumours is zero for total doses up to 2 Gy (i.e., the dose administered in the period of two months to the lung that was not implanted and in which no lung tumours were observed); the frequency is 5 to 15 percent for doses in the range of 7 to 110 Gy and 42.5 percent for the dose range of 13 to 215 Gy. Whether the dose rate might influence these results is not yet clear.

Table I

LUNG DOSES AND DOSE RATES AS A FUNCTION OF DISTANCE FROM THE ISOTOPES

	^{192}Ir (0.47 mCi)¤			^{125}I (0.4 mCi)¤¤		
distance (mm)	dose rate* ($Gy \cdot h^{-1}$)	dose** (Gy)	dose*** (Gy)	dose rate* ($Gy \cdot h^{-1}$)	dose** (Gy)	dose** (Gy)
2.5	0.17	215	436	0.075	110	222
5	0.042	54	109	0.019	27	55
10	0.011	13	27	0.005	7	14
25	0.0017	2	4.3	0.0008	1	2.2

¤ the T1/2 of ^{192}Ir is 74.5 d.
¤¤ the T1/2 of ^{125}I is 60.2 d.
* dose rate at the time of implantation.
** dose in the first 2 month of implantation.
***total dose in the observation period.

Kal, H.B. et al., p. 173. In: Annual Report of REP-TNO (1979).

SECONDARY TUMOURS AFTER SUPRALETHAL CHEMO-RADIOTHERAPY AND ISOLOGOUS BONE MARROW TRANSPLANTATION IN ACUTE LEUKAEMIA OF MALE BN/BiRij RATS

C. Zurcher, A.C.M. Martens, B. Lopes Cardozo, M.J. van Zwieten and A. Hagenbeek

Patients with hematological malignancies or aplastic anaemia treated with high dose chemo-radiotherapy and bone marrow transplantation. A combination of cyclophosphamide (Cy) and supralethal total body irradiation (TBI) is used in many centres.

Both radiation and Cy are known carcinogenic agents. Reports on the occurrence of secondary tumours in human patients successfully treated for their malignancy with total body irradiation (TBI) and Cy have been few. As the latency period of such tumours in man may be very long, the long-term effects of such a combined treatment modality for acute myeloid leukemia was studied in a rat model (Van Bekkum and Hagenbeek, 1977). Male BN/BiRij rats of 2 to 3 months of age received 10^7 Brown Norway myeloid leukaemic (BNML) cells on day 0 and were subsequently treated with Cy at 100 $mg.kg^{-1}$, high dose TBI (9 Gy gamma irradiation) and 10^8 normal isologous bone marrow cells. As a leukaemia relapse generally occurs within a period of 100 days, only those rats surviving past that time were followed for the occurrence of secondary tumours.

Of 110 rats treated, 46 survived for more than 100 days. The number of tumour bearing rats and the number and types of tumours observed in those dying within different posttreatment time intervals are given in Table I. Sixty-eight per cent of the surviving rats had one or more tumours. Only a few tumour bearing rats died before 300 days following treatment and a rapid increase in tumour incidence to 100% was found thereafter. For comparison, the tumour incidence in untreated male BN/BiRij rats of comparable age is estimated to be less than 5% (Burek, 1978). The treatment also resulted in decreased survival times for those surviving the initial period: median survival age of 500 days (age at start of treatment, 75 days; posttreatment interval, 425 days). In untreated male BN/BiRij rats, the median survival age is about 900 days. While some tumour types such as myelomonocytic leukaemia and urothelial tumours may occur in comparable frequencies, although much later, in untreated aging male BN/BiRij rats (11% and 35% with age ranges of 17 to 26 months and 16 to 43 months, respectively), malignant tumours of peripheral nerves and pheochromocytomas are uncommon in untreated BN/BiRij males (0% and 8%, respectively) (Burek, 1978).

Table I

TUMOUR INCIDENCE IN MALE BN/BiRij RATS SURVIVING $\geq$ 100 DAYS AFTER CYCLOPHOSPHAMIDE ($100\ mg \cdot kg^{-1}$), HIGH DOSE TBI (γ) AND ISOLOGOUS BM TRANSPLANTATION

time interval after treatment (days)	Nd	Ne	Nt	number of rats with: peripheral nerve tumour	pheochromocytoma	urothelial cell tumour	lymphohaemopoietic tumour	other tumours
100-199	2	0						
200-299	11	9	2		1	1	1	
300-399	9	8	4	2	1			2
400-499	8	7	7	3			1	3
500-599	9	8	8	1	1	3	1	3
600-699	6	5	4	1	1			3
700-799	1	1	1		1			1
total	46	38	26	7	5	4	3	12
%		100	68	18	13	11	8	31

Nd: number of rats dying in specified interval.
Ne: number of rats examined histologically.
Nt: number of tumour bearing rats.

Table II

TUMOUR INCIDENCE IN MALE BN/BiRij RATS SURVIVING $\geqslant$ 100 DAYS AFTER CYCLOPHOSPHAMIDE

				number of rats with:				
time interval after treatment (days)	Nd	Ne	Nt	peripheral nerve tumour	pheochromocytoma	endomyocardial sarcoma	granular cell tumour	other tumours
100-199	8	4						
200-299	34	20	5	3	2			
300-399	9	7	4	4				
400-499	5	5	5	5	1	2	1	1
500-599	4	4	4	3	1		1	
600-699	2	1	1	1				
700-799	3	3	2	2	1			
800-899	2	1	1		1			
total	67	45	22	18	6	2	2	1
%		100	49	40	13	4	4	2

Nd: number of rats dying in specified interval.
Ne: number of rats examined histologically.
Nt: number of tumour bearing rats.

As a preponderance of the latter two tumour types was also found in another series of BN/BiRij rats treated with 200 mg. kg^{-1} Cy (Hagenbeek and Martens, 1979), we compared the data reported above with the incidence of secondary tumours in 96 BNML leukaemic rats treated only with 100 mg.kg^{-1} Cy. The results of gross and microscopic examination of 45 rats which survived more than 100 days are given in Table II.

It appears that the median survival time in this group is even shorter (365 days; age at start of treatment, 75 days; posttreatment interval, 290 days) than in the rats treated with Cy combined with TBI, while the number of tumour bearing rats increases to 100% between 300 and 400 days. The incidences of peripheral nerve tumours and pheochromocytomas are also much higher in this group (accounting for more than 80% of all tumours) than in untreated aged male BN/BiRij rats. Of the other tumours, endomyocardial sarcomas and granular cell tumours are also most probably of neurogenic origin, leaving only one nonneurogenic tumour (an ameloblastic fibro-odontoma) out of the 29 tumours observed in this group. The latter figure contrasts sharply with a total of 19 out of 31 tumours being of nonneurogenic origin in the rats receiving TBI combined with 100 mg.kg^{-1} Cy.

It is concluded that, in this rat model, the carcinogenic potential of Cy is reflected mainly in the high incidence of tumours of neural tissue origin, while the TBI (gamma irradiation) induces tumours in a variety of tissues. Cy alone did not lead to an increased frequency of urothelial or lymphoreticular tumours in this series, in contrast to reports in man (IARC monograph, vol. 26) and rats (Schmähl and Habs, 1979). The overall incidence of tumour bearing rats is somewhat higher in the combined treatment group, but this may be related to the somewhat longer survival times of this group.

Finally, the important life-shortening associated with cachexia, also observed in rats without tumours and found with both treatment modalities, remains unexplained. It might be related to the high incidence of severe periodontitis in these rats. Fatalities directly attributed to severe infections were rare.

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A UBIQUITOUS TUMOUR-ASSOCIATED ANTIGEN DETECTED BY SERA FROM IRRADIATED RATS

P.A.J. Bentvelzen, M.A. Dubbeld, W.H. Koornstra, J.J. Broerse and M.J. van Zwieten

Molecular hybridization studies have revealed the presence of proviral sequences in normal rat DNA which are partially homologous to the genome of the murine mammary tumour virus (MuMTV) (W.H. Koornstra, this Annual Report, p. 57). In an earlier study, we found that no antigens related to structural proteins of MuMTV could be detected by means of a sensitive radioimmunoassay in a large number of radiation-induced or spontaneous rat mammary tumours (Bentvelzen et al., 1981). We therefore tentatively concluded that the endogenous MuMTV-related proviruses did not play a role in carcinogenesis of the rat mammary gland. However, Imai et al. (1981) reported that sera from several neutron-irradiated W/Fu rats contained antibodies reacting with MuMTV as detected by a haemagglutination inhibition test and immunofluorescence on mouse mammary tumour cells. The transient expression of endogenous virus in rats possibly evokes the antibody response. It might also be instrumental in the initiation of the carcinogenic process.

We have examined sera from neutron-irradiated female BN/BiRij and WAG/Rij rats for the presence of antibodies reacting with MuMTV by means of a solid phase radioimmunoassay. In this assay, wells of a microtiter plate containing solubilized MuMTV are first incubated with the test serum and thereafter with a radioiodinated rabbit antiserum to rat immunoglobulins which reacts with IgM and all IgG subclasses. In this system, antibodies to MuMTV can be detected in BN/BiRij rats which have been infected with that virus (Fig. 1). None of the 67 sera from irradiated rats reacted with MuMTV. Four sera were also tested on the highly purified viral proteins p28, p12, gp52 and gp36 and no reaction was found with either.

All of the sera were also tested by means of immunofluorescence on the mouse mammary tumour line C3HMT/cl11 which produces MuMTV. As a control, the syngeneic normal fibroblast line C3H10T1/2 was used. As can be concluded from Table I, 33 of the 67 sera stained the cytoplasm of the tumour cells. The reaction is not due to a viral antigen, however, since incubation of the sera with solubilized MuMTV did not drastically reduce the reactivity with the tumour cells. Several control sera reacted at the very low titer of 1:10, except for sera from four tumour-bearing animals which reacted at a titer of 1:80. It seems that radiation

Table I

REACTIVITY OF SERA FROM IRRADIATED RATS WITH MOUSE MAMMARY TUMOUR CELLS IN AN IMMUNOFLUORESCENCE TEST

	group	radiation treatment	no. of animals	time of serum collection after irradiation (months)	no. positive on C3HMT/cl11	endpoint titers
BN	I	0.4 Gy X rays	7	4	4	1:20 (2x); 1:80; 1:160
	II	0.4 Gy X rays	4	12	1	1:80
	III	1 Gy X rays	3	23	1	1:160
	IV	1.6 Gy X rays	5	3	2	1:40; 1:80
	V	4 Gy X rays	2	24	1	1:20
	VI	0.15 Gy 15 MeV neutrons	4	10	1	1:80
	VII	0.5 Gy 15 MeV neutrons	3	25	1	1:160
	VIII	5 x 0.02 Gy 0.5 MeV neutrons	4	10	2	1:40; 1:160
			32			
WAG	I	0.08 Gy X rays	4	9	2	1:10; 1:20
	II	0.4 Gy X rays	4	8	1	1:40
	III	1.6 Gy X rays	8	8	4	1:20 (2x); 1:40; 1:80
	IV	10 x 0.04 Gy X rays	4	8	3	1:40 (2x); 1:160
	V	20 x 0.02 Gy X rays	4	7	3	1:20; 1:40 (2x)
	VI	0.1 Gy 0.5 MeV neutrons	4	7	3	1:20 (2x); 1:40
	VII	0.15 Gy 15 MeV neutrons	3	25	1	1:40
	VIII	10 x 0.01 Gy 0.5 MeV neutrons	4	7	3	1:20 (2x); 1:160
			35			

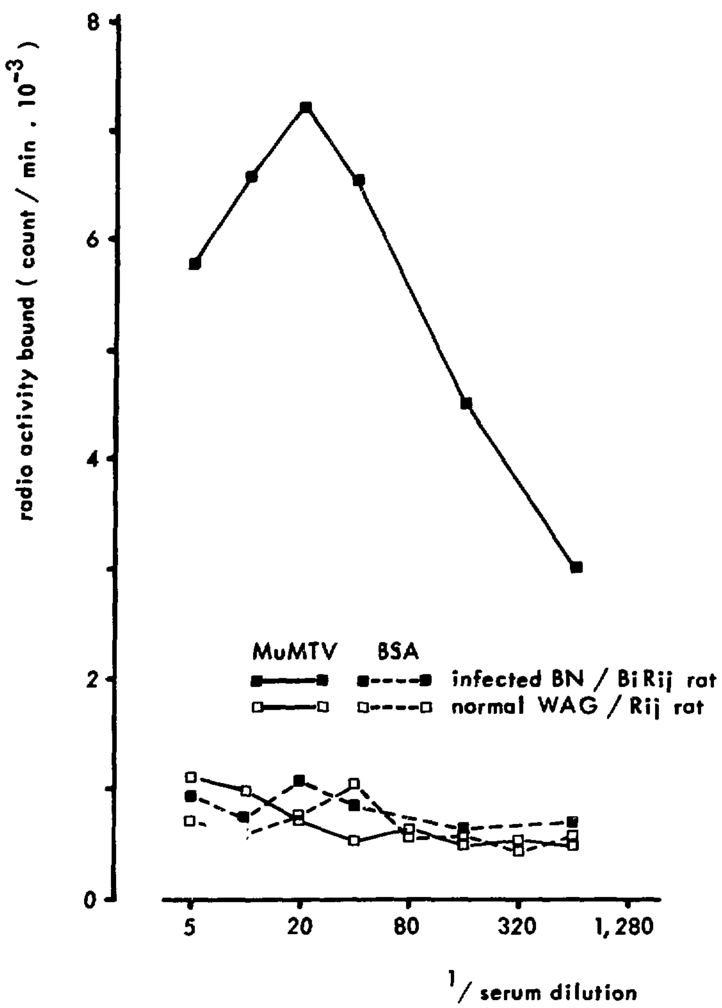

Figure 1: **Solid phase radioimmunoassay for the detection of antibodies reacting with the murine mammary tumour virus in a BN/BiRij rat infected with MuMTV.**

or the presence of a tumour potentiates the reactivity to the antigen(s) in the tumour cells.

All sera which reacted with the C3HMT/cl1l line also stained cells from two mammary tumours, a skin carcinoma, urether carcinoma and a rhabdomyosarcoma, all of rat origin. They also stained BALB/3T3 cells which had been transformed after incubation with Sprague Dawley rat DNA. Neither the parental BALB/3T3 cells nor normal rat fibroblasts were stained by these sera. The rat sera which did not react with C3HMT/cl1l cells did not stain any of the other tumour cell lines. Many sera from irradiated rats seem to contain antibodies reacting with ubiquitous tumour antigens. These do not seem to be the well-known 53K oncofoetal antigens, since these are located in the nucleus (Dippold et al., 1981). The human tumour antigen detected by a murine monoclonal antibody (McGee et al., 1982) could be related to the one detected by the rat sera used in this study.

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EFFECT OF GAMMA IRRADIATION ON RELEASE AND BIOLOGICAL ACTIVITY OF TRANSFORMING DNA FRAGMENTS

R.R. Jonker, V. Krump-Konvalinkova and K.J. van den Berg

The gene transfer-misrepair hypothesis of radiation carcinogenesis (Van Bekkum, 1975) proposes that DNA fragments released from disintegrating irradiated cells are incorporated into the genome of sublethally damaged cells. Fragments of DNA containing cellular transforming genes (analogous to oncogenes of tumour viruses) might be incorporated without putative controlling sequences. The recipient cells could thus be oncogenically transformed and become capable of uncontrolled proliferation.

The hypothesis is straightforward and the various steps and assumptions involved can be experimentally tested. One of the very first assumptions is that irradiation of cells leads to release of DNA fragments. This was studied in mouse thymocytes because of the highly radiation sensitive character of these cells. We prepared a thymocyte suspension from BALB/c mice. Dead cells and fragments of cells were removed by filtering the suspension through a thin layer of cotton wool and then centrifugation. This procedure considerably lowered the background level of DNA. The suspension was gamma irradiated and incubated at 37°C for 0 to 24 hours. Cells were removed by centrifugation and the DNA content of the supernatant was determined with the fluorescent dye bisbenzimide (Hoechst 33258). This method is sensitive enough to detect DNA concentrations as low as 0.1 $\mu g.ml^{-1}$. The DNA level in the supernatants increased with the radiation dose and the incubation time (Fig. 1). After 100 Gy irradiation followed by incubation of 24 hours at 37°C, up to 32% of the total cellular DNA was released. Sixty percent of it was due to the irradiation treatment, since the DNA level of unirradiated thymocytes also increased with the incubation time. DNA in the supernatant was isolated and analysed on an agarose gel. Most of it turned out to be of very low molecular weight (1 kbp). Only a small fraction had a molecular weight ranging from 1 to 12 kilobase pairs (kbp). This fraction may be active in a transfection experiment, since a 2 to 7 kbp fraction of EcoRI-digested BALB/c DNA was found to possess high transforming activity. The transforming capacity of the irradiated and released DNA will be further investigated.

The assumption that normal cells contain potential oncogenes which upon transfection can induce oncogenic transformation of recipient cells was supported by the finding that fragmented normal mouse DNA can transform murine fibroblasts (Cooper et al.,

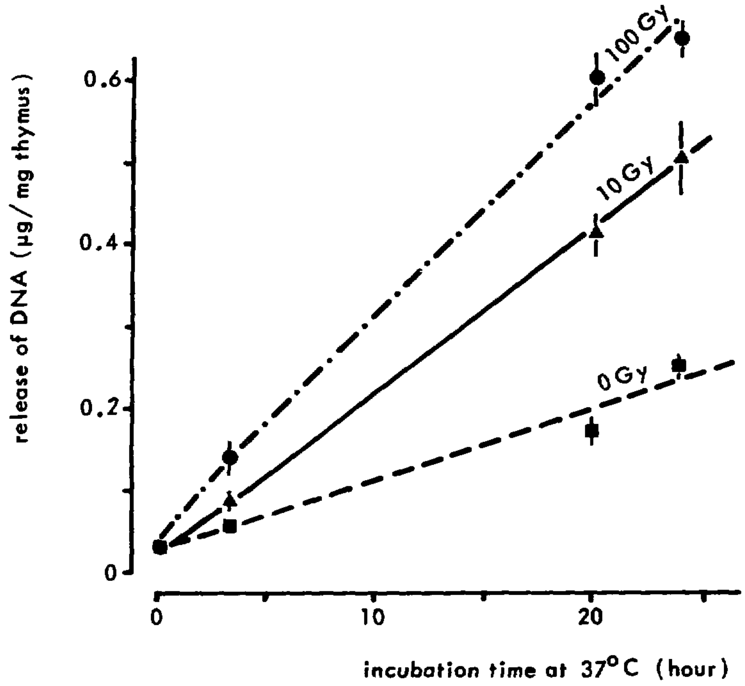

Figure 1: Release of DNA from thymocytes as a function of the dose of gamma irradiation.

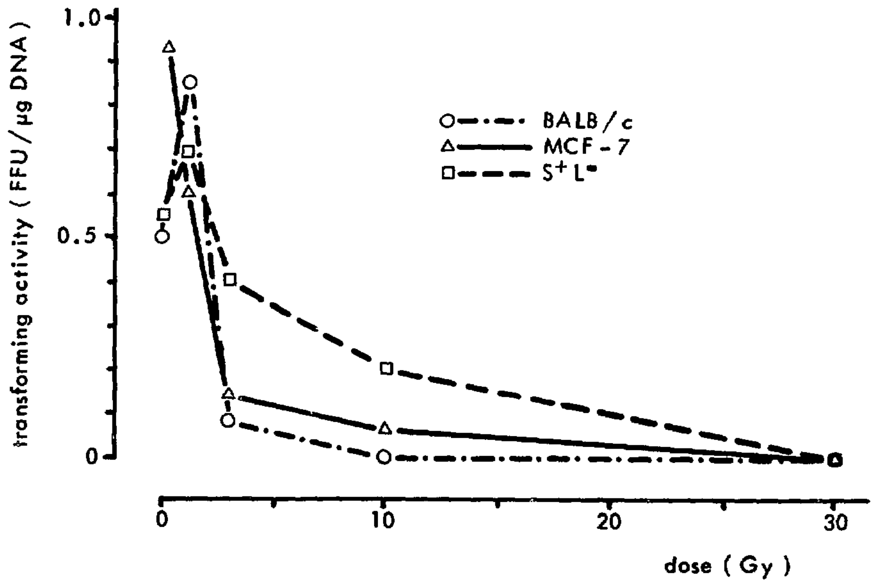

Figure 2: Transforming activity of DNA as a function of the dose of gamma irradiation. Recipient cells were NIH/3T3 cells in the case of S^+L^- and MCF-7 DNA and NIH/3T3 cells preinfected with Moloney murine leukaemia virus in the case of normal BALB/c DNA.

1980; Krump-Konvalinkova and Van den Berg, 1980).

We investigated whether irradiation of DNA could also lead to activation of a silent oncogene. High molecular weight DNA (ca. 50 kbp) was isolated from normal BALB/c mouse liver, MCF-7 cells and S^+L^- cells. MCF-7 is a human breast carcinoma cell line containing an activated oncogene (Lane et al., 1981); S^+L^- is a murine transformed BALB/3T3 subline containing multiple copies of the Moloney sarcoma virus oncogene. These DNAs were irradiated with 0 to 100 Gy of gamma rays and used in transfection experiments. NIH/3T3 cells were incubated with a calcium phosphate DNA coprecipitate for 5 hours at 37°C and then divided into 10 subcultures. Cells were freshly nourished with growth medium twice a week. After 8 days, medium with only 5% serum was used to select for transformed cells. Foci of transformed cells were observed at two to four weeks after DNA treatment. The results of these experiments are presented in Fig. 2. Both BALB/c and S^+L^- DNA exposed to the lowest level of gamma irradiation (1 Gy of gamma rays) seem to be activated in transforming capacity. Subsequent experiments with lower doses of irradiation indicate that activation is already quite effective at a dose of 0.3 Gy of gamma rays. Inactivation of controlling sequences might be responsible for this effect, since no detectable shearing of high molecular weight DNA takes place at this dose level. Higher doses of irradiation lead to a rapid and total inactivation of the transforming capacity of the tested DNAs. Single strand breaks may be involved in the inactivation effect, as the molecular weight of the isolated DNA fragments is only slightly decreased from ca. 50 kbp to ca. 20 kbp after 100 Gy gamma irradiation, indicating that the inactivation is not caused by fragmentation to a too low molecular size. The S^+L^- DNA is considerably less sensitive to inactivation, probably because these cells contain several copies of activated transforming genes.

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ONCOGENIC TRANSFORMATION INDUCED BY NORMAL BALB/c MOUSE DNA

V. Krump-Konvalinkova and A.G.M. Haaksma

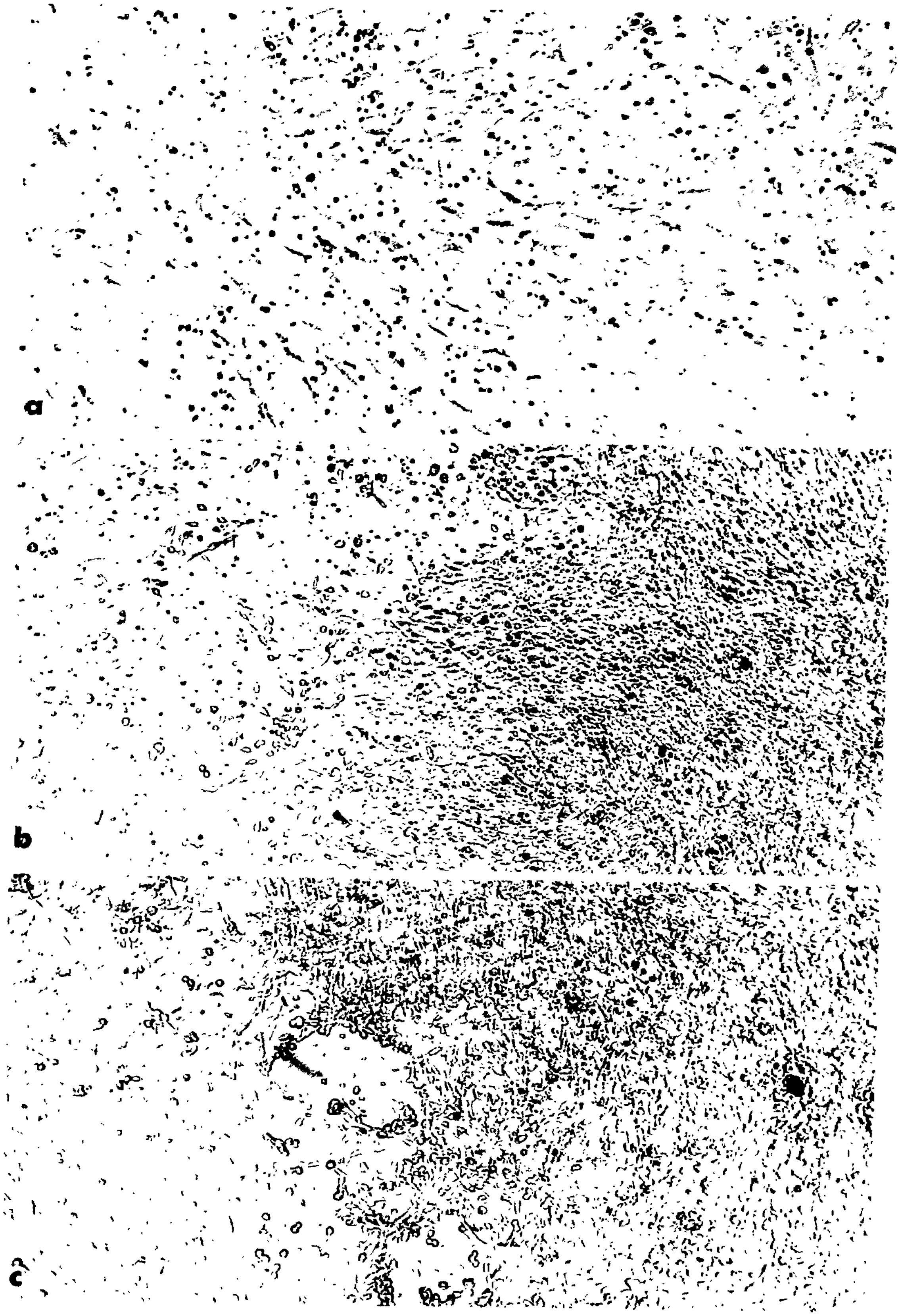
We have previously observed that normal BALB/c mouse DNA fragments induce oncogenic transformation of NIH/3T3 fibroblasts. The efficiency of transformation was found to be significantly enhanced when leukaemia virus preinfected fibroblasts were used as recipients (Krump-Konvalinkova and Van den Berg, 1980).

Investigation of the oncogenic activity of cellular genes is dependent on easy and reliable recognition of foci of phenotypically transformed cells in our assay. NIH/3T3 cells are susceptible to spontaneous changes in phenotype. Density inhibited cultures form accumulations of dead cells which can be mistaken for "real" foci. The best results were obtained when the DNA treated cells were grown in the normal serum concentration of 10% for about a week and then, as soon as the monolayer was confluent, substituting medium containing 5% serum. At this concentration, the normal cells cease to divide (Fig. 1a) and the foci of growing transformed cells are very easily detectable (Fig. 1b, 1c).

We have attempted to gain more insight into the promoting role of leukaemia virus by treating the NIH/3T3 cells with a mixture of normal mouse DNA fragments and plasmid pMlsp (McClements, 1980). This plasmid carries molecularly cloned long terminal repeat (LTR) of the Moloney murine sarcoma virus (Mo-MuSV) known to contain transcriptional control signals. The NIH/3T3 cells were transformed to a neoplastic phenotype at the high frequency of about 7 foci per μg mouse DNA per 10^6 recipient cells (Table I). This transformation efficiency was comparable with that induced by DNA from the human breast carcinoma cell line MCF-7 together with pMlsp DNA.

Individual cells in normal DNA induced foci were morphologically similar to those induced by MCF-7 DNA, but the foci were generally smaller and less dense than those induced by MCF-7 DNA (Fig. 1b, 1c). Interestingly, even calf thymus DNA sheared to the size of about 15 kb when used together with pMlsp DNA induced morphological transformation of NIH/3T3 cells at relatively high efficiency.

In these experiments, the presence of pMlsp completely simulated the enhancing effect of leukaemia virus preinfection. Our working hypothesis is that the pMlsp DNA might become cointegrated with the BALB/c DNA fragments in recipient genomes and provide an efficient transcriptional promotor to normal DNA transforming


a
b
c

fragments. Similar cointegration of foreign DNA in "competent" cultured cells has been observed by Perucho et al. (1980). The ligation of both eukaryotic and prokaryotic DNA fragments in the recipient cell has been demonstrated by Anderson et al. (1982). The ligation of LTR sequences to both the mouse DNA sequence homologous to the transforming gene of Moloney sarcoma virus (c-mos) and the rat DNA sequence homologous to Harvey sarcoma virus (c-ras) before transfer to NIH/3T3 cells resulted in the activation of the transforming potential of these normally non-oncogenic sequences (Blair et al., 1981; DeFeo et al., 1981).

The cotransfection of mouse and plasmid DNA should make the isolation of the transforming fragment of normal mouse DNA feasible. The plasmid sequences integrated together with the mouse transforming DNA fragment can serve as a sequence "tag" necessary to detect the presence of foreign DNA in recipient cells and possibly as a vector for molecular cloning of the transforming fragment. Foci of transformed cells were picked up by the cloning cylinder technique. The cells were grown in 5% serum to mass culture and their DNAs were analysed for the presence of plasmid pBR322 sequences by means of the Southern blot procedure (Southern, 1975; Jeffreys and Flavell, 1977). Autoradiograms revealed that all tested "primary transformants" contain several copies of plasmid DNA. At least one of these plasmid sequences is expected to be cointegrated with the mouse DNA transforming fragment. This result also provides evidence that, having acquired the transformed phenotype, the cells have incorporated exogenous DNA.

By means of "secondary transfection" with high molecular weight DNA from "primary" transformants we hope to transmit "plasmid-LTR-transforming fragment" complex and to dilute out the other plasmid sequences. The DNA of these secondary transfectants may therefore be used as starting material for molecular cloning of the transforming fragment of normal mouse DNA.

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Figure 1. A. A monolayer of salmon sperm DNA treated NIH/3T3 cells after 24 days in 5% serum.

B. A focus of MCF-7 + pM1sp DNA treated NIH/3T3 cells.

C. A focus observed on NIH/3T3 cells treated with a DNA mixture of normal BALB/c fragments and plasmid pM1sp.

Table I

TRANSFORMATION OF NIH/3T3 FIBROBLASTS WITH BALB/c MOUSE DNA IN THE PRESENCE OR ABSENCE OF PLASMID pM1sp CARRYING AN INSERT OF LONG TERMINAL REPEAT OF Mo-MuSV

DNA	frequency of transformation* foci.μg^{-1} DNA (foci.μg^{-1} "transforming" DNA)
BALB/c (sheared)	0 (foci observed after 6 subcultures only)
calf thymus (sheared)	0
salmon sperm (sheared)	0
salmon sperm (sheared) + pM1sp**	0
BALB/c + calf thymus (ca. 40 kb) + pM1sp	1.6 (7.7)
calf thymus (ca. 40 kb) + pM1sp	0.5
calf thymus (sheared) + pM1sp	1.4 (1.4)
MCF-7*** + calf thymus (ca 40 kb) + pM1sp	1.8 (7.6)

* 1 ml of DNA precipitate containing 7.5 ug of "transforming" (BALB/c, calf thymus-sheared, MCF-7) DNA with or without 22.5 ug of "carrier" (calf thymus or salmon sperm) DNA with or without 1 ug pM1sp/HindIII was distributed on about 6 x10^5 fibroblasts. Transformation frequency is calculated on the basis of the total amount of DNA. Between brackets calculated on the basis of only the amount of BALB/c DNA.

** pM1sp = one copy of Mo-MuSV-LTR + mink sequences flanking the Mo-MuSV integration site, cloned in the EcoRI site of pBR322.

***MCF-7 = human breast carcinoma cell line.

A COMPARISON BETWEEN A TRANSFORMING GENE IN NORMAL CELLULAR DNA AND RETROVIRAL ONC GENES

M.J. Vaessen, J.M. van der Maaden, V. Krump-Konvalinkova and P.A.J. Bentvelzen

It is commonly believed that normal vertebrate cells contain sequences which, when provided with appropriate control elements, can be oncogenic. A widely held theory states that these sequences have been inserted into genomes of rapidly transforming retroviruses during the course of evolution.

The hypothesis that normal cell genes can induce oncogenic transformation has been supported by transfection experiments in which normal cell DNAs have been found to induce transformation of nonneoplastic mouse fibroblast lines (Cooper et al., 1980; Krump-Konvalinkova and Van den Berg, 1980). In order to detect transforming activity, cell DNAs had to be fragmented. High molecular weight DNAs of normal cells appeared to lack detectable transforming activity.

Such DNAs of some neoplasms, however, induce transformation with considerable efficiency. It appears that these cells contain activated transforming genes. In some tumours these genes have proved to be homologous to onc genes of acute transforming retroviruses (for a review, see Cooper, 1982).

We have investigated whether the transforming sequence in normal cellular DNA is homologous to a retroviral onc gene. If so, the cellular homologue of the onc gene should show an altered restriction pattern in the genome of transformed cells. It would very likely be duplicated.

Total cellular DNA was isolated from four different BALB/3T3 sublines (designated as Baba) which had been transformed by normal BALB/c mouse DNA fragments. Following digestion with a restriction endonuclease, the DNA was submitted to gel electrophoresis. DNA from the parental BALB/3T3 cell line was used as a negative control. After transfer of the DNA to nitrocellulose filters, hybridization was carried out using ^{32}P labelled DNA of molecular clones of various retroviral onc genes. We have screened the Baba cell lines with the viral onc genes src, myb, myc, erb, mos, Ha-ras, Ki-ras, abl, fes and sis. With none of these probes was any difference detected between the Baba cell lines and the parental BALB/3T3 line. Therefore, the transforming effect of BALB/c DNA is not due to transfer of a DNA fragment homologous to the tested retroviral onc genes.

Krump-Konvalinkova et al. (1981) have investigated the sensitivity of the transforming activity of BALB/c DNA to cleavage with different restriction endonucleases. Analysis of the published nucleotide sequence of the long terminal repeat (LTR) of Moloney murine sarcoma virus (Dhar et al., 1980) revealed that none of the enzymes tested that have a recognition site in the LTR was found to leave the transforming activity of normal cellular DNA intact (Table I). On the other hand, not every enzyme capable of inactivating the transforming activity had a recognition site in the LTR.

Long terminal repeats of retroviruses, which have highly conserved sequences, have been reported to contain promotor and transcription enhancer sequences which can induce expression of genes in their proximity (Levinson et al., 1982). This led us to the hypothesis that the potentially transforming gene in normal mouse DNA may be adjacent to an endogenous LTR.

Kirschmeier et al. (1982) found in several radiation and chemically transformed sublines of the C3H10T1/2 mouse fibroblast cell line cotranscripts of an LTR and nonviral sequences. We have hybridized the LTR of Moloney murine sarcoma virus to total cellular RNA of several BALB/3T3 sublines which have been transformed either by means of transfection with DNA fragments from different mouse strains or spontaneously. As controls were used normal BALB/3T3 cells and cells infected with Moloney leukaemia virus or transformed by Moloney sarcoma virus. The results obtained from such an experiment are shown in Fig. 1. We have found that all

Table I

COMPARISON OF RECOGNITION SITES FOR RESTRICTION ENZYMES WITH DESTRUCTION OF TRANSFORMING ACTIVITY

restriction enzyme	recognition site in LTR (of Mo-MuSV)	destruction of transforming activity
AvaI	+	+
BalI	-	+
BamHI	-	+
EcoRI	-	-
HindIII	-	+
KpnI	+	+
PvuII	+	+
SphI	-	-
XbaI	+	+
XhoI	-	+

Figure 1: RNA blot analysis showing hybridization of the LTR probe to total cellular RNA from the following cell lines:

a) BALB/3T3 infected with Mo-MuLV;
b) normal BALB/3T3;
c) spontaneously transformed BALB/3T3;
d) BALB/3T3 transformed by C3Hf DNA fragments;
e) BALB/3T3 transformed by CBA DNA fragments;
f) Baba 4;
g) Baba 10;
h) $S^{+}L^{-}$ (BALB/3T3 transformed by Mo-MuSV).

transformed sublines examined thus far contain transcripts homologous to the retroviral LTR. Although the genome of BALB/c mice is known to contain at least 30 copies of endogenous retroviral LTR, no transcripts homologous to the LTR were detected in untransformed BALB/3T3 cells.

The fact that not only DNA transformed cell lines but also spontaneous transformants contain transcripts which hybridize to the LTR could indicate that the complex of an endogenous LTR with a cellular onc gene plays a general role in oncogenic transformation.

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SEQUENCES RELATED TO THE PUTATIVE MAM GENE OF THE MURINE MAMMARY TUMOUR VIRUS IN CELLULAR DNA OF SOME MAMMALIAN SPECIES

W.H. Koornstra, R.H. Dubbes, J.P.M. Jore,
M.J. Vaessen and P.A.J. Bentvelzen

When relaxed hybridization conditions are applied, sequences which are partially homologous to structural genes of the murine mammary tumour virus (MuMTV) can be detected in cellular DNA of rat, mink or human origin (Drohan and Schlom, 1979; Callahan et al., 1982). We have additionally tested cellular DNA of different vertebrates with the Southern blot hybridization technique for reactivity with the cloned proviral long terminal repeat (LTR) of this virus with different degrees of stringency. This LTR has been molecularly cloned. For hybridization studies the LTR insert was isolated from the plasmid PAT153 carrying this sequence.

This LTR is characterized by its glucocorticoid responsiveness with respect to enhancement of transcription. It can induce the expression of a cellular oncogene in its vicinity, as has been demonstrated in many virally induced mammary tumours (Nusse and Varmus, 1982). This LTR also contains a large open reading frame which in vitro can produce a protein with a molecular weight of 37,000 daltons (Dickson et al., 1981). This product is thought to play an initiating role in mammary carcinogenesis. The gene coding for this protein is therefore called <u>mam</u> (Bentvelzen et al., 1981). In mouse mammary tumours arising without involvement of an exogenous virus, the LTR of an endogenous provirus is activated (Breznik and Cohen, 1982).

Under highly stringent hybridization conditions (low salt, high formamide concentrations), an expected strong reaction was found between the MuMTV LTR and DNA from a BALB/c mouse. Rather unexpectedly, DNA isolated from a calf thymus also gave a strong reaction with the MuMTV LTR. A weak reaction was found with rat DNA (Fig. 1). Under relaxed conditions (high salt, low formamide concentrations), the intensity of the bands produced by rat DNA increased, while very thick bands were produced by the bovine DNA (Fig. 2). Faint bands appeared in the lanes with DNA from mastomys, the rabbit, the cat, the mink and the dog. No reaction was found with DNA samples of chicken, marmoset, rhesus monkey or human origin. The human samples were obtained from a placenta and the well-known breast cancer cell line MCF-7.

The finding of sequences which are partially homologous to the LTR of the murine mammary tumour virus in cellular DNA of several mammalian species indicates that the sequences related to struc-

Figure 1: Southern blot hybridization under stringent conditions using the long terminal repeat of the murine mammary tumour virus as a probe on DNA samples from a: rat strain BN; b: rat strain SD; c: mastomys; d: human MCF-7; e: rabbit; f: cat; g: marmoset; h: mink; i: human placenta; j: rhesus monkey; k: dog; l: calf; m: chicken; n: mouse BALB/c.

Figure 2: Southern blot hybridization under relaxed conditions with the probe and DNA samples as in Fig. 1.

tural genes of this virus, as found by Callahan et al. (1982), have been derived from an infectious retrovirus. This does not hold, however, for such sequences in human DNA. As the blotting pattern of laboratory rats contains more discrete bands than that of the African rat Mastomys, it may be concluded that the germ line of the laboratory rat has been infected with a virus related to the murine virus after separation of its ancestor from Mastomys. As the patterns of three genetically different inbred rat strains resemble each other, it may be concluded that this infection has taken place in an early stage of the domestication of the laboratory rat. The differences among the three strains may be due to additional infections, amplification following occasional expression of endogenous virus or translocation of the proviruses.

The very thick bands found with bovine DNA indicate the presence of numerous sequences related to MuMTV LTR. In bovine DNA, the presence of tens of thousands of copies of an LTR-like sequence with an open reading frame has been reported by Streeck (1982). Comparison of the nucleotide sequence of this LTR-like structure with that of the MuMTV LTR, however, did not reveal significant homology.

Although no expression of antigens related to structural proteins of MuMTV could be detected in radiation induced mammary tumours of rats (Bentvelzen et al., 1981), it would be worthwhile to look for expression of the sequence related to the LTR of MuMTV in such tumours. This sequence may be regarded as the endogenous <u>mam</u> oncogene, which would play an important role in various forms of mammary carcinogenesis.

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ENDOGENOUS RETROVIRAL SEQUENCES IN MAN

R.H. Dubbes, J.P.M. Jore and P.A.J. Bentvelzen

Most retroviruses that have been reported to be isolated from human leukaemic cells are closely related to the Simian Sarcoma Associated Virus/Gibbon ape Leukaemia Virus (SiSAV/GaLV) group (Nooter and Bentvelzen, 1980). We have therefore searched for the presence of SiSAV/GaLV related sequences in cellular DNAs from patients with neoplasms of mesenchymal origin by means of the so-called blotting hybridization technique (Southern, 1975). Briefly, cellular DNA is cut with a restriction endonuclease into fragments that are both species and enzyme specific. The fragments are separated according to size by electrophoresis on an agarose gel and blotted onto a nitrocellulose filter. After hybridization to a radioactive labelled probe only fragments that show homology to the probe will be visible upon autoradiography.

Using molecularly cloned SiSAV proviral DNA as a probe (a gift of Dr. R.C. Gallo, National Cancer Institute, Bethesda, Md., U.S.A.) we were not able to detect any fragments which specifically hybridized with the proviral DNA in any of the 31 samples tested. The stringent conditions of hybridization applied - high formamide content at a reduced temperature - should permit detection of as little as one proviral copy per 20 cells (Dubbes and Jore, 1981). The DNA samples were obtained from 6 patients with acute myelogenous leukaemia, 10 with acute lymphatic leukaemia, 6 with chronic myelogenous leukaemia, 6 with lymphosarcomas, 1 with a T cell sarcoma, 1 with a Burkitt-like lymphoma, 1 with a non-Hodgkin lymphoma, 1 with a malignant histiocytoma, 1 with a rhabdomyosarcoma and 1 with a chondrosarcoma. The negative results make it highly unlikely that SiSAV is present in human neoplasms of mesenchymal origin.

Lowering the formamide concentration - at the same reduced temperature - permits detection of distantly related sequences, as has been reported by Howley et al. (1979). Under these conditions - allowing for a 20 to 30% mismatch - we found 15 to 20 hybridizing fragments in these 34 human DNA samples as well as in DNAs from two normal individuals. These fragments were arranged in a constant pattern, suggesting that insertion of the multiple proviral sequences took place before speciation of man (Fig. 1). Thereafter, no new infections of the germline by either exogenous or endogenous viruses have taken place. The conservation of SiSAV related sequences in man suggest a certain - hitherto unknown - biological function of these sequences. The DNA from bone marrow

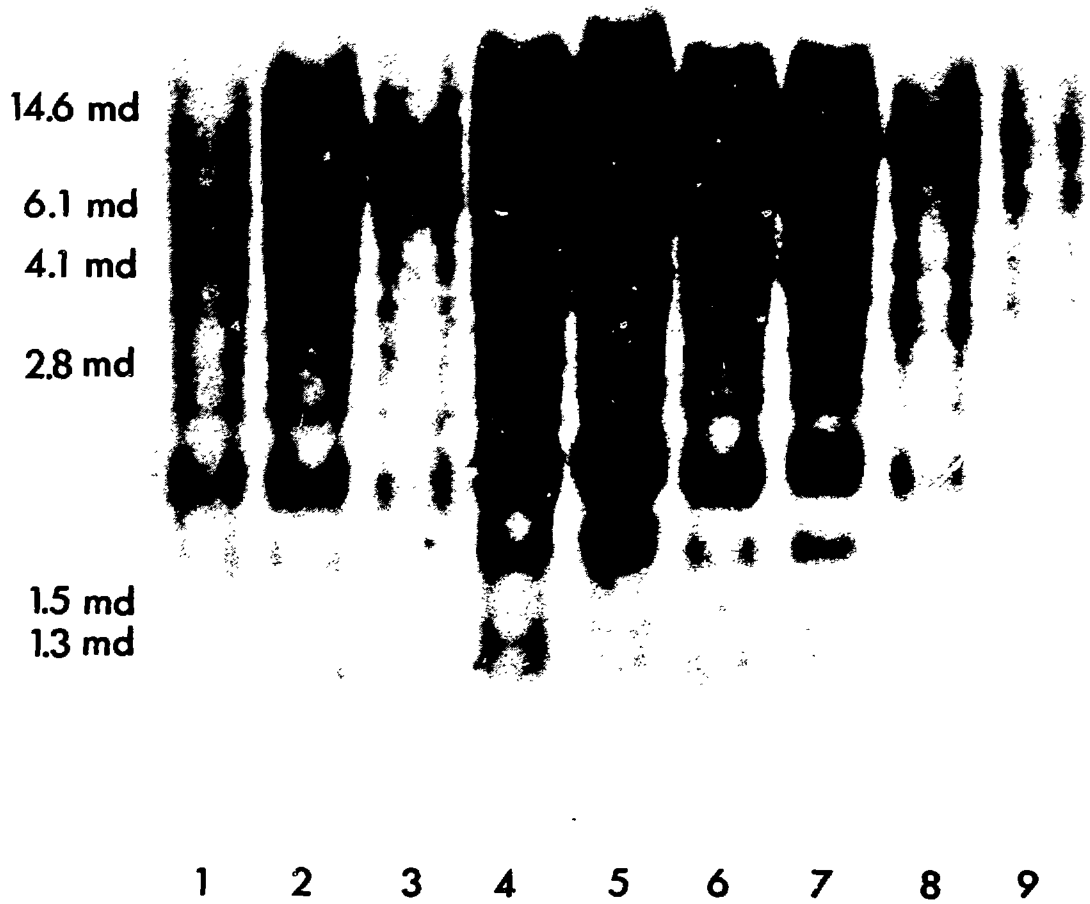

Figure 1: Southern blot hybridization pattern of nine human DNA samples, cut with the restriction enzyme HindIII and probed with ^{32}P-labelled SiSAV proviral DNA.

cells of a child with recurrent T cell sarcoma (lane 8) produces a fragment which is not found in the other DNA samples. This is probably due to translocation of a SiSAV related sequence. It remains to be established whether this translocation is associated with oncogenic transformation.

SiSAV related sequences have been found by us in several other vertebrates. In addition to DNA samples from various mammalian species (Jore et al., 1981), such sequences were also found in DNA from chickens but not from tobacco plants. DNA of a chimpanzee - man's closest relative - showed a restriction pattern that was different from that obtained with human DNA, indicating that integration of SiSAV related sequences in man took place after the separation of the ancestors of *Homo sapiens* from those of chimpanzees.

Using other retroviral probes and employing related hybridization conditions, retroviral sequences have been found in human DNA by Martin et al. (1982) and Callahan et al. (1982). We therefore compared the restriction patterns obtained these authors using murine mammary tumour virus proviral DNA as a probe (Callahan et al., 1982) and those obtained using AKR murine leukaemia virus/baboon endogenous virus reactive DNA from African green monkey DNA as a probe (Martin et al., 1982) with the restriction patterns obtained by us. From this comparison, it became clear that the three different probes detect different endogenous retroviral sequences in man. It can be concluded that human cellular DNA contains a great number of proviruses.

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REDUCED LIVER STORAGE AND PLASMA TRANSPORT OF RETINOIDS IN MICE AFTER EXPOSURE TO 3,4,3',4' TETRACHLOROBIPHENYL

A. Brouwer, A. Kukler and K.J. van den Berg

Polychlorinated biphenyls (PCBs) are prototypes of a group of halogenated aromatic hydrocarbons with wide industrial applications and sharing distinct features of toxicity (e.g. reduced body and thymus weight, chloracne, induction of aryl hydrocarbon hydroxylase (AHH) activity, etc.) and environmental contamination (biological stability, bioaccumulation, etc.) (Kimbrough, 1980). Several of these compounds show a definite carcinogenic potential, mainly at the level of promotion, especially concerning the liver of rodents (Kimbrough, 1980). Following accidental poisonings in rhesus monkey colonies due to polyhalogenated aromatic hydrocarbons, dermal symptoms resembling a vitamin A deficiency were noted (Van Putten et al., 1970; McConnell and Moore, 1979). It is possible, therefore, that the risk of an individual for cancer may increase via a decreased level of vitamin A which is now considered to be a natural inhibitor of carcinogenesis (Bollag, 1979).

In the study reported here, the effects of PCBs on vitamin A levels and the role of drug metabolizing enzymes (like AHH) were investigated. The induction of AHH activity, i.e., an enzyme of the mixed function oxidase (MFO) system, is a sensitive parameter which has been reported to correlate with the toxicity of PCBs and related compounds. In mice, the induction of MFO enzyme systems is under genetic control, i.e., segregation with the AH locus. This offers the attractive possibility of investigating the involvement of MFO enzymes in the reduction of retinoid levels by using either responsive (C57BL/Rij) or nonresponsive (DBA/2) strains of mice (Poland et al., 1979).

We have started to investigate effects of the PCBs Aroclor 1254, a commercial PCB mixture, and 3,4,3',4' tetrachlorobiphenyl (TCB), a toxic type of PCB, on retinoid levels in the liver and serum and on several parameters of toxicity in mice. We found before that low doses (5 $mg.kg^{-1}$ i.p. at weekly intervals) of Aroclor 1254 and 3,4,3',4'TCB induced a substantial decrease in retinol levels in the liver of C57BL/Rij mice after five weeks of exposure. 3,4,3'4'TCB, however, induced an additional decrease in the retinol levels in the serum. At this dosage, none of the toxicity parameters (e.g., reduced body and thymus weight, increased liver weight and induction of AHH activity) was markedly altered

(Van den Berg and Brouwer, 1981). An extensive experiment was therefore set-up to more thoroughly investigate a possible correlation between toxicity, induction of MFO enzyme systems and the effects of PCBs on retinoid levels.

C57BL/Rij and DBA/2 mice were weekly exposed to an i.p. dose of 1.5 to 100 $mg.kg^{-1}$ 3,4,3',4'TCB. After 4 weeks of exposure, toxicity parameters were investigated and retinoid levels in the liver and serum were analysed by high performance liquid chromatography (HPLC). The results were markedly different in the two strains of mice. In C57BL/Rij mice, 3,4,3',4'TCB induced a dose dependent decrease in liver retinol (up to 60%) and retinyl palmi-

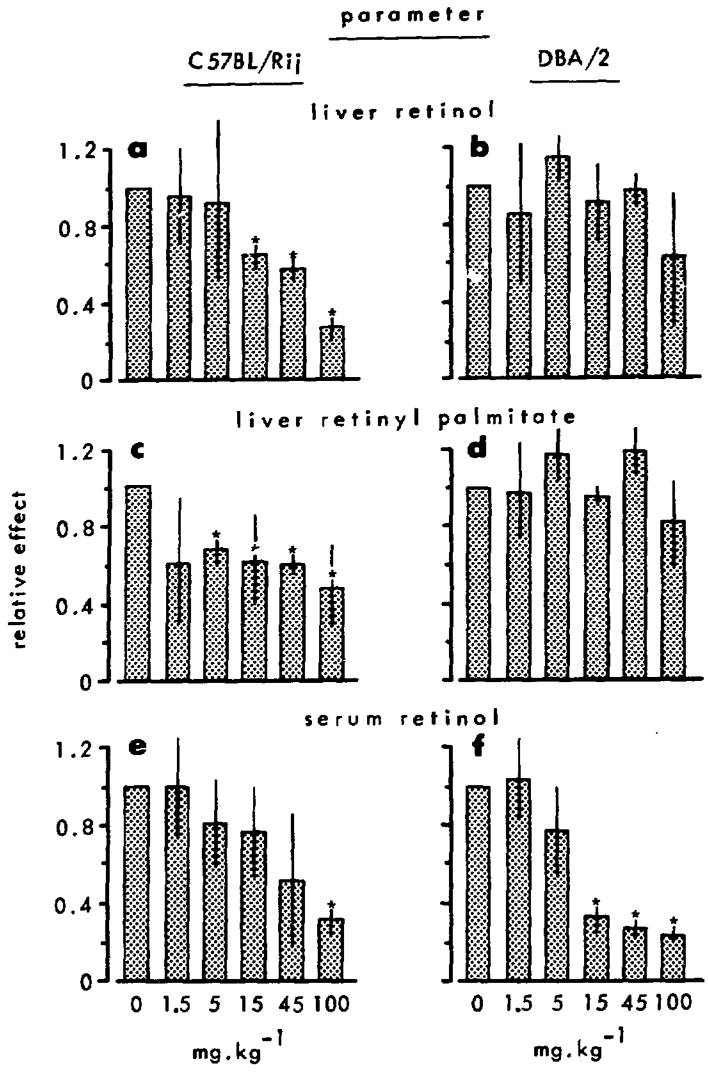

Figure 1: Effects of 3,4,3',4' TCB on retinoids in the liver and serum in C57BL/Rij (left graphics) and DBA/2 mice (right graphics) after 4 weeks of exposure.
* differs significantly from controls ($p < 0.05$).

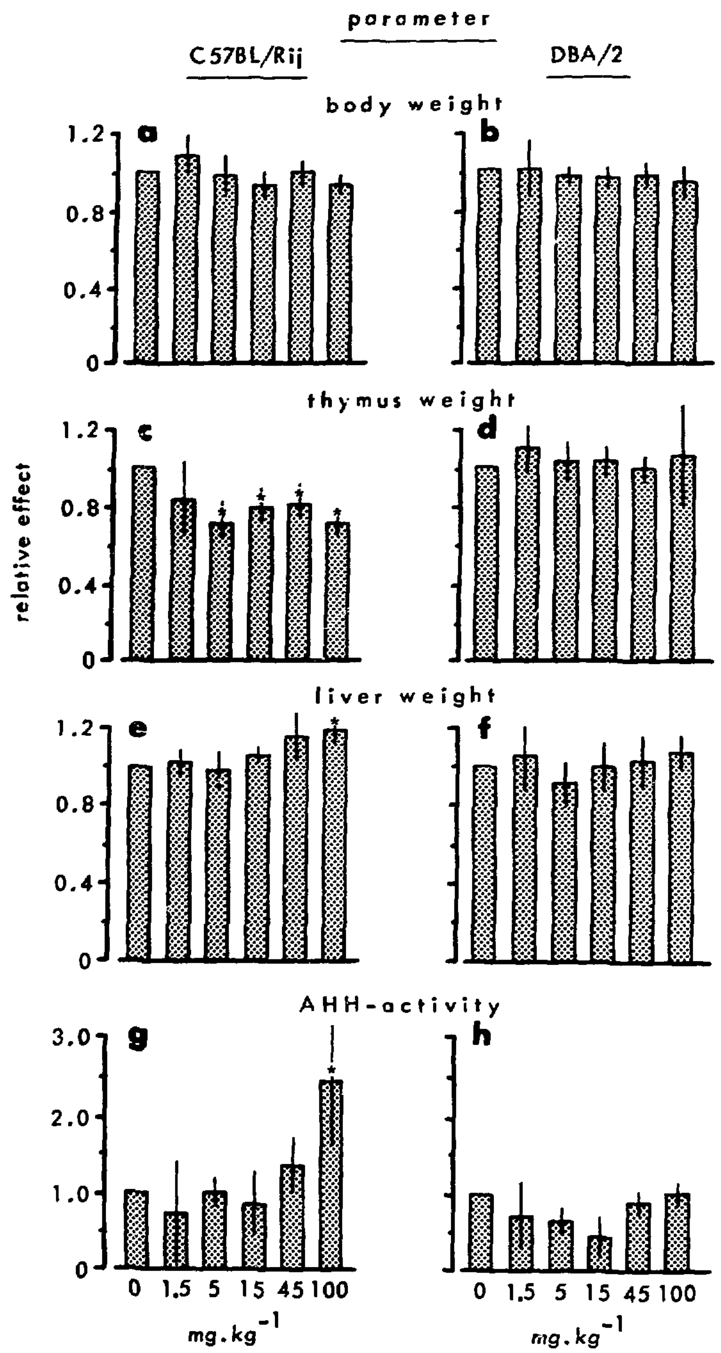

Figure 2: Effects of 3,4,3',4' TCB on toxicity parameters in C57BL/Rij (left graphics) and DBA/2 mice (right graphics) after 4 weeks of exposure. * differs significantly from controls ($p < 0.05$).

tate (up to 40%) and in serum retinol (up to 80%). In the DBA/2 mice, no significant changes in the major liver retinoids were observed; a dose dependent decrease in serum retinol (up to 80%) only was determined (Fig. 1).

In the C57BL/Rij mice, a reduction in thymus weight (by 30%), an increase in liver weight (up to 125%) and the induction of AHH activity at the highest dose of 100 $mg.kg^{-1}$ only were observed (Fig. 2). These parameters did not change in the DBA/2 mice (Fig. 2).

Since high doses of 3,4,3',4'TCB induced AHH activity in the C57BL/Rij mice, it is conceivable that enzymes from the MFO systems might be involved in the decrease in liver retinoids, for instance, by using the retinoids as endogenous substrates. The reduction in liver retinoids would then lead to a concomitant decrease in serum retinol. The effects of 3,4,3',4' TCB on retinoid levels, however, are more pronounced than those on toxicity parameters. This suggests that AHH or similar types of enzymes are not directly involved, but the participation of yet unknown inducible enzymes of retinoid catabolism cannot be excluded. Furthermore, this suggests that reduction in retinoid levels might be a very sensitive indicator for the toxicity of PCBs.

No decrease in liver retinoids is observed in the DBA/2 mouse, while the serum retinol concentration is markedly reduced. This suggests that there is a second level at which 3,4,3',4' TCB affects vitamin A, i.e., by lowering serum retinol levels only. This is probably not related to the induction of MFO enzymes, as DBA/2 mice represent a nonresponsive strain.

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THERAPY

TIME-TEMPERATURE RELATIONSHIP FOR THE INDUCTION OF THERMAL DAMAGE TO TUMOUR MICROCIRCULATION

H.S. Reinhold and A. van den Berg-Blok

Prolonged exposure of tumours to elevated temperatures has a deleterious effect on the tumour microcirculation, especially in the central parts of the tumours. Such a treatment is generally called hyperthermia and the thermal destruction of those parts of tumours which are likely to be hypoxic (and therefore resistant to radiation therapy) make hyperthermia the perfect supplement to radiation therapy. In earlier reports, we have shown that, at the moderate elevated temperature of 42.5°C, the time required to induce microcirculatory stoppage in 50% of tumours of the type Rhabdomyosarcoma BA1112 is about 150 minutes (Reinhold et al., 1978). Addition of glucose, 5-D-thioglucose or misonidazole resulted in a more efficient destruction of the microcirculation (Reinhold and Van den Berg-Blok, 1981).

It is of utmost importance to have available information on to what extent the treatment time can be shortened when the treatment temperature is raised. For cell culture systems, a rule of the thumb is that, when the temperature is raised by one degree C, the treatment time can be halved for obtaining the same level of damage. If the destruction of the tumour microcirculation were to follow the same trend, then hyperthermic treatment time might indeed be shortened by increasing the temperature.

In recent experiments, this factor was investigated in the system reported on earlier, i.e., Rhabdomyosarcoma BA1112 in "sandwich" observation chambers. In this system, the tumour temperature can be accurately controlled for 3 hours and the tumour microcirculation can be continuously observed microscopically. In this way, it is possible to determine at what time the microcirculation in the tumour ceases.

From the results shown in Fig. 1, it indeed appears that the destruction of the tumour microcirculation also follows the same kinetics as found in other systems and the conclusion seems to be justified that considerable shortening of the treatment time can take place when tumours are treated at higher temperatures and that, therefore, clinical application of hyperthermia as an adjunct to radiation therapy may become more efficient.

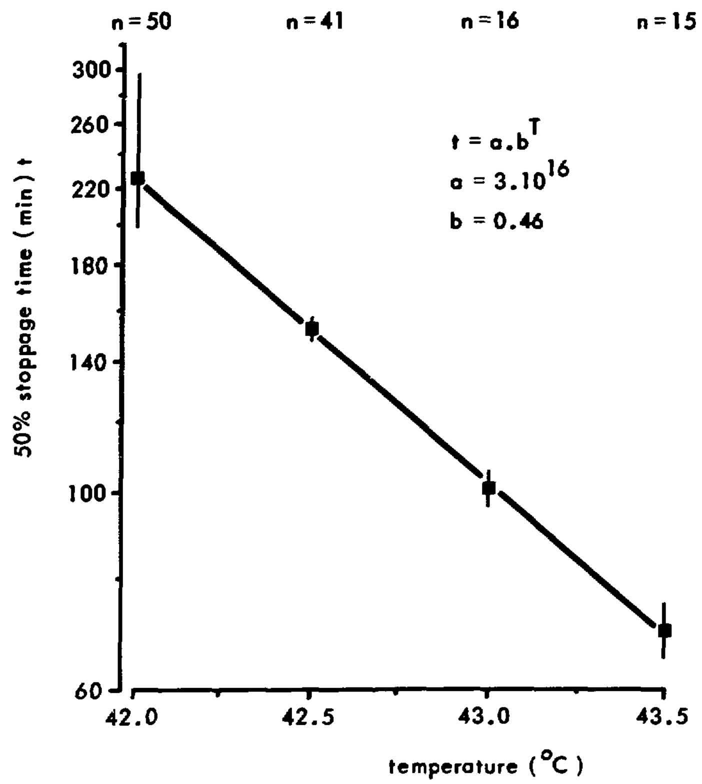

Figure 1: Relationship between treatment temperature and time for inducing (50%) microcirculation stoppage in Rhabdomyosarcoma BA 1112 in "sandwich" chambers.

Reinhold, H.S. et al., p. 231. In: Cancer Therapy by Hyperthermia and Radiation (ed. C. Streffer), Urban and Schwarzenberg, Baltimore-Munich (1978).

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COMPARISON OF RADIATION RESPONSES DETERMINED BY DIFFERENT ENDPOINTS FOR THE R-1,M RHABDOMYOSARCOMA GROWN IN THE SUBCUTANEOUS SPACE AND IN THE GASTROCNEMIUS MUSCLE OF WAG/Rij RATS

A.F. Hermens, H.T. Madhuizen and J. Deurloo

Considerable differences in growth characteristics and/or response to therapy may exist among different types of malignant tumours. These differences can be largely explained by variation in expression of the capacity for unlimited proliferation of clonogenic tumour cells. Proliferation, together with cell elimination determine to a large extent the variations in tumour growth and metastasis formation. Variations in probability for cure after therapy are mainly related to the intrinsic sensitivity of the clonogenic cells to therapy.

The capacity for cell proliferation and eventually its alteration by therapy can be influenced by many factors, e.g., by nutrients and oxygen supplied by blood vessels. Interactions of the tumour cells with elements from the stroma and/or surrounding normal tissue may either support or limit tumour cell proliferation. However, there is still little information about the importance of such interactions with respect to tumour growth and curability.

A study of this problem was performed with the transplantable R-1,M rhabdomyosarcoma system. Tumour material was implanted at two anatomical sites in the female inbred WAG/Rij rat: cultured R-1,M cells were introduced either deep inside the gastrocnemius muscle or into the subcutaneous space of the hind leg situated just above this muscle. Growth characteristics were determined from tumour growth curves constructed by using data on volume measurements determined at regular time intervals after transplantation of tumours. At volumes of about 1000 to 2000 mm^3, the growth rate of R-1,M tumours in the gastrocnemius muscle was slightly slower than that of such tumours growing in the subcutaneous space, i.e., a volume doubling time of T2 = 3.5 d as compared with T2 = 2.5 d, respectively.

Two types of responses to irradiation with single doses of 300 kV X rays were studied. 1) The dose required for obtaining cure in 50 percent of the irradiated tumours (TD-50) (Table I) was assessed and 2) the fraction of surviving tumour cells was determined with the in vitro plating technique described by Barendsen and Broerse (1969). From data on cell survival, presented in Fig. 1, it can be concluded that in euoxic tumours the probability for survival of cells present in R-1,M tumours in the subcutaneous

Table I

COMPARISON OF NUMBERS OF SURVIVING CELLS AFTER TREATMENT OF R-1,M TUMOURS WITH RADIATION DOSES REQUIRED FOR OBTAINING 50 PERCENT TUMOUR CURE WITH CELL NUMBERS REQUIRED FOR 50 PERCENT TUMOUR TAKE IN BIOASSAYS

tumour location in WAG/Rij rats	TD-50 dose (Gy)	number of surviving cells* euoxic conditions	hypoxic conditions	number of cells required for TT-50** R-1,M cells alone	R-1,M + F(R-1,M)◇
subcutaneous space of the hind leg	55	10 to 20	20 to 30	1.2×10^4	1.4×10^3
gastrocnemius muscle of the hind leg	63	1	30	3×10^4	not done

* Tumours were irradiated with 300 kV X rays generated by a Philips-Muller X-ray generator at a dose rate of about 1.15 $Gy.min^{-1}$. Due to geometrical conditions, the dose distribution over the tumour ranging from the side proximal to the X-ray tube to the opposite side may show a maximum variation of 15 percent of the dose administered to the tumour centre. During irradiation, the animals were kept anaesthetized with Nembutal. Their bodies were shielded with a 5 mm thick lead mold, except for the hind leg with tumour, allowing animals to breathe freely.

TD-50 dose: dose required for obtaining 50 per cent tumour cure.

The number of surviving cells at the TD-50 level were calculated for tumours measuring 2 cm^3 in volume, i.e. containing 10^9 cells per tumour, and using data on fractional surviving cells extrapolated from the curves shown in Fig. 1.

**TT-50: represents the number of tumour cells per inoculation required for obtaining 50 percent yield of tumours as determined by the bioassay technique described by Hewitt and Wilson (1959).

◇ F(R-1,M) feeder cells were prepared by irradiating cultured R-1,M tumour cells in vitro with 40 Gy of 300 kV X rays. Data on TT-50 have been described earlier (Hermens and Madhuizen, 1981).

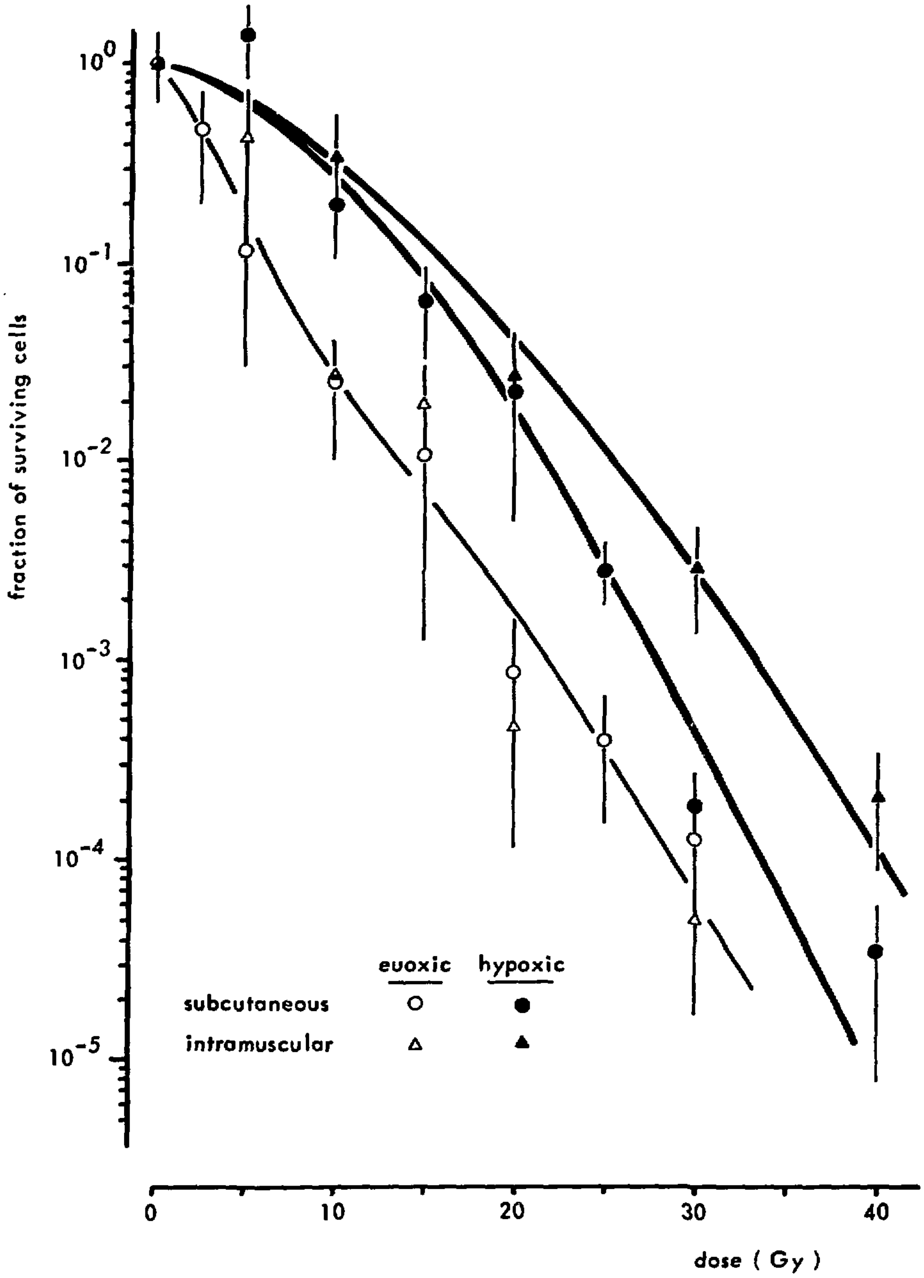

Figure 1: Dose-survival curves for tumour cells isolated from R-1,M tumours growing in the subcutaneous space and in the gastrocnemius muscle of syngeneic female WAG/Rij rats. Tumours were irradiated in the living animal or in animals killed at 15 minutes before irradiation. Each symbol with a bar represents the mean $\pm$ standard error of the mean calculated from results obtained in two to three independent experiments.

space is not significantly different from that for cells present in tumours situated in the gastrocnemius muscle. However, the slope of the curve obtained for hypoxic cells of tumours located in the muscle is less steep in the dose region between 0 and 25 Gy than the slope of the corresponding curve constructed for cells of tumours in the subcutaneous space.

The doses required for obtaining 50 percent tumour cure were determined in experiments in which animals were observed for more than 100 days after irradiation of tumours. The TD-50 for tumours growing in the subcutaneous space was 55 Gy, while the TD-50 for R-1,M tumours growing intramuscularly was 63 Gy. Using values for fractional surviving cells extrapolated from Fig. 1, it can be calculated that at the TD-50 dose level the number of surviving cells determined for tumours in the subcutaneous space varies from 10 to 20 cells for euoxic tumours and from 20 to 30 cells for completely hypoxic tumours. For tumours growing in the muscle, these values are less than or equal to 1 cell and about 30 cells, respectively. These small numbers of cells which give rise to recurrent tumour growth when irradiated with TD-50 doses are in sharp contrast with the fifty to more than one hundred times larger numbers of cells that are required for obtaining 50 percent tumour take (TT-50) after transplantation of R-1,M cells into either of the two anatomical locations (Table I). The conditions in the tumours after irradiation seem to be more suitable for cell proliferation than the local conditions provided by the environment of cells after transplantation of tumour cells.

The somewhat higher radiation dose required for obtaining a 50 percent cure of tumours growing in the muscle can be explained by assuming that the cells of tumours growing in the muscle are slightly more radioresistant than the cells present in tumours in the subcutis. Conversely, the damage induced by radiation, especially in hypoxic cells of tumours growing in the subcutaneous space, may be increased by factors which limit the rate of cell proliferation, such as poor nutrient and oxygen supply or toxic agents concentrating in the treated tumour. On the other hand, it is possible that the properties of the milieu developing in the irradiated muscle adjacent to a shrinking tumour may be more suitable for supporting the proliferation of even a single cell than that of the irradiated subcutis.

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MONOCLONAL ANTIBODIES AGAINST THE BN MYELOCYTIC LEUKAEMIA

A.C.M. Martens, A.H. Mulder and A. Hagenbeek

A great problem in the treatment of human leukaemia is the recognition of the leukaemic cell population when it is decreasing in size as a result of the remission induction therapy. Below the level of 5% leukaemic cells in the bone marrow the classical haematological methods generally become unreliable. The effect of the therapy cannot be quantified and therefore cannot be used to make decisions about ending or continuing the therapy. The same problem of recognition and quantification of leukaemic cell numbers occurs with respect to early detection of recurrent disease. Since the establishment of monoclonal antibody (MCA) technology, this situation has changed in a more hopeful direction. If leukaemia specific antigens exist and MCAs can be raised against them, they might offer means, in combination with advanced cell sorting techniques, to lower the detection limit of leukaemic cells in the bone marrow. The exact determination of the lower limit was the subject of this study.

In our institute, we are focusing on acute myelocytic leukaemia in a Brown Norway rat leukaemia model (BNML), the characteristics of which are strikingly similar to those of human acute myelocytic leukaemia. A number of MCAs have been raised against the leukaemia cells two of which (RM124 produced at Johns Hopkins University, Baltimore, Md., U.S.A.; and RijRl, produced in our own laboratory) have been tested with respect to the specific recognition of the BNML cells. Promising MCAs should meet either one of two criteria: 1) selective recognition of a specific antigen on leukaemia cells; or 2) binding to a nonspecific antigen that makes leukaemic cells distinguishable from normal cells either by a higher or lower degree of binding of the MCA to the antigen.

All determinations described were made on the FACS II cell sorter where, after a two-step labelling method, the amount of fluorescence of the cells was taken as a measure for the presence and density of the antigen on the cell membrane. The binding of RM124, reacting with leukaemia cells, to normal cells was not completely absent, but was very low and restricted to monocytes and nonselectively to a minority of granulocytes.

The second MCA (RijRl) appeared to recognize a myeloid differentiation antigen which is present in very low amounts on CFU-S (the haemopoietic stem cell). During myeloid differentiation, the antigen increases in density. The highest antigen density determined with RijRl was found on the BNML cells, therefore making this antibody useful in the detection of the leukaemia cells in mixtures, although it does not recognize a leukaemia specific antigen.

Both MCAs were used to label cell populations which consisted of artificial mixtures of leukaemic and normal bone marrow cells. The fluorescence intensity profiles of a cell mixture containing 1% leukaemic cells is shown in Fig. 1. In both cases, a small population of cells showing a higher fluorescence intensity is present and amounts to about 1%. For the RijR1 MCA, the position of the extra peak coincided with the peak position of a pure leukaemic cell population, suggesting that this peak represents the small number of leukaemic cells present in the mixture. In the case of the RM124 MCA, the position of the small extra peak was found in higher fluorescence intensity regions than was observed for pure leukaemic marrow.

The cells present in these extra peaks were sorted for morphological determination. They proved to be of leukaemic origin. At this stage of the study, the detection limit could already be

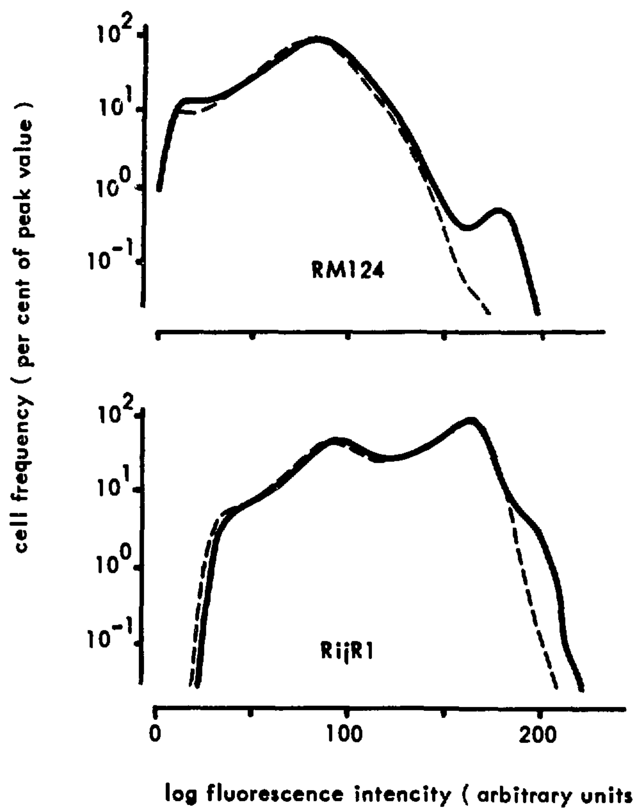

Figure 1: Fluorescence intensity profiles of a mixture of 1 leukaemic cell per 100 normal bone marrow cells compared with 100% normal bone marrow after labelling with RM124 and RijR1.

Table I

RESULTS OF CELL SORTING AFTER LABELLING OF VARYING MIXTURES OF LEUKAEMIC AND NORMAL BONE MARROW CELLS WITH RijR1 OR RM124

RijR1			RM124		
mixture LC/NC (percentage)	percent LC* after sorting	concentration factor	mixture LC/NC (percentage)	per cent LC* after sorting	concentration factor
5	76	15	ND	ND	-
2	84	42	ND	ND	-
1	ND	-	1	99	99
0.2	ND	-	0.2	91	455

LC: leukaemic cells (BNML)
NC: normal bone marrow cells
* : morphological discrimination

decreased by a factor of 15 to 455 (Table I). Obviously, this concentration factor depends mainly on the initial percentage of leukaemic cells. The cell sorter generally yields 90 to 100% purity if the target cells can be fully discriminated.

The number of cells required to perform a reliable morphological cell preparation analysis determines the total number of cells which have to be analysed, e.g., if 10 sorted cells are required from a mixture of 1 in 10^5, 1 million cells must be analysed. However, as we have shown that all cells in these extra peaks belong to the leukaemic population, the profile as such is indicative for the presence and quantity of the leukaemic cells. So far, we have examined mixtures ranging from 1 in 10 up to 1 in 1000 (leukaemic/normal cells) and in all instances an extra peak of fluorescence has been detected with the RM124 MCA, indicating that the detection limit can be decreased even further. Prior concentration of the leukaemic cells by, e.g., the density centrifugation method, will probably yield even better results. This subject remains a topic of ongoing research.

IMMUNOTHERAPY OF MINIMAL RESIDUAL DISEASE IN A RAT MODEL FOR HUMAN ACUTE MYELOCYTIC LEUKAEMIA

A. Hagenbeek and A.C.M. Martens

One of the major problems in today's treatment of acute leukaemia is to maintain a complete remission. The crucial question is how to eradicate minimal residual disease (MRD) after successful remission induction chemotherapy. In human acute myelocytic leukaemia (AML), high-dose chemoradiotherapy followed by bone marrow transplantation applied during the first complete remission seems to be a promising approach. It is expected that approximately 50% of the patients treated with this combined modality will be long-term disease-free survivors (e.g., Thomas et al., 1982). However, death from relapsing leukaemia after this treatment procedure remains a significant obstacle, the incidence ranging from 5 to 45% in the various clinical studies reported.

This report describes the results of nonspecific immunotherapy as "adjuvant" therapy after bone marrow transplantation in a rat model for human AML (BN acute myelocytic leukaemia; BNML). Two agents were studied: BCG and rat alpha interferon.

a) BCG

Brown Norway (BN) rats were inoculated with 10^7 BNML cells at day 0 (8 rats per group). At day 13, 100 mg cyclophosphamide per kg body weight was injected i.p. In one group, 7.0 Gy total body irradiation (TBI) was given 24 h later (X rays; Philips-Müller 300, 300 kV, 10 mA, HVL of the beam 3.0 mm Cu, dose rate 0.7 Gy per min). Subsequently, 6 x 10^7 normal isologous BN marrow cells were injected i.v. In the cyclophosphamide-TBI group, daily injections of 1 ml of packed cells were given as additional supportive care at days 17 to 21. At day 15, 3.0 mg BCG was injected i.v. Cyclophosphamide-TBI-treated rats and nontreated BNML rats served as controls.

Results are shown in Fig. 1. It appears that 90% of the rats which received BCG after prior cyclophosphamide-TBI treatment were long-term disease-free survivors. The one rat dying at day 84 died from cachexia (loose teeth) and showed no signs of leukaemia at autopsy. The survival times of rats treated with cyclophosphamide-TBI alone was significantly prolonged as compared with those of leukaemic controls, i.e., a median survival time of 56 versus 20 days. This increase in lifespan of 36 days corresponds with a 9 log leukaemic cell kill. Thus, the tumour load present at day 13, i.e., 5 x 10^9 cells, is reduced to 5 leukaemic cells. From

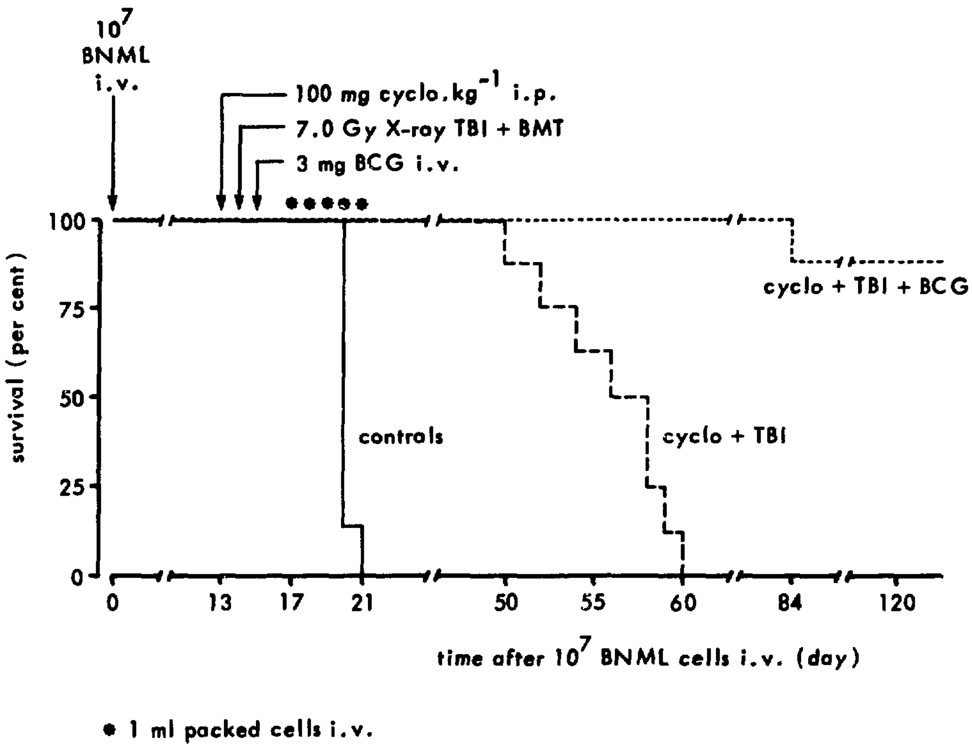

Figure 1: The efficacy of BCG in eradicating residual leukaemia after high dose chemoradiotherapy.
cyclo: cyclophosphamide; TBI: total body irradiation; BMT: bone marrow transplantation (6×10^7 normal isologous BN bone marrow cells, i.v.).
Each group consisted of 8 rats.

previous studies in nontreated leukaemic rats (Martens and Hagenbeek, 1978), it is known that maximal immunostimulation occurs around day 14 after BCG injection. If the tumour load is less than 10^3 cells at that time, cure is observed. This is confirmed in the present experiment, with the addition that BCG induces effective immunostimulation in the heavily pretreated, bone marrow transplanted rat. Apparently, the appropriate target cells responsible for this reaction are (still) present (T lymphocytes, macrophages, natural killer cells?).

b) Interferon

The efficacy of rat alpha interferon (type 1; obtained from Dr.H. Schellekens, Primate Center TNO) against MRD was tested in a similar experiment (Table I). The effect of repeated injections of interferon was much less pronounced as compared with that of BCG: 9% of the rats were cured. However, deaths from leukaemia relapse occurred at the same times as in the appropriate control group

Table I

EFFICACY OF RAT ALPHA INTERFERON (TYPE 1) IN THE TREATMENT OF MINIMAL RESIDUAL DISEASE AFTER HIGH DOSE CHEMORADIOTHERAPY FOLLOWED BY BONE MARROW TRANSPLANTATION

(BN acute myelocytic leukaemia)

	MST* (days)	range (days)	cures/total
cyclo + TBI	56	49-62	0/12
cyclo + TBI + interferon	58	55-90	1/11 (9%)
BNML controls	18	18-19	0/ 7

day 0: 10^7 BNML cells i.v. (7-12 rats per group)
day 13: cyclophosphamide, 100 $mg.kg^{-1}$ i.p.
day 14: 7.0 Gy X rays total body irradiation followed by 10^8 normal isologous bone marrow cells i.v.
Starting at day 15: daily 25,000 U interferon i.p. five times a week for 4 weeks; total dose: 500,000 U.
*MST: median survival time of rats dying from leukaemia.

(median survival time: 58 and 56 days, respectively). Possibly, modifications in the interferon treatment regimen (more frequent injections; higher dosages) will improve the results obtained so far.

It may be concluded that nonspecific immunotherapy (in particular - so far - BCG) to eradicate small numbers of surviving leukaemic cells proves to be effective even if applied after high dose chemoradiotherapy followed by bone marrow transplantation. These findings warrant further clinical investigations.

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ISOBOLOGRAMS IN CANCER COMBINATION CHEMOTHERAPY

J.H. Mulder

When two agents applied in a combination interact, the terms enhancement, augmentation, sensitization, potentiation, supra-additivity, therapeutic synergism, etc., are frequently used to indicate a positive interaction. The most common synonym for a negative interaction is antagonism. The simple concept of additivity (Valeriote and Lin, 1975) is based on either multiplication or summation of the effects caused by the agents separately. The effect of an A+B combination is the product of their individual survival fractions, provided that their dose response curves are straight lines where cell survival on a logarithmic scale is plotted against dose on a linear scale and provided that each agent acts independently of the other. In practice, however, dose response curves for cytostatic agents are generally not represented by exponential or linear curves. To decide whether or not the effects of drugs in a combination are additive is difficult and may even lead "the unsophisticated worker into errors" (Scott, 1961).

A more sophisticated way to compare the observed effect of a drug combination with the expected effect calculated by multiplication or summation of the effects of its constituents is to determine what dosage level of each drug provides the same quantitative effect (tumour cell kill or death of the animal). Then, in a X-Y plane with axes representing the dosages of each drug on a linear scale, a line can be constructed joining one dose of A and one of B and the combination of A + B that should have equal effect. Such an equieffective line calculated and constructed from dose response curves is termed an isobole. With agents giving linear dose response curves and which do not interact in a combination treatment, an isobole will be a straight line indicating that an additive effect of the drug combination should be expected. As many dose response curves are <u>non</u>linear, Steel and Peckham (1979) proposed an isobolar method that entails constructing an envelope of additivity rather than a single additive line. The envelopes in an isobologram represent equieffective regions.

In the experiments reported here, additivity envelopes were constructed for the combination of 5-fluorouracil (5-FU) and methotrexate (MTX) in the treatment of mice bearing L1210 leukaemia. The previously claimed antagonism of 5-FU + MTX combination chemotherapy on the basis that the determined combined response was smaller than the algebraic sum of the individual responses of the agents is questioned (Mulder et al., 1979).

In Fig. 1, an additivity envelope is constructed for an isoeffect of an increase of life span (ILS) of 6 days as follows. The choice of 6 days is based on the ILS obtained after treatment with a 5-FU + MTX schedule. Using the dose response curves of 5FU and MTX, five 5-FU + MTX combinations that would produce an ILS of 6 days, assuming that 5-FU and MTX did not interact, were compared (Table I). The calculated values for these combinations were used for constructing the envelope in Fig. 1.

As an example, a dose of 90 $mg.kg^{-1}$ 5-FU was selected from the dose response curve. This dosage increased the mouse survival by 4 days. We then determined from the dose response curve of MTX the dose of MTX that would alone give an ILS sufficient to make up the difference between the 4 days of 5-FU and the 6 days selected. So, a dose of MTX corresponding to 2 days had to be "found". As the

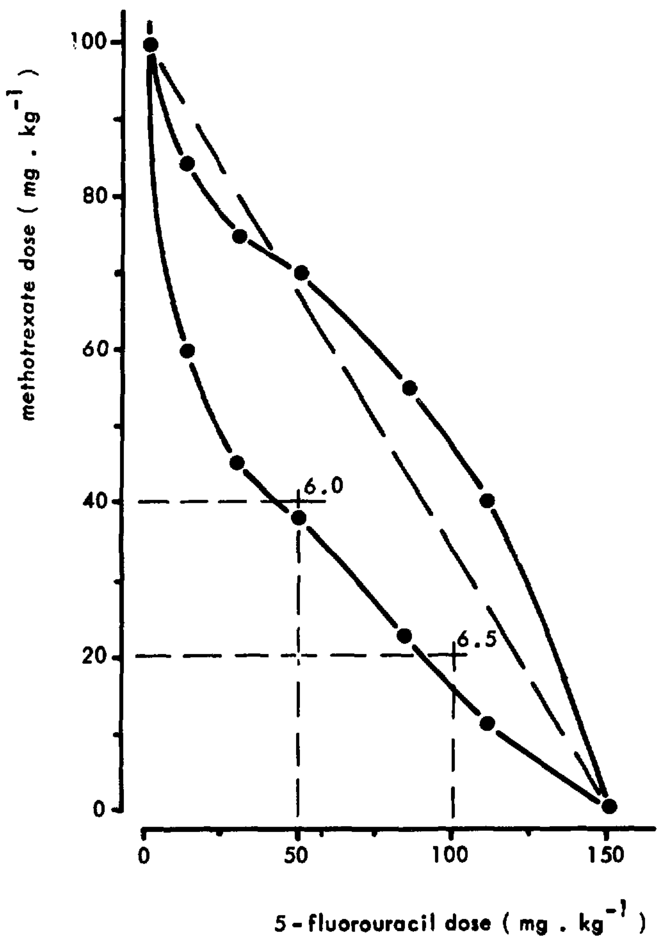

Figure 1: An additivity envelope is constructed for an isoeffect of 6-day ILS. The observed effects of two applied 5-FU + MTX combination therapies fall within this envelope, indicating an additive interaction.

dose response curve of MTX shows a plateau, the determination of the required complementary dose of MTX depends on which part of the MTX dose response curve is used. A low value for MTX dosage is obtained if the complementary dose needed for an ILS of 2 days is taken from the low dose range of the curve, in contrast to a high value obtained in the high dose (plateau) region of the curve. As presented in Table I, the two MTX dosages needed for a 2-day ILS were 23 and 55 $mg.kg^{-1}$, respectively.

Table I

DETERMINATION OF 5-FU AND MTX VALUES FOR CONSTRUCTING ONE ADDITIVITY ENVELOPE INDICATING AN ISOEFFECT REGION OF 6 DAYS ILS

5-FU dosages ($mg.kg^{-1}$) with corresponding ILS (days)	complementary dosages of MTX corresponding with complementary ILS values
150 results in 6 days ILS	no MTX needed
110 " " 5 " "	12 and 40 needed for 1 additional day ILS
90 " " 4 " "	23 and 55 " " 2 " " "
50 " " 2.5 " "	39 and 70 " " 3.5 " " "
30 " " 2 " "	45 and 75 " " 4 " " "
12 " " 1 " "	60 and 85 " " 5 " " "

For any selected dose of 5-FU (see Table I), there are two values for MTX. Thus, over a range of five values of 5-FU, the corresponding two values for MTX define an envelope-shaped area, and all 5-FU + MTX combinations represented by points within this envelope are said to be additive. The next step was to indicate in this isobologram the in vivo dosages tested in the combination which resulted in an ILS of 6 days. Combinations of 5-FU + MTX resulting in vivo in an ILS of 6 days and lying above the constructed envelope of additivity are said to be antagonistic, as they required in vivo more of the agents to give an ILS of 6 days.

From the data presented in Fig. 1, the observed effect of the two tested combinations lie within or close to the lower edge of the 6-day isoeffect envelope. Consequently, the determined effect of 5-FU + MTX therapy in the dosages tested should be regarded as additive. By applying the simple concept of additivity, an antagonistic effect of 5-FU + MTX therapy was incorrectly suggested (Mulder et al., 1979), as both determined effects resulted in effects far less than were obtained by summation of the effects.

In conclusion, the earlier claimed 5-FU/MTX antagonism can not be validated when the more sophisticated method of analysing the experimental data as described by Steel and Peckam (1979) is used.

The therapeutic significance of interactions between cytostatic agents observed in animal experiments require careful consideration before extrapolations to clinical therapy are made. When increase in the life span of tumour bearing animals is the endpoint of drug effectiveness, both tumour cell kill and critical normal tissue damages may be included in a single parameter. Goldin's "therapeutic" synergism, i.e., the ability of the drugs in combination to produce a therapeutic response superior to the maximum response to either drug alone, and the above-discussed interaction between different agents with respect to a single specified effect are two very different terms, since, even with 5FU + MTX combination chemotherapy, therapeutic synergism may be found regardless of whether or not both 5-FU and MTX show interaction at the cellular level (Goldin et al., 1954).

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PHARMACOKINETIC STUDIES OF DAUNOMYCIN IN THE RAT

K. Nooter, P. Sonneveld, J. Deurloo and R. Oostrum

Anthracycline antibiotics, originally isolated from Streptomyces species, exhibit a high cytotoxic activity against normal and neoplastic cells. At present, two such drugs, daunomycin and adriamycin, are widely used clinically in cancer chemotherapy. For purposes of devising rational drug dosimetry schedules in patients, the pharmacokinetic parameters of daunomycin in the laboratory rat were studied. Daunomycin and its major metabolite daunomycinol were determined by High Performance Liquid Chromatography technology (Baurain et al., 1979). BN rats were treated with daunomycin according to the EORTC-LAM 6 protocol for acute myeloid leukaemia (one i.v. bolus injection of 7.3 $mg.kg^{-1}$ body weight for 3 successive days). According to Freireich et al. (1966), a dose of 7.3 $mg.kg^{-1}$ body weight in the rat is comparable with a dose of about 40 $mg.m^{-2}$ in man. At different time intervals after administration of the drug, biological fluids (plasma, urine and bile) and organs (heart, lung, liver, kidneys, skeletal muscle, thymus, spleen and bone marrow) were collected.

The lungs and kidneys showed the highest peak concentration of daunomycin per gram of tissue (30 to 40 µg) after one i.v. bolus injection, followed by the heart, liver and spleen (18 to 24 µg per gram). The smallest amount of drug was found in the bone marrow, thymus and muscle (4 to 10 µg per gram). Drug induced congestive heart failure is a well-known limiting factor in patient treatment with anthracyclines. Our data show that this cardiotoxicity is not caused by excessive drug uptake by the heart muscle as compared with other tissues.

Plasma concentration time curves were characterized by a very rapid distribution phase. At two hours after injection, the plasma levels for both daunomycin and daunomycinol were in the 100 $ng.ml^{-1}$ range. A second and third injection caused only a moderate increase in daunomycin and daunomycinol plasma steady state concentrations. The 24-hour urinary excretions for daunomycin for the first, second and third injections were 3.4%, 3.7% and 3.8% of the total administered dose, respectively and those for daunomycinol were 4.7%, 5.6% and 5.7%, respectively. Apparently, the urinary drug excretion is only slightly affected by successive daunomycin administrations.

The 24-hour bile excretions for the first, second and third injections were for daunomycin 12%, 28% and 19% of the total administered dose and for daunomycinol 13%, 15% and 17%.

A typical example of a series of drug concentration time relationships in kidney tissues after one, two and three i.v. bolus injections of daunomycin is shown in Fig. 1. Two successive administrations of the drug led to higher peak and steady state concentration of daunomycin. This accumulation phenomenon is seen in all organs studied and might be explained by, e.g., a facilitated cellular drug uptake induced by the first daunomycin injection. But, then it is expected that urinary and/or billary excretions are decreased after the second and third drug administrations. This is clearly not the case. As the third daunomycin injection results in only a slight increase in drug concentration, it is more likely that an unknown saturable drug depot from which drug release is very slow plays a role. At the time of the second daunomycin injection, this depot can no longer be filled and the drug concentration to be taken up by the other organs is much higher than after the first injection. In addition to the aforementioned organs, we also determined drug concentrations in the gut, intraperitoneal fat and brain tissue. The brain does not take up daunomycin and when taken up by the gut and intraperitoneal fat

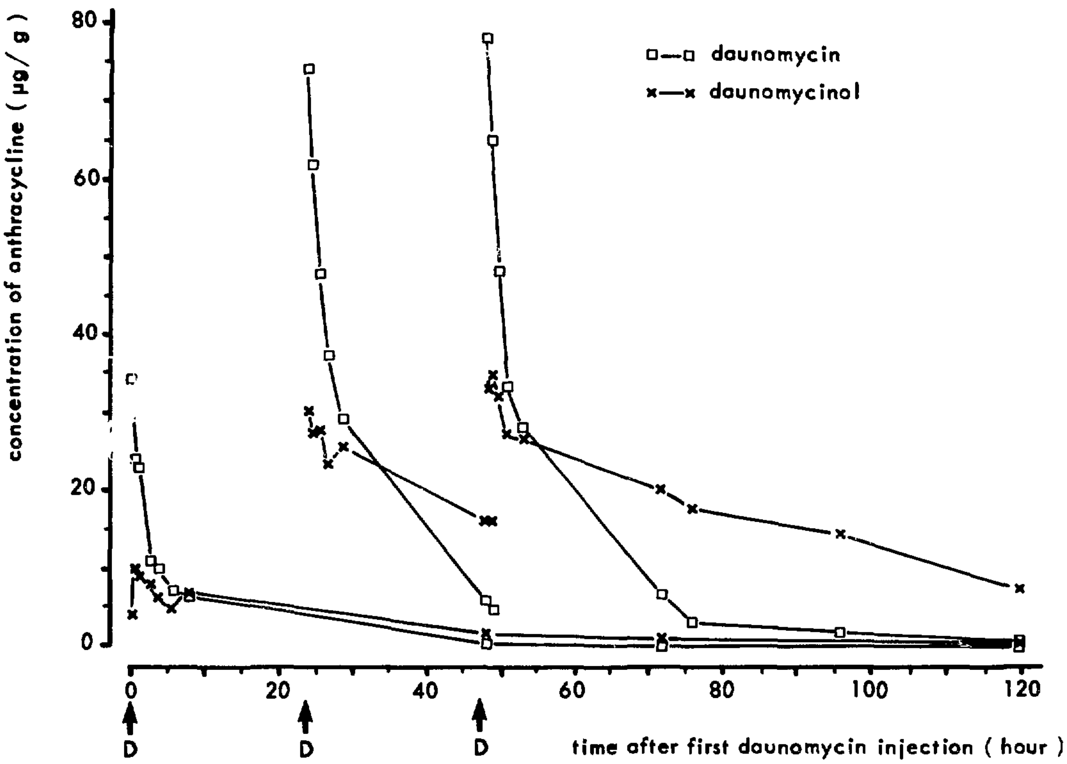

Figure 1: Time concentration relationship in kidney tissue after one, two and three successive i.v. bolus injections (7.3 mg per kg body weight) of daunomycin in the rat. The arrows indicate time of daunomycin injections.

daunomycin shows a disappearance behaviour comparable with that found for, e.g., kidney tissue.

The shape of the disappearance curves as shown for kidney in Fig. 1 is typical for organs such as the heart, lungs and liver. Fig. 2 shows a typical time concentration curve for haemopoietic tissues such as the spleen, bone marrow and thymus. Remarkable are the daunomycin uptake phase in the first hours after injection and the long lasting daunomycinol accumulation, which could be responsible for the myelotoxicity generally found in anthracycline treated patients. The data obtained so far are encouraging for further study on the pharmacokinetic behaviour of anthracyclines in the laboratory rat.

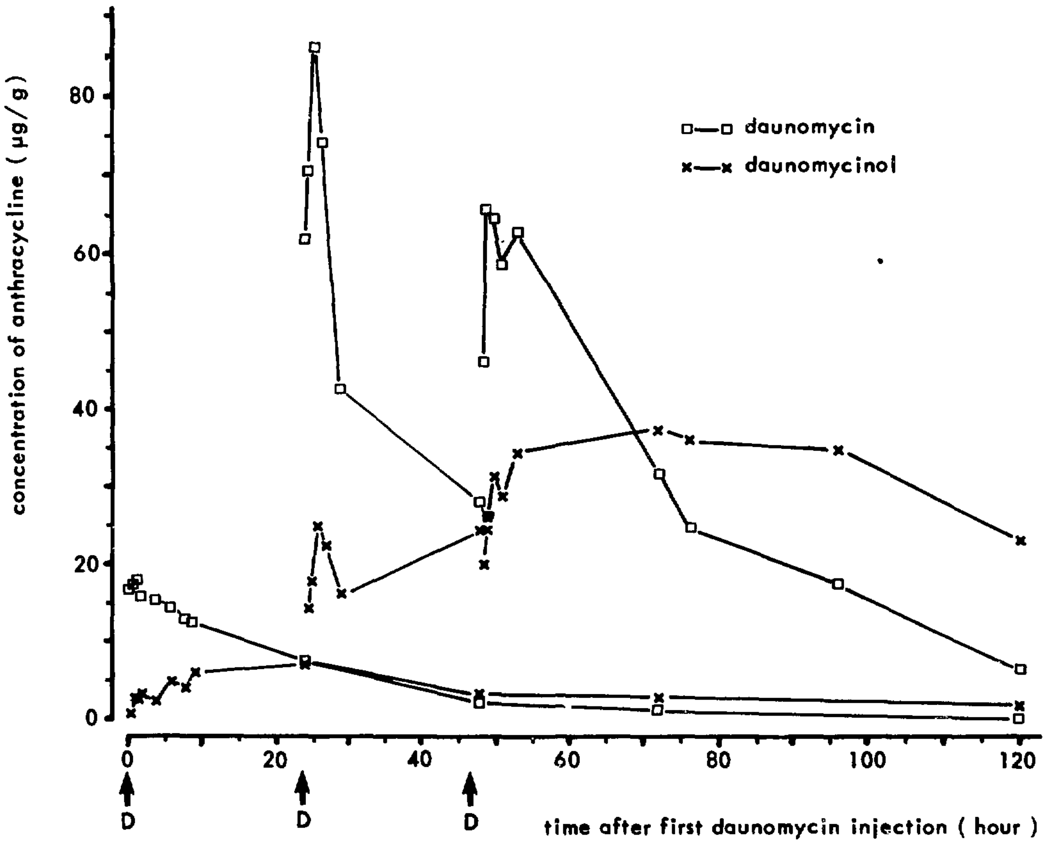

Figure 2: Time concentration curve in spleen tissue. The arrows indicate time of daunomycin injections.

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EVALUATION OF COMPARTMENTAL MODELS FOR THE DISTRIBUTION OF ADRIAMYCIN AND DAUNOMYCIN IN THE RAT

F.W. Schultz, J.A. Mulder, K. Nooter and P. Sonneveld

Information on the magnitudes and durations of cytostatic drug levels in various organs is important with respect to toxicity problems and therapeutic effectiveness. Both normal BN/BiRij rats and rats suffering from myeloid leukaemia (BNML) were given a single i.v. dose of adriamycin (ADM) or daunomycin (DAU) (7.5 $mg.kg^{-1}$). The concentrations of the drugs and their metabolites were determined in several organs and body fluids by means of a high performance liquid chromatography method (Baurain et al., 1979).

A drug can be transported and/or metabolized along indicated paths (Fig. 1), presumably according to first order kinetics. The amount of drug going from one compartment to another per unit time must be proportional to the amount of drug present in the former. Drug transport is described by a set of first order linear differential equations in which the proportionality constants are the parameters. By solving these model equations simultaneously for any set of parameter values, a model output is calculated: drug concentration-time curves for each compartment. The determined drug concentration-time data points represent the system output. An algorithm has been designed that searches for the optimum parameter values by minimization of a maximum likelihood criterion function [the performance index PF (Sonneveld et al., 1981)]. It takes into account measurement errors that are distributed at random according to a gaussian probability density function. The extent of the correspondence is expressed by the Total Correlation Coefficient (TCC), which is a measure of goodness of fit of a model response to a given set of observations.

<u>ADM in the BN rat (0 to 24 hours after bolus dose)</u>

Metabolism is ignored, as no ADM metabolites could be detected in significant quantities. Enterohepatic recirculation is not modeled either (Fig. 1a). It was at first assumed that drug excretion into urine is dependent on kidney concentration. Then, the optimum model output (Table I; run LC41) fits the organ data sufficiently well (TCC = 0.989 to 0.994) but predicts plasma concentrations that lie far above the data points (TCC = 0.259). A model structure, substituting the kidney-urine for the plasma-urine path, results in a considerable improvement of the plasma fit (TCC = 0.625) and a slightly better urine fit, while the pre-

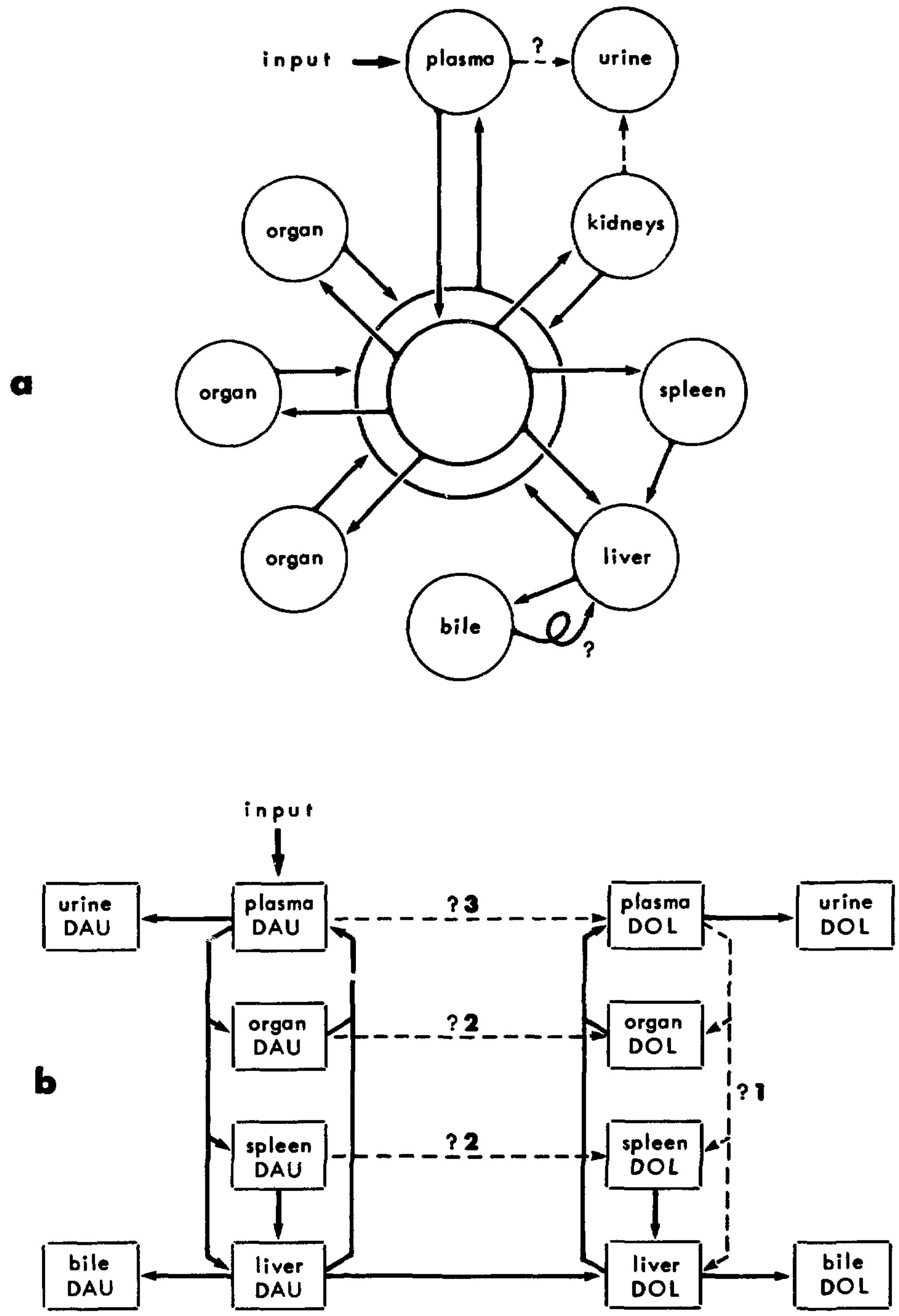

Figure 1: Model structure for a) ADM distribution; b) DAU and DOL distribution.

Table 1

COMPARISON OF MODEL OUTPUTS

run	LC41	MS02	BN02	BN55	ML10	DN11		DN35	
PF_{min}	85.00	88.37	95.00	94.96	87.49	1690.74		133.94	
TCC						DAU	DOL	DAU	DOL
whole model	0.986	0.985	0.902	0.957	0.916	0.926		0.948	
1 plasma	0.259	0.625	0.503	0.554	0.850	0.832	0.000	0.782	0.000
2 liver	0.989	0.982	0.986	0.992	0.993	0.978	0.663	0.977	0.952
3 spleen	0.994	0.992	0.994	0.992	0.970	0.986	0.828	0.989	0.925
4 heart	0.990	0.990	0.991	0.990	0.994	0.994	0.871	0.985	0.884
5 kidneys	0.991	0.981	0.989	0.993	0.962	0.997	0.908	0.997	0.932
6 lungs	0.980	0.980	0.979	0.980	0.982	0.907	0.868	0.901	0.921
7 tissue	-	-	-	-	-	-	-	-	-
8 bile	-	-	0.986	0.978	0.980	-	-	-	-
9 urine	0.966	0.974	0.980	0.970	0.986	0.947	0.000	1.000	0.981
10 bone marrow	-	-	0.896	0.973	0.839	0.895	0.600	0.905	0.942
11 muscles	-	-	-	-	-	0.929	0.909	0.945	0.963

LC41 BN ADM 0 to 24 hours, based on relation : kidney concentration-urine excretion.
MS02 BN ADM 0 to 24 hours, based on relation : plasma concentration-urine excretion.
BN02 BN ADM 0 to 24 hours, plasma volume as in MSo2: 5.70 ml
BN55 BN ADM 0 to 24 hours, adjusted plasma volume : 28.50 ml
ML10 BNML ADM 0 to 24 hours
DN11 BN DAU 0 to 48 hours, 1 metabol. rate; pathways 2 only
DN35 BN DAU 0 to 48 hours, various rates; all pathways 1, 2, 3 included

Fig. 1b

$$TCC = \mathrm{sqrt}\ [1 - (x_{obs} - x_{model})^2 / (x_{obs})^2]$$

TCC = 0: no fit at all; TCC 0.98: acceptable fit; TCC = 1: perfect fit diction for the other compartments remains about constant (run MS02).

Considering free exchange with total extracellular body water the volume of the plasma compartment was accordingly enlarged. Comparison of runs BN02 and BN55 reveals that the optimum model output for the latter run shows better fit for liver, kidneys and lungs but poorer fit for spleen and heart, a somewhat poorer fit for urine and bile, but a much better respectively slightly better fit for the bone marrow and plasma compartments.

ADM in the BN rat with myeloid leukaemia (0 to 24 hours after bolus dose)

The optimum output of the same model, that predicts the ADM distribution in the BNML rat (run ML10), agrees rather well with the determined data points in most organs. However, the prediction starts too low for the plasma data and the measurements in the bone marrow increase with time while the prediction decreases.

DAU in the BN rat (0 to 48 hours after bolus dose)

To explain the determined DAU data points, the same model as used for ADM was applied, except that a separate compartment was introduced for muscles. Furthermore, significant metabolism to Daunomycinol (DOL) takes place in the BN rat. DOL was detected in all organs. Therefore, DOL compartments had to be added (Fig. 1b).

A few questions are: a) do all organs metabolize or only the liver?; b) do all this to the same extent?; c) does it occur in plasma?; d) does redistribution, plasma DOL-organ DOL, take place? As an illustration, Table I shows the optimum model response for two cases: run DN11 (metabolism with equal rates in all organs, not in plasma; no redistribution) and run DN35 (metabolism in all organs, plasma included, with different rates; redistribution). With the latter model better fits are achieved. However, with neither model could a satisfactory fit to plasma DAU data and to DOL data in general, be obtained. The assumption of metabolism being a zero order process, i.e., a constant amount is transformed per unit time, was tested, but this led to even poorer results. The conclusion so far is that the drug distribution processes are mainly first order ones. Some other mechanisms must be included to improve the fit in the plasma and the metabolite compartments.

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PHARMACOKINETICS OF CYCLOSPORIN A IN KIDNEY ALLOGRAFT PATIENTS

P. Sonneveld, K. Nooter and W. Geerlings*

Cyclosporin A (CyA) is a powerful immunosuppressive drug which is presently used in many clinical protocols to prevent allograft rejection (kidney transplants) or to antagonize Graft versus Host disease (bone marrow transplants). The therapeutic use of CyA, however, is severely limited by the nephrotoxicity of the drug which results in progressive renal insufficiency. This effect in particular limits its applications in kidney transplantation.

CyA is a lipophilic cyclic undecapeptide which is metabolized to approximately 28 metabolites and is excreted in the urine. Therefore, a reliable method is necessary to quantify the pharmacokinetics of CyA and to assure that plasma concentrations remain within certain limits. Up to now, a radioimmunoassay (RIA) has been mostly used for monitoring the plasma disappearance of CyA. However, this assay does not discriminate between CyA and its

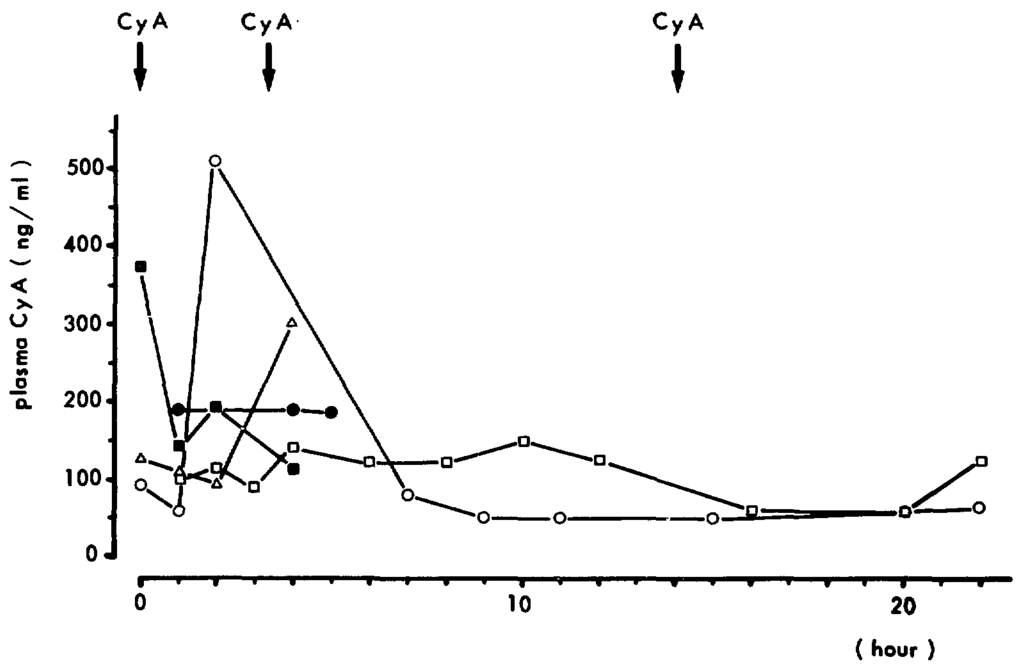

Figure 1: Plasma concentrations of Cyclosporin A in kidney transplant patients during treatment, as determined by HPLC.

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metabolites. Therefore, a High Pressure Liquid Chromatography (HPLC) method was developed for routine plasma monitoring of CyA in humans. Four patients were studied and the results were compared with RIA determinations. All patients had received kidney allografts and had been treated with CyA three times daily for at least one week.

Following oral administration of CyA (150 mg, 3 times daily), an ample variation in plasma concentrations with peak levels of 120 to 500 $ng.ml^{-1}$ was observed (Fig. 1). CyA rapidly disappears from the plasma, with variable plasma half lives. During the elimination phase, plasma concentrations of 50 to 200 $ng.ml^{-1}$ were found. The mean plasma concentration during a 24-hour observation period was 151 $ng.ml^{-1}$, which is well within the range of "optimal" therapeutic plasma levels (100 to 400 $ng.ml^{-1}$). However, considerable nephrotoxicity which could only be attributed to CyA occurred in these patients. Under CyA treatment, a deterioration of renal function was observed in all patients. Thus, the "optimal" plasma concentration in these patients must be reevaluated.

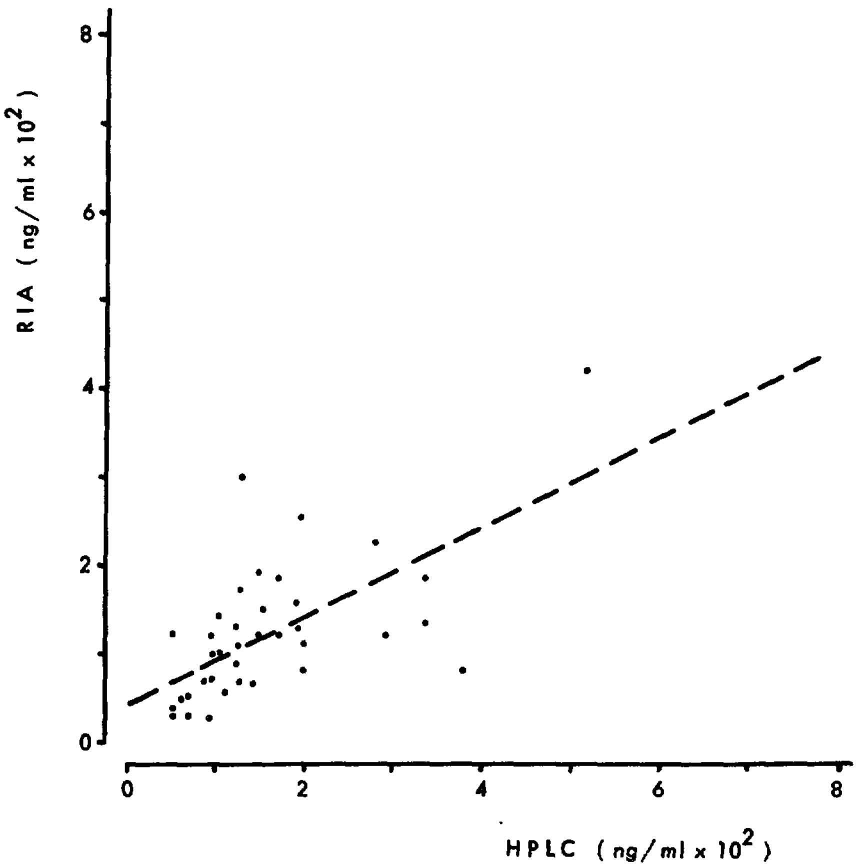

Figure 2: Plasma concentrations of Cyclosporin A as determined by HPLC and compared with radioimmunoassay determination.

The results of HPLC analysis were compared with those of RIA determinations. No correlation between the two methods was evident (Fig. 2), although HPLC seemed to measure higher concentrations than did RIA. The lack of correlation can be attributed to two mechanisms: firstly, RIA determines simultaneously both parent drug and metabolites, while HPLC determines each compound separately; secondly, RIA determines only the free non-protein-bound plasma fraction of CyA, while HPLC determines the total plasma concentrations. In view of the considerable toxicity of CyA, the mechanism of which is unclear, it seems desirable to monitor patients with HPLC instead of RIA, since metabolites can also be analysed in that way. In addition, since a physical method of analysis is used, this method is more reproducible.

ANTITUMOUR EFFECTIVENESS OF 1-HOUR VERSUS CONTINUOUS DRUG EXPOSURE DETERMINED BY AN IN VITRO CLONOGENIC ASSAY

P. Lelieveld, W. Akkerman, J.H. Mulder
and B. van der Vecht-de Jong

The in vitro clonogenic assay in soft agar medium can be a valuable tool in predicting the sensitivity or resistance of tumour cells to cytostatic drugs. As a standard procedure, tumour cells are exposed to drugs for 1 hour, then washed to remove the drug, resuspended in soft agar medium, plated in Petri dishes and cultured in order to grow into colonies (Salmon et al., 1978). The drug exposure time of 1 hour has been an arbitrary choice. This time period, however, may be suitable for, e.g., alkylating agents, while, for cell cycle phase specific drugs, longer exposure times may be necessary to determine an effect on clonogenicity. Continuous drug exposure, i.e., drug exposure during the entire period of incubation for colony formation, has the advantage of simplicity. We have compared the effect of 1-hour versus continuous exposure to determine the drug sensitivity of the experimental mouse tumours leukaemia L1210, osteosarcoma C22LR and Lewis lung. The following drugs were tested against all three tumours using both time durations of exposure: melphalan, BCNU, adriamycin, cis-platin, 5-fluorouracil, methotrexate, vinblastine and vincristine.

Drug exposure for 1 hour was carried out as follows. One million tumour cells were incubated at 37°C for 1 hour with appropriate concentrations of the drugs in Hepes-buffered Hanks' balanced salt solution without serum (L1210) or with 5% fetal calf serum (osteosarcoma, Lewis lung). In order to remove extracellular drug the drug-incubated cells were diluted at least 1000 times. Then, 100 L1210, 1000 osteosarcoma and 500 Lewis lung cells were plated in each Petri dish. Continuous exposure to the drugs was achieved by adding appropriate concentrations of the drugs to the cells in Petri dishes containing soft agar medium. The culture medium consisted of Dulbecco's medium supplemented with 60 $\mu mol.l^{-1}$ 2-mercaptoethanol, 20 $mg.l^{-1}$ L-asparagine, 0.3% bacto-agar (Difco), 15% (Lewis lung) or 20% horse serum (L1210, osteosarcoma) and, except for L1210, rat erythrocytes in a final dilution of 250 times from whole rat blood. The dishes were incubated at 37°C in an atmosphere of 10% CO_2 in humidified air (L1210) or 5% oxygen (osteosarcoma, Lewis lung). The colonies were counted after 8 (L1210) or 14 days (osteosarcoma, Lewis lung). Plating efficiencies for cultures of these lines are 60, 15 and 30% for

L1210, osteosarcoma and Lewis lung cells, respectively. For each drug, the dose which caused 50% inhibition (ID50) of the number of colonies relative to colonies arising from untreated cells was calculated.

The results for adriamycin are given in Table I. A 13 to 15-fold higher drug concentration is necessary to produce the same inhibition of colony formation for the 1-hour as compared with the continuous exposure. For the other drugs, only the ratios are indicated (Table II). High to very high ratios were obtained for antimetabolites such as 5-fluorouracil and methotrexate and the spindle poisons vinblastine and vincristine. These cell cycle phase specific drugs exert their main action in a restricted phase of the cell cycle. Consequently, an exposure of cells to these drugs for 1 hour limits their effectiveness. Continuous exposure to these drugs should be more effective, as was demonstrated.

Table I

INHIBITION OF COLONY FORMATION IN VITRO AFTER 1-HOUR AND CONTINUOUS EXPOSURE OF THREE EXPERIMENTAL TUMOURS TO ADRIAMYCIN

	50% inhibition dose ($ng{\cdot}ml^{-1}$)*		
tumour	1-hour exposure	continuous exposure	ratio
leukaemia L1210	356	24	15
osteosarcoma C22LR	257	19	14
Lewis lung	222	17	13

*median values from at least 2 experiments per tumour.

The ratios for melphalan and for BCNU on L1210 are smaller than 1, indicating that continuous exposure is less effective than a 1-hour exposure time. For melphalan, this discrepency may be explained by differences in assay procedure between 1-hour and continuous drug exposure. Immediately after preparation of the melphalan solution, the tumour cells in the 1-hour series are exposed to the drug. This is in contrast to the procedure for the continuous exposure series, where this takes place at about 90 min later. During this period, rapid hydrolysis might have reduced the quantity of active drug (Furner and Brown, 1980). To test this hypothesis, a solution of melphalan was incubated at 37°C for 90 min and then used for a 1-hour treatment of leukaemia L1210 cells.

Table II

RATIOS OF ID50 VALUES AFTER 1-HOUR AND CONTINUOUS EXPOSURE OF THREE EXPERIMENTAL TUMOURS TO CYTOSTATIC DRUGS*

drug	leukaemia L1210	osteosarcoma C22LR	Lewis lung
	1-hour ID50 / continuous ID50		
melphalan	0.2	0.1	0.7
BCNU	0.9	7	14
adriamycin	15	14	13
cis-platin	23	37	28
5-fluorouracil	47	54	157
methotrexate	3797	296	1470
vinblastine	1146	614	651
vincristine	366	324	663

*for each exposure time, at least two experiments were carried out for each drug and each tumour.

Now, a ratio of 1.6 was obtained, supporting the theory of rapid inactivation of melphalan by hydrolysis.

There is good general agreement between the results obtained with the three tumour lines. Only for BCNU is the value for L1210 deviating. For L1210, the 1-hour exposure is carried out without serum, while, for the other tumours, serum is present during the 1-hour drug treatment. This may be a reason for a low value for L1210, although preliminary data do not support this, as addition of serum to the assay for L1210 did not significantly change the ratio.

In summary, the clonogenic soft agar medium assay is a widely used technique for predicting the sensitivity or resistance of human tumours. The standard technique employs drug exposure for 1 hour. Using experimental mouse tumours, we have shown that this exposure time may be too short for antimetabolites and other cell cycle phase specific drugs. Continuous exposure of tumour cells to drugs of these two classes seems a good alternative. An additional advantage of continuous exposure is the simplicity of the assay, which also suggests its preferential use for screening new drugs in vitro.

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HUMAN TUMOUR RESPONSE TO CIS-PLATIN ANALOGUES IN THE MODIFIED SALMON ASSAY

J. de Ruiter, R.J.F. Middeldorp and L.M. van Putten

The cloning of human tumour cells in soft agar in order to predict sensitivity to chemotherapy is presently performed in a large number of oncology centres in the world. The general tendency is that about thirty percent of the tumours taken directly from a patient will give both a sufficient number of tumour cells to test a few cytostatic drugs and clonogenic growth.

The aim of our study was to compare the in vitro sensitivity of several human tumours to a number of cis-platin analogues. For that reason we started to study the in vitro growth of a panel of human tumours which were growing in nude mice. The method as described by Hamburger and Salmon (1977) was used, with some slight modifications. Initially tumour growth was poor and we therefore tested some possible growth promoting additions. Courtenay and Mills (1978) described the growth promoting effect of the addition of rat erythrocytes in their clonogenic assay. We therefore tested whether a similar growth promotion was obtained in the clonogenic assay according to Hamburger and Salmon. The results after addition of 0.006 ml of erythrocytes of BN/Bi rats in the underlayer of the cell cultures are shown in Table I. For these

Table I

EFFECT OF ADDITION OF RAT ERYTHROCYTES TO TUMOUR CLONOGENIC GROWTH IN THE SALMON ASSAY

type of human tumour	no. of experiments	ratio: plating efficiency with ery's / plating efficiency without ery's
ovarian tumours P, L and S	9	2.5 (0.6 - 5.9)*
colon tumours H and K	5	1.7 (0.8 - 5.0)

The median plating efficiency was 0.09% and varied between 0.02% and 1.4%

* not included in the data are the results of three experiments in which no growth was observed in control dishes. In two of these experiments tumour growth was observed if erythrocytes were added.

experiments, one primary colon tumour was used and the other tumours were obtained after 3 to 6 passages in nude mice. No growth in control dishes without erythrocytes was observed in three experiments with ovarian tumours. In two of these three experiments the tumour did grow after the addition of erythrocytes. In most other experiments, the addition of erythrocytes increased the plating efficiency. Although the increase in plating efficiency was not sensational, we decided to use the addition of erythrocytes as a routine in our assay.

The next point to verify was whether drug sensitivity should be tested with the drug continuously present in the culture dish during the two weeks of culture or just during one hour of incubation before culture. The results as presented in Table II were analysed by determination of the ID70 dose, i.e., the drug dose which decreased the growth of colonies by 70 per cent. It is evident that the ID70 values are much higher for 1 hour incubation than for continuous exposure, as could be expected. A number of investigators conclude that the tumour is sensitive to a certain drug if the ID50 or ID70 is lower than one-tenth of the peak plasma concentration. If we use this way of analysis, it can be concluded that, with the continuous exposure, both tumours are sensitive to all three drugs tested. After 1 hour incubation, both tumours are insensitive to all three drugs. Since no data on the clinical response of the tumours are available, it is difficult to

Table II

ID70 ($ng{\cdot}ml^{-1}$) DOSE OF CIS-PLATIN ANALOGUES FOR TWO HUMAN TUMOURS IN THE SALMON ASSAY

drug	continuous exposure		1 hour incubation		1/10th peak plasma concentration
	colon tumour	lung tumour	colon tumour	lung tumour	
CHIP	$4x10^2$ (+)	$6x10^2$ (+)	$2x10^4$(-)	10^5(-)	10^3
CBDCA	$2x10^3$ (+)	$2x10^3$ (+)	$3x10^4$(-)	$2x10^5$(-)	$5x10^3$
Cis-platin	90 (+)	$4x10^2$ (+)	$3x10^4$(-)	$6x10^4$(-)	10^3
TNO-6	$3x10^2$ (+)	$2x10^2$ (+)	-	-	$3x10^2$

(+) ID70 dose   1/10th peak plasma concentration.
(-) ID70 dose   1/10th peak plasma concentration.

conclude which method provides the best prediction of drug sensitivity. The choice of the height of the threshold concentration (in most studies, one-tenth of the peak plasma concentration) is of course very important for the conclusion of sensitivity or not. Although the threshold concentration used might theoretically be the best approximation to the condition in patients, a more pragmatically obtained threshold might provide a better prediction. In the pharmacokinetic analysis of platinum derivatives, it has been established that platinum occurs in the plasma in two chemical forms: protein-bound and free. The protein-bound platinum is not biologically active; therefore, there is some doubt whether the parameter should be based on total platinum rather than on free platinum, which is present in a lower concentration. Such an observation makes it difficult to use the peak plasma concentration as an unambiguous parameter. Additional data on the merits of short versus continuous exposure to Cis-platin are discussed on p. 99 We plan to obtain in vitro results on more tumours and compare these results with in vivo data in the subrenal capsule assay (p. 107 and p. 111).

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COMPARATIVE EVALUATION OF NORMAL MICE, PRETREATED IMMUNOCOMPETENT MICE AND NUDE MICE IN THE SUBRENAL CAPSULE ASSAY; OPTIMALIZATION OF THE METHOD

M.B. Edelstein and T. Smink

The subrenal capsule assay developed by Bogden et al. (1981) for use in normal mice offers several advantages in the determination of human tumour drug sensitivity. It is rapid, providing an answer within 6 days, it is reliable, giving information in ca. 75% of tumours tested, and it provides accurate information (Griffin et al., in press) in ca. 80% of cases studied so far. In addition, it may provide more relevant information than in vitro assays, since the effects of drug metabolites can be determined. Recently, however, we have found that the use of normal mice presents problems in the evaluation of tumour growth. Infiltration by host cells (lymphocytes, macrophages and fibroblasts) which is apparent histologically makes measurement of the "tumour" mass difficult not only for the untreated control but also for groups treated with adriamycin (ADM) and cis-platinum (cis-Pt) (Edelstein et al., in press). Because cyclophosphamide (Cyclo) treated groups demonstrated no host cell infiltration, pretreatment of the normal mice was begun with various forms of immunosuppression. Two types of control were used: immune competent normal mice and immune deficient nude mice. As immunosuppressive treatments, Cyclo, 4.0 Gy gamma whole body irradiation, cortisone acetate and silica were used at 24 hours before tumour implantation under the renal capsule. X-ray irradiation (4.5 Gy) at 4 hours before tumour implantation was also tested. A control group, an ADM and a cis-Pt treated group were used for the testing of two ovarian tumours coded XOv.P and XOv.S and carried as early passage tumours in nude mice.

The results of the different forms of pretreatment in the various treated groups can be seen in Table I, with relative tumour size (surface area) and the estimated distribution of component cells in a cross-sectional coupe of the tumour-kidney complex indicated for each group. Experimental groups consisted of 5 animals, per control group 10 mice were used. Using microscopic analysis, estimations were made by comparing the fractions of the various cell types. Unpretreated normal mice exhibited tumours which showed consistently (ca. 80%) host cell infiltrate, consisting of immune cells (I) and fibroblasts (F) in contrast to the same tumours in nude mice, where ca. 90% of the mass of tissue measured was tumour (T). Silica pretreatment does not differ from

Table I

EFFECTS OF DIFFERENT FORMS OF PRETREATMENT ON THE TUMOUR SIZE AND CELLULAR COMPOSITION OF TWO OVARIAN TUMOURS IMPLANTED UNDER THE RENAL CAPSULE

			XOv.P				XOv.S			
			relative tumour*	per cent			relative tumour*	per cent		
mouse	pretreatment	treatment	size	T**	I**	F**	size	T	I	F
nude BALB/c	none	control	1.67	90	-	10	2.21	90	-	10
		Cyclo	1.59	90	-	10	2.10	70	-	30
		cis-Pt	1.12	90	5	5	1.27	90	-	10
		ADM	1.40	90	5	5	1.53	90	-	10
normal BCBA/F1	none	control	1.78/2.05	15	85	-	3.08/1.75	-	100	-
		cis-Pt	1.75/2.55	10	90	-	2.53/1.33	-	100	-
		ADM	2.01/2.36	10	90	-	2.83/1.50	30	70	-
	- 24 hours	control	1.70	90	-	10	1.18	50	-	50
	200 $mg.kg^{-1}$	cis-Pt	0.81	85	-	15	1.31	80	-	20
	Cyclo i.v.	ADM	0.88	90	-	10	0.93	90	-	10
	- 24 hours	control	2.18	60	40	-	2.07	60	35	5
	4.0 Gy whole body	cis-Pt	1.02	85	5	10	1.55	70	-	30
	gamma irradiation	ADM	1.36	90	5	5	0.95	90	10	-
	- 4 hours	control	2.74	85	15	-	-	-	-	-
	4.5 Gy whole	cis-Pt	2.48	70	30	-	-	-	-	-
	body X-ray irradiation	ADM	1.42	90	10	-	-	-	-	-
	- 24 hours	control	1.55	80	20	-	1.21	90	-	10
	2.5 mg cortisone	cis-Pt	0.96	90	5	5	1.15	100	-	-
	acetate s.c.	ADM	1.12	85	10	5	1.44	90	10	-
	- 24 hours	control	1.46	10	90	-	1.67	20	80	-
	2.5 mg silica	cis-Pt	1.30	20	80	-	1.69	30	70	-
	i.v.	ADM	1.44	10	90	-	1.61	5	95	-

no treatment; cortisone, Cyclo and 4.0 Gy irradiation all effectively reduce most cell infiltration, but not better than 4.5 Gy administered at 4 hours before tumour implantation (all ca. 75% tumour). Differences between drug treatments and untreated tumour are apparent, but, since 4.5 Gy at 4 hours before implantation is at least as effective as other regimens and more practical for primary tumours, we have chosen it to be our routine pretreatment. Since employing preirradiation, 4/4 and 4/4 primary ovarian tumours have been found to be sensitive to cis-Pt and ADM, respectively, in comparison with 1/19 and 1/19 before our decision to use pretreated animals.

We feel that only pretreated normal mice should be used for testing either primary tumours or early passage human tumours.

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Footnote to Table I.

* Relative tumour size was calculated with respect to the tumour size on the day of implantation (= 1.0).

**T = tumour; I = immune cells; F = fibroblasts/fibrosis/stroma.

Per control group, 10 mice were used. There were 5 mice in each treated group. Values are the means for the group. Repeated experiments are indicated by the multiple values.

COMPARISON OF PLATINUM DERIVATIVES FOR EFFECTIVENESS AGAINST HUMAN TUMOURS IN VIVO

L.M. van Putten and R.J.F. Middeldorp

The subrenal capsule assay for drug sensitivity testing of human tumours has been used to compare four platinum derivatives. The technique for the subrenal capsule assay as developed by Bogden was modified by Edelstein and Smink (see page ..). Using the modified technique, four platinum derivatives were compared for their antitumour effect. The four platinum derivatives were cis-diammine dichloroplatinum (cis-platin), cis-1,1-di(aminomethyl) cyclohexane platinum sulphate (TNO-6), cis-diammine-1,1-cyclobutane dicarboxylate platinum (CBDCA) and cis-dichloro, trans-dihydroxy bis-isopropylamine platinum (CHIP). As a standard treatment, 2/3 of the LD50 was given i.v. on day 1 and day 5 after grafting of human tumour under the kidney capsule. The tumours were 4th to 6th passage transplants in nude mice of human tumours of the ovary, colon or lung (except for one lung tumour used at passage 10). Generally, 5 to 10 tumours in control mice were compared with 4 to 5 tumours in treated mice. Tumour measurements in two diameters (a and b) were made on the day of grafting (day 0) and on day 6. Results were evaluated by statistical significance of tests. Since no human tumour sensitivity data in vivo were available for the drugs used the endpoint is not an absolute threshold of efficacy of the drug, but only a relative endpoint for comparing the frequency of statistically significant (Student's t test $p < 0.05$) differences in tumour volume, ($a \times b^2$), area ($a \times b$) or diameter $1/2(a + b)$ between treated and control tumours.

After measurement of the diameter of the tumour in two dimensions, the change in mean diameter $1/2(a + b)$ between day 0 and day 6 could be calculated. Alternative parameters were change in tumour area ($a \times b$) or tumour volume ($a \times b^2$), but these showed significant differences in a lower frequency than the use of the mean diameter as originally proposed by Bogden (1981). Results were considered as undependable due to inadequate testing if no growth occurred in control tumours or if less than four mice per treated group survived treatment to permit tumour measurement on day 6.

In attempting to evaluate the validity of this testing system, the preliminary results (Table I) suggest that cis-platin is active on more tumours than each of the other three derivatives, which appear equally effective. However, this suggestion is valid only for the very limited number of tumour types tested so far.

Table I

FOUR PLATINUM DERIVATIVES TESTED IN DUPLICATE TESTS IN HUMAN TUMOURS AT EQUITOXIC DOSES
(subrenal capsule technique)

no. of tumours giving in two tests	CBCDA	CHIP	TNO-6	cis-platin
2 significant results (++)	2	2	2	5
1 significant result (+-)	3	3	3	2
0 significant result (--)	3	3	3	2
total no. of tumours tested	8	8	8	9
drug dose per injection (2/3 LD50)	90	30	4.5	8 $mg.kg^{-1}$ (x2)
++ tumours were lung	2/3	1/3	1/3	1/3
++ " " colon	0/2	0/2	1/2	1/2
++ " " ovary	0/3	1/3	0/3	3/4

Bogden, A.E. et al., Cancer 48 (1981) 10.

INFLUENCE OF THE TREATMENT WITH APD-BISPHOSPHONATES ON THE BONE LESIONS IN THE MOUSE 5T2 MULTIPLE MYELOMA

J. Radl, J.W. Croese, M. Vieveen, R.J. Brondijk, M. Kazilova, J.J. Haaijman, C. Zurcher, P.H. Reitsma* and O.L.M. Bijvoet*

One of the most serious and debilitating complications in patients suffering from multiple myeloma (MM) is bone destruction due to the tumour. While the mean survival of these patients is increased by 1 to 2 years as a consequence of therapy with cytostatics and steroids, no progress has been recorded up to now in the prevention or cure of the bone lesions. New prospects were offered recently by the discovery that bisphosphonates retard growth and dissolution of hydroxyapatite crystals in vitro and inhibit bone resorption in tissue culture and in vivo. A major success has already been achieved in the treatment of the Paget's disease of bone with bisphosphonates. These findings stimulated pilot clinical trials in the treatment of osteolytic bone disease due to breast cancer and multiple myeloma. However, a preclinical experimental study in a suitable animal model is warranted in order to establish whether bisphosphonates will prevent or retard the bone destruction due to these tumours and, if so, the other potential effects of such a treatment (e.g., the toxicity for normal and malignant cells, influence on the dissemination pattern of the tumour, etc.). In this study, the effects of the treatment of the 5T2 mouse multiple myeloma, which very closely resembles the human MM, including the typical bone lesions (Radl et al., 1981) with APD-bisphosphonates (APD) (Reitsma, 1982) were investigated.

In the first experiment, the dose effect of APD was tested in three groups of male C57BL/KaLwRij mice. Each group received standard food pellets containing APD in amounts of 500, 2000, and 10,000 p.p.m. (mg.kg^{-1}), respectively. Urine from each individual mouse was collected and investigated for calcium and hydroxyproline content for one week before and for 2 weeks after introduction of the APD treatment. Because each of the doses was found equally efficient in decreasing the calcium and hydroxyproline excretion, 2,000 p.p.m. was chosen for further experiments. However, mice of the 3 individual groups were kept on the original diet and were followed-up in order to detect any potential side

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effects of this treatment. While all animals receiving 500 and 2000 p.p.m. of APD have remained without overt symptoms of disease up to the present (i.e., for 21 months), all mice receiving the highest dose of APD showed at 6 to 7 months after initiation of the treatment a progressive anasarca, anaemia, thrombocytopenia and leukocytopenia and they died within 2 months after the first symptoms appeared. The histopathological examination of these mice demonstrated focal necrosis of bone marrow and liver cells.

In the second experiment, the influence of APD treatment on the mouse 5T2 MM was investigated. Five groups of male C57BL/KaLwRij mice were included: group 1 consisted of normal control untreated mice. In the 2nd group, mice were each transplanted with 1 x 10^6 bone marrow cells from a pool of four donors bearing the 5T2 myeloma; these mice were not treated. Group 3 consisted of mice transplanted with the 5T2 MM as in the second group; two months later, when the MM was clearly developed (as evaluated by the serum concentration of the myeloma protein), treatment with APD was started. In group 4, normal control mice were treated with APD in the same way as those in group 3. Each of these four groups consisted of 16 mice. In the fifth group, five mice were transplanted with the 5T2 MM cells as in group 3, but the APD treatment was started directly from the day of transplantation. After six weeks, i.e., 3.5 months after implantation of the 5T2 MM, all mice were killed and subjected to necropsy and histological examination. In addition, X-ray pictures of the carcasses were made on a Faxitron X-ray machine. On these, the osteolytic lesions in the skeleton were evaluated by scoring their presence and extent in individual bones and expressed in arbitrary units. As shown in Fig. 1, the differences between the treated and untreated groups with 5T2 MM were highly significant. A similar difference between groups 2 and 3 was seen when measuring the surface area of metaphyseal bone directly below the epiphyseal plate of the distal part of the right femur by microradiography. These results show that the treatment of the 5T2 MM with APD could protect the mice against a loss of bone to a significant extent.

Another question is whether the APD treatment influenced the growth pattern and extent of the tumour. To study this problem, the concentration of 5T2 myeloma protein in the serum and the dissemination pattern of the 5T2 MM cells in the different organs of each tumour bearing mouse was investigated. The concentration of the 5T2 MM proteins in the serum was determined by radioimmunoassay. A two-step incubation method was applied, using anti-5T2-idiotype and anti-IgH1^b allotype monoclonal antibodies.

Preliminary results indicate that the treatment with APD did not significantly influence the extent of the tumour mass as reflected in the serum 5T2 MM protein concentration. This was confirmed by histopathological examination. It showed no clear-cut

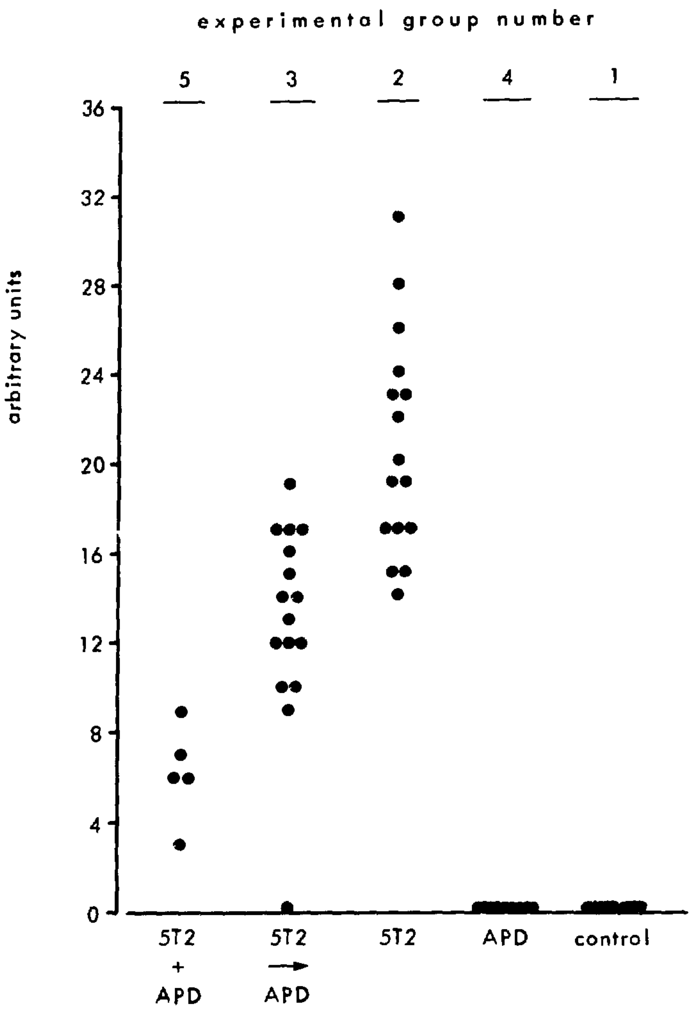

Figure 1: Scoring of the bone lesions caused by the 5T2 multiple myeloma in untreated and APD treated C57BL/KaLwRij mice.
The presence and extent of the lesions in individual bones were evaluated from the X-ray pictures and expressed in arbitrary units. Statistical evaluation among the three groups with untreated and APD treated MM demonstrated significant differences, with P values < 0.001.

differences in the extent or the distribution of the diffusely growing myeloma cells in the bone marrow, the spleen, the lymph nodes and other tissues.

The main effect of APD treatment was reflected in less destruction of cortical bone and in a higher number of small bone trabecula in the bone marrow cavity as compared with the untreated MM mice. The most readily recognizable locations of these changes were the sternal ends of the ribs, the distal parts of the femurs and the proximal parts of the tibiae. In some cases, an increase in numbers of small trabecula was observed even when compared with untreated normal control mice. In some of the MM mice which received APD immediately after transplantation, extensive areas of necrosis in the bone marrow cavity of the ribs, femurs and tibiae were observed. This finding suggests that the APD treatment might also exert some cytotoxic effect on the MM cells.

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A NEW TYPE OF ANTITHYROID AUTOANTIBODY IN *PRAOMYS* (*MASTOMYS*) *NATALENSIS*

J. Coolen, H.A. Solleveld, C. Zurcher
and J.J. Haaijman

Autoantibodies directed against various components of thyroid tissue are frequently encountered in sera of aging man. They are often determined by using an indirect fluorescence technique on slides of thyroid tissue from hyperthyroid patients. A number of fluorescence patterns may be observed: 1) nuclear fluorescence due to antinuclear factors, also reacting with nuclei in other tissues; 2) cytoplasmic fluorescence caused either by antibodies directed against mitochondrial antigens (these autoantibodies will also react with mitochondria from tissues other than thyroid) or microsomal antigens (thyroid specific); 3) fluorescence confined to material in the lumen of the thyroid follicles. The different staining patterns are most likely caused by different autoantibodies, as combinations of the various patterns may be found with sera of some patients.

The intralumenal fluorescence can be distinguished in two patterns: one in which the fluorescence has a "cloudy" appearance and is confined to material lining the luminal side of the follicular cells and another in which the fluorescence is more or less evenly distributed throughout the entire lumen. The autoantibody responsible for the former is thought to be reactive with thyroglobulin, whereas the latter appears to react with another antigen present in colloid. This antigen was designated as CA II (Pinchera et al., 1980). In man, the presence of antithyroglobulin and antimicrosomal antibodies is often associated with clinical autoimmune thyroiditis (Burek et al., 1982), whereas the significance of the anti-CA II antibodies is not yet clear.

The rodent *Praomys* (*Mastomys*) *natalensis* (henceforth called Mastomys) is considered a suitable animal model for age-related autoimmune phenomena (Coolen et al., 1978). In earlier reports, use was made of sera from moribund animals. It was shown that those sera contained autoantibodies directed against thyroid antigens in a significant frequency (Coolen et al., 1981). The autoantibodies reacting with the thyroid secretory substance all gave an even fluorescence distribution and were reminiscent of the anti-CA II mentioned above in human patients. However, a clear correlation was shown to exist between the presence of these antibodies in the serum of individual animals and a lymphoplasmocellular thyroiditis in these animals (Solleveld et al., 1982a, b). It was thus concluded that in Mastomys the anticolloid autoantibodies were pathogenic.

A longitudinal approach was taken to gain a better insight into the temporal relationship between the appearance of antithyroid autoantibodies and ensuing thyroid pathology (see this Annual Report p. 121). In studying these sera for autoantibodies

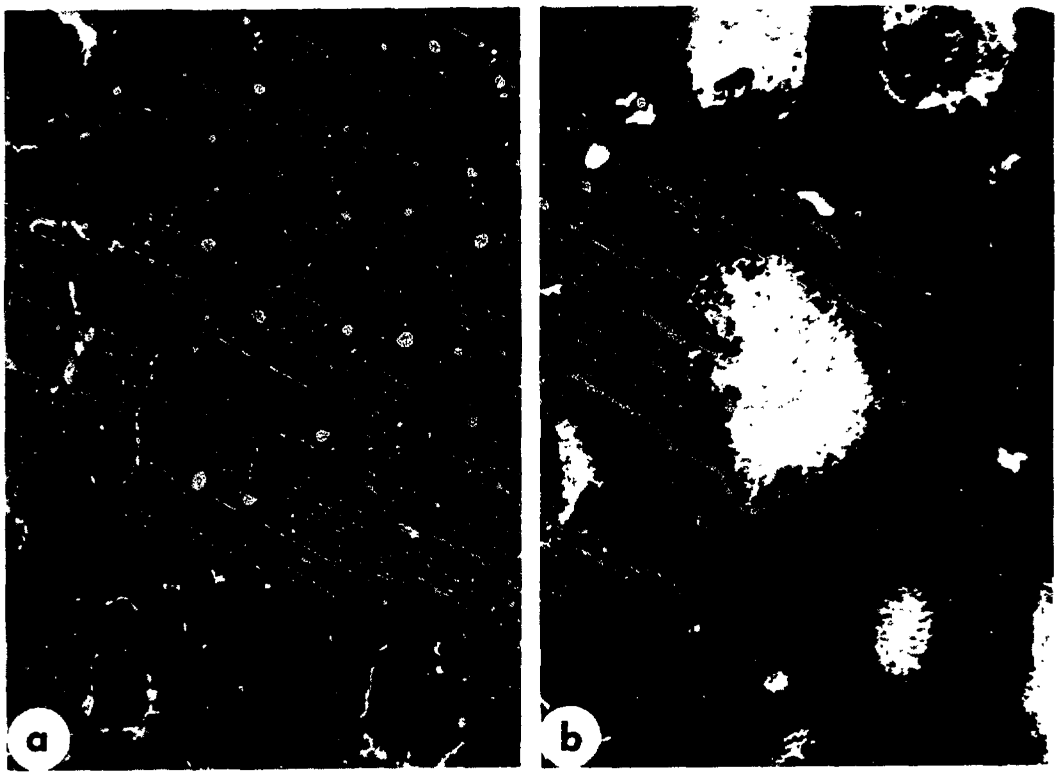

Figure 1: Anti-TX and anticolloid autoantibodies.
Cryostat sections from snap-frozen rat thyroid were incubated with 1:10 diluted serum from a 9-month-old Mastomys (left-hand figure) and from a 32-month-old Mastomys (right-hand figure). Bound antibody was visualized with a double-sandwich as described previously (Coolen et al., 1981). The left-hand side of the figure demonstrates the anti-TX (fine linear deposit along the follicular cells) and the right-hand side demonstrates anticolloid reactivity (even fluorescence throughout entire follicular lumen). Magnification: 170x.

to thyroid antigens, we discovered a fluorescence pattern which, to our knowledge, has not yet been described in human thyroid immunopathology.

The pattern is shown in the left-hand panel of Fig. 1. It consists of a fine linear deposit directly adjacent to the apical poles of the follicular cells. It does not resemble human anti-thyroglobulin reactivity, as the pattern is sharply linear and not thickened into a cloudy rim. Nor does the pattern resemble anti-microsomal antibodies, because the deposits are located extra-cellularly. As the target antigen of this new autoantibody is unknown, we therefore call it anti-TX. In Fig. 2, the occurrence

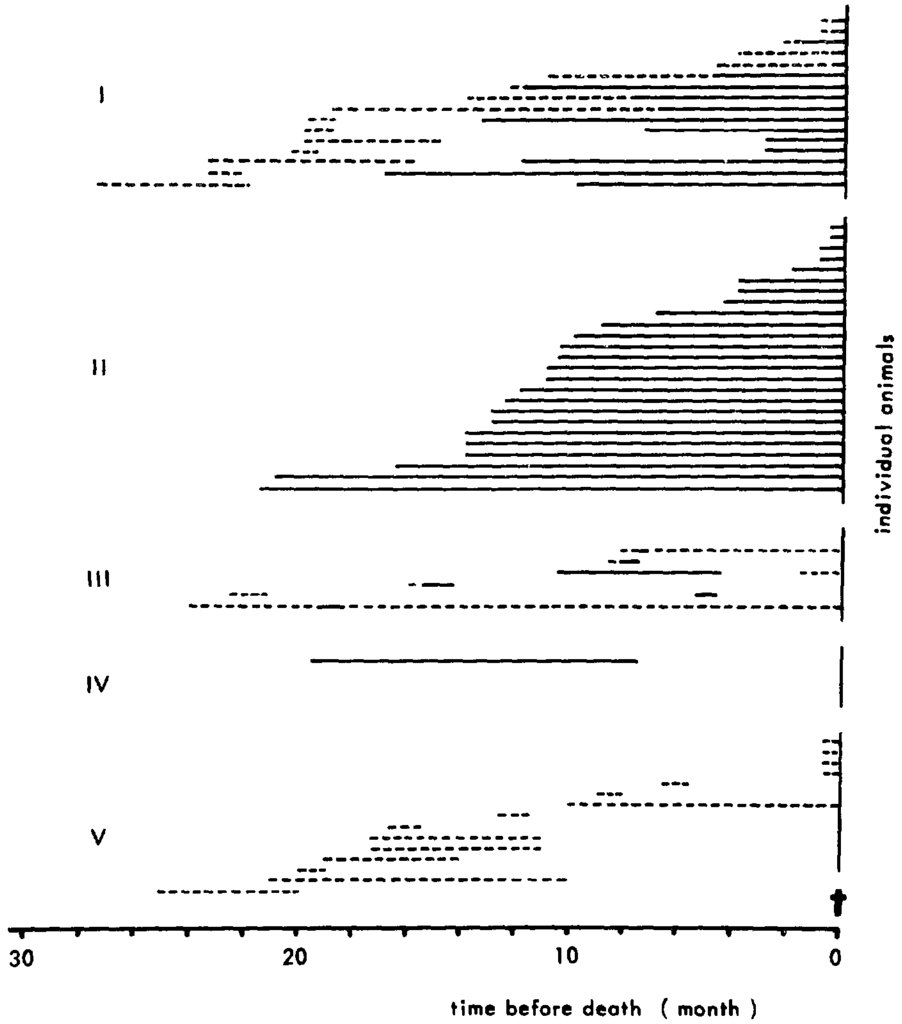

Figure 2: Temporal relationship between the presence of anti-TX and anticolloid antibodies and time to death.
Serial serum samples were obtained from Mastomys of various ages. The sera were screened, among others, for the presence of anti-TX (-----) and anticolloid (——), respectively. Data for individual animals are plotted as a function of time preceding the death of the animals.

of anti-TX and anticolloid (CA II) in sera of *individual* animals is plotted. There appears to be no direct relationship between the two autoantibodies. The data of Fig. 2 were ordered into 5 groups. Groups II and V show only anticolloid and only anti-TX, respectively. In group I, anti-TX precedes anticolloid and, in group III, anti-TX was observed together with a transient appearance of anticolloid. In the single animal of group IV, anti-colloid was transiently present and no anti-TX was evident.

The anti-TX antibody is most likely not pathogenic for the thyroid: none of the animals from group V developed thyroiditis, while this occurred in most animals of groups I, II, III and IV. The expected survival time appears not to be negatively influenced by the presence of anti-TX. The mean age of the animals was 27 months for group I and 28 months for group II. Anti-TX appeared relatively early in life: the mean life expectancy after the first appearance of anti-TX was in the order of 14 months (group I, 14 months; group III, 15 months; group V, 12 months).

Anti-TX illustrates a well-known but still little understood phenomenon of the aging immune system. The ability to distinguish self from nonself is gradually lost. However, most autoantibodies in the elderly cause no distinct pathology. An explanation could be that the cytotoxic T cell effector arm of the immune system is already weakened before the onset of autoantibody production.

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AUTOANTIBODIES TO THYROID COLLOID AND THYROIDITIS IN UNTREATED, NEONATALLY THYMECTOMIZED AND ORCHIECTOMIZED RANDOM BRED MALE *MASTOMYS*: A LONGITUDINAL STUDY

H.A. Solleveld, J. Coolen, J.J. Haaijman and C. Zurcher

In *Mastomys* as in man, the occurrence of autoantibodies increases with age (Coolen et al., 1981; Solleveld et al., 1982a, b). As some problems related to autoimmunity are difficult to solve by studies in man, some longitudinal experiments were performed in *Mastomys*. We wished to determine: a) the time of first appearance of autoantibodies; b) the persistance of autoantibodies; c) the temporal relationship between the presence of autoantibodies and autoimmune disease; and d) the effect of neonatal thymectomy (nTx) and prepubertal orchiectomy (Orx) on autoantibody formation.

Although autoantibodies to a variety of tissue antigens were tested, this report will be restricted to the occurrence of autoantibodies to thyroid colloid. Autoantibodies to thyroid colloid were determined in serial serum samples of individual random bred male *Mastomys* by the indirect immunofluorescence technique using frozen sections of rat thyroid gland (Coolen et al., 1981). Moribund animals were euthanized and a complete necropsy was performed (Solleveld, 1981). All major tissues, including the thyroid gland, were evaluated histologically. The severity of the thyroiditis was assessed as grade 1, 2, 3 or 4 using a standard scoring system (Kite et al., 1969). To investigate the relationship between the occurrence of thyroiditis and the presence of autoantibodies, all treated and untreated animals which showed one or more positive samples during their lifetimes were studied histologically for the presence of thyroiditis. The temporal relationship between the presence of anticolloid antibodies and thyroiditis is given in Fig. 1. The data presented confirm the earlier observation of a strong correlation between the presence of anticolloid antibodies and thyroiditis in this species (Solleveld et al., 1982a, b). In the few animals without thyroiditis, autoantibodies appeared shortly before death or were present only transiently. Autoantibodies to thyroid colloid were first detected at the age of 9 months. All samples investigated at 4 months were negative. In most cases, the anticolloid antibodies persisted throughout life. Only in 9 of a total of 57 positive *Mastomys* were autoantibodies found to be only transiently present. In 7 of these cases, thyroiditis was present at death. Possibly, free antibodies were no longer detectable because of antigen-antibody binding in the

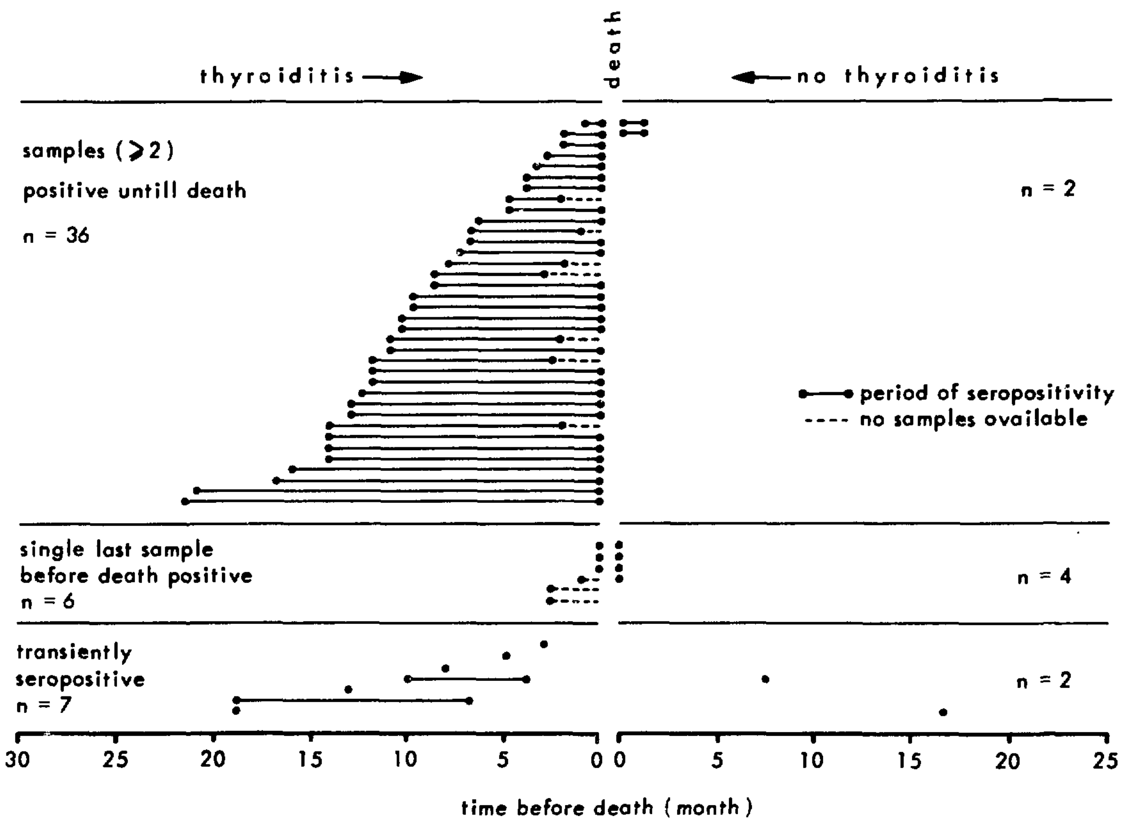

Figure 1: Time of appearance of autoantibodies and period of seropositivity.

thyroid itself or because of the formation of immune complexes in the serum (Solleveld et al., 1982b). Another possibility is that autoantibodies are not the only factors in the pathogenesis of thyroiditis in Mastomys; cellular immune responses may also play a role.

The effects of nTx and Orx on the incidence of anticolloid antibodies are shown in Fig. 2. A surprising finding was the high incidence (64% at 27 months) of anticolloid antibodies in the untreated males, since the incidence in a previous study (about 5 years earlier) was only 16% (Solleveld et al., 1982b). Genetic drift is the most likely explanation for this difference in incidence. The incidence of anticolloid antibodies in nTx (21% at 27 months) and Orx (44% at 27 months) animals was lower than in the controls, while there was no difference in 90, 50 and 10% survival among the three groups (Hollander et al., 1981). Also the incidence of thyroiditis in nTx males was lower than in the controls (30% and 60%, respectively: $p < 0.05$). In addition, the severity of thyroiditis was less in nTx animals than in the

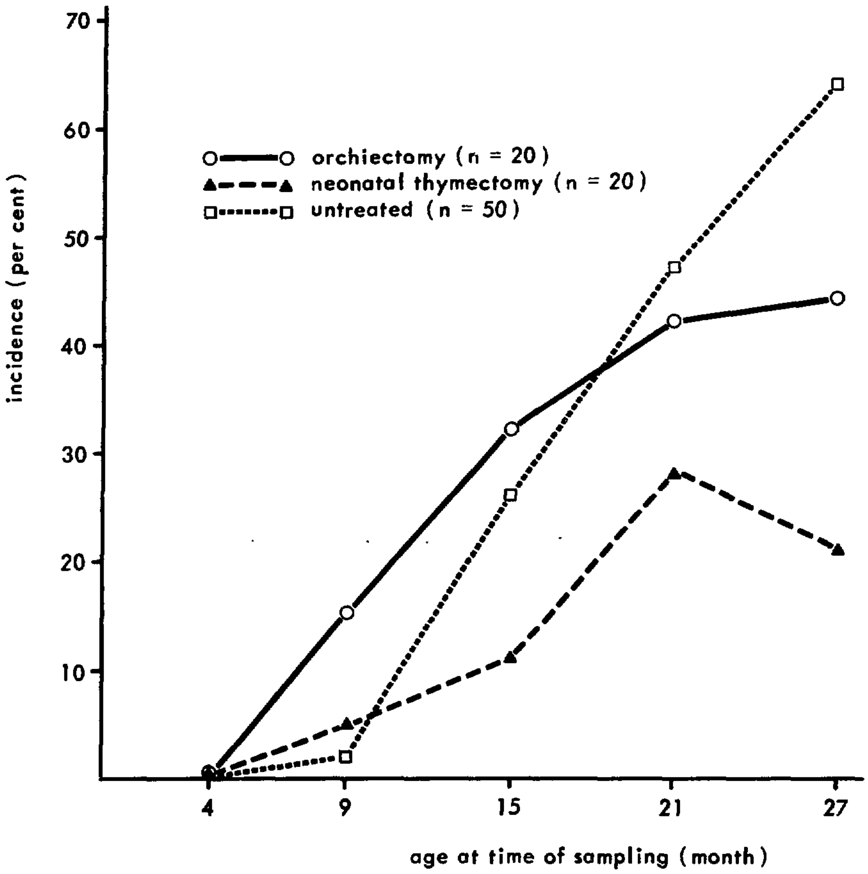

Figure 2: Incidence of autoantibodies to thyroid colloid in cohorts of neonatally thymectomized, orchiectomized and untreated male *Mastomys* during a longitudinal study.

controls. The mean scores for severity were 1.00 and 1.55, respectively. Since thyroid antigens are already present in the circulation prenatally and neonatally, it is possible that a major portion of T cells capable of recognizing these antigens is eliminated or inactivated by nTx, resulting in the prevention or a decrease in the severity of thyroiditis (Lipscomb et al., 1979). The incidence of autoantibodies in Orx males was also lower than in controls (44 against 64% at 27 months), but this difference was not statistically significant. This may indicate that the incidence of anticolloid antibodies is not influenced by sex hormone concentrations in the serum. On the other hand, however, thyroiditis was more severe in Orx animals than in controls. The mean severity was 2.33 and 1.55, respectively ($p < 0.05$). This suggests an ameliorating effect of androgens on the severity of the thyroiditis. The effects of nTx and Orx on the incidence of autoimmune thyroiditis are different from those found in other spontaneous

(Noble et al., 1976; Wick et al., 1981) and experimental (Penhale and Ahmed, 1981) models of autoimmune thyroiditis; it is worthwhile, therefore, to study autoimmune thyroiditis in Mastomys in more detail along with the other models.

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THE DISTRIBUTION OF IgA1 AND IgA2 CONTAINING CELLS IN VARIOUS LYMPHOID ORGANS OF MAN

J.J. Haaijman, J. Coolen, C. Deen, C.J.M. Kröse,
J.J. Zijlstra and J. Radl

Human IgA is divided into two subclasses: IgA1 and IgA2. Although immunochemically distinct, the difference in function of the two subclasses has not yet been adequately defined. Both are able to form dimers with a J-chain connection and both dimers can be excreted through epithelia at the secretory sites via binding with secretory component. IgA2 was originally considered as the "mucosal" IgA because it was mostly found in external secretions such as, e.g., saliva, tears, milk and bile. IgA1 is more abundant in serum and was thought to be synthesized primarily in the bone marrow.

IgA and its subclasses are still elusive immunoglobulins. More than 50% of the immunoglobulin containing cells (C-Ig cells) in the bone marrow are of the IgA class. The bone marrow has been shown to be the major source of all circulating Ig's (Benner et al., 1981); however, after antigenic stimulation, antibody activity in appreciable quantities was seldom demonstrated within the IgA class. Likewise, it is unexplained why peripheral blood cells synthesize and secrete IgA so frequently after mitogenic stimulation.

We therefore began a survey of the localization of C-IgA1 and C-IgA2 cells in different lymphoid organs of man, using our newly prepared set of mouse monoclonal antibodies directed against the isotype specific determinants of IgA, IgA1, IgA2, IgM and IgG.

Autopsy material from the digestive and respiratory tracts, spleen and lymph nodes was obtained through the courtesy of the Department of Pathology, University of Leiden (Prof.Dr. F. Eulderink). Only those patients who did not die from overt immunologically related diseases were selected. Tissues were processed for immunohistology as described by Eikelenboom et al. (1982). Immunoglobulin containing cells were identified by the indirect method using peroxidase-labelled rabbit antiserum directed against mouse immunoglobulins as the second step and diaminobenzidine as the developing agent. The results are described in semiquantitative terms (Table I), since our technique allows the demonstration of only one class of C-Ig cells per slide and also because our collection of material is not yet extensive enough to draw final conclusions.

Table I

Ig ISOTYPE DISTRIBUTION OF VARIOUS LYMPHOID ORGANS OF MAN

organ	no/n**	IgM	IgG	IgA1	IgA2
		estimated frequency*			
trancheal mucosa	3/5	±	±	++++	+
digestive tract:					
duodenum	3/3	±	±	++++	++
ileum	3/3	±	±	++++	++
colon	5/5	±	±	++	++++
spleen	3/3	+	++	+	±
lymph nodes					
mediastinal	5/20	++	+++++	+	±
mesenteric	2/5	-	++	±	-
adaortal	3/17	+	++	±	±
axillar***	2/7	++/-	+++++/+	±/-	±/-

* the number of "+'s" indicates a rough scoring of the number of C-IgG cells. ± stands for sporadic, scattered cells.

** no = number of patients; n = number of different specimens taken per site.

***for these nodes, an extreme variation in the total number of C-Ig cells was observed.

The surface of the trachea is lined with epithelium below which numerous mucinous and serous glands are present. The acini of the glands are lined with plasma cells mostly of the IgA1 subclass (Fig. 1). An estimated 10% of the C-Ig cells belongs to the IgA2 subclass, whereas IgM and IgG plasma cells are scarce. A similar picture was seen in the duodenum and ileum. Abundant numbers of C-IgA plasma cells are situated directly below the epithelium within and between the villi. In these portions of the digestive tract, the IgA1 constitutes more than 60% of the C-IgA cells. C-IgM and C-IgG cells were again seen only sporadically. In colon material taken from 5 patients, on the other hand, C-IgA2 cells were represented more frequently than C-IgA1 ones. The ratio between IgA2 and IgA1 cells in the colon was about 3 to 2. The present conclusion based on the available material is that IgA1 appears to be the major IgA subclass of the upper respiratory and upper digestive tracts and IgA2 has a similar prevalence in the lower digestive tract.

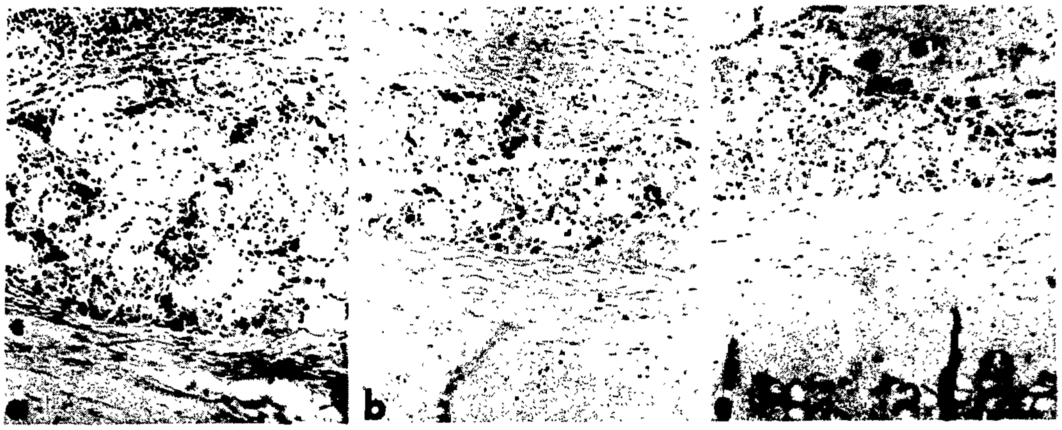

Figure 1: Cytoplasmic immunoglobulin containing cells of tracheal mucosa. Slides were first stained with mouse monoclonal antibodies directed against IgA (a), IgA1 (b) and IgA2 (c) and then with a peroxidase conjugated rabbit antimouse Ig.

In the spleen, the C-Ig cells are localized on the borders of the red and white pulps. The number of C-Ig cells is relatively small. They are mostly of the IgG class. In contrast to, e.g., laboratory mice, C-IgM cells are scarce. Over 80% of the few C-IgA cells in the spleen belongs to the IgA1 subclass.

Lymph nodes were obtained from various sites. In all of these studied so far, the vast majority of the C-Ig cells belong to the IgG class. The second most observed Ig class was IgM (about 20% of the number of C-IgG cells). C-IgA cells were found only rarely and mostly in the subcapsular sinus. The total C-Ig cell expression varied greatly even between nodes taken from nearby locations. C-Ig positive cells were found most frequently in the lymph nodes from the mediastinal region.

Our preliminary results show that monoclonal antiimmunoglobulin reagents can be used to study the normal distribution of intracellular Ig isotypes in various lymphoid tissues in man. Extension of these studies may lead to a better understanding of the role of the different Ig isotypes and can form the basis for an evaluation of age-related immunological dysfunctions.

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DNA REPAIR IN FIBROBLASTS OF AGING RATS

J. Vijg, E. Mullaart, Adr. Brouwer,
P.H.M. Lohman* and D.L. Knook

Inbred rodents are characterized by a great interindividual heterogeneity in life span (Burek, 1978). This interindividual variability is reflected in differences in the rate of functional decline of the organ systems as well as in different patterns of pathological changes (Hollander, 1981; Zurcher and Hollander, 1982). Therefore, the rate at which the aging process proceeds differs from individual to individual. The question arises of whether it is possible to determine the rate of aging by means of a biomarker, a "clock" to determine the extent to which organ and bodily functions have aged in a given individual.

One possible biomarker might be the quality of one's DNA repair systems. The necessity of maintaining the integrity of the genetic information encoded in the DNA of pro- and eukaryotes makes DNA repair a function of primary importance in all species. It is not surprising therefore that much attention is presently being paid in aging research to a possible correlation between the rate of aging and the quality of DNA repair. The observation that agents which damage cellular DNA lead in time to certain (but not all) aspects of senescence lends support to this hypothesis.

The rationale for the present investigation is twofold: 1) to gain insight into the heterogeneity within inbred rat strains in relation to DNA repair variables; 2) to investigate whether specific repair parameters correspond to differences in life span and can possibly be used as prognostic indicators for the life span of individual animals.

The first parameter studied was the autoradiographic detection of the incorporation of ^{3}H-thymidine into the DNA of tissue cultures damaged by UV light or by UV mimetic chemicals. This factor (unscheduled DNA synthesis or UDS) is commonly used in repair studies and is an indication of the capacity of an organism to remove the piece of DNA containing the damaged site and to subsequently fill the gap (excision repair).

The cells used for determinations of UDS were fibroblasts isolated from a piece of skin (5 x 10 mm) excised from the tail of female WAG/Rij rats. After growing and passaging the cells for about two weeks the quantity of cells was sufficient for dose response and time course studies. For this purpose, 2 x 10^5 cells

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were seeded in 3.5 cm ø Petri dishes into which 11 mm diameter cover slips had been previously placed. After 24 hours, the medium was removed and the cells were rinsed twice with PBS and exposed from above to 254 nm radiation supplied by a low pressure mercury vapour lamp at a fluence rate of 0.38 $J/m^{-2}/s^{-1}$. Exposed and unirradiated control samples were both incubated at 37°C with ^{3}H-thymidine in a humidified CO_2 incubator for various periods of time (up to 30 hours). At the end of the incorporation period, the cells on the coverslips were fixed and prepared for autoradiography by standard procedures. Assessment of UDS is possible by counting the number of autoradiographic grains per nucleus in the lightly labelled cells such as shown in Fig. 1 (the easily discriminated heavily labelled cells are S phase cells). More reproducible values can be attained by measuring the grain surface area per nucleus. For this purpose, we applied an automated device (Artek, model 380) that measures the grain surface area per nucleus of the images of the cells projected on a television screen. In this way, UDS of at least 30 cells was quantitated. The data obtained were displayed as histograms and as average numbers per nucleus.

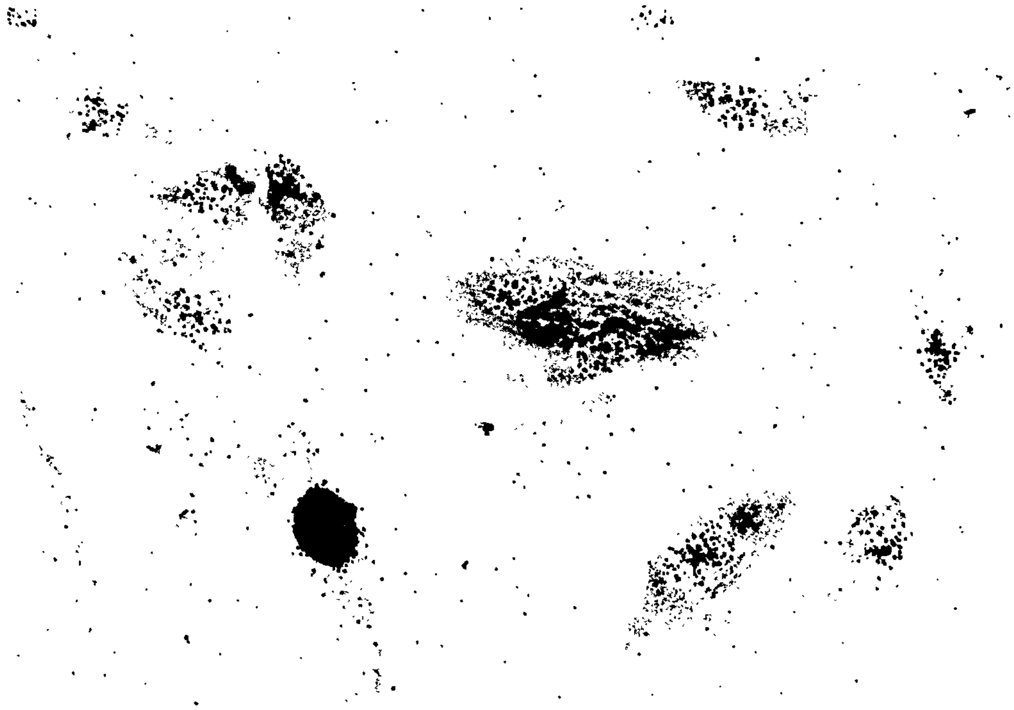

Figure 1: Autoradiogram of rat fibroblasts after irradiation with 10 $J.m^{-2}$ UV light and labelled for 3 hours with ^{3}H-thymidine. Repairing cells (lightly labelled) are clearly distinguished from mitotic cells (heavily labelled).

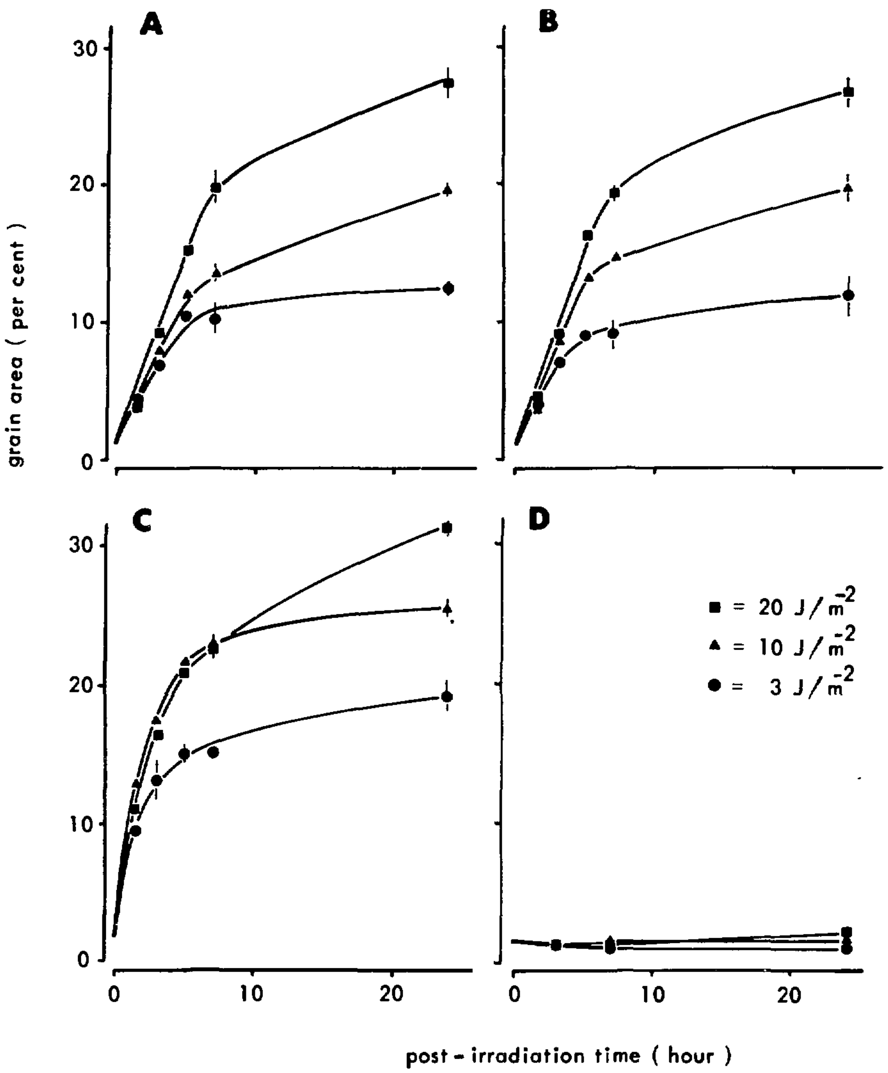

Figure 2: Kinetics of UV induced unscheduled DNA synthesis in fibroblasts of (A) a 3-month-old rat; (B) a 40-month-old rat; (C) a normal human; (D) a xeroderma pigmentosum patient.

Data are obtained from the average of the individual histograms representing measurements on 30 cells. Bars indicate the difference between measurements on two individual cultures.

An important prerequisite for our study is that the capacity for UDS is not lost or diminished during passaging of the cells in vitro. Although it has been found in several other studies that rodent cells lose their capacity for UDS upon cultivation in vitro, no significant decrease in UDS with the passage number was observed in our system.

The results of a dose response and time course study with fibroblasts isolated from a young (3-month-old) and an old (40-month-old) rat are shown in Fig. 2A, B as plots of repair synthesis after different doses versus the time of repair. For comparison, the same determinations were simultaneously made in normal human fibroblasts (HAN) (Fig. 2C) and xeroderma pigmentosum (XP2CA) fibroblasts (Fig. 2D). Excision deficient XP2CA cells are characterized by decreased rates of UDS; our results are in agreement with this.

The rate of repair synthesis in human cells is biphasic (Paterson et al., 1973). Our results now make it clear that rat cells do not differ from human cells in this respect. An initially high rate of UDS is followed after about 7 hours by a second process with a decreased slope. It has been suggested that this last part of the curve could be due to less accessible lesions in the DNA (Paterson et al., 1973). A further comparison of the curves of rat and human cells shows that the initial rate of UDS is much higher in humans. This is in accord with earlier findings of Hart and Setlow (1974), who reported a positive correlation between the initial rate of UV induced UDS and the maximum species lifespan. However, no difference can be observed between the UDS curves of the old and the young rat. Although UDS values from more animals should be obtained, the present finding is an indication that there may be neither a great interindividual nor a great age-dependent variation in inbred rats with respect to this specific repair parameter.

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CULTURES OF RAT LIVER FAT-STORING CELLS

A.M. de Leeuw, S.P. McCarthy and D.L. Knook

Fat-storing cells (FSC) present in the liver are now being regarded as a separate cell population (see for a review: Wake, 1980) from the other main cell types of the liver, the parenchymal, endothelial and Kupffer cells.

At the ultrastructural level, FSC have a dual appearance: 1) they contain a varying number of lipid droplets in their cytoplasm, indicating a function in lipid storage. The presence of vitamin A in these lipid droplets has established a role for FSC in the vitamin A metabolism in the liver. 2) the presence of a well-developed Golgi system and swollen cisternae of the rough endoplasmic reticulum, filled with flocculent material - fibroblastic characteristics - has suggested a role for FSC in the synthesis of components of the extracellular matrix of the liver, for example, collagens. Disturbances in the metabolism of these extracellular matrix components may lead to liver fibrosis and FSC may play an important role in these processes.

However, detailed investigations into the functions of this cell type have been hampered by the lack of an *in vitro* system. A recently developed isolation and purification procedure for FSC (Knook et al., 1982) enables investigation of the role of FSC in vitamin A metabolism and collagen synthesis.

Preliminary results with this cell isolation system have already indicated that under physiological conditions about 80% of total liver vitamin A is present in the FSC and that with increasing age of the animal the total vitamin A content also increases.

The study of isolated cells *per se* is, however, of limited value. Investigations of some cellular processes require the possibility to study cells over a longer period of time than possible with freshly isolated cells under well-defined conditions. In this report, the basic behaviour of purified FSC in a culture system will be described. The interaction of the three main sinusoidal liver cell types in coculture (endothelial, Kupffer and fat-storing cells) has been described recently (De Leeuw et al., 1982).

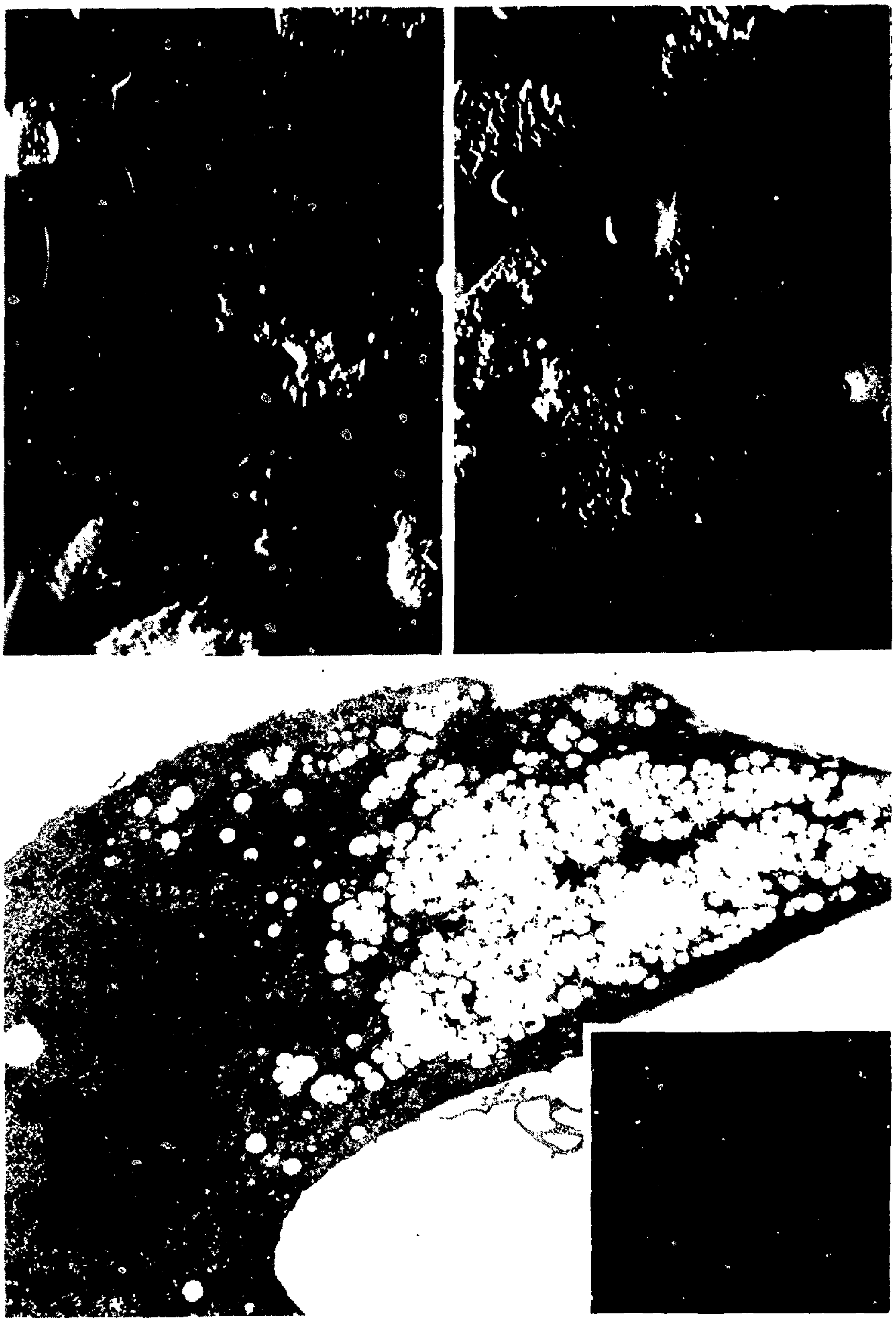
Rat liver FSC were isolated and purified as described previously (Knook et al., 1982) and established in culture in both plastic culture dishes (0.25×10^6 per 1.5 cm^2 dish) and on glass coverslips in Dulbecco's modification of Eagles medium supplemented with penicillin, streptomycin and 20% fetal calf serum. The

Figure 1: A. Nomarski contrast. FSC after 3 days in culture. The cell body contains a very prominent nucleus with sharply outlined edges and surrounded by lipid vacuoles. Long flat processes extend from the cell body (1,750 x).

B. Nomarski contrast. FSC after 6 days in culture. Cell flattening has become more pronounced as compared to A. Cell processes show distinct filamentous structures (1,750 x).

C. Transmission electron micrograph. FSC 9 days in culture. Retention of lipid droplets. Note the Golgi system and the cisternae of the rough endoplasmic reticulum (3,000 x).

Insert: Vitamin A autofluorescence of FSC cultured for 4 days, after irradiation with light of 328 nm. Fluorescence is confined to lipid droplets.



medium was refreshed every two days. The behaviour of the cells was followed by both light and electron microscopy.

Directly after isolation, FSC as viewed by phase contrast microscopy mostly resembled bunches of grapes, due to their content of lipid droplets. Although the cells attached to the substratum within 24 hours, cell spreading became pronounced only after 2 to 3 days. At that time, they showed a very distinct morphology, which made it possible to clearly distinguish them from contaminating Kupffer cells. From the cell body, long flat processes (Fig. 1A) which sometimes contained lipid vacuoles extended. The cell body contained the large, prominent nucleus (in most cells, showing two nucleoli) surrounded by varying numbers of lipid vacuoles (Fig. 1A). During the following days, cell flattening proceeded and an increase in the number of FSC was clearly observed. After 6 days, the cells had an extremely flat appearance, their processes often showing a large number of filamentous structures (Fig. 1B).

At the ultrastructural level, the main change during the first day in culture was a loss of the long thin cytoplasmic processes characteristic of freshly isolated cells and the development of broad flat processes containing microfilaments, with which the cells attached to the substratum. The chromatin pattern changed from rather clumped heterochromatin to well-dispersed euchromatin. From day 1 to 6, progressive cell flattening was accompanied by a large increase in the volume of the cells. Lipid droplets remained present but seemed to be split into smaller ones (Fig. 1C). Vitamin A as determined by autofluorescence after irradiation with light of 328 nm remained associated with the lipid droplets, at least during one week in culture (Fig. 1C: inset). The cells also contained a well-developed Golgi system and a network of swollen cisternae of the rough endoplasmic reticulum filled with flocculent material (Fig. 1C).

Confluency occurred at 5 to 7 days after starting the cultures. Trypsinization followed by replating (dilution 1:2) gave rise to cultures of dividing cells with the same morphology as the primary cultures.

Further experiments are now being planned for a further characterization of these cells by typing of the elements of the cytoskeleton and of secretion products, e.g., collagens.

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AGE RELATED CHANGES IN THE CLEARANCE CAPACITY OF KUPFFER AND ENDOTHELIAL LIVER CELLS

Adr. Brouwer, R.J. Barelds and D.L. Knook

The reticuloendothelial system (RES) is a heterogeneous one of scattered macrophages and other phagocytic cells which make an important contribution to natural resistance to disease agents. The functions of the RES include the clearance of a large variety of potentially harmful substances from the plasma, catabolism of macromolecules, participation in the immune response and the secretion of various effector substances. The function of the RES is generally assessed both clinically and experimentally by the plasma clearance rates of intravenously injected colloidal test substances.

In previous studies, it was shown that both Kupffer and endothelial cells among the sinusoidal liver cells constitute an important part of the RES. Several reports have described an age related decline in the clearance function of the RES in both man and experimental animals (Brouwer and Knook, 1981). The capacity for in vitro endocytosis of ^{125}I-labelled heat aggregated colloidal albumin (CA^{125}I) by cultured rat Kupffer cells was found to gradually decline with age, with a maximum decrease of about 40% at between 3 and 36 months of age (Brouwer and Knook, 1981). In this study, we have assessed the in vivo endocytic capacity of both Kupffer and endothelial liver cells with the same RES test substance.

Female BN/BiRij rats of various age groups were injected with a saturating dose of CA^{125}I (2.5 mg per 100 g body weight, which was determined in pilot experiments) to ensure maximum uptake of the substrate by the liver cells. The preparation of sinusoidal cells was started by perfusion of the liver at about 6 min after injection of CA^{125}I. For the preparation of sinusoidal cells, a new method which is carried out entirely at below 8°C to prevent the degradation of the ingested radiolabelled protein was employed (Praaning-van Dalen et al., 1982). In brief, the liver is perfused with medium containing 0.2% pronase for 10 min at 8°C. The liver is then excised, thoroughly minced and filtered. The resulting liver cell suspension is centrifuged to harvest the sinusoidal cells, which are then purified by density centrifugation and separated into Kupffer and endothelial cell fractions by centrifugal elutriation using a Sanderson type separation chamber.

Table I

UPTAKE OF INTRAVENOUSLY INJECTED $CA^{125}I$ BY LIVER AND SPLEEN OF FEMALE BN/BiRij RATS OF VARIOUS AGES

		liver			spleen		
age (month)	n	organ weight (gram)	uptake (% dose/liver)	uptake ($\mu g\ CA \cdot g^{-1}$ liver)	organ weight (gram)	uptake (% dose/spleen)	uptake ($\mu g\ CA \cdot g^{-1}$ spleen)
3	5	5.12 ± 0.13	8.54 ± 1.04	66.2 ± 9.3	0.35 ± 0.02	0.92 ± 0.09	104.6 ± 4.4
6	5	5.02 ± 0.15	7.37 ± 0.89	57.1 ± 6.9	0.30 ± 0.02	0.80 ± 0.05	103.2 ± 8.2
18	3	6.48 ± 0.60*[a]	8.42 ± 0.37	65.0 ± 2.9	0.36 ± 0.03	0.67 ± 0.07	93.6 ± 7.2
30	4	7.20 ± 0.31*[a]	9.27 ± 0.89	71.8 ± 6.9	0.48 ± 0.05	0.74 ± 0.05	86.0 ± 3.4*
36	5	7.09 ± 0.32*[a]	8.88 ± 1.24	68.8 ± 9.6	0.50 ± 0.05*[a]	0.56 ± 0.06*[a]	61.5 ± 4.4*[a,b,c]

Values are given as means ± SEM.
n: number of individual animals tested.
% dose: percentage of total amount injected.
*: differs significantly ($p < 0.05$) from 3-month-old rats.
a: differs significantly ($p < 0.05$) from 6-month-old rats.
b: differs significantly ($p < 0.05$) from 18-month-old rats.
c: differs significantly ($p < 0.05$) from 30-month-old rats.

The plasma disappearance and the uptake of $CA^{125}I$ in the spleen, the whole liver and in Kupffer and endothelial cell fractions were assessed by radioactivity counts. Because a saturating dose of $CA^{125}I$ was injected, only about 15% of the injected dose was cleared from the plasma within the 6 min incubation period. The liver and spleen, which were the main tissues involved in this clearance, contained about 9 and 1% percent of the injected dose, respectively (Table I). In the spleen, the uptake per whole organ decreased from 0.9 to 0.6% of the injected dose at between 3 and 36 months of age. The amount of $CA^{125}I$ per gram of spleen tissue decreased simultaneously (Table I).

The uptake in the whole liver did not change with the age of the rat. Within the liver, the sinusoidal cells are the main sites of uptake of $CA^{125}I$ (Praaning-van Dalen et al., 1981). Expressed per 10^6 cells, Kupffer and endothelial cells take up $CA^{125}I$ to about the same extent (Table II). With increasing age, no significant changes were observed for the endocytosis of $CA^{125}I$ by endothelial cells (Table II). For Kupffer cells, the endocytic capacity appears to be maximal at 6 months of age. The uptake by these cells is significantly lower in both 3- and 36-month-old rats as compared with 6-month-old ones (Table II). When the uptake is expressed per mg of cellular protein, the differences between 6- and 36-month-old rats are even greater, due to an increase in Kupffer cell size with age (not shown). These in vivo results correlate well with in vitro studies which showed an age-related decrease in the capacity of cultured Kupffer cells to endocytose $CA^{125}I$ (Brouwer and Knook, 1981). There is an apparent discrepancy

Table II

IN VIVO ENDOCYTOSIS OF INTRAVENOUSLY INJECTED $CA^{125}I$ BY KUPFFER AND ENDOTHELIAL LIVER CELLS, DETERMINED BY ISOLATION OF LIVER CELLS AT LOW TEMPERATURE (8°C)

age (month)	n	Kupffer cells	endothelial cells
3	5	32.4 ± 5.3	53.5 ± 8.7
6	5	80.1 ± 10.5*	43.3 ± 4.8
18	3	63.2 ± 5.3	59.6 ± 5.8
30	4	55.7 ± 4.1	69.2 ± 4.9
36	5	48.4 ± 6.7[a]	44.7 ± 7.7

Values are given as ng per 10^6 cells per min (mean ± SEM).
Symbols as in Table I.

between the observed decrease in endocytosis by Kupffer cells and the unchanged clearance by the whole liver. This may, at least partly, be a reflection of the fact that the liver contains fewer Kupffer cells (12×10^6 per g of liver) than endothelial cells (42×10^6 per g of liver) (Praaning-van Dalen, et al., 1981). As a result, Kupffer cells account for only about 15 to 20% of the total liver uptake.

The results obtained in these studies indicate that the maximum endocytic capacity of Kupffer cells declines with age, while that of endothelial cells remains constant during the life span. The decreased endocytic capacity of Kupffer cells may be partly responsible for the decreased clearance capacity of the RES with age. Another part of this decrease may be due to reduced splenic clearance.

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AGE-RELATED CHANGES IN ALBUMIN ELIMINATION IN FEMALE WAG/Rij RATS

G.J.M.J. Horbach, S.H. Yap* and C.F.A. van Bezooijen

Age-related changes in protein synthesis have been extensively studied over the last ten years, but little information is available on the age-related changes in protein elimination.

We have investigated changes in the albumin elimination rate in vivo with age in female WAG/Rij rats. Plasma clearance studies with radioiodinated albumin were performed in 3-, 12-, 24- and 36-month-old rats. Each rat was injected intravenously with 5 µCi of ^{125}I-albumin and TCA precipitable counts were determined in 20 µl plasma at several time-points during five days. From the plasma activity curves obtained in this way, the following characteristics were calculated:

1. the elimination half-life, t1/2(el), an index for the relative elimination rate;
2. the apparent volume of distribution, Vd, an index for the whole body albumin pool;
3. the clearance, Cl, an index for the absolute elimination rate.

Plasma clearance studies were made in each rat of the different age groups using two different albumins. Each rat was injected with albumin isolated from 3-month-old rats. In this approach, observed age-related changes in the elimination of the albumin were due to physiological changes in the animal. The same rats were injected with albumin isolated from rats of the same age as the recipient. This approach revealed age-related changes in the elimination of albumin due to a combination of physiological changes in the animal and changes in the albumin molecule.

The results in Table I show that, after administration of albumin isolated from 3-month-old rats, no changes in the elimination half-life are seen. However, the values for clearance and apparent volume of distribution are higher at 24 months than those at 3 and 12 months of age. At 36 months of age, only a significant increase in clearance as compared with the 3-month value is seen.

After administration of albumin isolated from rats of the same age as the recipients, again no changes in elimination half-life with age are seen. The values for clearance and apparent volume of distribution are higher at 24 months of age as compared with those at 3 and 12 months of age. The higher values for apparent volume

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Table I

AGE-RELATED CHANGES IN THE METABOLIC CHARACTERISTICS FOR ALBUMIN ELIMINATION IN FEMALE WAG/Rij RATS

age of the recipient (months)	age of the rats from which the albumin was isolated: 3-month-old rats			rats of the same age as the recipient			36-month-old rats		
	t1/2(el) (h)	Cl(ml.h^{-1})	Vd (ml)	t1/2(el) (h)	Cl (ml.h^{-1})	Vd (ml)	t1/2(el) (h)	Cl (ml.h^{-1})	Vd (ml)
3	44.3±3.9	0.226±0.028	14.4±1.6	44.3±3.9	0.226±0.028	14.4±1.6	42.9±4.8	0.277±0.050	16.4±3.9
12	43.6±7.9	0.263±0.047	15.9±2.0	46.2±6.1	0.244±0.038	16.7±2.9	-	-	-
24	41.5±4.0	0.356±0.063[ab]	20.6±4.4[ab]	39.6±4.7	0.359±0.067[ab]	20.2±2.7[ab]	-	-	-
36	39.0±5.3	0.308±0.071[ad]	17.3±3.4[d]	40.3±4.6	0.420±0.056[abc]	24.0±4.1[abc]	-	-	-

The characteristics were calculated from the individual plasma activity curves.

a: value differs significantly ($p < 0.05$) from 3-month value.

b: value differs significantly ($p < 0.05$) from 12-month value.

c: value differs significantly ($p < 0.05$) from 24-month value.

d: value differs significantly ($p < 0.05$) from the value obtained when using albumin isolated from rats of the same age as the recipient.

Table II

INFLUENCE OF AGE ON PLASMA ALBUMIN CONCENTRATION, TOTAL ALBUMIN ELIMINATION, URINARY EXCRETION AND GASTROINTESTINAL PROTEIN LOSS IN FEMALE WAG/Rij RATS

age (months)	plasma albumin concentration ($mg \cdot ml^{-1}$)	total albumin elimination (mg per 24 h)	urinary albumin excretion (mg per 24 h)	ratio urinary excretion versus total elimination (%)	gastrointestinal clearance (ml/h)
3	36.3 ± 2.1	197 ± 12	0.22 ± 0.08	0.12 ± 0.06	0.161 ± 0.062
12	39.4 ± 2.8	235 ± 53	0.20 ± 0.11	0.09 ± 0.07	0.184 ± 0.084
24	40.6 ± 2.2	353 ± 74^{ab}	2.4 ± 1.4^{ab}	0.7 ± 0.5^{ab}	0.130 ± 0.035
36	38.2 ± 1.2	430 ± 70^{abc}	16 ± 16^{abc}	2.4 ± 1.9^{abc}	0.113 ± 0.021

Plasma and urinary albumin concentrations were determined in the same rats as used in the elimination experiments; gastrointestinal clearance in other rats.
Total albumin elimination was calculated by clearance x plasma, albumin concentration x 24.
Values are expressed as mean ± S.D. for a minimum of six determination.
a: value differs significantly ($p < 0.05$) from 3-month value.
b: value differs significantly ($p < 0.05$) from 12-month value.
c: value differs significantly ($p < 0.05$) from 24-month value.

of distribution and clearance at 36 months of age as compared with those at 24 months are seen only when albumin isolated from rats of the same age as the recipients is used. Therefore, the higher values for the clearance and apparent volume of distribution at 24 months of age as compared with 12 months of age cannot be due to changes in the albumin but to changes in the physiology of the animal. Since the administration of albumin isolated from 36-month-old rats does not induce such changes in the elimination characteristics in 3-month-old rats as in 36-month-old ones, the increase in these characteristics between 24 and 36 months of age can be due to a combination of a change in the albumin molecule and a change in the physiology of the rat between these two ages.

The urinary albumin excretion was determined in rats of the different age groups by radial immunodiffusion and the gastro-intestinal protein loss by the $^{51}CrCl_3$-method according to Van Tongeren and Reichert (1966). This in order to establish the cause of the increase in albumin elimination with age. Table II shows an enormous increase in the urinary albumin excretion at between 12 and 36 months of age. However, the age-related changes in albumin elimination cannot be attributed to this increase in urinary albumin excretion, since the high excretion rate at 36 months of age contributes only 2.4% to the total elimination of albumin. The gastrointestinal clearance remains constant with age. However, no estimation of the albumin clearance through the gastrointestinal wall is made in this experiment. After injection of $^{51}CrCl_3$, at least albumin and transferrin are labelled with a selection for transferrin. Therefore, one must bear in mind that the ^{51}Cr clearance results can be extrapolated to albumin clearance only when no distinction is made between different plasma proteins by the intestinal wall during leakage (Van Tongeren and Reichert, 1966). The cause of the increase in albumin elimination with age is unknown at present.

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DETERMINATION OF THIOPENTAL IN THE PLASMA OF PATIENTS AFTER GENERAL ANAESTHESIA

C.F.A. van Bezooijen, A.N. Sakkee and J. Nauta*

Thiopental is frequently used during general anaesthesia. It is known as a short-acting barbiturate. However, the term "short-acting" is misleading, as can be judged from the following description of the fate of thiopental in dogs. After intravenous injection, thiopental disappears quickly from the blood into well-perfused tissues. Thereafter, redistribution takes place to less perfused tissues and finally to fat tissue. The rate of redistribution of thiopental determines the time of the narcosis. The fate of the narcotic following the redistribution is determined by metabolism of the drug by the liver. A possible metabolite of thiopental is the active narcotic pentobarbital. The objective of this study is to determine the pharmacokinetics of thiopental and the possible occurrence of the metabolite pentobarbital in the plasma of patients after general anaesthesia.

A method was developed to separate and quantitate thiopental and pentobarbital using High Performance Liquid Chromatography (HPLC). A C-18 reversed phase column was used. The mobile phase consisted of distilled water : acetonitrile : methanol (23:14:3). The detection was performed with a UV absorbance detector at 254 nm as well as at 280 nm, which is the optimum UV wavelength for detecting thiopental. As an internal standard, butobarbital was used. Secobarbital, which was administered to the patients before the general anaesthesia, could be separated from butobarbital, thiopental and pentobarbital (Fig. 1A).

By making use of a postcolumn ionisation in which the pH of the mobile phase was adjusted to 10, the detection of butobarbital, pentobarbital, secobarbital and thiopental at 254 nm became 30 times more sensitive. The detection limits are 0.5, 0.7, 1.0 and 0.2 $ng.ml^{-1}$ plasma for butobarbital, pentobarbital, secobarbital and thiopental, respectively. The barbiturates were extracted by a dichlormethane procedure.

The yield of this extraction method was 85 to 90%. Addition of the standards butobarbital and thiopental to blank plasma followed by extraction results in a chromatogram such as shown in Fig. 1B. Although peak number 2 appears just in front of butobarbital, it does not interfere with the peak height of butobarbital; the same

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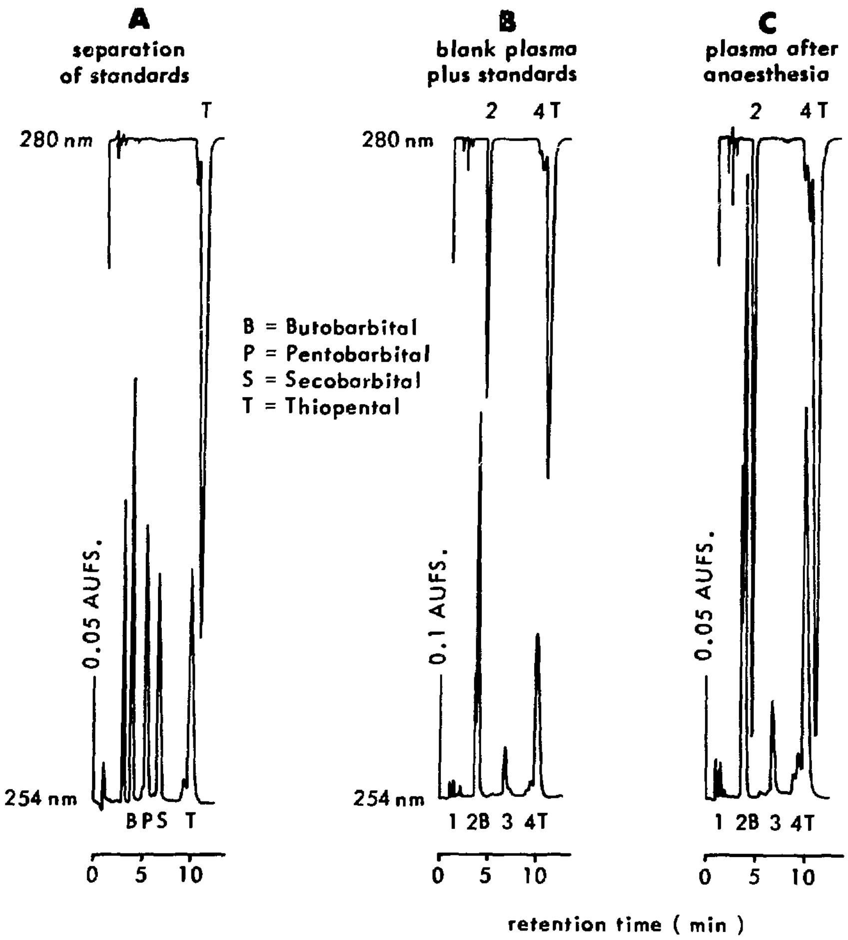

Figure 1: A. HPLC chromatogram after separation of butobarbital, pentobarbital, secobarbital and thiopental. AUFS is the unit of sensitivity of the UV detector.

B. HPLC chromatogram after extraction of blank plasma to which the standards butobarbital and thiopental are added.

C. HPLC chromatogram of patient plasma after extraction. The numbers 1-4 are the plasma peaks observed in Fig. 1B.

is true for peak number 4 which appears in front of the thiopental peak.

In addition, spiking blank plasma with thiopental and butobarbital and thereafter extraction resulted in a linear relationship between the ratio of the peak height of thiopental and the peak height of butobarbital and the thiopental concentration. From this finding, it could be deduced that thiopental and butobarbital were extracted by this procedure with the same efficiency and independently of the thiopental concentration.

Samples were taken from patients after 1, 3, 5, 7, 10, 15, 20, 30, 45, 60 and 90 min and 2, 3, 4, 5, 6, 7, 8, 10, 12, 24 and 28 hours. An example of a chromatogram of patient plasma after general anaesthesia is shown in Fig. 1C.

No traces of secobarbital and the possible metabolite pentobarbital could be detected. The plasma concentration curves for thiopental in man reveal that its pharmacokinetics are those of a four-compartment model drug (Fig. 2). From these data, it can be

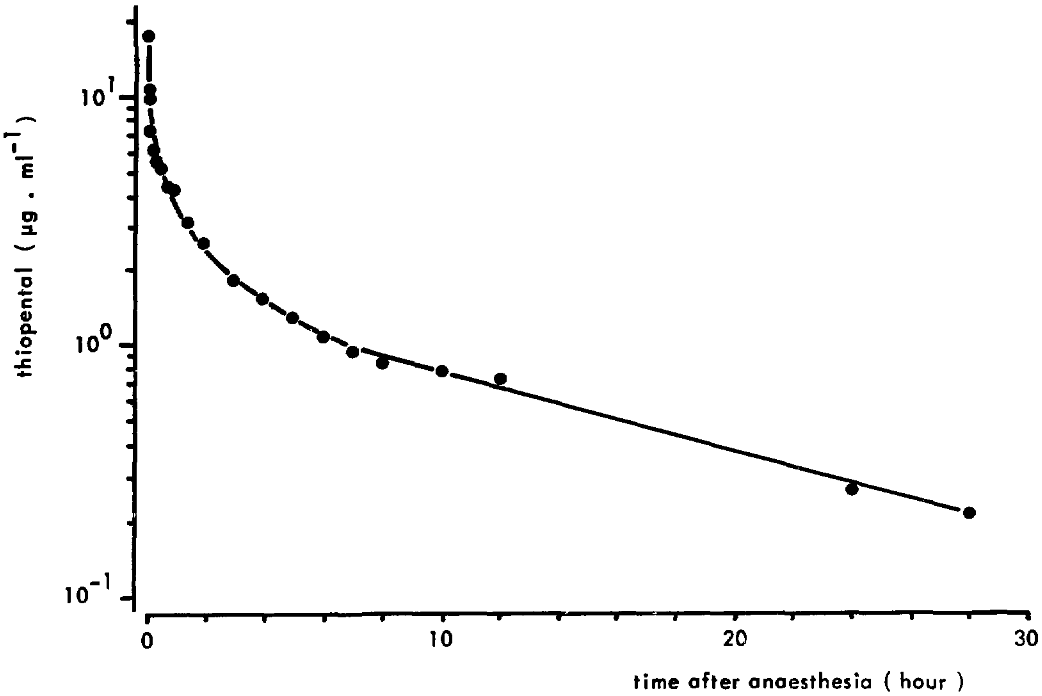

Figure 2: Plasma disappearance curve of thiopental of a 22-year-old male patient.

derived that thiopental in man indeed followed the three distribution phenomena mentioned above for dogs. By applying this technique to plasma of patients of different ages after general anaesthesia, it will be determined whether the pharmacokinetics of thiopental might change with age.

IMMUNOLOGY AND TRANSPLANTATION BIOLOGY

MONOCLONAL ANTIBODIES SPECIFIC FOR RHESUS MONKEY LYMPHOCYTES

G. van Meurs and M. Jonker

Monoclonal antibodies (MCAs) identifying different subsets of lymphocytes can be useful in studies of immunological diseases, organ and bone marrow transplantation and the regulation of the normal immune response. In addition, such reagents can also be used to manipulate the immune response.

Although a number of MCAs specific for human lymphocytes crossreact with rhesus monkey cells, many do not. To use the rhesus monkey as an optimal model for preclinical testing of MCAs, it was necessary to produce MCAs which react with rhesus monkey lymphocytes. BALB/c mice were immunized with rhesus monkey peripheral blood lymphocytes. Three days after the last injection, spleen cells were fused with a HAT-sensitive mouse myeloma cell line (SP2/O). Hybrids were selected in HAT medium. The culture supernatants from growing hybrids were tested in an ELISA technique (Lansdorp et al., 1980) using a pool of rhesus monkey lymphocytes as targets. Positive supernatants were subsequently tested in an indirect immunofluorescent technique using flow cytometry (FACS II, Becton and Dickinson). The MCAs produced were tested on peripheral blood cells, thymocytes and bone marrow cells. By setting the appropriate scatter windows (forward light scatter and 90° light scatter), the fluorescence of lymphocytes, granulocytes and "blast" cells could be determined separately. Table I shows the reaction patterns of eight rhesus specific MCAs.

To investigate whether these MCAs reacted with a well-defined subpopulation of peripheral blood lymphocytes, doublestaining experiments were performed using commercially available MCAs. Cells were incubated with the MCA under investigation and a B, T4 or T8 lymphocyte defining MCA either alone or with a mixture of the two antibodies and subsequently incubated with the second step antibody. From the reaction patterns obtained the proportion of B, T4 or T8 lymphocytes was approximated. As reference antibodies were used OKT4 and Leu3a for the identification of T4 cells, OKT8 and Leu2a for the identification of T8 cells and B8.12 (reactive with DR antigen) and a rabbit antimonkey Ig for the identification of B cells. The results of these experiments are summarized in Table II.

TABLE I

REACTIVITY PATTERNS OF MONOCLONAL ANTIBODIES SPECIFIC FOR RHESUS MONKEY LYMPHOCYTES

		expression on rhesus monkey cells						chimp	human
MCA	Ig class	PBL ly %	PBL gr %	thymocytes %	bm ly %	bm gr %	bm bl %	PBL ly %	PBL ly %
GM1	IgG1	60	50	35	20	60	25	50	50
GM5	IgG1	80	45	0	60	25	70	80	90
GM6	IgG1	0	0	0	60	15	60	0	0
GM9	IgG1	40	0	70	10	0	10	0	0
GM10	IgG2b	50	60	0	10	0	0	0	0
GM11	IgG2a	20	20	0	20	0	0	0	0
GM12	n.d.	20	0	0	0	0	0	0	0
GM13	IgG1	15	0	0	0	0	0	20	0

% : per cent of cells stained (approximate)
PBL: peripheral blood
bm : bone marrow
ly : lymphocytes
gr : granulocytes
bl : blasts (based on forward and 90° scatter characteristics)

TABLE II

REACTIVITY PATTERNS OF MCAs WITH PBL SUBSETS

MCA	B cells	T4 cells	T8 cells
GM1	0%	50%	60%
GM9	0%	0%	100%
GM10	0%	20%	70%
GM11	100%	0%	0%
GM12	100%	0%	50%
GM13	0%	0%	30%

B cells : as defined by α-DR and/or α-Ig antibodies
T4 cells : as defined by OKT4 and Leu3a antibodies
T8 cells : as defined by OKT8 and Leu2a antibodies.
Percentages of reactive cells are approximations.

GM1 reacted with $\pm$ 60% of peripheral blood lymphocytes (both T4 and T8) and also with granulocytes but not with B cells. The proportion of bone marrow lymphocytes reactive with GM1 was much lower, while the proportion of bone marrow granulocytes was approximately the same as in blood. GM1 reacted only weakly with thymocytes (Table I).

GM5 reacted weakly with all blood lymphocytes. The reaction was stronger with granulocytes and lymphocytes in the bone marrow; thymocytes were negative. The antigen recognized by this antibody may be expressed more strongly on granulocytes than on other haemopoietic cells.

GM6 reacted exclusively with cells from bone marrow and most likely reacts with an antigen present only on immature cells. Monkeys receiving an allogeneic bone marrow graft after total body irradiation had $GM6^+$ cells in the circulation shortly after transplantation. Whether these cells were derived from the host or the donor remains to be established. This marker may become useful when monitoring bone marrow graft recipients.

GM9 reacts exclusively with lymphocytes in the blood, thymus and bone marrow. The reactive cells are the T8 cells (Table II).

Although GM9 identifies the same T cell subset as OKT8 and can also inhibit the binding of OKT8 (FITC labelled) to rhesus monkey cells, it does not react with human or chimpanzee lymphocytes. Thus, the determinant recognized by GM9 must be different from that recognized by OKT8, but is most likely located on the same molecule.

GM10 reacts with a proportion of blood and bone marrow lymphocytes and also weakly stains peripheral blood granulocytes. GM10 does not react with B cells, but stains a proportion of both T4 and T8 cells. The GM10 reactive cells are not the same as the GM1 reactive cells, although these two populations show some overlap.

GM11 reacts with ca. 20% of blood and bone marrow lymphocytes. A proportion of blood granulocytes is also positive. Since the definition of the granulocyte population is based on scatter characteristics, it is quite possible that the monocytes are contained in this population and that GM11 reacts with those. In the blood lymphocyte population, GM11 reacts with B cells exclusively. When added to mixed lymphocyte cultures, most combinations were inhibited by GM11. These results suggest that GM11 reacts with an Ia-like antigen.

GM12 reacts with a subset of blood lymphocytes and not with any other cell populations tested. All B cells and ca. 50% of the T8 cells were stained by GM12.

GM13 reacts with a small subset of blood lymphocytes and with no other tested cell population. GM13 stained only a proportion of the T8 cell population. It is interesting to note that some rhesus monkeys carrying an allogeneic organ (kidney or liver) had very low numbers of GM13 positive cells (1% or less) with normal levels of T8 cells, while those recovering from a sublethal dose of whole body irradiation (after rejection of an allogeneic bone marrow graft) had very high numbers of GM13 positive cells with normal levels of T8 cells. Perhaps, the GM13 positive cells play an important role in graft rejection and are suppressed in animals which carry an allograft without signs of chronic rejection and are abundant in monkeys that recently rejected a graft.

Only two of the MCAs were reactive with human peripheral blood lymphocytes and three with chimpanzee lymphocytes. This shows that most of the antibodies produced thus far are species specific, although it is clear that in some cases molecules are recognized which have a homologous counterpart in man. The antibodies which react with both human and rhesus leukocytes and those which identify species specific determinants of homologous molecules expressed on the same cell population in man and rhesus monkeys will be used for further testing.

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THE IMMUNOSUPPRESSIVE EFFECTS OF MONOCLONAL ANTIBODIES SPECIFIC FOR HUMAN LYMPHOCYTE SUBSETS IN RHESUS MONKEYS

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A large number of monoclonal antibodies (MCA) specific for antigens expressed on subpopulations of human lymphocytes have become available in the past few years. Apart from their value as reagents for studying lymphocyte differentiation and cellular interactions in the immune response, these MCAs will most likely replace antilymphocyte sera (ALS) as immunosuppressive agents in organ transplantation. The major advantage of MCAs over ALS is the selective kill of cell populations responsible for graft rejection, while the populations responsible for other immune responses are not affected. Since many of the antihuman MCAs also react with lymphocytes of nonhuman primates, macaques are good subjects for the testing of such reagents for efficacy and toxicity. Recent studies in cynomolgus monkeys have shown that OKT4, an MCA specific for human "helper T cells", can prolong kidney graft survival (Cosimi et al., 1981). In the study reported here, the immunosuppressive effect of several antihuman MCAs was investigated in rhesus monkeys receiving skin allografts. A number of animals was injected i.v. with MCAs specific for different T-cell subsets (see Table I). These MCAs defined the subset containing mostly helper T cells (T4 cells), the subset containing mostly cytotoxic and suppressor T cells (T8 cells) and all T cells comprised of both T4 and T8 cells.

Injection of a MCA did not always result in the elimination of the T cells of the relevant subset from the circulation. Two other effects on the subsets were observed: "coating" and "modulation". As can be seen in Table I, the coating phenomenon occurred with three of the tested MCAs: OKT4 and OKT4A, both reactive with T4 cells and B9.8 and B9 pool, MCAs reactive with T8 cells. An example of coating is given in Fig. 1A. Upon injection of OKT4A, the relative numbers of T4 and T8 cells were monitored using an indirect immunofluorescence technique and FACS analysis. No significant decrease in T4 cells was observed. However, an increase in numbers of cells stained with GM/F (goat antimouse Ig; FITC labelled) was found during the first 10 days of treatment, indicating that T4 cells coated with MCA were present in the circulation. After 10 to 13 days of treatment, the relative numbers of

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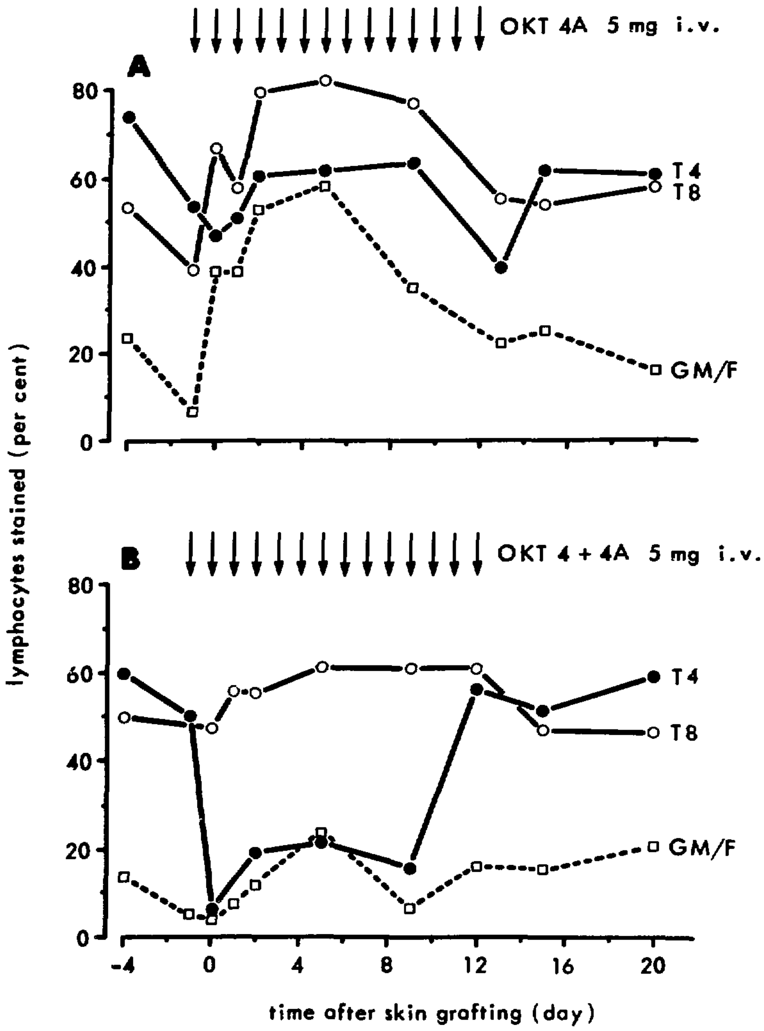

Figure 1: Effect of intravenous injection of OKT4 (A) and OKT4+4A (B) MCAs on the relative numbers of T4 and T8 cells in peripheral blood lymphocytes. The vertical arrows indicate the timing of the injections.
T4 and T8: relative number of "helper" and "cytotoxic/suppressor" T cells as defined by an indirect immunofluorescence test with OKT4A and OKT8A MCAs and FACS analysis.
GM/F: relative number of cells stained with the second step antibody alone.

cells stained with GM/F returned to normal values, indicating that the T4 cells were no longer coated with antibody in spite of continued treatment. This was most likely due to the formation of antimouse IgG antibodies which neutralized the injected MCA. No significant change in the relative numbers of T8 cells was observed

TABLE I

THE EFFECT OF TREATMENT OF RHESUS MONKEYS WITH MONOCLONAL ANTIBODIES AGAINST T-CELL SUBSETS

MCAs used for treatment	immuno-globulin class	reactive cell population (as defined in humans)	daily antibody dose	effect on relevant T-cell subset	skin graft survival (days)
OKT4A	IgG2a	helper T cells	5 mg per animal	coating	16 , 19
OKT4	IgG2b	"	"	coating	11.5 , 18
OKT4+OKT4A	IgG2b+IgG2a	"	"	elimination	13 , 16
OKT8A	IgG2a	cytotoxic/suppressor T cells	5 mg per animal	slow elimination	10 , 11
B9.8	IgG2a	"	0.5 mg.kg^{-1}	coating	11.5 , 12.5
B9 pool	IgG1,IgG2a,IgG3	"	0.5 mg.kg^{-1}	coating or elimination	14,15, 14
OKT11A	IgG2a	all periferal T cells	5 mg per animal	"modulation"	13.5 14
WT1	IgG2a	"	1.5 mg.kg^{-1}	"modulation"	15 17
42 untreated** control animals					10.3 ± (range: 8.5 - 13)

* Each number represents one skin graft survival time in a rhesus monkey.
**These were related and unrelated donor/recipient combinations differing for a single RhLA-A or B-locus.

(after subtraction of the cells stained with GM/F). Interestingly, when OKT4 and OKT4A were injected simultaneously, the T4 cells were eliminated from the circulation (Fig. 1B). Thus, when both the OKT4 and OKT4A antibody molecules are bound to the different antigenic determinants recognized by these antibodies, an efficient elimination of the lymphocytes by the lymphoreticuloendothelial system can occur.

Treatment with OKT11A or WT1 had an effect on the OKT11A or WT1 reactive cells which we have called "modulation". After treatment with these MCAs, the relative numbers of OKT11A and WT1 reactive cells decreased to background values, while those of T4 and T8 cells did not change. Since the T4 and T8 cells are contained in the OKT11A population, this indicates that the cells were not eliminated but that the OKT11A antigen was internalized or shed from the cells (antigen modulation).

The daily injected MCA dose (Table I, column 4) was in all instances sufficient to maintain significant MCA levels in the serum of treated monkeys until the next injection was given 24 hours later. However, after 10 to 13 days of treatment, the injected MCA was neutralized by antimouse IgG antibodies formed by the treated monkeys in all cases. Skin graft survival time was significantly prolonged (p 0.005, Mann-Whitney U-test) when the animals were treated with anti-Th MCAs (OKT4, OKT4A) irrespective whether T4 cells were eliminated from the circulation or were just coated with anti-T4 antibody. These data suggest that the T4-cell population at least is important for skin allograft rejection. MCAs reactive with T8 cells (OKT8A, B9.8) injected separately did not prolong skin graft survival times. However, when a pool of antibodies reactive with the T8 cells (B9 pool) was injected, a slight prolongation in skin graft survival time was observed (p 0.01). The failure to observe prolongation of skin graft survival time with separate antibodies might be due to poor crossreactivity of these antibodies with rhesus cells. MCAs reactive with all T cells (OKT11A, WT1) resulted in a modest but significant prolongation (p 0.01, Mann-Whitney U-test) of skin graft survival times. This is most likely because the cells responsible for graft rejection could not exert their immunological function when antigenic modulation occurred.

It can be concluded that MCAs reactive with T4 cells and in some cases those reactive with T8 or all T cells can cause immunosuppression. It is not absolutely necessary to eliminate the cells, since coating with MCA or antigen modulation by MCA also interferes with the immunological function. One of the major drawbacks might be the rapid appearance of antimouse Ig antibodies. A further investigation of monoclonal antibodies identifying perhaps even more specific subsets of immunocompetent cells may be required to achieve more vigorous immunosuppression, including suppression of antibody formation against the foreign mouse protein. This should increase the immunosuppressive potential of MCAs in experimental and, hopefully, clinical transplantation.

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THE INFLUENCE OF ALLOIMMUNIZATION ON MLC REACTIVITY BETWEEN RhLA IDENTICAL RHESUS MONKEY SIBLINGS

F.J.M. Nooij, M. Jonker, A.A. van Es and H. Balner

Results of experiments performed in the early 1970's with rhesus monkeys indicated that, as in other mammalian species, an MHC-linked D locus exerts an overriding influence on MLC reactivity (Neefe et al., 1973; Balner and Toth, 1973). Besides the D locus, there are strong indications for the existence of other loci, the products of which can stimulate allogeneic lymphocytes in culture (Balner and Toth, 1973). In 1979, van Es and Balner found that unrelated D identical monkeys which were initially MLC negative showed strong mutual reactivity in MLC after an exchange of skin grafts and booster injections of blood. This effect was attributed to gene products of a locus called D'. This D' locus was thought likely to be linked to the MHC (in the rhesus monkey, called RhLA), since RhLA identical siblings showed only marginal and infrequent MLC reactivity against each other after a similar course of mutual immunizations. However, when a limited number of RhLA identical siblings were further immunized with skin and blood from haploidentical halfsibs (i.e., third party monkeys), strong MLC responsiveness among the RhLA identical sibling pairs could sometimes be provoked. Thus, it was postulated that factors not controlled by the RhLA region can also exert an influence on the MLC under certain conditions (van Es and Balner, 1979).

It seemed interesting to investigate whether the apparent difference in response after mutual immunization between unrelated D identical individuals and RhLA identical siblings could be quantified. An attempt was made to induce MLC responsiveness in initially MLC negative pairs of RhLA identical siblings by increasing the number of mutual immunizations. Cross immunizations were performed among 4 pairs of RhLA identical siblings which had never been alloimmunized prior to the experiments. Each animal received skin grafts and several series of booster injections with T lymphocytes from its RhLA identical partner. After each immunization, the response of the lymphocytes was tested in the MLC.

Table I shows the effect of mutual alloimmunization on the MLC reactivity among the 4 pairs of RhLA identical siblings. None of the monkeys showed any reaction against the RhLA identical partner before alloimmunization. An exchange of skin grafts and one series of mutual booster injections (moderate immunization) did not lead to clear responsiveness in MLC against the partner. Monkey LH was an exception, in that MLC reactivity against cells of his partner OI became clearly positive. Results obtained after a second or a third series of mutual booster injections (extensive immunization)

THE EFFECT OF MUTUAL ALLOIMMUNIZATION ON THE MLC REACTIVITY AMONG FOUR PAIRS OF RhLA IDENTICAL SIBLINGS

RhLA identical sibling combination	MLR before immunization	MLR after moderate immunization	MLR after extensive immunization
MG ⟶ UZ*	-	-	-
UZ ⟶ MG*	-	-	+
LH ⟶ OI*	-	+	+
OI ⟶ LH*	-	-	+
QV ⟶ UA*	-	-	+
UA ⟶ QV*	-	-	+
JN ⟶ PS*	-	(-)	-
PS ⟶ JN*	-	(-)	-

reveal that 5 of the 8 animals had become MLC positive against their partner.

It was postulated that the RhLA linked D' locus is the analogue of the human HLA-linked SB locus (Shaw et al., 1980). However, it seems that the D' locus is mapping outside the presently known RhLA loci, since at least 1 of the 8 RhLA identical siblings became responsive immediately after the first alloimmunization, which might be due to recombination. This would be in agreement with the finding that the human SB locus also maps outside the presently known MHC loci.

Current data show that extensive mutual immunization of RhLA identical sibling pairs leads to conversion of the MLC reactivity in 63% of the cases. To explain this observation, the following two hypotheses are proposed:

1. D' is an RhLA linked immunization locus (or loci) which enhances responsiveness to weak MLC stimulating determinants, including those which are not defined by the MHC. Thus, after moderate immunization, rhesus monkeys could be made MLC responsive against an RhLA identical sibling partner only if exposure to a different D' antigen (i.e., tissue from a third party monkey) had also taken place. As RhLA identical siblings share the entire MHC region, it has been assumed that such animals react against MLC antigens which are controlled by a locus mapping outside RhLA and which is designated by van Es and Balner (1979) as D". However, when RhLA identical siblings were mutually alloimmunized in a more vigorous manner, 5 of the 8 animals became responsive to their RhLA identical part-

ner. It is assumed that extensive sensitization to different D" products is, after all, sufficient to convert a negative MLC into a positive one.

2. Alternatively, all positive MLC reactions after mutual immunizations among both unrelated pairs of D identical monkeys and also among RhLA identical sibling pairs may be due to differences concerning products of minor MLC loci. These loci need not necessarily be linked to RhLA. Products of these loci would exert their influence only after alloimmunization and differences for these products may have a quantitative effect on MLC responsiveness. *Unrelated D identical partners are* likely to share fewer of these minor MLC loci products than do RhLA identical sibling partners. For that reason, conversion of the MLC response requires fewer mutual immunizations in the unrelated D identical monkeys than in RhLA identical siblings.

We believe that the *first* hypothesis is the more likely one: extensive immunization against disparate D" products can lead to MLC responsiveness among RhLA identical siblings. As mentioned above, the human SB locus is linked to HLA. The D' locus in rhesus monkeys is most likely the analogue of this SB locus. In both primate species, products of this locus exert their influence only after alloimmunization and they have a priming role. Thus, it is very reasonable to assume that there is also an MHC linked priming locus in nonhuman primates, which makes the second hypothesis rather unlikely.

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PLATELET TRANSFUSIONS HAVE A POSITIVE EFFECT ON KIDNEY ALLOGRAFT SURVIVAL IN RHESUS MONKEYS AND INDUCE VIRTUALLY NO CYTOTOXIC ANTIBODIES

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and H. Balner

Pretransplant blood transfusions have a strong positive effect on kidney allograft prognosis in monkeys and man. The administration of transfusions to patients awaiting kidney transplantation has now become an established procedure. However, the risk of sensitization remains one of the handicaps of deliberate transfusions. Therefore, it was interesting to investigate in a primate model whether a positive effect on graft survival could be achieved without the concomitant induction of cytotoxic antibodies if platelets were transfused instead of whole blood. An effect of platelet transfusions on kidney graft survival could also shed some light on the mechanism of the transfusion effect, since platelets carry SD antigens but lack some of those (e.g., DR) carried by leukocytes and erythrocytes. Thus, a controlled preclinical study was started in rhesus monkeys, a species phylogenetically close to man and which in the past has been shown to provide experimental results which can be extrapolated to man.

Young unrelated male and female rhesus monkeys (Macaca mulatta) weighing 3 to 5 kg and born at the Rijswijk Primate Center were used. The animals had not been alloimmunized prior to the transfusions. All host/donor combinations were mismatched for one or two DR antigens and had two or three disparities for A and B locus antigens. Recipients received 2 $mg.kg^{-1}$ azathioprine and 1 $mg.kg^{-1}$ prednisolone which, starting on the day of transplantation, were administered i.m. daily for a maximum period of 45 days. Transfusates consisted of platelet suspensions separated from 20 ml fresh unpooled blood and were given three times at biweekly intervals before transplantation, the last at 2 to 3 weeks prior to operation. Transfusates consisted of ca. $4x10^9$ washed platelets and contained less than 1 leukocyte per $5x10^4$ platelets. Transfused control animals were given three transfusions of 20 ml fresh unpooled whole blood according to the same timing protocol. Screening for pretransplant alloantibodies in the recipients' sera was done after each transfusion and just prior to transplantation.

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The survival times observed in the experimental animals are shown in Fig.1. Nontransfused control monkeys (group I) had a mean survival time (MST) of 12.8 $\pm$ 1.2 days, with survival times ranging from 9 to 22 days. In order to have a parameter besides the MST for evaluation, animals surviving 22 days were considered to show "prolonged" graft survival. The administration of three transfusions of whole blood had a beneficial influence on graft survival (group II).A MST of 33.6 $\pm$ 4.1 days was found, with 14 of 23 animals (61%) showing a prolonged survival time ($p < 0.005$ by Mann-Whitney U test). Three transfusates consisting of platelets only (group III) also had a clearly positive effect on graft prognosis. The MST of this group was 34.7 $\pm$ 5.5 days, with 10 of 15 recipients (66%) showing a prolonged survival time ($p < 0.005$).

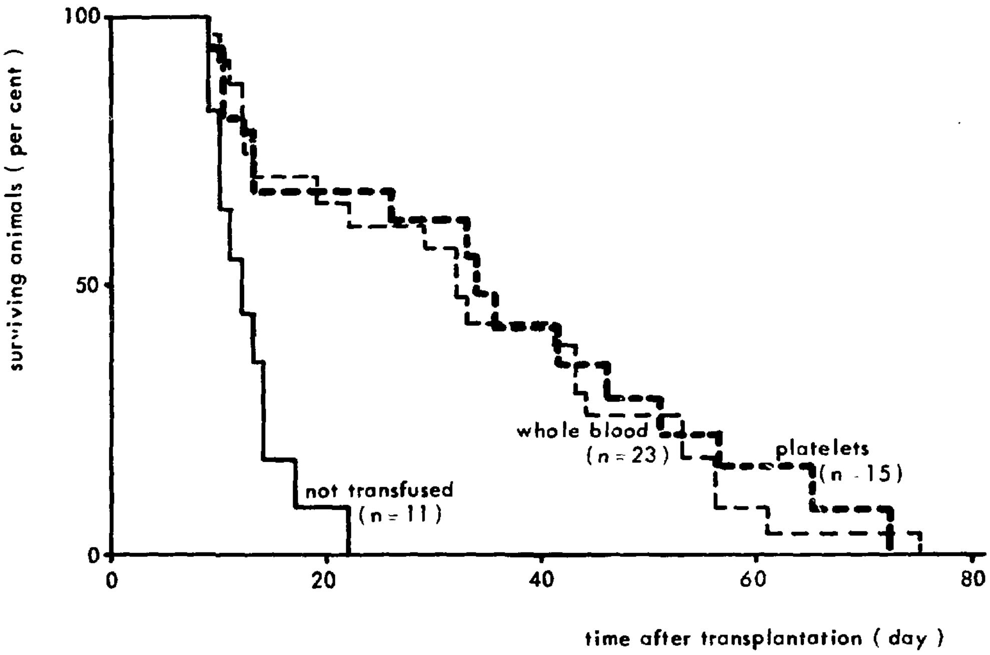

Figure 1: **Actuarial analysis: influence of three pretransplant transfusions of platelets or whole blood on kidney allograft survival in rhesus monkeys.**

In all animals, death due to terminal graft rejection was proved at necropsy by the histological characteristics of acute or chronic rejection or intermediate stages. The histological patterns of rejection in the animals receiving platelet transfusions were in no way different from those observed in monkeys dying at comparable intervals after transplantation and having received whole blood transfusions.

The monkey data presented here clearly demonstrate that third party platelet transfusions have a beneficial influence on kidney graft survival in an outbred species. Animals receiving platelets had a MST which was significantly longer than that of nontransfused controls. In terms of MST, the effect of platelet transfusions is comparable with that obtainable with transfusions of whole blood.

In addition, differences between the influence of platelets and whole blood were noted, which if confirmed could make the administration of platelet transfusions a more attractive pretransplant procedure than giving transfusions of whole blood.

Only a small number of the recipients given platelet transfusions produced conventional cytotoxic antibodies against unseparated lymphocytes. In comparison, the sera of animals given transfusions of whole blood usually contained strong and "broad" cytotoxic antibodies against T lymphocytes.

In patients awaiting kidney transplantation, the incidence of alloantibodies directed against potential kidney donors is increased by multiple blood transfusions, which makes it more difficult to find a suitable donor for such multitransfused patients. The lower incidence of antibodies following platelet transfusions could therefore be a considerable advantage of the application of this type of transfusate for clinical purposes.

THE INFLUENCE OF CO-TRIMOXAZOLE ON LEUKOCYTE REGENERATION IN MICE AFTER BONE MARROW TRANSPLANTATION

P.J. Heidt, A. Klepke and C.P.J. Timmermans

The occurrence of bacterial and fungal infections is one of the most life-threathening complications in patients with a severe and long-lasting granulocytopenia (Bodey et al., 1966). One of the methods for preventing these infections is selective decontamination (SD) of the gastrointestinal (GI) tract. This aims at elimination of potential pathogenic microorganisms while leaving the anaerobic microflora unaffected (Van der Waaij and Berghuis-de Vries, 1974). Preservation of these anaerobes is of vital importance, since this part of the microflora is responsible for the so-called colonization resistance which prevents exogenous microorganisms from colonizing the GI tract (Van der Waaij et al., 1971).

One of the drugs frequently used for SD is co-trimoxazole (SxT), a combination of trimethoprim and sulfamethoxazole (Heidt, 1979; Sleyfer et al., 1980; Dekker et al., 1981). Sometimes, during treatment with SxT or with one of the components of this drug combination, a haematotoxicity such as neutropenia and thrombocytopenia is observed (Bradley et al., 1980; Sheenan, 1981; Dekker et al., 1981). Since a possible haematotoxicity can be a contraindication to the use of SxT for SD after bone marrow transplantation (BMT) or during remission induction therapy for leukaemia, we investigated the influence of this drug combination on leukocyte regeneration in female C3H mice after isologous BMT.

For this experiment, we used germfree mice to rule out a possible stimulation of the regenerating bone marrow by components from the microflora in the control group. The animals were housed in macrolon cages (5 animals per cage) in a laminar flow isolator and were fed sterile food and drinking water <u>ad libitum</u>. The experimental group (25 animals) was given SxT in the drinking water, to which the nonabsorbable antibiotics Neomycin ($1\ g.l^{-1}$), Streptomycin ($1\ g.l^{-1}$), Bacitracin ($1\ g.l^{-1}$) and Pimafucin ($100\ mg.l^{-1}$) were also added to prevent possible exogenous contaminations as was sugar ($50\ g.l^{-1}$) to improve the taste. The control animals (n = 25) received no SxT in their drinking water to which the same nonabsorbable antibiotics and sugar were added. After SxT was given for one week to the experimental group, all animals were irradiated under germfree conditions (7.225 Gy); this was followed by an isologous BMT (3×10^6 nucleated cells per recipient) on the next day. At regular intervals after transplantation, a group of 3 experimental animals and 3 control animals were sacrificed after blood was collected from the orbita under complete anaesthesia. This blood was used for a total and a differential count of the leukocytes. The results, which are summarized in Fig. 1, show that there is a delay in the regeneration of the leukocytes in the SxT treated group and that this delay is due to a slower regeneration of both the granulocytes and the lymphocytes.

Our findings confirm the in vitro findings of Golde et al. (1978) who found a 50% inhibition of the formation of human erythroid and granulocytic colonies by trimethoprim. Their finding that this inhibition could be reversed by folinic acid suggests that SxT can interfere with haemopoiesis by inhibition of tetrahydrofolate synthesis. A possible reversal of granulopoiesis inhibition by folinic acid in mice treated with SxT after BMT will be the subject of future studies.

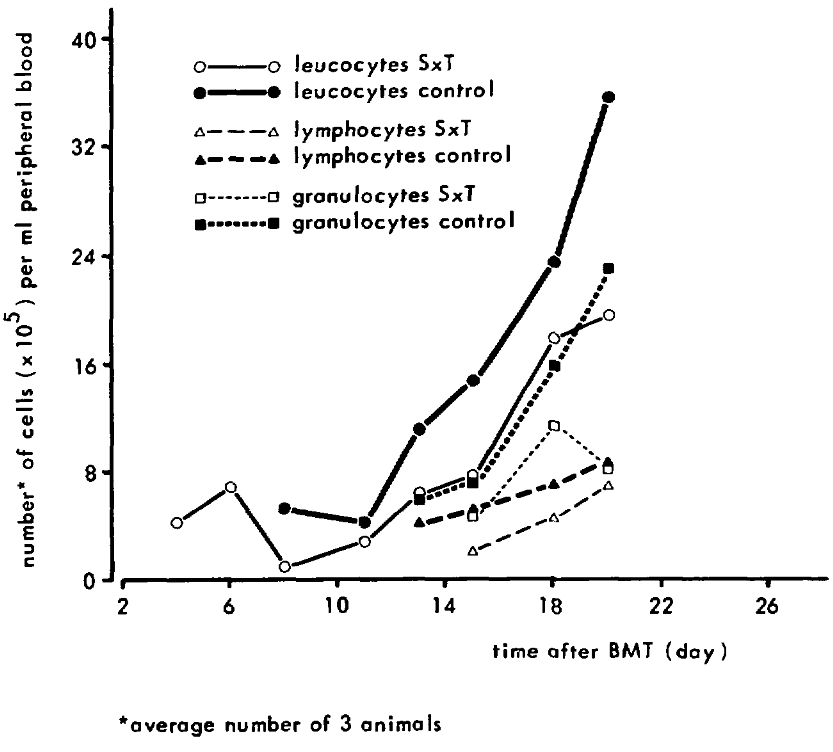

Figure 1: Number of leukocytes, lymphocytes and granulocytes in germfree SxT treated and control C3H mice after total body irradiation and isologous bone marrow transplantation.

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A SOLUBLE FACTOR INVOLVED IN THE SUPPRESSION OF THE GRAFT VERSUS HOST REACTION AND MIXED LYMPHOCYTE CULTURES

S. Knaan-Shanzer and D.W. van Bekkum

The acute Graft versus Host disease (GvHD) occurring in lethally irradiated mice after transplantation of allogeneic bone marrow and spleen cells can be prevented by adding thymocytes or spleen cells of neonatal mice to the transplanted cells. We previously showed that the cell subpopulation in the neonatal thymus involved in the GvH suppression consists of PNA^- T lymphocytes ($Thy\text{-}1^+$) carrying a Lyl^+2^- cell surface phenotype (Van Bekkum and Knaan, in press).

The mechanism by which those nonspecific suppressor T cells exert their immunoregulatory activity is not yet fully known. One explanation is that the suppressor cells act on effector cells by cell to cell contact. Shou (1980) and Argyris (1981) reported, however, that lymphocytes cultured in vitro, without further stimulation, release into the supernatant a soluble factor which suppresses the proliferation of fresh lymphocytes.

The present study concerns the question of whether the suppressive effect of neonatal thymus cells on GvHD in lethally irradiated mice is mediated by a soluble agent. In vivo testing for GvHD inhibition was not feasible because the amount of supernatant from thymocyte cultures is a limiting factor. Therefore, spleen cells used for inducing GvHD were incubated with the test preparations in vitro prior to grafting.

Supernatants to be tested for the presence of suppressor factor (SF) were prepared from cultures of thymocytes or spleen cells of neonatal mice or rats. As control preparations, we used supernatants collected from similar cell cultures prepared from adult animals. All cultures were carried out in 12 ml Falcon tubes containing 1 ml of cells (24×10^6 cells.ml^{-1}) in RPMI 1640 medium supplemented with 20 mM Hepes, 5×10^{-5} mM 2-mercaptoethanol and 10% fetal calf serum, in a humidified atmosphere of 5% CO_2/95% air at 37°C. After 4 days, the supernatant was separated from the cells by centrifugation, pooled and stored at -20°C. SF prepared on different days were not pooled.

For estimating the suppressive activity of the preparations on GvHD inducer cells, supernatants from 6×10^6 cells (neonatal or adults) were added to aliquots of 1×10^6 CBA spleen cells (GvH inducer cells). The mixture was incubated for 60 min at 37°C. The incubated cells were then inoculated along with 1×10^6 CBA bone marrow cells into lethally irradiated (C57BL x CBA)F1 mice. Seven of eleven supernatants from neonatal cell cultures tested suppressed the GvH reaction, while no activity was shown by the ten control preparations.

Table I shows the effect of different supernatants on GvH mortality and on the CFU-S in the marrow. The effect on the CFU-S was determined by in vitro incubation of bone marrow cells with supernatant prior to inoculation into lethally irradiated syngeneic mice. The incubation conditions here were identical to those of the spleen cells in the GvHD assay.

The results suggest that the SF is not species specific, since factors taken from cultures of either mice or rat cells suppress the in vivo GvH reaction in mice to the same extent. Moreover, the activity of the SF is specifically directed against T cells (i.e., the GvH inducer subpopulation). It does not affect the pluripotent haemopoietic stem cells in the bone marrow. This property of the suppressor factor is important, since, in humans, the marrow is the source of stem cells as well as T lymphocytes responsible for GvH disease.

Table I

SUPERNATANTS TAKEN FROM CULTURES OF NEONATAL SPLEEN AND THYMUS CELLS SUPPRESS GvHD INDUCTION BY CBA SPLEEN CELLS BUT DO NOT AFFECT THE CFU-S IN CBA BONE MARROW

supernatants tested for suppressor factor		spleen cells tested in the GvH reaction		bone marrow tested in the CFU-S assay	
no.	source of cultured cells	survival after 20 days	per cent suppression	CFU-S per 10^6 cells inoculated	per cent of decrease
-	-	0/10	0	140	0
	neonatal				
72	CBA spleen	10/10	100	110	21
73	WAG thymus	8/10	80	132	6
74	WAG spleen	8/10	80	142	-1
	adult				
75	CBA spleen*	0/10	0	140	0
87	WAG thymus*	2/10	20	113	19
90	WAG spleen*	2/10	20	97	30

*Control preparations.

For the purification of the suppressor factor, an assay which detects small amounts of the agent is needed. One obvious in vitro system is the mixed lymphocyte reaction (MLR). The MLR with mouse lymphocytes was found to yield relatively low values and the results varied considerably among different mouse batches. With rat lymphocytes, the results were excellent and the MLR was found to be inhibited by cocultured neonatal thymus and spleen cells but not by cocultured adult thymus cells. Therefore, the supernatants were assayed for inhibition of the MLR of rat lymph node cells. Most of the supernatants from cultures of both rat and mouse neonatal tissues cause reproducible inhibition of the MLR (Table II). In the in vivo GvHD model, the proportion of supernatants with suppressor activity was lower than in the MLR. This could indicate

Table II

SUPPRESSION ACTIVITY OF SUPERNATANTS IN THE IN VITRO RAT MLR AND IN THE GvH

source of cells cultured	reactivity of supernatants in MLR*	reactivity of supernatants in GvH
neonatal		
mouse thymus	1/4**	1/4
mouse spleen	5/5	3/5
rat thymus	6/6	1/2
rat spleen	7/8	2/3
adult		
mouse spleen	0/3	0/3
rat thymus	0/4	0/4
rat spleen	0/3	0/3

* The MLR consist of 4×10^5 WAG/Rij lymph node cells cocultured with equal numbers of irradiated (34 Gy) lymphocytes from either BN (allogeneic cultures - ABx) or WAG/Rij rats (syngeneic cultures - AAx). To 150 ul MLR cultures 50 ul of the different supernatants were added.

$$\% \text{ suppression } [1 - \frac{\text{ABx-AAx) in the present of supernatant}}{\text{(ABx-AAx) control}}] \times 100$$

**number of supernatants inducing more than 50% suppression/number of supernatants tested.

a higher sensitivity of the in vitro test system or might be due to differences between mouse and rat lymphocytes. In neither assay did any of the control preparations show suppressor activity. These results indicate that the MLR can be considered as a most suitable system for the screening of preparations for suppressor activity.

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RAPID EXPANSION OF HUMAN CYTOTOXIC T-CELL CLONES: GROWTH PROMOTION BY (A) HEAT-LABILE SERUM COMPONENT(S) AND BY VARIOUS TYPES OF FEEDER CELLS

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S.J.L. Bol and R.L.H. Bolhuis

Culture conditions that allow prolonged and rapid expansion of cloned T cells are a prerequisite for extensive functional characterization and analysis of surface markers of these cells and for in vivo testing. In attempts to improve culture conditions for human T-cell clones with cytolytic activity, several culture parameters were studied, including allogeneic feeder cells, monocytes, stimulator cells, geometry of culture vials, lectins, indomethacin and fresh versus heat-inactivated human serum.

These studies were performed using various human T-cell clones with different cytolytic activities. Peripheral lymphocytes were primed against an EBV transformed B-cell line (B-LCL). Clones were obtained by limiting dilution methods in the presence of Interleukin-2 (IL-2) and irradiated feeder cells. After about 10 days of culture, a number of clones were obtained, expanded and tested for cytolytic activity. Two clones with different cytolytic activities against the stimulator cell and one without cytolytic activity were selected and used to study the culture conditions required for expansion.

It was found that some continuously growing cell lines can exert strong growth promoting activity when combined with allogeneic "feeder" lymphocytes (Fig. 1). This apparently synergistic effect between the irradiated feeder cells was <u>not</u> restricted to the original B-LCL stimulator cell (BSM), since an unrelated B-LCL (APD) exerted similar growth promoting activity and these cells also induced growth of noncytotoxic clones. Two T-cell lines tested (CEM and MOLT-4) did not show growth promoting activity, whereas the myeloid cell line K562 showed a moderate effect. None of these feeder cells had any effect on cytolytic activity or target cell specificity of the clones. Furthermore, depletion of monocytes from PBL feeder cells resulted in threefold lower culture results. On comparison of different culture vials used for the expansion of clones, it was found that cell yields obtained in 96-well round bottom microtiter plates exceeded those obtained with 16 mm ø Cluster wells or 75 cm^2 flasks.

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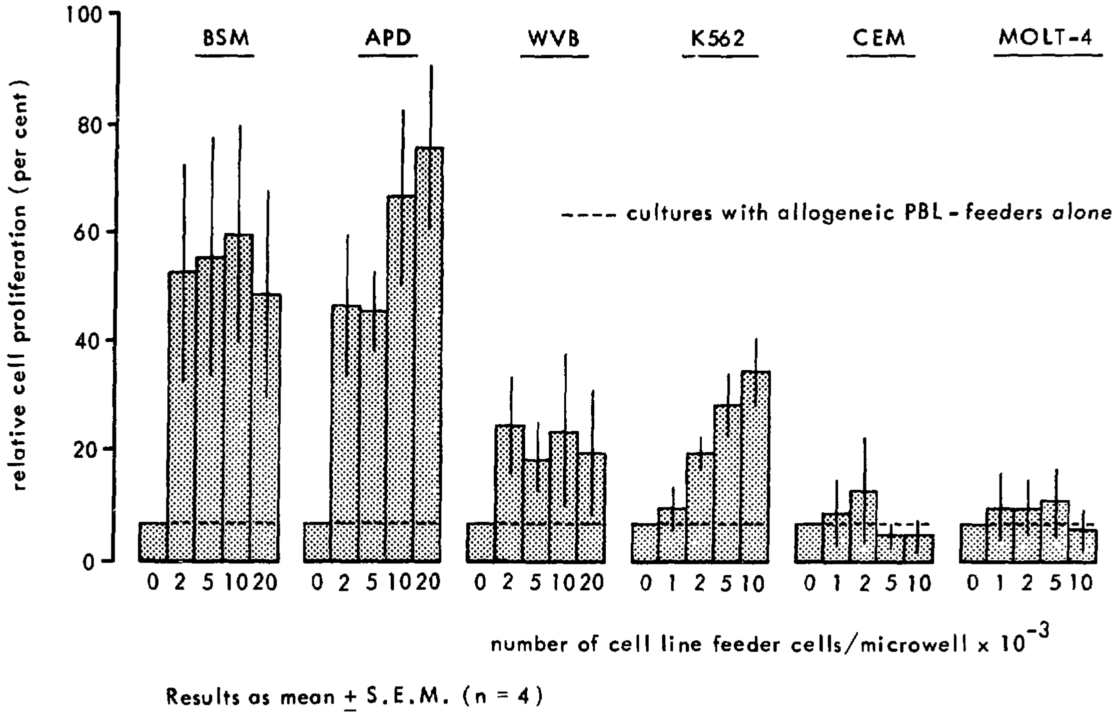

Figure 1: Growth stimulating capacity of different irradiated (20 Gy) continuous cell lines on the proliferation of three human cloned T cells. Results presented as mean ± S.E.M. (n = 4). Cultures were performed in round bottom microtiter plates (Greiner) for 7 days. RPMI-Hepes supplemented with 10% heat-inactivated human serum, leukoagglutinin (1 $\mu g \cdot ml^{-1}$), indomethacin (1 $\mu g \cdot ml^{-1}$), 5% IL-2 and 2×10^4 irradiated (40 Gy) allogeneic PBL per well was used.

The importance of a heat-labile serum component in human serum was suggested on comparison of fresh versus heat-inactivated human serum (10%) added to cultures of T-cell clones. Considerable improvement in cell yield (Table I) was found for all clones tested by using medium supplemented with fresh, not heat inactivated serum. Recent results indicate that this effect is probably not due to scavenge of toxic radicals from the culture medium, since the addition of glutathione (250 µM) or mercaptoethanol did not restore the inhibiting effect induced by heat inactivation of human serum. It should be noted that the benefit of fresh human serum may be species restricted, as a murine IL-2 dependent T-cell line was inhibited by the same fresh serum batch that stimulated the human clones. Optimal cell yields were obtained when all of the factors mentioned above were combined. With the method described, we were able to expand one cytotoxic T cell up to 10^9 cells in about one month and with full retention of cytolytic activity and target cell specificity. This is important for immune

Table I

DECREASED CELL PROLIFERATION WITH HEAT-INACTIVATED SERUM

clone	serum	serum (batch)* 1		serum (batch)* 2	
		exp. 1	exp. 2	exp. 3	exp. 4
A	fresh	141	27	58	180
	inactivated	24	1.8	21	15
B	fresh	-	-	9.1	45
	inactivated	-	-	6.0	5
C	fresh	-	97	15.6	52
	inactivated	-	82	8	17

Results expressed as cell yield (x 10^3) per well after 7 days of culture under standard culture conditions (see legend to Fig. 1) + 10^4 irradiated (20 Gy) BSM (B-LCL) feeder cells per well. Cultures were established with 500 cells per well.
*Pooled serum of 4 healthy AB individuals.

therapeutic proposals because large amounts of cells might be needed within a limited time for effective treatment. In the coming period, we will focus on the generation, selection and expansion of cytotoxic T cells with high cytolytic activity against fresh human tumour cells.

DIFFERENT TYPES OF CYTOTOXIC EFFECTOR CELLS ARE INVOLVED IN LECTIN DEPENDENT CELL MEDIATED CYTOTOXICITY

R.L.H. Bolhuis

Allospecific cytotoxic T cells (CTL) are generated in a mixed lymphocyte culture. We have earlier reported that lymphoid cells which are activated in a mixed lymphocyte culture (MLC-AK cells) can lyse in vitro grown auto-, allo- and xenogeneic tumour cells, even when these target cells are natural killer (NK) cell resistant. Moreover, MLC-AK cells may exert lectin dependent cellular cytotoxicity (LDCC), i.e., lysis of NK-resistant tumour or lymphoid target cells which are precoated with, e.g., the lectin phytohaemagglutinin (PHA). Even autologous PHA treated lymphoid cells are lysed by MLC-AK cells. We have investigated the influence of OKT-8 and FK-18 monoclonal antibodies on CTL, MLC-AK and LDCC activity. OKT-8 is a monoclonal antibody directed against a T cell subset including suppressor and cytotoxic precursor cells; FK-18 is a monoclonal antibody directed against antiallocytolytic T cells. Lytic activity of cytotoxic cells was determined in a 4 h ^{51}Cr release cell mediated lympholysis (CML) assay. It was demonstrated that monoclonal anti-CTL antibodies (FK-18) do not inhibit the MLC-AK killing of tumour cells or the LDCC, but they do inhibit the lytic activity of the allospecific CTL. The inhibition of lysis by FK-18 is summarized in Table I. From these observations, we concluded that a subpopulation of the CTL's which are involved in allospecific lysis is not active in the lysis of NK resistant or target cells by MLC-AK and also not in exerting LDCC. Some of the CTL, however, can exert LDCC. This might be explained by a different membrane structure being involved in allospecific CML as compared with LDCC, since allospecific CTL effector function is blocked by anti-CTL MCA (FK-18), while LDCC effector cell function is not.

The various effector cells are functionally defined and we therefore investigated whether the allospecific CTL, the MLC-AK cells and the lymphoid effector cells exerting LDCC represent a) different, b) partially overlapping or c) even identical cell populations. Cell separations were performed using the "panning" technique in which anti-CTL MCA (OKT-8) is coated on a plastic surface. Cells bearing the relevant cell surface marker were bound to these surfaces and subsequently isolated by overnight incubation allowing the cells to detach. A representative experiment is presented in Table II. It can be concluded from these data that an enrichment of allospecific CTL and cells exerting LDCC in the

Table I

THE EFFECT OF THE FK 18 MCA ON CTL, MLC-AK AND LDCC LYTIC ACTIVITIES*

lytic activity target cells	CTL Ly**	MLC-AK K 562	LDCC Ly-PHA***
FK-18¤ absent	26.0 ± 12.2 (n = 13)¤¤	54.1 ± 16.8 (n = 12)	23.1 ± 18.9 (n = 13)
FK-18 present	9.8 ± 11.0	53.2 ± 13.3	20.3 ± 14.9
% difference	16.2 ± 12.6	0.9 ± 6.4	2.8 ± 9.4
	$p < 0.001$	$p < 0.4$	$p < 0.3$

* lytic activity is expressed as % lysis at the effector : target cell ratio of 40 : 1

** Ly: normal lymphocytes.

***PHA: precoated targets.

¤ FK-18: monoclonal antibody directed against antiallo-CTL.

¤¤ number of donors tested.

Table II

THE EFFECT OF ISOLATION BY "PANNING" OF OKT-8^+ CELLS ON ALLOSPECIFIC (CTL), ACTIVATED KILLER (AK) AND LECTIN DEPENDENT CELLULAR CYTOTOXICITY (LDCC)

lytic activity	CTL	MLC-AK	LDCC
target cells	Ly	K 562	Ly-PHA-bl*
	percent specific ^{51}Cr-release¤		
unfractionated	14	47	27
OKT 8^-**	6	51	11
OKT 8^+***	30	23	47

* Ly-PHA-bl: 3-day blast cells; 10 $\mu g.ml^{-1}$ per 10^6 cells.
** OKT-8^- cells: percent yield, 45.
***OKT-8^+ cells: percent yield, 23.
OKT-8^+ : T cell fraction containing suppressor/cytotoxic cells.
¤ Effector : target cell ratio 40 : 1

OKT-8 positive fraction is obtained, whereas MLC-AK activity is found primarily in the negative fraction. Similar results were obtained by using anti-CTL MCA (FK-18) plus complement to eliminate the allospecific CTL by lysis. A reduction in allospecific CTL activity as well as in LDCC activity was observed, whereas the MLC-AK activity remained unaffected.

Cytolytic T cells were isolated by first labelling the cells with FK-18 followed by a goat antimouse FITC-conjugated antiserum as a second layer and sorting them in a fluorescence activated cell sorter. The FK-18 positive cell fraction showed LDCC as well as anti-allo activity. The FK-18 negative cell fraction, among which are the MLC-AK cells, still showed some LDCC activity. This could also be concluded from experiments where cells were cloned by limiting dilution and after expansion of the cloned cells tested for lytic activity in the various lytic assays. Cloned allospecific CTL also exerted LDCC activity. This confirms the data presented here obtained with uncloned CTL separated from the MLC responder population on the basis of their cell surface markers.

Thus, some of the individual CTL can possibly exert two different forms of lysis, involving different receptors on the individual effector cell and different target structures on the target cells. This was found even when autologous PHA treated effector cells were used as targets. An interesting cloned T cell is the one that shows lytic effects in LDCC, MLC-AK and allospecific CML. We will report details on this in the near future.

MORPHOLOGY AND SURFACE MARKERS OF HUMAN NK CELLS

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and R.L.H. Bolhuis

Natural killer (NK) cells have been defined as lymphoid cells exerting cytotoxicity on a variety of tumour cells without prior immunogenic stimulation. The present study deals with the characterization and purification of human NK cells. Starting with Ficoll separated and monocyte depleted peripheral blood lymphocytes, NK cells can be enriched in the low buoyant density fractions of discontinuous Percoll gradients (Saksela and Timonen, 1980). These low density fractions contain high percentages of large granular lymphocytes (LGL) (Timonen et al., 1979). In addition, cells which resemble LGL but lack the characteristic granules are detected: large agranular lymphocytes (LAL). Although the highest concentrations of LGL and LAL are found in the same low density fraction, the modal density of LAL is higher than that of LGL. These data confirm the association between NK lytic activity and the occurrence of LGL. However, the coenrichment of LGL and LAL indicates the possible involvement of (part of) the LAL population in the NK phenomenon.

In addition to this morphological characterization, the monoclonal antibodies HNK-1* (Abo and Balch, 1981) and B73.1** (Perussia et al., 1983) have been described as selectively recognizing human NK cells. These antibodies were used to further analyse the composition of the NK cell population. In peripheral blood lymphocyte preparations an average of 14% HNK-1 positive cells can be detected. HNK-1 positive (POS) and negative (NEG) cells were obtained by FACS II cell sorting and tested for NK activity (cytotoxicity against K562) and morphology (Table IA). With cells from donor 1, almost all NK activity was present in the HNK-1 POS cell fraction, which is in agreement with data of Abo and Balch (1981). However, the samples from donors 2 to 4, cells with NK activity could not be separated on the basis of HNK-1 antigen expression. Morphological analysis of the fractions showed enrichment for LGL and LAL in the HNK-1 POS fraction. The HNK-1 NEG fraction was devoid of LGL, but the frequency of LAL was similar to that found in unfractionated material. In another series of experiments NK cells were enriched in the low density

* provided by Dr. T. Abo, University of Alabama, Birmingham, Ala., U.S.A.

**provided by Dr. G. Trinchieri, Wistar Institute of Anal. and Biol., Philadelphia, Pa., U.S.A.

Table I

CYTOTOXICITY AGAINST K562 AND FREQUENCY OF LGL AND LAL IN HNK-1 POS AND NEG LYMPHOCYTE FRACTIONS

Cell fraction	donor	% of total lymphocytes	% HNK-1 POS	% CTX* to K562	% LGL¤	% LALe¤¤
A. Without pre-separation						
unfractionated peripheral	1	100	19	43**	n.d.¤¤¤	n.d.
blood	2	100	17	41	n.d.	n.d.
lymphocytes	3	100	19	17	9	6
	4	100	13	9	18	13
FACS II HNK-1 NEG	1	81	2	4	1	21
	2	83	0.5	23	n.d.	n.d.
	3	81	0.5	18	2	6
	4	87	1	7	0	8
FACS II HNK-1 POS	1	19	86	70	64	32
	2	17	98	44	n.d.	n.d.
	3	19	99	25	58	21
	4	13	91	13	61	14
B. Pre-separation on Percoll density gradient						
unfractionated peripheral	5	100	17	21***	13	8
blood	6	100	n.d.	66	19	13
lymphocytes	7	100	11	14	4	13
	8	100	5	34	12	0
Percoll fraction 2 + 3	5	25	24	26	14	9
	6	10	33	65	25	19
	7	14	25	28	17	22
	8	15	17	54	32	11
FACS II HNK-1 NEG	5	18	1.5	7	9	30
	6	6	1	69	15	13
	7	9	4	32	4	21
	8	11	1	57	11	88
FACS II HNK-1 POS	5	6	97	33	99	0
	6	3	97	67	83	15
	7	3	92	50	91	8
	8	3	92	60	97	4

* cytotoxicity of NK against the human myeloid leukaemia cell line K562, determined in the ^{51}Cr-release assay.
** percentage specific lysis of K562 at a lymphocyte to target ratio of 10.
*** percentage specific lysis of K562 at a lymphocyte to target ratio of 40.
¤ Large Granular Lymphocytes.
¤¤ Large Agranular Lymphocytes.
¤¤¤ not determined.

fractions (no. 2 + 3) of Percoll gradients (Table IB). As was observed without pre-separation, FACS II separation of cells from the NK enriched cell suspension on the basis of the HNK-1 marker again showed that NK cells could not be separated in 3 of 4 experiments (donor 5, enrichment of NK in HNK-1 POS fraction; donors 6 to 8, no separation). In comparison with unfractionated material, NK activity was increased in both the HNK-1 POS and NEG fractions. In contrast to the results obtained with cells without pre-separation on Percoll, the HNK-1 POS fraction contained primarily LGL, whereas the frequency of LAL in this fraction was the same or less than in unfractionated material. In the HNK-1 NEG fraction, the frequency of LGL was similar to that in unfractionated cell suspensions, but, interestingly, this fraction was enriched for LAL.

Our results allow the following conclusions: a) HNK-1 is not an absolute marker for NK cells, since NK activity can be found in both HNK-1 POS and NEG cell fractions. In addition, high frequencies of HNK-1 POS cells are detected in the high density Percoll fractions lacking NK activity (data not shown); b) NK lytic activity is associated with the occurrence of LGL and (in part) with LAL; c) The largest proportion of LGL expresses the HNK-1 marker, but a small subpopulation of HNK-1 NEG LGL can be detected. Both types of LGL are enriched in the low density Percoll fractions 2 + 3; d) LAL are predominantly HNK-1 NEG. Part of the HNK-1 NEG LAL is preferentially enriched in the low density Percoll fractions.

Thus, a HNK-1 NEG and POS subpopulation of NK cells can be distinguished. The HNK-1 POS NK cells are correlated with the occurrence of LGL. In the HNK-1 NEG NK cell fraction, small lymphocytes, LAL and only few LGL can be detected. Since small lymphocytes have been reported not to be the prime effector cells in conjugate formation with tumour target cells (Timonen et al., 1979), the LAL might be good candidate for the HNK-1 NEG NK cell.

In further studies on the properties of human NK cells, the B73.1 monoclonal antibody was used. Preliminary results show that the frequency of B73.1 POS cells is lower than that of HNK-1 POS cells and that NK cells are separated by FACS II sorting in the B73.1 POS fraction. Morphological analysis of the sorted fractions showed increased frequencies of both LGL and LAL in the B73.1 POS fraction and decreased frequencies in the B73.1 NEG fraction in comparison with unfractionated material. A combination of HNK-1 and B73.1 labelling may be used to separate the subpopulations of LAL from LGL, so that the proposed NK activity of LAL can be tested. Moreover, the possibility to segregate NK cells with LGL and LAL morphology would be of advantage in the study of the relationship between LGL and LAL with respect to function and maturation state.

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PATIENTS WITH CHRONIC T CELL LYMPHOCYTOSIS: RELATION BETWEEN THE CELL SURFACE CHARACTERISTICS, ENZYME ACTIVITIES, MORPHOLOGY AND THE NATURAL KILLER AND KILLER CELL ACTIVITY

C. Ronteltap, M.A.W. de Rooy-Braam, A.M. de Leeuw, C.J.M. Melief*, D. Roos*, R.L.H. Bolhuis and R.J. van de Griend

Patients with chronic T-cell lymphocytosis were found to comprise a strongly heterogenous group of individuals with regards to their clinical progress. Two major groups can be distinguished on the basis of lymphocyte cell surface markers. The lymphocytes of one group are mainly of the Tγ type, are virtually OKT4 negative and OKT8 positive (4 of 6). Clinically, this group has a good prognosis, in contrast to the second group of patients who are suffering from chronic T-cell malignancies of the T-non-γ type, which have a poor prognosis (Rümke et al., 1982).

We have recently found that lymphocytes of patients of these two groups can also be distinguished by analysis of three enzymes involved in purine metabolism: adenosine deaminase (ADA), purine nucleoside phosphorylase (PNP) and 5'nucleotidase (5'NT). The 5'NT activity of lymphocytes of Tγ lymphocytosis patients was within the range of that of normal Tγ lymphocytes (which is much lower than that of normal T-non-γ lymphocytes) (Table I). In chronic T-cell malignancy, 5'NT activity was either low, normal or highly elevated in comparison with normal T-non-γ cells. ADA and PNP activity were within the normal range in both groups, in contrast to T-ALL and "Null" ALL patients, which show strongly elevated levels of ADA (Van de Griend et al., in press). In chronic T cell malignancy (T-non-γ), the ratio between ADA and PNP activity is always < 1.0 (not shown) but is > 1.5 in 4 of 5 patients with Tγ lymphocytosis. This value is also found for normal Tγ lymphocytes (Table I). The isoenzyme pattern of lactate dehydrogenase (LDH) in the Tγ patient group reflects that of normal Tγ cells and thymocytes.

Functional activities of lymphocytes derived from Tγ or T-nonγ patients were also different. Patients with chronic T-cell malignancy show no natural killer (NK) or antibody dependent cytolysis (ADCC or K cell) activity at all. Those with Tγ lymphocytosis always have a high K cell activity but a strongly reduced NK activity in 2 of 5 patients as compared with normal lymphocytes (Table II). We have investigated whether or not this reduced NK cell

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Table I

SURFACE MEMBRANE MARKERS AND BIOCHEMICAL PROPERTIES OF LYMPHOCYTES FROM PATIENTS WITH T γ LYMPHOCYTOSIS

			monoclonal antibodies (% binding)		enzyme activity (nmoles per 10^8 cells·min^{-1})				
patient	E	EA-RFC	OKT4	OKT8	5'NT	ADA	PNP	ratio ADA/PNP	LDH (H/M ratio)
1	81	81	11	73	0.9	73	127	0.6	1.25
2	85	85	4	5	n.t.	n.t.	n.t.	n.t.	n.t.
3	87	74	17	82	1.2	92	58	1.6	1.07
4	67	57	10	87	1.2	117	55	2.1	0.89
5	95	84	5	2	3.7	92	43	2.1	1.06
6	60	63	15	75	2.6	132	89	1.5	1.29
normal T γ cells		85	14	35	1.1	86	59	1.5	1.02
± SD		±17	±9	±16	±0.8	±63	±17	±0.7	±0.14
normal T-non γ cells		< 3	60	34	6.1	80	71	1.1	2.21
± SD			±6	±3	±2.3	±20	±23	±0.3	±0.23

activity can be correlated with particular cell surface marker characteristics as detected by binding of monoclonal antibodies (MCA) or by morphological criteria. It was found that 35 to 75% of lymphocytes from T γ patients who exhibited NK cell activity were HNK-1 positive (HNK-1 is reported to recognize an antigen on NK cells; Abo and Balch, 1981). However, T γ patients without NK activity also showed up to 50% HNK-1 positive cells. Thus, the presence of HNK-1 positive cells in patients with T γ lymphocytosis does not necessarily reflect NK cell activity. This discrepancy between HNK-1 positive cells and NK cell activity confirms our finding in normal individuals (p. 181). Furthermore, a proportion of the lymphocytes, similar to the percent HNK-1 binding cells, have the morphology of large granular lymphocytes (LGL). Electron microscopic examination revealed that variable numbers of lymphocytes in these patients have parallel tubular arrays (PTA) (Table II).

Another antigen that was shown to be present on T γ cells and (part of) NK cells is recognized by the OKM1-MCA. The patients whose lymphocytes exhibited NK cell activity showed 50 to 90% OKM1

Table II

MONOCLONAL ANTIBODIES AND MORPHOLOGICAL CRITERIA IN RELATION TO NATURAL KILLER CELL ACTIVITY

patient	% specific ^{51}Cr-release NK		ADCC		morphology		monoclonal antibodies (% binding)		
	40*	10	40	10	LGL**	PTA***	HNK-1	OKM1	B73.1
1	52	22	73	64	+++¤	+++	75	50	75
2	53	31	55	31	+++	n.t.	73	84	n.t.
3 A¤¤	3	1	61	61	±	-	19	10	2
B	20	7	83	83	+	+	27	5	6
C	2	0	n.t.	n.t.	n.t.	+	30	26	4
4	1	0	78	12	±	-	51	11	n.t.
5	59	59	84	81	+	-	35	90	37
normal lympho-cytes ± SD	46 ±13	24 ±9	39 ±21	21 ±12	+	±	15 ±7	16 ±6	15 ±9

* number of effector cells per target cell
** large granular lymphocytes
***parallel tubular arrays
¤ one + means present in normal amounts.
¤¤ this patient was tested three times at 3-month intervals.

positive cells and those lacking NK cell activity had low levels of OKM1 positive cells. Furthermore, the B73.1, an MCA also reported to be specific for NK cells (Perussia et al., in press) was also used for analysis of our patients. No B73.1 positive cells were found in a patient that lacked NK cell activity, whereas up to 75% B73.1 positive lymphocytes were found in T γ (cell lymphocytosis) patients who indeed had NK cell activity. We have tested a number of markers which have been reported to be related to some extent with NK cell activity in patients with T lymphocytosis. Although the limited number of data available do not allow quantitative conclusions, they indicate that the MCA B73.1 is probably the best marker for NK cells available. We conclude that in T γ lymphocytosis as in chronic T-cell malignancies the major feature of the lymphocytosis is the absence of cells which exhibit NK cell activity. This correlates with the absence of cells with antigens recognized by B73.1 and OKM1.

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THE CLONING AND EXPRESSION OF A RAT ALPHA INTERFERON GENE

R. Dijkema*, P.H. Pouwels* and H. Schellekens

Interferons are proteins produced by cells after induction of an antiviral state by viral or nonviral agents. Three types of interferons (IFNs), alpha, beta and gamma, are now well characterized. Which of the different types is produced depends on the host cell and the type of inducer.

The conventional procedure of rat alpha IFN production is the infection on a large scale of rat embryo cells (Ratec) with Sendai or New Castle Disease virus. This procedure yields moderate amounts of a mixture of different rat alpha IFN subtype proteins. The rat alpha IFN gene family is composed of many related alpha IFN subtype genes. Large amounts of specific subtype rat alpha IFN proteins can be produced by the molecular cloning of the different genes by recombinant DNA technology and the subsequent expression of these genes by prokaryotic (e.g., the bacterium E.coli) and eukaryotic (e.g., yeast) cells.

A collaborative research project was started at the Medical Biological Laboratory and Primate Center of TNO aimed at the molecular cloning and expression of rat IFN genes in order to study the antiviral and antitumour activity(ies) as well as the toxicity of separate subtypes in a species specific model.

We first isolated and characterized the rat alpha IFN mRNAs from virus-induced Ratec cells. A cDNA library was constructed by use of reverse transcriptase of the IFN-enriched mRNA population. Screening of transformed bacteria containing hybrid (cDNA/plasmid DNA) vectors was performed by using human and mouse alpha IFN genes as probes. As a result, three positive clones containing a partial copy of the same rat alpha IFN subtype gene was isolated. Using this rat alpha IFN cDNA as a probe, we were able to detect at least 12 related rat alpha IFN subtype genes present in chromosomal liver DNA of the rat.

Since the rat alpha IFN genes comprise a family of nonspliced genes, we then investigated a rat gene library (Sargent et al., 1979) constructed in bacteriophage . Screening of this library for rat alpha IFN sequences resulted in the expected number of 12 rat alpha IFN subtype genes per total genome. By use of differential hybridization with probes comprising different parts of the gene, we were able to identify the rat alpha IFN subtype gene that was predominantly expressed upon viral infection of Ratec cells. This gene was characterized by DNA sequence analysis according to Maxam and Gilbert (1980) and was shown to be homologous to the

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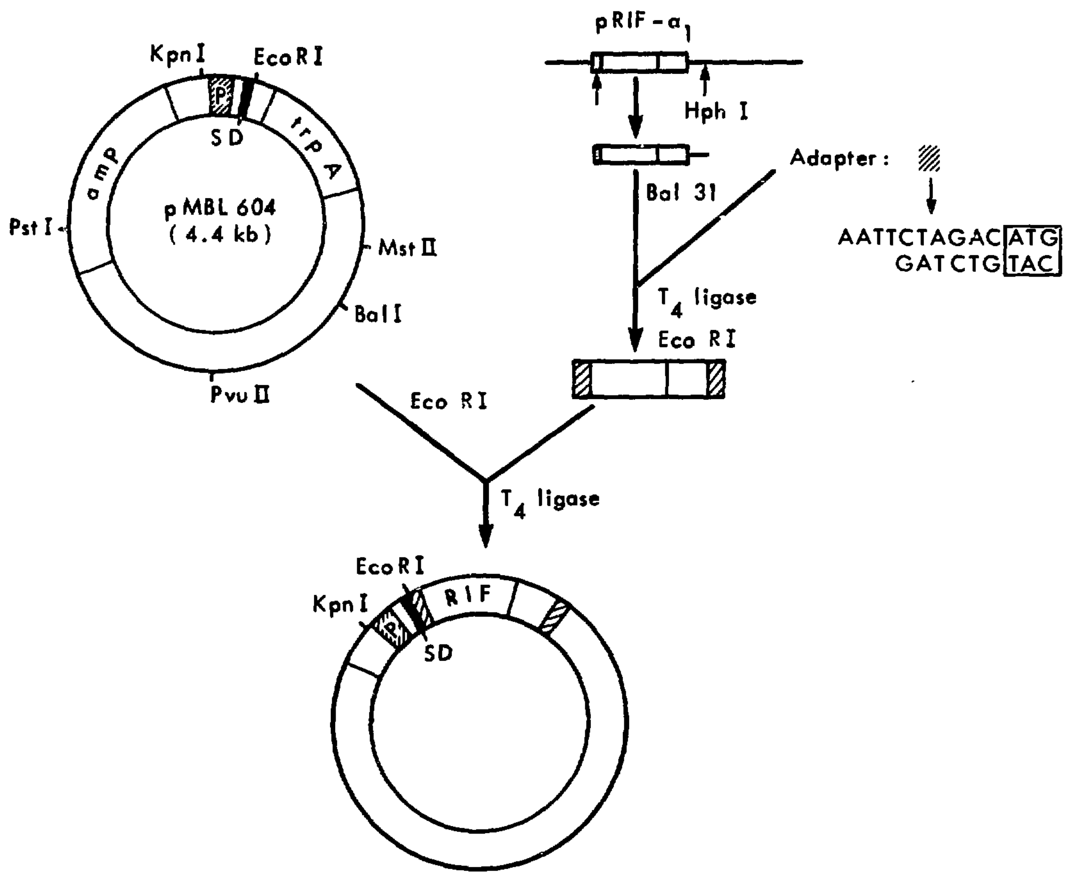

Figure 1: Construction of a prokaryotic expression vector for rat alpha interferon production.

human and mouse alpha IFN counterparts (HuIFN alpha 1/MuIFN alpha 1, respectively).

Since eukaryotic proteins (such as IFNs) and prokaryotic proteins are regulated in different ways, it is necessary that the gene organization is manipulated in order to achieve maximal expression of a eukaryotic protein in prokaryotic organisms like the bacterium E.coli.

Therefore,

a) the regulatory signals for transcription of the IFN gene had to be removed and replaced by prokaryotic regulatory signals of a particular gene (e.g., the tryptophane operon);

b) the coding sequence of the leader peptide of the IFN gene responsible for transport over the eukaryotic cell membrane had to be removed, since a prokaryotic organism cannot handle this signal sequence in a proper way;

c) as a consequence of the removal of the signal sequence, the initiator methionine triplet was also removed and had therefore to be artificially added in the form of a synthetic "adapter" containing a methionine start codon.

In this way, a hybrid vector was constructed for the expression of rat alpha 1 IFN in E.coli involving the E.coli regulatory signals for transcription followed by an artificial start codon directly attached to the coding sequence of the IFN gene (Fig. 1). This vector introduced into susceptible E.coli was able to direct the synthesis of rat alpha 1 IFN. In the future, this E.coli-produced rat alpha 1 IFN will be purified by polyclonal or monoclonal antibodies directed against naturally derived and recombinant rat alpha IFNs.

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EXPERIMENTAL HAEMATOLOGY

CLONAL ASSAYS OF MOUSE HAEMOPOIETIC PROGENITORS IN SERUM FREE CULTURES

S. Merchav, L. Witvliet* and G. Wagemaker

The proliferation and maturation of haemopoietic progenitors in vitro in semisolid or viscous cultures requires stimulation by specific glycoprotein molecules such as erythropoietin for erythroid cells and colony stimulating factors for progenitors of granulocytes and macrophages. Apart from these factors, haemopoietic cells have nutritional requirements which promote growth in a nonspecific manner. Culture media are usually supplemented with serum which as a result of its complex nature is a source of many undefined and uncontrolled variables and which has been incidentally shown to contain specific haemopoietic regulators. The complete replacement of serum with chemically defined substances is therefore imperative for accurate analysis of the specific growth requirements of individual haemopoietic precursors.

A serum free culture system for the assay of mouse macrophage progenitors has been developed. It consists of the α-modification of Dulbecco's medium (containing 4.5 $g.l^{-1}$ glucose), supplemented with 1% (w/v) delipidated, deionized bovine serum albumin, 4×10^{-6} M Fe-saturated transferrin, 10^{-7} M Na_2SeO (Guilbert and Iscove, 1976), 10^{-4} M 2-mercaptoethanol, 10^{-3} $g.l^{-1}$ nucleosides, 1.5×10^{-5} M cholesterol and 1.5×10^{-5} M linoleic acid. Using macrophage colony stimulating factor (M-CSF) purified from pregnant mouse uteri as the stimulus for macrophage colony formation, the number of colonies resulting was directly proportional to the number of cells cultured, thus ruling out a possible contribution of products by the cells to the serum free culture system.

Fig. 1 demonstrates the critical role of cholesterol in these cultures. Fig. 2 shows the dose response curve for M-CSF under serum free culture conditions as compared with serum supplemented controls (5%). Serum free cultures were found to be superior to serum supplemented cultures for growth of macrophage progenitors

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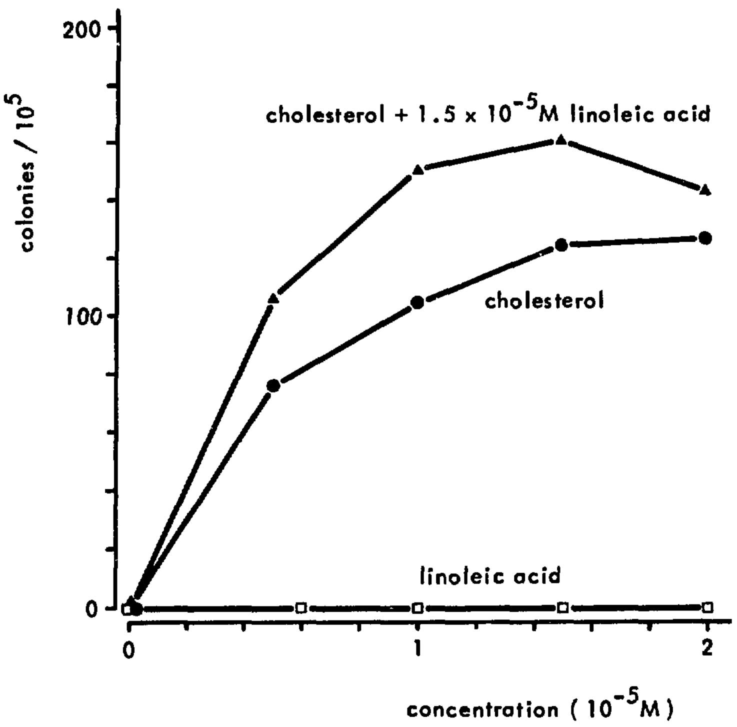

Figure 1: Effect of cholesterol and linoleic acid in cultures stimulated with M-CSF.
Titration curves for linoleic acid or cholesterol alone and for cholesterol supplemented with 1.5×10^{-5} M linoleic acid. Cultures containing 5×10^4 bone marrow cells per ml were incubated at 37°C in 7.5% CO_2 in air of 100% humidity. Colonies were scored after 7 days. The apparently saturating concentration of the M-CSF used was 0.13 $\mu g.ml^{-1}$

(38 ± 7 percent increase; mean of 10 experiments), the colony size remaining unchanged. This could not be attributed to either an alteration in responsiveness to M-CSF or an increase in background colony counts.

Serum free cultures as developed for macrophage progenitors were also found to support growth of developmentally early (BFU-E) and late (CFU-E) erythroid progenitors and megakaryocyte progenitor cells. The serum free culture conditions described here can thus serve as a basic system common to several haemopoietic progenitors, to which a variety of components (i.e. nutrients, hormones, etc.) may now be added for analysis of their effects on haemopoietic cell differentiation.

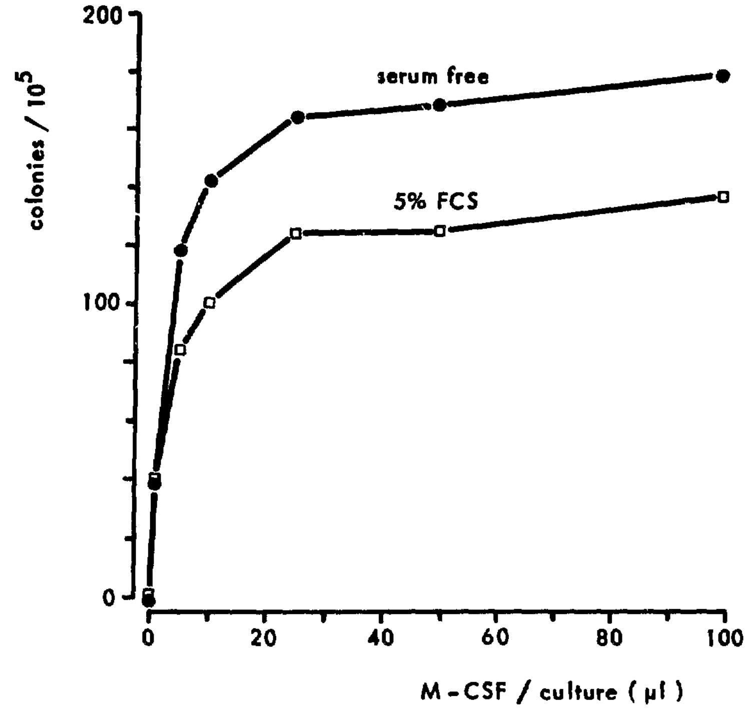

Figure 2: Dose response curve for M-CSF in defined cultures with or without fetal calf serum. Culture conditions as described in the legend to Fig. 1.

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SPECIFICITY OF GRANULOCYTE/MACROPHAGE COLONY STIMULATING FACTORS

G. Wagemaker and H. Burger

Certain mouse bone marrow cells generate colonies of neutrophilic granulocytes and macrophages in viscous cultures which contain "colony stimulating factors" (CSFs). These granulocyte/macrophage progenitors, termed GM-CFU, are bipotential cells. The CSFs can be broadly subdivided into two types. One type has been isolated from human urine, L-cell conditioned medium (Stanley and Heard, 1977) and pregnant mouse uteri and has a molecular weight ranging from 60 to 70 kdaltons. Under serum free culture conditions (p. 193), it exclusively stimulates the formation of macrophage colonies and has consequently been termed M-CSF. Another type has been isolated from the supernatant of lungs of endotoxin treated mice (Burgess et al., 1977). This GM-CSF has a molecular weight of 29 kdaltons and stimulates the formation of colonies composed of both granulocytes and macrophages. M-CSF and GM-CSF appeared to stimulate mainly the same GM-CFU population, since colony numbers induced by one factor cannot be increased by also adding the other, while the buoyant density distributions of GM-CFU stimulated by either factor were found to be identical. In addition, colony formation initiated by one factor can be propagated by the other (Johnson and Burgess, 1978; Metcalf et al., 1982).

The supernatant of lectin stimulated spleen cells contains a variety of haemopoietic factors and also stimulates growth of granulocyte and macrophage colonies. Its colony stimulating activity is distributed in two distinct peaks when subjected to ion exchange chromatography (Fig. 1). As determined by polyacrylamide gel electrophoresis (PAGE), the size and charge characteristics of the rapidly eluted CSF were indistinguishable from those of M-CSF purified from pregnant mouse uteri, whereas the more slowly eluted peak coincided with GM-CSF purified from mouse lung conditioned medium. The slowly eluted peak was isolated and shown to stimulate colony formation of granulocytes and macrophages. This GM-CSF had a molecular weight of 28 kdaltons as determined by gel filtration and did not increase the number of colonies induced by M-CSF from pregnant mouse uteri or GM-CSF from mouse lungs. It also did not stimulate early (BFU-E) or late (CFU-E) erythroid progenitor cells.

Using serum free methylcellulose cultures (p. 193), colony growth stimulated by M-CSF could not be improved by adding fetal calf serum. Colonies induced by GM-CSF, however, failed to reach

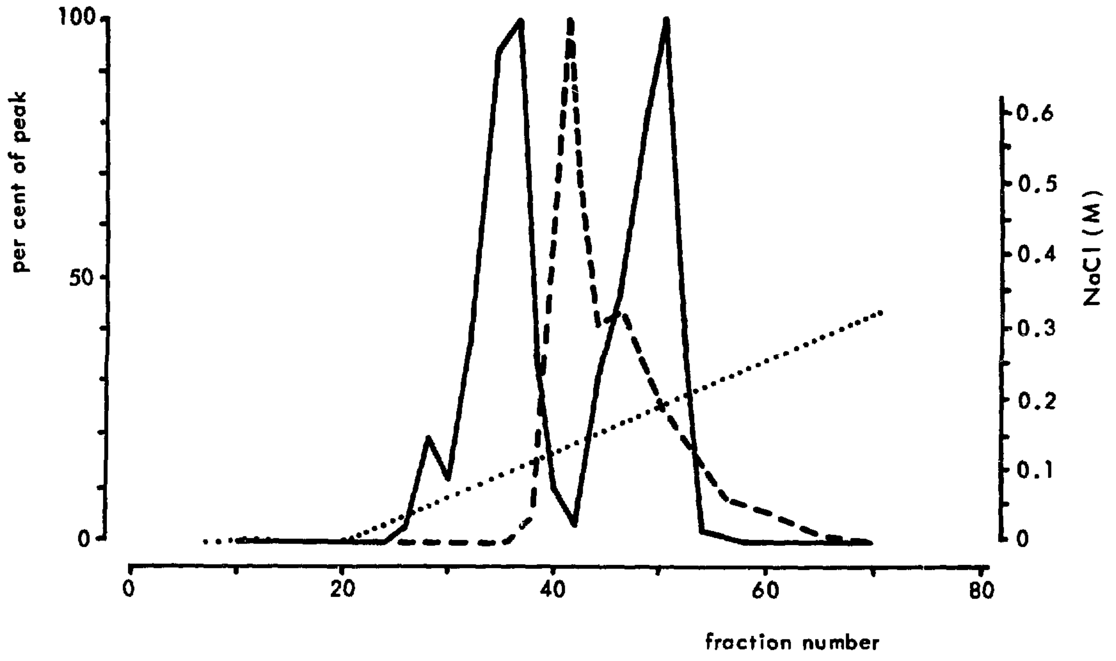

Figure 1: Colony stimulating factors in Concanavalin A stimulated spleen cell conditioned medium, fractionated by ion exchange chromatography on DEAE-Sepharose, pH = 8.5.
The fractionated material was enriched for haemopoietic factors by prior adsorption to Con A-Sepharose and gel filtration. Interrupted line: protein distribution; dotted line: NaCl gradient.

numbers obtained in cultures supplemented with 5% fetal calf serum. A variety of compounds was subsequently tested for their capacity to meet the growth requirements for GM-CSF stimulated colony formation. Among those, hydrocortisone at a concentration of 10^{-6}M proved to be effective (Fig. 2). At this hydrocortisone concentration, serum was ineffective in further increasing colony formation, suggesting that serum under the described culture conditions only served as a source of corticosteroids. The promoting action of hydrocortisone was specific for GM-CSF stimulated GM-CFU, since colony formation stimulated by M-CSF (Fig. 2) or colony growth from early erythroid progenitor cells was inhibited. The nature of the GM-CFU stimulated by GM-CSF in hydrocortisone supplemented cultures remains to be determined.

These observations demonstrate the specificity of regulatory factors acting within a single differentiation pathway and provide evidence for a physiological difference between M-CSF and GM-CSF. Increased corticosteroid levels (as occur in infectious diseases, burns, etc.) appear to result at the haemopoietic progenitor cell level in a selectivity for neutrophilic leukocyte production at the expense of the monocyte/macrophage and erythroid differentiation pathways.

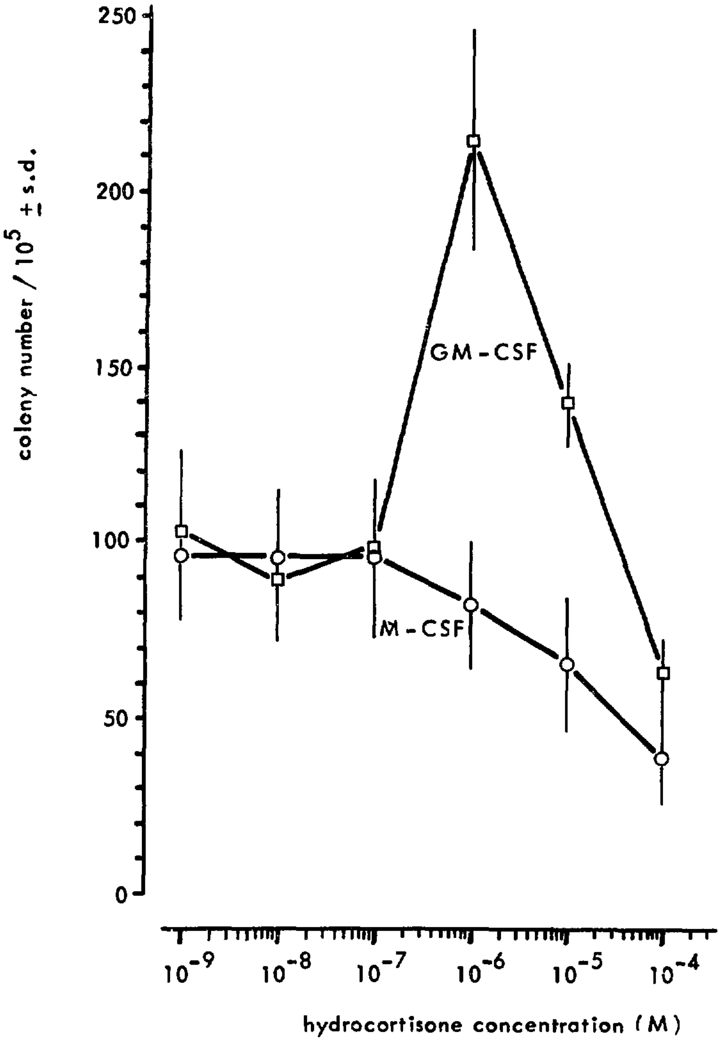

Figure 2: Effect of hydrocortisone on colony formation in serum free cultures stimulated with M-CSF purified from pregnant mouse uteri or GM-CSF purified from spleen conditioned medium.

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PURIFICATION OF MOUSE PLURIPOTENT HAEMOPOIETIC STEM CELLS

J.W.M. Visser, J.G.J. Bauman and A.H. Mulder

In 1981, we described a two-step procedure for obtaining 60 to 100-fold enriched suspensions of mouse pluripotent haemopoietic stem cells (CFU-S) from normal mouse bone marrow (Visser and Bol, 1981). In brief, cells were first separated by equilibrium density centrifugation and then labelled with fluoresceinated wheat germ agglutinin (WGA-FITC) and separated by a light-activated cell sorter. The enriched stem cells behaved normally with regard to their capacity to protect lethally irradiated mice and their homing *in vivo*.

During the last year, several attempts were made to further purify the haemopoietic stem cells. One of these series of experiments combined the above described two-step procedure with monoclonal antibody labelling and cell sorting. Since both the antibodies and the WGA were labelled with FITC, the WGA-FITC had to be washed off the cells after the two-step procedure prior to labelling with antibody by incubation in 0.2 M isotonic N-acetyl-D-glucosamine (15 min, 37°C). This incubation efficiently removed the WGA-FITC fluorescence from the cells without affecting the CFU-S viability and homing. In addition, control experiments with unfractionated cells indicated that the use of biotin-avidin for conjugation of FITC to antibody provides shielding of the Fc part of the antibody from macrophage recognition so that the CFU-S assay is not affected by the labelling. Therefore, in the experiments described here, the sorted cells were subsequently labelled with antibody-biotin and avidin-FITC.

From earlier work, it is known that the membrane of CFU-S contains the H2K antigen, whereas many other bone marrow cell types are H2K negative (Van den Engh et al., 1978; Trask and Van den Engh, 1980). As shown in Fig. 1, three subpopulations of cells with different fluorescence intensities can be distinguished with an enriched stem cell fraction which is labelled with anti-H2K-biotin avidin-FITC (Becton Dickinson, Sunnyvale, CA). Cells from the subpopulation with the brightest fluorescence were sorted and injected into lethally irradiated syngeneic recipients to enumerate CFU-S. The enrichment of CFU-S in this subpopulation was found to be 150 to 200-fold as compared with unfractionated bone marrow (Table I). The recovery in the present three-step enrichment procedure at least approaches 40%.

The number of macroscopically visible spleen colonies at 8 days after transplantation of the purified stem cells was three-

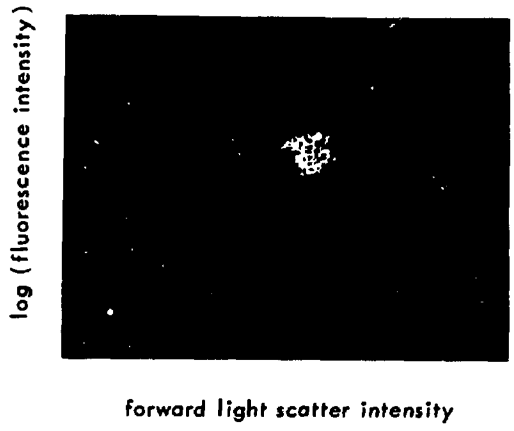

Figure 1: Dot plot of fluorescence versus forward light scatter intensities of a mouse bone marrow subpopulation which is enriched for CFU-S by equilibrium density centrifugation and light-activated cell sorting for WGA binding cells as described previously (Visser and Bol, 1981). The cells are labelled with anti-H2K-biotin and avidin-FITC after removal of the WGA.

Table I

ENRICHMENT OF CFU-S BY COMBINING EQUILIBRIUM DENSITY CENTRIFUGATION AND LIGHT-ACTIVATED CELL SORTING FOR WGA AND H2K POSITIVE MOUSE BONE MARROW CELLS

	number of CFU-S per 10^5 cells	
	day 8	day 12
unfractionated	25 $\pm$ 2	30 $\pm$ 2
low density cells	-	148 $\pm$ 8
low density, WGA^+	-	1770 $\pm$ 100
low density, WGA^+, $H2K^+$	1500 $\pm$ 200	4500 $\pm$ 300

fold lower than that counted at day 12 (Table I). The number of spleen colonies after transplantation of unfractionated bone marrow is 1.2 to 1.3-fold higher at 12 days than at 8 days. This indicates that the sorting procedure purifies the CFU-S subpopulation which has been designated by Magli et al. (1982) as the delayed type spleen colony forming cell. Baines and Visser (in press) have demonstrated by light-activated cell sorting of H33342 stained cells that this CFU-S type is probably the quiescent subpopulation of the stem cells.

In earlier experiments (Visser and Eliason, in press) we have observed that the f-factor was basically similar for unfractionated cells and for highly enriched stem cells from the density cut/WGA sorting separation. Since by the use of the biotin-avidin method the H2K labelling procedure of unfractionated cells did not affect the spleen colony assay, it is likely that the f-factor for the stem cells sorted by the present three-step procedure is also similar to that of unfractionated bone marrow CFU-S, viz., 0.05 (Lahiri et al., 1970). Therefore, from the incidence of 4 to 5 spleen colonies per 100 cells injected, it can be calculated that the concentration of cells capable of spleen colony formation (CFC) is 80 to 100%. This indicates that the three-step procedure yields almost pure stem cell suspensions.

The sorted stem cells can be grown in vitro by a number of techniques. Routinely, the stem cells were deposited at one cell per well in Terasaki culture trays during the last sorting step. Before sorting, each well was filled with 10 µl of liquid serum-free culture medium provided with haemopoietic regulators. This cloning technique facilitates the direct observation of the stem cells from the beginning of the culture, so that the earliest events of haemopoietic differentiation can be studied directly in vitro. Up to now, 25 to 30% of the cloned stem cells has been triggered to produce progeny in vitro by the addition of PMUE and 18-hours-PES (cfr. Bol et al., 1979). The first divisions were found to occur at 24 to 48 hours after cloning. Rather heterogeneous growth kinetics were observed during the first days. All wells which contained more than four cells on the third day after the cloning of single cells generally showed extensive growth for at least two weeks. From day 3 on, heterogeneous morphology was observed. Although all cells at day 3 were undifferentiated blast cells as judged after May Grünwald Giemsa staining, about 25% of them showed fibroblastoid stretching and adherence to the bottom of the well, while the other cells remained spherical. The early differentiation steps will be further studied by using monoclonal antibodies directed against differentiation antigens.

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HIGH ENRICHMENT OF NORMOBLASTS BY FLOW SORTING USING ANTI-T200 MONOCLONAL ANTIBODIES

J.G.J. Bauman and M.G.C. Platenburg

Differentiation of the pluripotent haemopoietic stem cell into the various blood cell types involves many cytochemical changes. A number of these changes are accompanied by expression of antigens on the cell surface of haemopoietic cells. These so-called differentiation antigens can be detected by using monoclonal antibodies directed against them. Three types of differentiation antigens are found to be present: conservative, transient and true differentiation antigens (Van den Engh et al., in press). The cell lineage specific antigens designated as conservative differentiation ones, like Pgp-1, are already expressed on the pluripotent stem cell and may increase in density on mature blood cells of one lineage, while the density decreases on others.

Transient differentiation antigens are present only during a certain phase of differentiation and are absent from mature blood cells. True differentiation antigens, like Thy-1 and MAC-1, are present only on mature cells of one lineage (Trask and Van den Engh, 1980).

The T200 antigen is an example of a conservative differentiation antigen with an interesting distribution. The T200 molecule is a member of a family of cell surface glycoproteins with molecular weights of 200 to 220 kilodaltons (Sarmiento et al., 1982) and it is called T200 because it is expressed in higher density on mature T cells. It is found on all leukocytes and is therefore sometimes called the common leukocyte antigen (Springer, 1980).

We have demonstrated that T200 is also present on the pluripotent stem cells (CFU-S) in the same density as on most nucleated haemopoietic cells. This was demonstrated with biotin-labelled anti-T200 monoclonal antibody (clone I3/2; Trowbridge, 1978) and avidin-FITC conjugates. The use of the biotin-avidin system allows quantitative recovery of CFU-S after labelling of the cell surface antigen. Mouse bone marrow cells [(C57BL x C3H)F1] stained for the T200 antigen in this way were fractionated on the fluorescence activated cell sorter (FACS II) according to three parameters: high forward light scatter (FLS; above the lymphocyte peak), low perpendicular light scatter (PLS; less than granulocytes) (Visser et al., 1980) and fluorescence intensity (Fig. 1). On the basis of their fluorescence, the cells in the FLS/PLS window, the so-called blast window, were separated into six fractions and analysed for

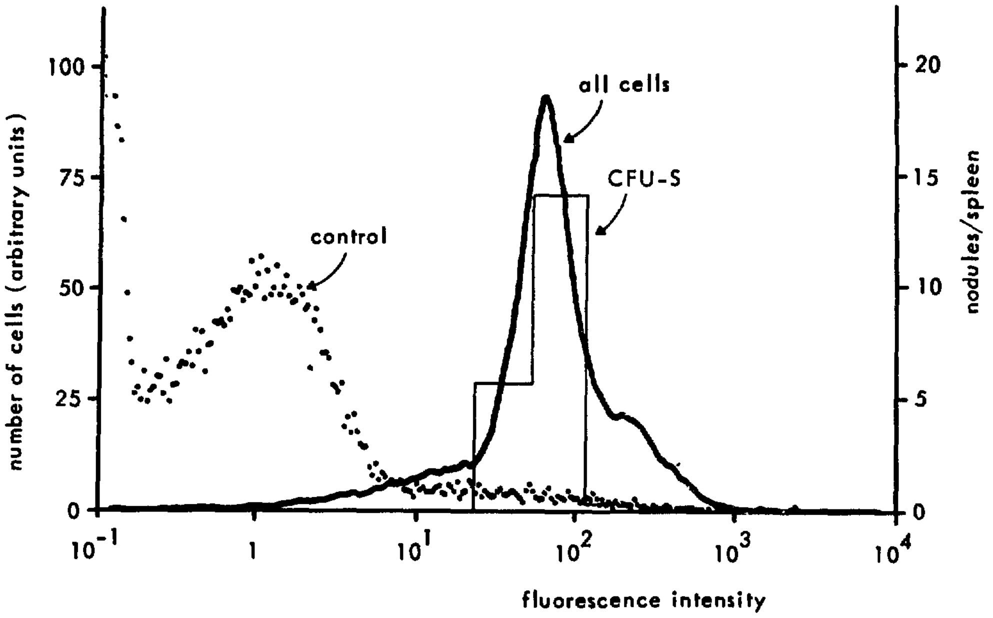

Figure 1: Fluorescence distribution of bone marrow cells in the high FLS-low PLS window after staining with biotin-labelled anti-T200 antibody (clone I3/2) and avidin-FITC. Cells of six fractions with different fluorescence intensities were sorted and analysed for CFU-S content. Each mouse received the equivalent of 5×10^{4} cells. The control was bone marrow cells stained with avidin-FITC alone.

CFU-S content. The CFU-S were found at the same fluorescence intensity as most other bone marrow cells.

In these experiments, we also found a variable proportion of nucleated cells (5 to 10%) in mouse bone marrow which were negative for the T200 antigen (Fig. 1; the broad low shoulder at left with fluorescence intensity below 2×10^{1}). We sorted these negative cells on the FACS. After May-Grünwald Giemsa staining, a differential count was performed on the sorted fractions (Table I). The differential counts of four fractions were compared with the counts of sorted but unfractionated bone marrow cells. The four fractions sorted were: from all nucleated bone marrow cells, the T200 negative and the T200 positive cells; and from the cells in the blast window (having a relative high FLS and low PLS), the T200 negative and the T200 positive cells. When T200 negative cells were selected from all nucleated cells, normoblasts were obtained at 80% purity. When such cells were selected from the blast window, normoblasts were obtained in a purity of 96%.

Mature erythrocytes are negative for the T200 antigen. Most normoblasts were also found to be T200 antigen negative. Some normoblasts were also found in the T200 positive fractions. The pluripotent stem cell, the BFU-E and all other nucleated haemopo-

Table I

DIFFERENTIAL COUNT OF SORTED FRACTIONS OF ANTI-T200 BIOTIN + AVIDIN-FITC LABELED MOUSE BONE MARROW CELLS*

sorted fraction	percent of total	differential count** of 200 cells in fraction: myeloid	lymph	mono	normo	other
all cells	100	40	35	6	18	1
live nucleated cells						
negative	12	11	8	0	81	0
positive	83	42	34	5	16	3
cells in blast window***						
negative	5	0.5	3.5	0	96	0
positive	2.2	6	43	25	26	0

* (C57BL/Rij x C3H/Rij)F1 female mice of 7 weeks of age were used as bone marrow donors.

** percentages

*** of the 20% cells which fall in the blast window, only the 5% most negative and the 2.2% most positive cells were sorted. The remaining about 13% of positive cells were mainly myeloid and some lymphoid cells, but no complete differential count was done.

ietic cells are T200 positive. Therefore, the question arises of at which differentiation stage do cells of the erythroid lineage loose the T200 antigen. We have attempted to determine whether the CFU-E contained the T200 antigen. However, we were unable to culture CFU-E colonies after anti-T200 biotin + avidin-FITC labelling and cell sorting. So far, it can be concluded only that the T200 antigen is lost in erythroid differentiation at a stage between the BFU-E and the normoblast.

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APPEARENCE OF THYMIC NURSE CELLS AFTER GAMMA IRRADIATION

A.H. Mulder and D.W. van Bekkum

Lymphostromal interactions in the thymus were recognized with the electron microscope some years before the isolation of thymic nurse cells (TNC). The isolation of these large epithelial cells which are called nurse cells because they enclose thymocytes (Wekerle et al., 1980) made these interactions accessible for investigation. It was soon recognized that the enclosed thymocytes were all viable and immature and that they could be ejected from the nurse cell in vitro. The nurse cell was found to have both H-2 and I-A antigens on its cell membrane and it was suggested that it was the site where thymocytes were "educated" to discriminate between self and non-self. This hypothesis still lacks experimental support.

Since prothymocytes home from the bone marrow to the thymus, it was tested in the mouse whether prothymocytes could be recaptured from thymic nurse cells. Bone marrow cells were labelled with the red fluorescing anthracycline daunomycin and varying numbers (up to 25 x 10^6 nucleated bone marrow cells) were injected into lethally irradiated recipients. At several time intervals after transplantation (up to 24 hours), thymuses were removed and the TNCs were isolated. No specific red fluorescence was found within the TNCs. These experiments were repeated with supravital compounds (bisbenzimide H33342 and fluorescein isothiocyanate) at concentrations which have been shown not to affect viability, homing pattern and function (Visser et al., 1981; Scollay et al., 1980). Again, no specific fluorescence was found in the TNC after transplantation of labelled bone marrow into irradiated mice.

The number of TNCs isolated in these experiments was increased as compared with unirradiated animals; therefore, the relationship between the dose of total body gamma irradiation and the time after irradiation was investigated. Maximal numbers of TNCs were found at 6 hours after irradiation with 4 Gy (Fig. 1). Eight to 12 hours after irradiation, the number of TNCs isolated decreased and had returned to preirradiation levels at 24 hours. The relation between TBI dose and the number of TNCs per thymus is shown in Fig. 2. The number determined at 3 hours increased with the dose to reach a maximum at 4 Gy.

We later studied the morphology of the TNCs isolated at 4 to 6 hours after irradiation. On electron microscopic examination, signs of degeneration and death of the enclosed thymocytes was detected. Bundles of tonofilaments were present in all TNCs, which

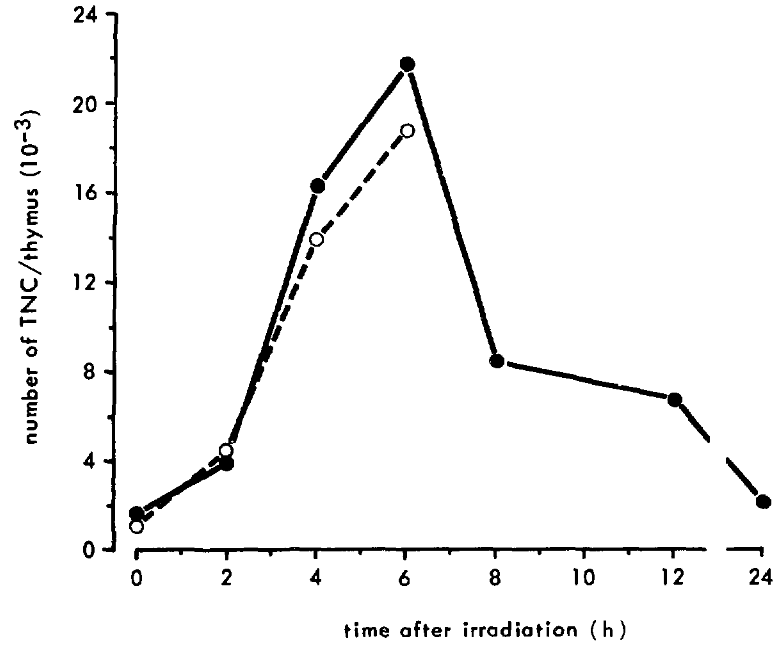

Figure 1: Number of TNCs per thymus isolated at various intervals after total body gamma irradiation (4 Gy). Two separate experiments are shown.

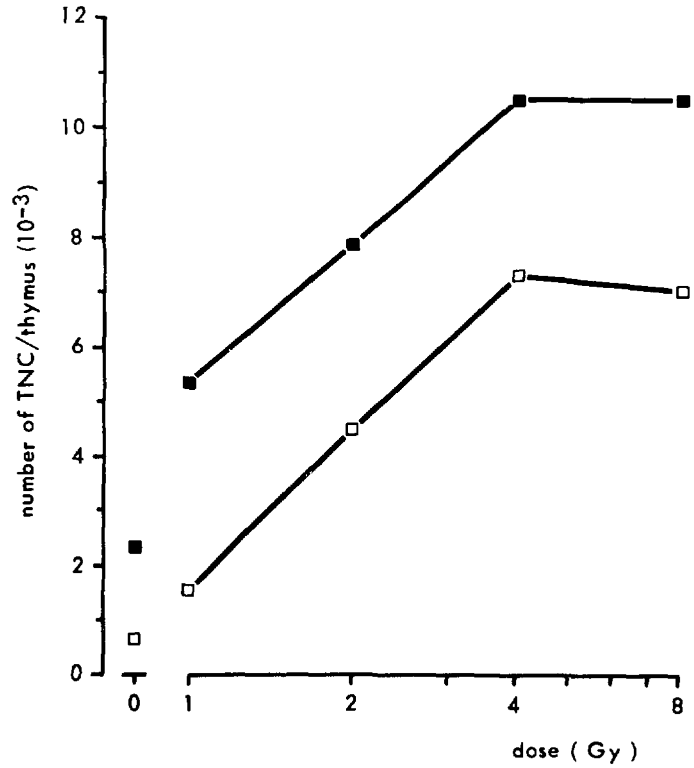

Figure 2: Number of TNCs per thymus isolated at 3 hours after varying doses of total body gamma irradiation. Two separate experiments are shown.

suggests the epithelial origin of the TNC. In addition with the use of a monoclonal antibody raised against prekeratin, the presence of related molecules in the cytoskeleton of the TNC was demonstrated. This proves that the TNCs appearing after irradiation are epithelial cells.

The studies presented here suggest that the TNCs have a function after the first differentiation steps have taken place, since bone marrow cells, which contain prothymocytes, do not home to the thymic nurse cell. The increase in TNCs after irradiation suggests that cells which are already differentiated into immature thymocytes are enclosed. The presence of areas with bundles of tonofilaments and many mitochondria means that the epithelial cells probably represent the active part in this interaction. Further experiments with metabolic inhibitors of the microtubules may give more insight into the mechanism of engulfment of the immature thymocytes by the TNC.

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THE EFFECT OF DEXAMETHASONE ON EARLY T-CELL PROGENITORS

W.J.A. Boersma

As reported by Basch and Kadish (1977), prothymocytes are relatively resistant to hydrocortisone. A modest enrichment of prothymocytes in hydrocortisone pretreated donor mice was even observed.

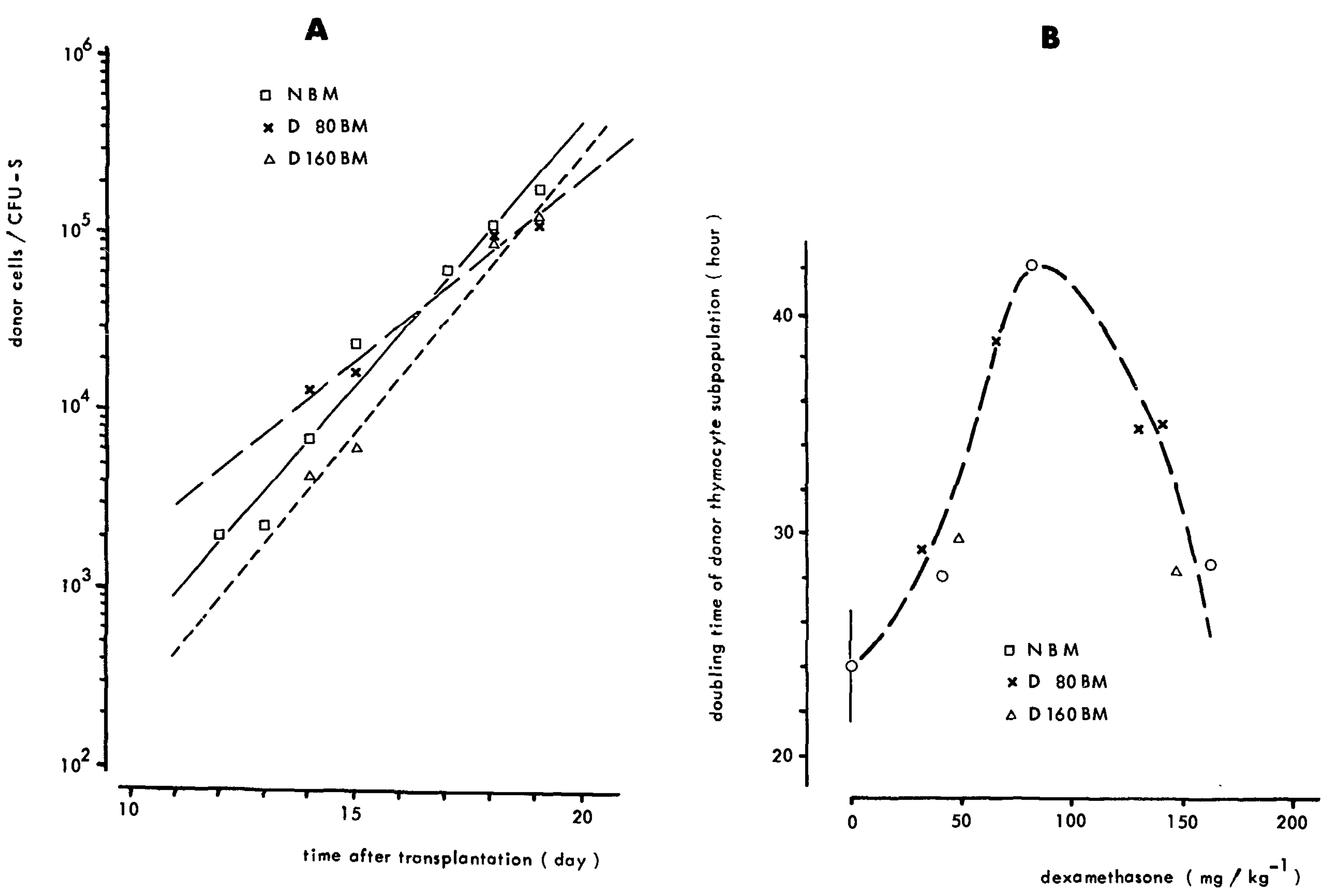
In the bone marrow of 6-week-old (C3H x AKR)F1 mice treated with dexamethasone three days before sacrifice, the number of CFU-S per 10^5 nucleated cells was changed in a dose dependent manner. In the bone marrow of dexamethasone treated mice, the proportion of CFU-S was increased up to 1.2 to 2.0 times (Table I). Since prothymocytes and CFU-S are closely related cell types, the assumed effect of dexamethasone on early T cell progenitors has to be corrected for the influence that pluripotent stem cells might have on T cell development in thymus regeneration studies.

Table I

CFU-S IN BONE MARROW AT 3 DAYS AFTER DEXAMETHOSONE TREATMENT

	dose $mg.kg^{-1}$	CFU-S per 10^5 cells	CFU-S per femur
expt. I	0	21.2 ± 3.4	2325
	64	26.9 ± 3.2	4700
	128	25.8 ± 3.9	3400
expt. II	0	25.4 ± 2.2	2718
	32	34.6 ± 2.2	3114
	64	40.7 ± 3.4	3744
	128	49.8 ± 2.8	4482
expt. III	0	25.6 ± 2.0	3315
	40	31.4 ± 3.5	2449
	80	37.4 ± 4.0	2768
	160	36.8 ± 3.0	2742
	240	36.7 ± 7.0	2496

Bone marrow donors were 6-week-old (C3H x AKR)F1 mice. Recipient C3H mice (8-week-old) were lethally irradiated with a dose of 8.65 Gy of gamma rays. Donor mice were pretreated with dexamethasone i.p. Controls were injected with saline i.p.


A
□ N B M
× D 80 BM
△ D 160 BM
donor cells / CFU-S
10^6
10^5
10^4
10^3
10^2
10
15
20
time after transplantation (day)
B
□ N B M
× D 80 BM
△ D 160 BM
doubling time of donor thymocyte subpopulation (hour)
40
30
20
0
50
100
150
200
dexamethasone (mg / kg^{-1})

Quantification of prothymocytes has been based on analyses of parallel growth curves of donor thymocytes after transplantation of different doses of progenitor cells into lethally irradiated recipients (Boersma et al., 1981; Boersma, 1982). After transplantation of bone marrow of (C3H x AKR)F1 mice pretreated with different doses of dexamethasone, the rate of development of donor derived thymocytes is changed in a dose dependent way. We expressed the growth of donor derived thymocytes per equivalent of bone marrow cells containing one CFU-S. After treatment of donor mice with a dexamethasone dose of 80 $mg.kg^{-1}$, the rate of development of donor derived thymocytes in lethally irradiated recipients was decreased. This suggests an enrichment of prothymocytes over CFU-S. After treatment of donor mice with a high dose of dexamethasone (160 $mg.kg^{-1}$), the rate of development of donor derived thymocytes returned to normal values but simultaneous a delay was observed after transplantation of bone marrow from untreated donors (Fig. 1A). In Fig. 1B, the relationship between the doses of dexamethasone and the change in doubling time of the donor derived thymocyte subpopulation is shown. The doubling time of the donor thymocytes increased with the dose to up to 80 $mg.kg^{-1}$. At higher doses, a decrease to normal values was observed. It might be concluded that prothymocytes which are even resistant to high doses of dexamethasone ($\geqslant$ 80 $mg.kg^{-1}$) represent about 50% of the number of prothymocytes present in normal bone marrow. Other prothymocytes are relatively dexamethasone sensitive.

After dexamethasone treatment ($\geqslant$ 32 $mg.kg^{-1}$), the number of donor derived thymocytes at the end of the regeneration period ($\sim$ day 20 after transplantation) is dependent of the dose. This indicates that calculations based on the relatively high number of donor derived thymocytes earlier during the regeneration period (day 12 to 14) may lead to an overestimation of the number of prothymocytes when extrapolation of the growth curve to the day of transplantation is used as a measure for the proportion of functional T cell progenitors in the bone marrow. In intact dexame-

Figure 1: A. Thymocyte regeneration after transplantation of bone marrow from normal (NBM) and from dexamethasone pretreated donors (80 $mg.kg^{-1}$: D80BM; 160 $mg.kg^{-1}$: D160BM).
Log-linear regression analysis showed that the slopes of the regeneration curves after transplantation of NBM and D80BM are different ($p = 0.044$). Regeneration curves after transplantation of NBM and of D160BM are parallel only (equal slope $p = 0.669$; different offsets). The delay indicates a decrease in prothymocytes of $\sim$ 50%.
B. Dexamethasone dependent changes in the doubling time of donor derived thymocytes after transplantation of bone marrow from dexamethasone pretreated donors (3 experiments).

thasone treated mice (60 $mg.kg^{-1}$), a biphasic regeneration of thymus cells consisting of a rapid phase from day 0 to day 12 and a slow phase thereafter has been described (Boersma et al., 1979). Therefore, definite conclusions with respect to the number of prothymocytes present in bone marrow of dexamethasone treated donors require analyses of thymocyte growth kinetics during the first ten days after transplantation.

The growth curve for donor derived thymocytes consists, in fact, of the combined growth curves for simultaneously developing "low" and "high" Thy-1^+ donor derived cells (Boersma et al., 1982). Therefore, the suggestion of whether the differences in the overall growth rate observed after transplantation of bone marrow from dexamethasone pretreated donors might reflect an enrichment of progenitor cells for the relatively slowly growing "low" Thy-1^+ cells which are enriched in "cortisone resistant" thymocytes was considered. However, preliminary results of the determination of the proportions of "low" and "high" Thy-1^+ cells in the donor-type thymocyte subpopulation gave no evidence in support of this.

Our results might be explained as follows. After treatment of donor mice with dexamethasone, the bone marrow becomes enriched in prothymocytes as a response to the demand created by the depletion of the thymus cell population. These prothymocytes, however, might have a reduced proliferative capacity due to the effect of dexamethasone. Also for cortisone resistant thymocytes, an inhibition of cell proliferation early after treatment has been described (Duval et al., 1977). From our studies, it has become apparent that dexamethasone pretreatment of donor mice leads only to a virtual enrichment of functional prothymocytes. For real estimates of prothymocytes, it is necessary to analyse the growth of donor derived thymocytes during the whole thymus regeneration period (up to day 20). Donor thymocyte determinations at one time point only might lead to erroneous interpretation of results (Basch and Kadish, 1977).

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A RAPID ENZYME BLOTTING METHOD FOR THE DETECTION OF NOVEL THYMIDINE KINASE ISOZYMES AFTER GENE TRANSFER

K.J. van den Berg and R.H. Merema-Bosma

The gene for thymidine kinase (TK) from Herpes simplex virus (HSV) is widely used in gene transfer experiments because it can convey resistance against several inhibitors of <u>de novo</u> DNA synthesis upon various TK^- cells. This allows positive selection of cells in which the transferred TK gene is efficiently expressed. Furthermore, when the TK gene is used in conjunction with other genes in co-transfer experiments, selection for TK positive cells often yields cells into which the nonselectable gene has also been transferred.

It has been suggested that mouse haemopoietic cells as well as tissue culture cells having an endogenous TK (TK^+ phenotype) can tolerate much higher levels of methotrexate after transfection with HSV-TK-DNA (TK^{++} phenotype) (Mercola et al., 1980). Such a system offers attractive possibilities for introducing foreign genes into normal mouse haemopoietic cells using the TK^{++} phenotype for selection of positive cells.

It is not clear, however, whether the resulting TK^{++} phenotype originates from the expression of the novel TK gene or from duplication of endogenous cellular genes. Multiplication of endogenous genes for dihydrofolate reductase, for instance, has been implicated in the resistance of cells against methotrexate.

In order to detect in an early phase the expression of the novel TK gene against a background of endogenous TK activity, we developed a rapid enzyme blotting method on the basis of Southern's procedure for DNA (Southern, 1975). In this procedure molecules are separated according to molecular weight by electrophoresis in agarose gel. The molecules are then transferred from the gel to filter paper by capillary blotting after which the identity of the molecules on the filter paper is established by binding to a radioactive probe and autoradiography.

In principle, such a procedure should also work for proteins and enzymes. Especially for TK is the use of ion-exchange filter paper advantageous, since at a relatively high pH the enzyme with its negative overall charge is tightly bound to the positively charged paper. In addition, when the paper containing bound TK is incubated with substrate (thymidine, iododeoxycytidine, etc.), the phosphorylated product is also firmly bound to the paper and thus can be used to ascertain the position of the bound enzymes. The assumption is that TK is still enzymatically active when bound to ion-exchange paper. This was tested first in experiments such as shown in Fig. 1. A small volume of enzyme extract was placed on ion-exchange paper in the form of a narrow band, incubated with radioactive labelled substrate, washed and autoradiographed. The figure clearly shows that the enzyme reaction is specific (lane A) and sensitive (lane B). Incubation with a tenfold higher label concentration (lane C) shortens the autoradiographic exposure time by a factor of ten but does not increase the overall sensitivity of the method.

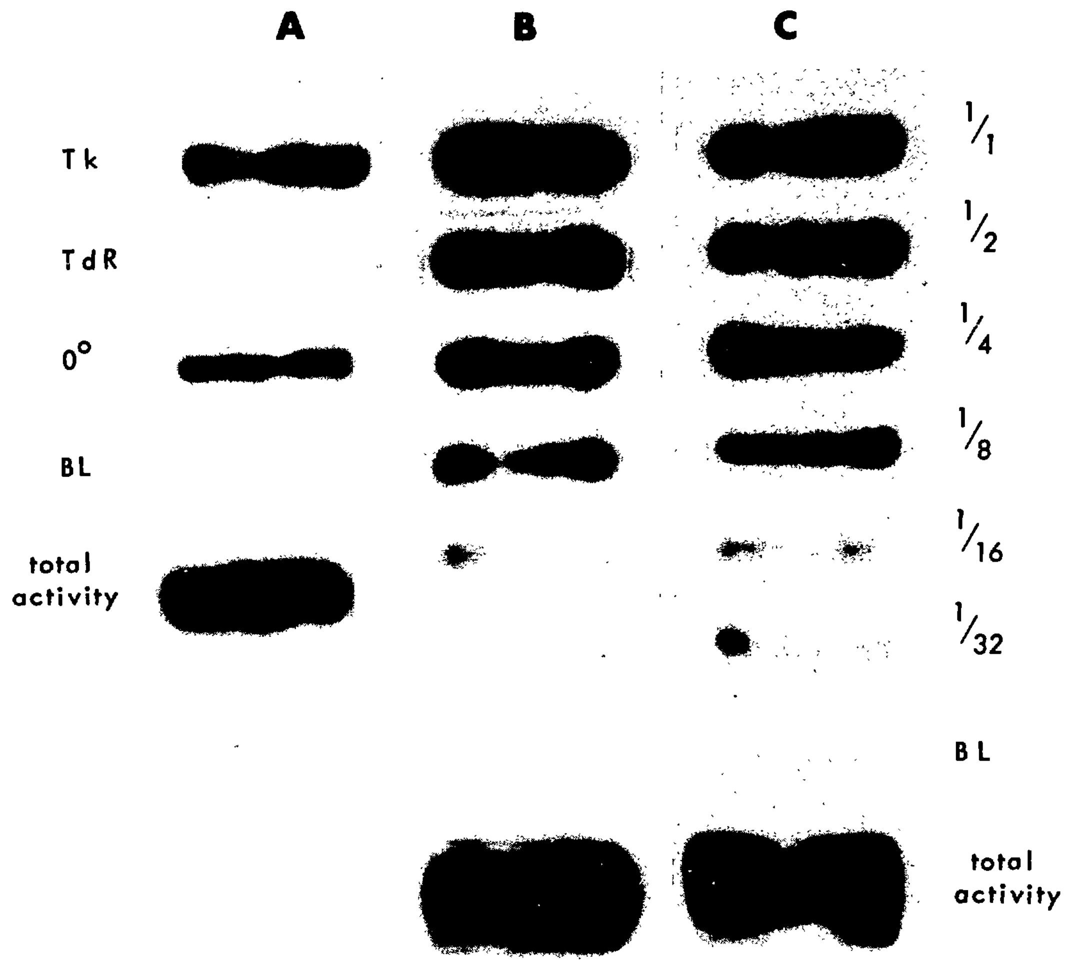

Figure 1: Enzymatic activity of TK bound to ion-exchange paper. Five microliters of enzyme extract (1.6 µg of protein) were placed as a narrow band on squares of DE81 paper. The squares were incubated with ^{125}Iododeoxyuridine (IdU; 1 µCi.ml^{-1}) for 1 h at 37°C, washed and autoradiographed. A. Specificity of the reaction; incubation under various conditions. B. Sensitivity of the reaction with serial dilutions of enzyme extract using 1 µCi.ml^{-1} IdU. C. Same as B with 10 µCi.ml^{-1} IdU.

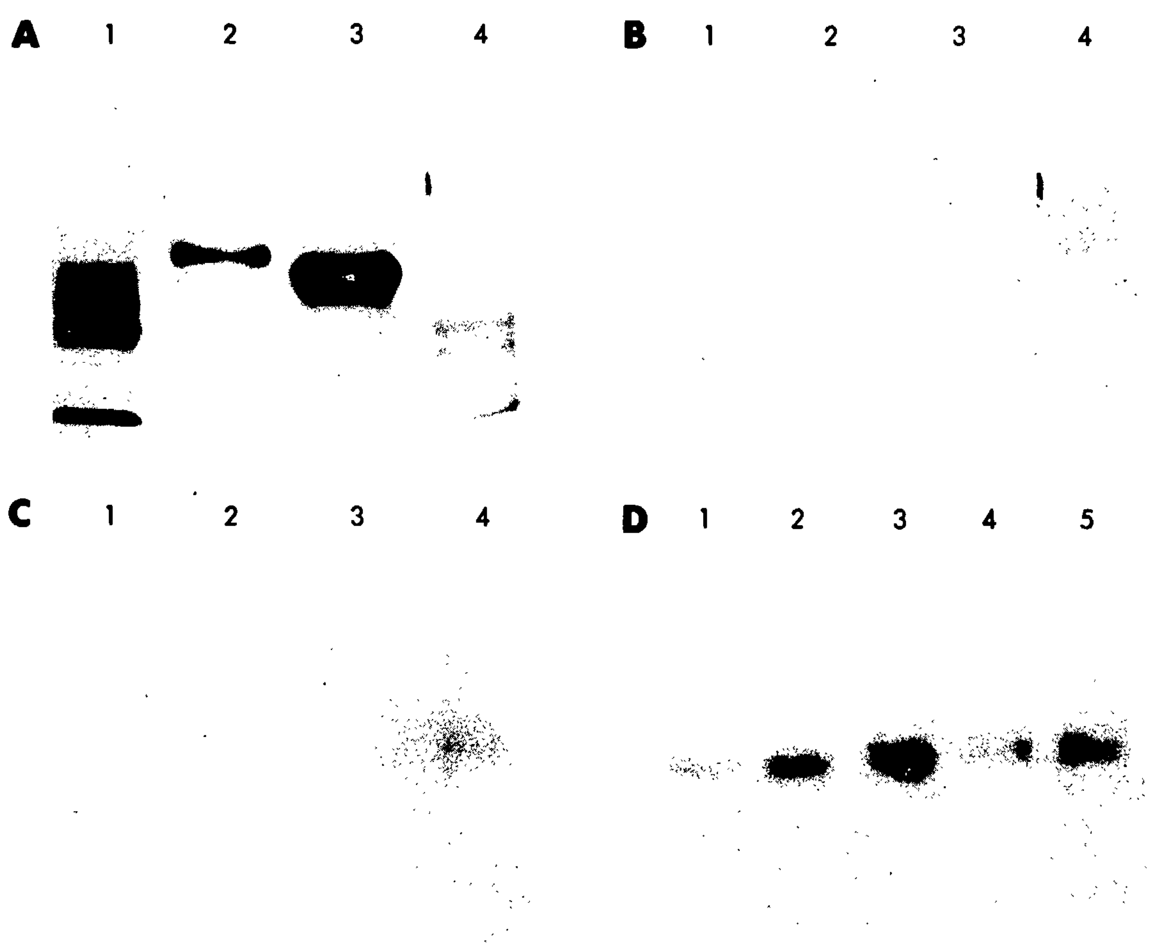

Figure 2 Electrophoresis and blotting of TK. Protein standards (lanes 1, 2 3) and extracts containing TK (lane 4) were subjected to discontinuous agarose electrophoresis Replicate gels were directly stained with Coomassie Blue (A) or first blotted and then stained (B). After blotting, the ion-exchange paper was incubated with radiolabelled substrate washed and autoradiographed (C). In (D) different amounts of TK extract were electrophoresed Slots 1 through 5 contain 3, 6 10. 13 and 16 µg of total protein, respectively.

Thymidine kinase isozymes can be separated by electrophoresis on the basis of known differences in molecular weight and iso-electric point (hence, charge). Although polyacrylamide gels give superior resolution of proteins, the medium of choice for blotting purposes is agarose. Because of the relatively high rate of diffusion in agarose gels, capillary blotting can be completed in a relatively short period of time (usually overnight). Unfortunately, the same property also affects electrophoretic resolution. This effect is much more pronounced for proteins than for DNA. Using a discontinuous electrophoresis system (Ornstein, 1964) considerably improved the resolution of proteins in agarose gels. Fig. 2A shows reasonably sharp bands of protein standards (lanes 1, 2 and 3) in a stained agarose gel before blotting. Lane 4 shows the protein pattern of an extract containing TK activity. Fig. 2B shows a replicate gel stained after blotting. Except for proteins of very high molecular weight at the top of the gel the lower molecular weight proteins have been well transferred to the paper. The phenomenon of poor transfer of high molecular weight molecules is also observed with DNA. An autoradiograph of the ion exchange paper after incubation with labelled substrate is shown in Fig. 2C. A band representing thymidine kinase activity is visible in lane 4. The sensitivity of this method is considerable, as TK activity can be spotted in a mixture of only a few microgram of total protein (Fig. 2D). This blotting method for TK will be of great help in detecting early expression of transferred TK genes.

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REDUCTION OF THE SEPARATION RESPONSE IN YOUNG RHESUS MONKEYS

H.Dienske and S.J. Suomi*

When a young monkey is temporarily isolated from its familiar social environment, it responds with a number of behavioural and physiological changes. The main behaviour responses are increased contact soliciting vocalizations and becoming either hyperactive or underactive. A physiological response is an increase in plasma cortisol levels in the course of a few hours (Suomi, 1982). These responses often are regarded as being a model for stress in humans.

It was investigated whether the degree of these stressful responses to separation could be reduced. Two 1 or 2-year-old rhesus monkeys were taken from a group of 10 infants and 3 adults and placed in two adjacent cages for 2 hours. The separation reaction was experimentally manipulated by permitting the infants to have contact through a wide grid. Temporary prevention of contact was achieved by closing an opaque sliding door. Contact and separation during experiments were continuously alternated with mean durations of 30 seconds and two minutes, respectively.

Behaviour was observed in the home cage (H), during a baseline separation (B, no contact provided) and three experiments (E1, E2, E3: alternating of contact and door closed) per pair. Cortisol levels were determined before and after baseline separation and after experiment 3 (E3) at 13.00 and 15.00 h. The 10 infants were matched pair-wise according to their separation behaviour during continuous separation. This led to one hyperactive, one underactive and three mainly vocalizing pairs (see Table I, pairs 1, 2 and 3 to 5, respectively; compare B with H).

Two questions can be raised: (a) Does access to a peer reduce the separation response? (b) Is there any learning effect, i.e., does a possible reduction also extend to the times during which the door is closed.

The results can be summarized as follows.

1) Although the full arm length could pass the grid and the infants spent much time in the open door, body contact was highly

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*Primate Laboratory, University of Wisconsin, Madison, Wisc., U.S.A. The experiments were carried out in this laboratory with the generous assistance of many persons.

TABLE I

SEPARATION RESPONSES OF 1 TO 2-YEAR-OLD RHESUS MONKEYS PLACED IN A STRANGE ROOM FOR 2 HOURS

designation of infant	vocalizations (per min)				locomotion (% of time)				cortisol (ug/100 ml)	
	H	B	E (open)	E (closed)	H	B	E (open)	E (closed)	B	E3
T1	0	0.5	1.5	4.7	14	44	7	41	68	82
C1	0.3	0.2	0.2	0.6	7	57	17	50	64	74
T2	0	0.6	0.2	0.6	14	5	6	8	38	64
C2	0	3.9	0.3	0.7	11	6	12	10	86	64
T3	0.1	2.8	0.6	3.2	13	29	8	9	62	66
C3	0	1.7	0.7	2.1	17	20	13	31	46	70
T4	0	12.8	8.6	10.8	5	26	5	14	50	45
C4	0	8.1	6.3	10.1	12	15	5	35	45	54
T5	0	7.6	6.2	14.6	16	4	9	4	56	64
C5	0	7.1	4.6	9.9	18	22	14	28	66	49
mean	0.0	4.5	2.9	5.7	13	23	10	23	58	63

H : Home cage.

B : Baseline (continuous) separation.

E : Experiments: the infants of one pair (1 to 5) could have contact with each other through a grid for about 30 seconds of every 2 min (averages) by opening and closing a door. Data are means of three experiments.

T : The door was opened when one infant (T) showed no or a mild separation response and closed again if the response was strong. The pair member (C) was the yoked control.

cortisol: plasma levels after 2 hours of separation at 15.00 h, during continuous separation (B) and the last experiment (E3).

exceptional during the experiments. This was unexpected, as it is known that unrestricted infant pairs often cling to each other. The contact apparently was only visual.

2) As long as the infants had access through the open door, contact soliciting vocalizations and locomotion time were more similar to the home cage values than to the baseline values (see Table I; Wilcoxon Matched Pairs Test applied to differences from H, $p < 0.05$ for both vocalize and locomotion). So, the visual contact with a peer during separation from the familiar environment indeed has a soothing effect.

3) Reduction of the separation response of one of the two infants by means of reinforcement was attempted. When the separation behaviour had been relatively mild, the door was opened as a reward. The door was closed if it acted in an undesired way. The other infant, which had the same regimen of door movements but not in connection with its behaviour, served as the yoked control. Reinforcement attempts were not successful; in only 7 of the 15 training sessions (i.e., experiments) did the trained animals show a milder separation response than their yoked controls as long as the door was closed.

4) No learning effect was achieved, as no decrease in the separation response was found in three subsequent experiments (Kruskal Wallis one-way analysis of variance).

5) Learning was also absent in the course of the experimental sessions. Neither vocalizations nor locomotion were correlated with the sequence number of a door movement (Kendall rank correlation, $p > 0.05$ in most cases).

6) The cortisol levels after the last experiments were not different from those after baseline separation (Table I).

It can be concluded that, during the experiments, the soothing effect of the sight of a peer did not extend to times when the partner was not visible. Their reactions to opening and closing the door were surprisingly immediate and insensitive to the repetitions provided. The effects on vocalizations and locomotion were different. During access, vocalization rate was reduced to about 50% and locomotion became normal.

In spite of these frequent and consistent reductions in the separation response, the infants did not generalize this mitigation to periods of time when the peer could not be seen. This is a quite remarkable result for a species with exquisite learning capacities. Apparently, separation stress cannot be easily suppressed.

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THE RELATIONSHIP BETWEEN ATTENTION AND ACTIVITY AND ITS DISORDERS IN CHILDREN

G. de Jonge, H. Dienske and J.A.R. Sanders-Woudstra

A substantial proportion of children has difficulties at school and home related to poor concentration and overactivity. The third "Diagnostic and Statistical Manual of Mental Disorders" (DSM-III, 1980) designated this as "Attention Deficit Disorder" and provided criteria for inattention, impulsivity and hyperactivity. Two subtypes are given: Attention Deficit Disorder with and without Hyperactivity. The DSM-III states, however, that "it is not known whether the subtypes are two forms of a single disorder or represent two distinct disorders." We studied this possible relationship between attention and activity by using behaviour measurements instead of the usual subjective ratings.

Data were collected from 101 children with various kinds of problem behaviour and 32 controls of 3 to 12 years of age. Their behaviour was measured during psychiatric investigation. Four attention parameters were determined; two describe attention to the psychiatrist and the other two concern tasks and toys.

1. Answered questions: appropriate response to the psychiatrist's questions (% of questions).
2. Complied with requests: performed what the psychiatrist requests (% of requests).
3. Completed tasks: number of tasks which were completed (% of all tasks with a clear endpoint, such as puzzles and drawings)
4. Functional play: playing with a toy or performing a task in a manner that is at least as good as the function for which it is designed (% of all time spent in handling of toys).

The degree of activity was derived from three measurements:

5. Walking (% of time).
6. Standing up, i.e., end of sitting (number per hour).
7. Toy changes: the child starts to play with a different type of toy (number per hour).

The relationship between attention and activity has two facets. One is whether it can be justified that the seven parameters belong to the two categories Attention and Activity as given above. The other, dependent of this justification, is whether children with a poor attention tend to be hyperactive as well.

We have four arguments in favour of our subdivision.

1. The common sense interpretation of our parameters, as follows from the given descriptions.

2. In problem children, values of the attention parameters are on the average lower than in control children, whereas the re-

verse is true for the activity parameters.

3. The values for the four attention parameters all sharply increase with the child's age, whereas the activity parameters considerably decrease.

4. A principal component analysis showed that the four attention parameters are distinct from the three activity ones. For, the cross correlations of the former are rather high and positive; those of the latter are also considerable but have the opposite sign (see Table I; correlations with the same sign demonstrate positive correlations; opposite signs, however, do not indicate negative correlations in this analysis). The results are quite similar for controls and problem children.

The coherence described would also result from the opposite changes in the attention and activity parameters with age. Therefore, we first made the parameters independent of age by transfor-

TABLE I

PRINCIPAL COMPONENT ANALYSIS OF SEVEN PARAMETERS APPLIED TO 101 CHILDREN WITH VARIOUS KINDS OF PROBLEM BEHAVIOUR AND 32 CONTROLS OF 3 TO 12 YEARS

parameters	cross correlations controls	cross correlations problem children
ATTENTION		
1. Answered questions	-.6	-.8
2. Complied requests	-.8	-.7
3. Completed tasks	-.3	-.6
4. Functional play	-.4	- 6
ACTIVITY		
5. Walking	+.6	+.7
6. Standing up	+.5	+.8
7. Toy changes	+.9	+.7
Total Explained Variance (%)*	38	48

* The chance level is 100/7 = 14%.

mation to the number of standard deviations a value differed from controls of 1 1/2 years older and younger.

The next question is whether children with a poor attention also were very active. The degrees of attention and activity per child are expressed as mean age-corrected values for parameters 1 to 4 and 5 to 7, respectively. Inattentive and hyperactive are defined as mean values beyond the 5% confidence interval. Of the problem children, 64% was inattentive and 48% hyperactive. The association between these two disorders (28% of the children) did not exceed chance level (viz., 31%). A disproportionate percentage of the children with both disorders had a low IQ (46% under IQ = 80; where this was 4 for the other problem children). So, inattention and high activity are possibly connected with developmental retardation in other respects, as children are less attentive and more active the younger they are. To check this, the parameter values were adjusted to mental instead of chronological age in the 82 children with a known IQ. This considerably reduced hyperactivity (from 48 to 25%, Chi-square test, p = 0.01) but the influence on inattention was not significant (decrease from 64 to 57%; p = 0.52). The association between the two disorders decreased accordingly from 34 to 19% and remained at chance level (chance levels 31 and 14%).

It is concluded that about half (44%) of the association can be ascribed to the developmental retardation that follows from the IQ and that inattention and hyperactivity are independent disorders that coincide at chance levels.

The severity of these disorders can be quantified by using a small number of simple, directly observable behaviour patterns.

Diagnostic and Statistical Manual of Mental Disorders, American Psychiatric Association, Washington D.C. (1980).

TREATMENT WITH SEROTONERGIC ANTIDEPRESSANT REDUCES STEREOTYPED LOCOMOTION IN RHESUS MONKEYS

C. Goosen, L.G. Ribbens, H.G.M. Westenberg* and K.J. v.d. Berg

Rhesus monkeys which have been housed separately for prolonged periods of time are likely to show idiosyncratic abnormal activities. These include locomotion stereotypies and self-directed, normally social behaviour. The diversity in the activities as well as the apparent distorted self-directed communication suggest an analogy with schizoaffective disorders in humans (Goosen, 1981). It is quite possible, therefore, that laboratory-reared rhesus monkeys can be used for preclinical assessment of therapeutic effects of antidepressive or neuroleptic drugs.

As a start to testing this hypothesis, we determined behavioural effects of clomipramine, a serotonin reuptake inhibitor which exerts antidepressant effects in about 70 per cent of human patients.

The subjects were 12 adult female rhesus monkeys which showed various abnormal activities. The experiment included five stages: (b1) baseline before treatment; (c) clomipramine treatment; (p) placebo treatment; (pp) placebo treatment while the other roommates also received placebo; (b2) baseline after treatment.
The experimental design was randomized as a double blind crossover, i.e., each individual was its own control and the observer did not know when a subject received drug or placebo. The drug (75 mg) or placebo was administered orally in capsules under manual restraint at 8.30 a.m. and 4.30 p.m.

On eight consecutive days starting at 10 to 12 days after the beginning of the different experimental periods, each subject was observed for 50 minutes. The subject's behaviour was recorded by means of an observer operated event recorder. The observer was concealed behind a one-way vision screen. Prior to the observations, the subjects were moved at 8.30 a.m. from their home cages to observation cages (dimensions, 160 x 80 x 80 cm) in a different room. Observations were made at between 9.30 and 12.00 a.m.; each subject was always observed at the same time of day. The amounts of time per thousand seconds observation that the subjects spent in various activities during the pp period are given in Table I. Arrows indicate increases and decreases attributed to clomipramine treatment and which were statistically significant (Mann-Whitney U test; $p < .05$). There were considerable differences in the amounts of time spent in the various activities. Ten of twelve subjects responded to clomipramine but the type of activity affected

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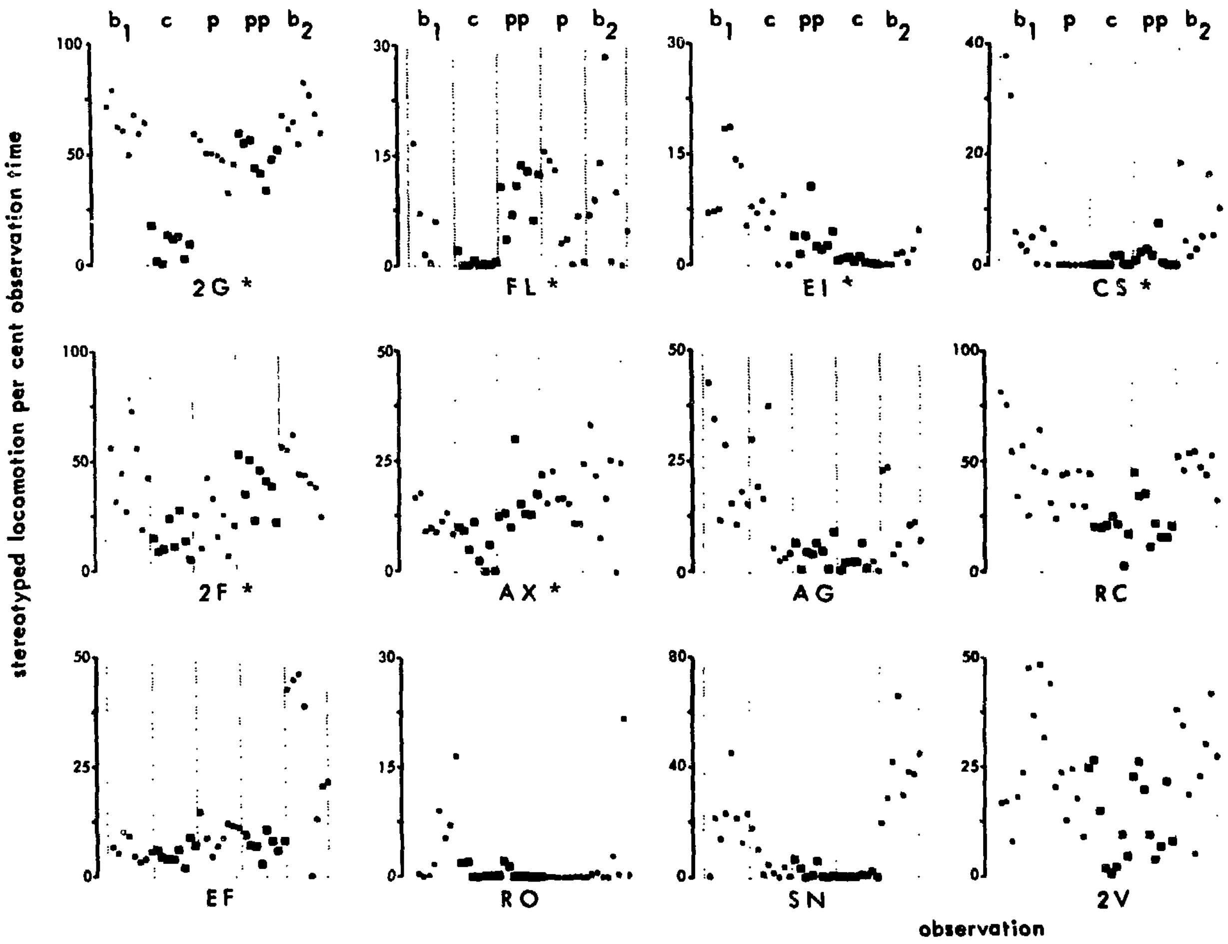

Figure 1: Amounts of stereotyped locomotion during the different experimental periods for the different subjects. * significant differences between c and pp, Mann-Whitney U-test $p < 0.05$.

Table I

AMOUNTS OF TIME PER THOUSAND SECONDS OF OBSERVATION SPENT IN THE DIFFERENT ACTIVITIES BY THE DIFFERENT SUBJECTS DURING THE PLACEBO (pp) PERIOD

subjects	2G	2F	RC	AX	2V	FL	EI	EF	AG	CS	SN	RO
stereotyped locomotion	488 ⇓	382 ⇓	256	155 ⇓	149	94 ⇓	79 ⇓	73	44	21 ⇓	19	4
self-aggression	0	4	2	5	0	11 ⇓	0	0	0	1 ⇓	5	21 ⇓
self-sex	1	0	·	0	0	0	0	0	0	10	1	0
self-holding	0	18	0	145 ⇓	0	11	89	127	0	·	37	172 ⇓
self-grooming	47	99	134 ⇓	81 ⇑	0	114	199	247	93	257	290	302
rub hand	2 ⇓	·	·	·	·	·	·	138 ⇓	·	·	·	·
caress	78	45	56	8 ⇑	0	44 ⇓	182	142	241 ⇓	17	3	51
infantile	24 ⇓	11	0	41 ⇓	0	460	0	1	2 ⇓	0	0	10
scratch	7	4	111	2	1	5	6	5	10 ⇓	8	6	4
manipulate	50 ⇓	157	47	37 ⇑	761	25 ⇑	38 ⇑	4	198	37	23	79
bounce	0	9	0 ⇓	5 ⇓	0	4 ⇓	0	0	1	·	0	1
walk	11 ⇓	9	16 ⇓	82 ⇓	2	17 ⇓	16	9 ⇑	15	7	6	14
stand	92 ⇓	35	126 ⇓	77	17	35 ⇓	74 ⇓	79 ⇑	70 ⇓	17	49	34
sit	149 ⇑	187 ⇑	240 ⇑	325	32	144	219	185 ⇑	277 ⇑	577	294	214 ⇑
sit crouched	0 ⇑	8	10	8	0	1	172	9	17	6	257	121

Arrows indicate the direction of significant changes (Mann-Whitney U test, $p < .05$) due to clomipramine.
⇑ increase, ⇓ decrease.
· means: pattern not shown by the subject.

differed among subjects. Two types of responses were seen:

a) decreases in self-directed activities and in stereotyped locomotion, which suggests a potentially therapeutic effect;

b) increases in sitting and or decreases in standing or walking without any other specific activity; this seems comparable to the mild sedation effect seen in humans.

The two types of effects are not fully correlated with each other in their distribution over the different subjects.

The abnormal activity most commonly affected by clomipramine among all subjects was stereotyped locomotion. In order to illustrate the variation within individuals as well as between individuals, Fig. 1 gives the daily amounts of stereotyped locomotion by each subject. The subjects are arranged according to the sequence of the experimental periods; asterixes indicate subjects responding to clomipramine. Decreased stereotyped locomotion due to clomipramine treatment tends to persist after termination of treatment, e.g., 2F and AG. Some subjects responded to drug as well as placebo treatment (SN, RC, 2V). Differences between placebo periods p and pp in animal EI also indicate the occurrence of a communication effect, i.e., the subject responded to effects of the drug in a roommate. These results indicate that clomipramine exerts behavioural effects in rhesus monkeys which are quite comparable with those in man. Further research under social conditions must show whether the behavioral effects are accompanied by increased social activity. If this holds true, the rhesus monkey can play a valuable role in the preclinical evaluation of newly developed antidepressant drugs.

Goosen, C., Biol. Psychiat. 16 (1981) 697.

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